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NAME OF WORK: PROVN OF RCC SHELTER ALONGWITH ASSOCIATED WORK SERVICES AT AF STN BAREILLY

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Drawing : 116 Sheets

(Signature of Contractor)

AAD (Contracts)
For Accepting Officer

*These documents are not attached with the tender and can be seen in the office of **CE (AF) Prayagraj / CWE (AF) Bareilly / GE (AF) Bareilly** during working hours.

Civ-Tele: 0532-2975213

Headquarters
Chief Engineer (AF)
Military Engineer Services
Prayagraj-12

958130/ 10 /E8

30 Dec 2024

M/S_____
_____
_____

TENDER FOR CANO: CE(AF)/ALD/BLY/ 28 OF 2024-25

NAME OF WORK: PROVN OF RCC SHELTER ALONGWITH ASSOCIATED WORK SERVICES AT AF STN BAREILLY

Dear Sir,

1. Tender documents in respect of above work are uploaded on the site www.defproc.gov.in. The tender is based on single stage two cover e-tendering system. The contents of **Cover I & Cover II** are specified in NOTICE OF TENDER.

2. Bids will be received online by ACCEPTING OFFICER upto the date and time mentioned in the NOTICE INVITING TENDER (NIT). No tender/bid will be received in physical form and any tender/bid received in such manner will be treated as non bona fide.

3 Bid will be opened on line on a due date fixed for opening as per critical dates given in the portal. Cover 1 will be opened first. Tenderers may see the result of the opening of Cover 1 on the tender portal. Date of opening of Cover 2 shall be decided after technical evaluation of Cover 1 keeping in view the various eligibility criteria given in the NIT and the same will be intimated to the tenderers while uploading the technical evaluation on the tender portal.

4. Your attention is also drawn to instruction on filling and submission of tender attached herewith. You may forward your points on tender documents through e-mail and/or depute your technical representative for discussion on tender/drawings and to clarify doubts, if any, prior one week of bid submission starts date. You are requested not to write piece meal points and forward your points duly consolidated in one go. You may also attend the pre bid meeting on the date given in the tender.

5. Un-enlisted contractors are required to submit the scanned copies (in pdf file) of documents required as per eligibility criteria mentioned in instructions for filling the tender documents and Appendix ‘A’ to NIT alongwith EARNEST MONEY DEPOSIT (EMD) and tender fee on procurement portal mentioned above and submit the physical documents in the office of Chief Engineer (AF) Prayagraj Zone within time limit specified in NIT. Inadequacy /deficiency of documents shall make the bid liable for rejection resulting in disqualification for opening of finance bid.

6. Contractor having not executed standing security bond and standing security deposit in any MES formation shall upload scanned copy of EARNEST MONEY DEPOSIT(EMD) mentioned in Notice of Tender and shall ensure receipt of hard copy of EMD in the office of tender issuing authority before date & time fixed for this purpose. In case of failure to abide by any of those two requirements, the finance bid will not be opened.

7. Enlisted contractors of MES shall submit the scanned copies (pdf file) of enlistment letter, tender fee and such other documents as mentioned in Appx ‘A’ to NIT on Defence e-Procurement Portal and submit physical documents in the office of Chief Engineer (AF) Prayagraj Zone before date & time fixed for this purpose.

8. The contractor must ensure that the tender/bid is uploaded on the tender portal using the authorized DSC in time as the Accepting Officer will take no cognizance of any quotations/offer received in any other electronic or physical form like email/fax/by hand/through post from tenderer/bidder.

9. Keeping in view delays due to system failure or other communication related failures, it is suggested that tender/bid may be uploaded sufficiently in advance of the last due date and time fixed on the portal.

FORWARDING LETTER (Contd…/-)

10. General Conditions of Contracts (IAFW-2249) (1989 Print) and errata and amendments thereto, Schedule of Minimum Fair Wages and MES SSR (Part-I and Part-II) are not enclosed with these documents. These are available for perusal in the Office of GE/AGE concerned and this office. Tenderer is also requested to keep the copy of these documents with him as same are available in the market.

11. **PERFORMANCE SECURITY DEPOSIT**

11.1 After acceptance of the Tender, the contractor will be required to lodge with the Accepting Officer **PERFORMANCE SECURITY DEPOSIT @ 5%** of contract amount. The amount is required to be lodged within 28(Twenty-Eight) days of the receipt by the contractor of notification of acceptance of tender/bid, failing which action as stipulated in Condition of GCC shall be taken.

12. **ANY TENDERER, WHICH PROPOSES ALTERATIONS TO ANY OF THE CONDITIONS, SPECIFICATIONS LAID DOWN IN THE TENDER DOCUMENTS OR ANY NEW CONDITION, WHATSOEVER, IS LIABLE TO BE REJECTED.**

Yours faithfully

(Signature of Contractor)

AAD (Contracts)
For Accepting Officer

INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER

1. EARNEST MONEY DEPOSIT (EMD)

1.1 Contractor(s) who are not enlisted with MES/who are enlisted but have not executed the Standing Security Bond shall submit Earnest Money Deposit as detailed in Notice of Tender in one of the following forms, along with their tender/bid:

(a) Deposit at Call Receipt from a Scheduled Bank in favor of GE/AGE(I) concerned.

(b) Receipted Treasury Challan, the amount being credited to the Revenue Deposit of GE/AGE(I) concerned.

1.2 It is advisable that Earnest Money is deposited in the form of deposit call receipt from an approved Schedule Bank for easy refund. In case the tenderer/bidder wants to lodge 'Earnest Money Deposit' in any other form allowed by MES, a confirmation about its acceptability will be obtained from the Accepting Officer well in advance of the bid submission end date and time. Earnest Money Deposit shall be submitted in the name of concerned GE/AGE(I).

NOTES: Earnest Money Deposit (EMD) in the form of cheque/Bank Guarantee etc will not be accepted. **Non-Submission of Earnest Money Deposit (EMD) (scanned copy alongwith Technical Bid & hard copy before the date & time fixed for opening of BOQ) will render the bid disqualified for opening of Cover -II (finance bid).**

2. GENERAL INSTRUCTIONS FOR COMPLIANCE

2.1 The bids received only in the electronic form will be considered. All bids shall be submitted on defproc.gov.in portal. Documents should be scanned and forwarded in 'pdf' format and 'xls' form as indicated. Bids shall be uploaded on 'defproc.gov.in' portal on or before the bid closing date mentioned in the tender. No tender/bid in any other electronic or physical form like date email/fax/by hand/through post will be considered.

2.2 The bid shall be DIGITALLY signed using authorized DSC. All pages of tender documents, drawings, corrections/alterations shall be signed/initialed by the lowest bidder after acceptance of tender for making original and CTC of the contract.

2.3 In the technical bid, a scanned copy of power of Attorney in favour of the person uploading the bid using his/her DSC shall be uploaded. In case the digital signatory himself is the sole proprietor, scanned copy of an affidavit on stamp paper of appropriate value to this effect sating that he has authority to bind the contractor (through partnership deed, general power of attorney or Memorandum and Articles of Association of the company) in all the matters pertaining to the contracts with Union of India including arbitration clause. A scanned copy of the documents confirming of such authority shall be attached with the tender/bid in "PDF" form, if not submitted earlier. The person uploading the bid on behalf of another partner(s) or on behalf of a firm or company using his DSC shall upload with the tender/bid a scanned copy (in "pdf" form) of /Power of Attorney duly executed in his favour by such other or all of the Partner(s) or in accordance with constitution of the company in case of company, stating that he has authority to bind such other person of the firm or the Company in case of company, stating that he has authority to bind such other person of the firm or the Company, as the case may be, in all matters pertaining to the contract including the Arbitration Clause.

2.4 Even in case of Firms or Companies which have already given Power of Attorney to an individual authorizing him to sign tender in pursuance of which bids are being uploaded by such person as a routine, fresh power of Attorney duly executed in his favour stating specifically that the said person has authority to bid such partners of the Firm, or the Company as the case may be, including the condition relating to Arbitration Clause, should be uploaded in "pdf" form with the tender/bid; unless such authority has already been given to him by the Firm or the Company. It shall be ensured that power of attorney shall be executed in accordance with the constitution of the company as laid down in its Memorandum & Article of Association.

2.5 Drawings, if issued in physical form, must be returned duly initialed by the tenderer/bidder in separate envelope indicating his name and address.

2.6 The tender shall be signed, dated and witnessed at all places provided for in the documents after acceptance. All corrections shall be initialed. The Contractor shall initial every page of tender and shall sign all drawings forming part of the tender.

INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER (Contd…/-)

2.6 Any tender/bid, which proposes alterations to any of the conditions whatsoever, is liable to be rejected.

3. **BOQ**

3.1 The tenderer shall quote his rates on the BOQ EXCEL FILE only as per guide line of e-procure web portal. No alteration to the format will be accepted and such bid will be disqualified.

3.2 In case any tenderer wishes to revise/modify the rates quoted in the BOQ file, he can do so only in the BOQ files before uploading the tender through https://defproc.gov.in site only before closing date & time.

3.3 After the uploading of tender, Department may upload the errata/ amendment through corrigendum. The tenderer/bidder should submit their offer considering the errata/amendment carried out through corrigendum issued from time to time.

3.4 While uploading the bid, the tenderers/bidders should specifically check whether any **revised BOQ** has been uploaded by department through corrigendum prior to **Bid submission start date**. Tenderers/bidders attention is specifically drawn to the fact that they should submit their offer on revised BOQ only. In case any tenderer/bidder submits offer on pre-revised BOQ in lieu of Revised BOQ**, it will be considered as a willful negligence by the tenderer/bidder and quotation shall be considered non-bonafide.**

4. Tenders/bidders who uploaded their priced tenders/bids and are desirous of being present at the time of opening of the tenders/bids, may do so at the appointed time.

5. Hard copies of all above documents shall be sent by the contractor to the Tender issuing authority well in advance to be received before the date & time fixed for the same.

6. The contractor shall employ Indian Nationals after verifying their antecedents and loyalty. Attention is also drawn to relevant provisions in Special Condition referred here-in-after and also Conditions 24 & 25 of IAFW-2249 (General Conditions of Contract).

7. **PERFORMANCE SECURITY DEPOSIT**

7.1 In case, the tender submitted by contractor is accepted, the contractor will be required to lodge **"Performance Security Deposit"** for an amount equivalent to **5% of the Contract Sum/Lump Sum** in favour of the Accepting Officer within 28 (Twenty-Eight) days of the receipt by the contractor of notification of acceptance of his tender/bid. The Performance Security shall be in the form of "Bank Guarantee", "FDR" or "Insurance Surety Bonds". Failure of the contractor to comply with the requirements of Performance Security shall constitute sufficient grounds for cancellation of the contract and award of work and forfeiture of the Earnest Money. In case of MES enlisted contractor who have not deposited the Earnest Money as per requirement, amount equal to the Earnest Money stipulated in the Notice Inviting Tender, shall be notified to the contractor for depositing the amount through MRO. Issue of tender to such contractor shall remain suspended till the aforesaid amount equal to the Earnest Money is deposited in Government Treasury {See Condition 19 of General Conditions of Contract (IAFW-2249) (as amended under amendment No. 47 to IAFW- 2249)}.

8. **REVOKATION/REVISION OF OFFER UPWARD/ OFFERING VOLUNTARY REDUCTION AFTER CLOSING OF BID SUBMISSION DATE & TIME**

8.1 In the event of tenderer/bidder revoking his offer or revising his rates upwards/offering voluntary reduction, after closing of bid submission date & time, his offer will be treated as revoked and the Earnest Money deposited by him shall be forfeited. In case of MES enlisted Contractor, the amount equal to the Earnest Money stipulated in the Notice of tender, shall be notified to the tenderer/bidder for depositing the amount through MRO. Bids of such Contractors/bidders shall not be opened till the aforesaid amount equal to the earnest money is deposited by him in Govt Treasury.

8.2 In addition, bids of such tenderer/bidder and his related firm shall not be opened in second call or subsequent calls. Reduction offered by the tenderer/ bidder on the freak high rates referred for review shall not be treated as voluntary reduction.

INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER (Contd…/-)

9. **CPM (Critical Path Method)**

9.1 The Project planning for work covered in the scope of tender is based on CPM.

9.2 The tenderer/bidder is expected to be fully conversant with the CPM technique and employ technical staff who can use the technique in sufficient details. Sufficient books and other literature on the subject are widely available in the market which the tenderer/ bidder may make use of.

9.3 The tenderer's/bidder's attention is drawn to special condition of the tender regarding preparation of the detailed network analysis and time schedule for the work and his liability for employing sufficient resources to adhere to this schedule. Any inability on the part of the tenderer/bidder in using the technique will be taken as his technical inefficiency and will affect his class of enlistment and future prospect/invitation to tenders for future works.

10. Department may issue amendments/errata in form of CORRIGENDUM to tender/ revised BOQ to the tender documents. The tenderer / bidder is requested to read the tender documents in conjunction with all the errata/amendments/corrigendum, if any, issued by the department.

11. These instructions shall form part of the contract documents.

AAD (Contracts)
For Accepting Officer

(Signature of Contractor)

IN LIEU OF IAFW-2162 (REVISED –1960) MILITARY ENGINEER SERVICES
(NOTICE INVITING TENDER)

1. A tender is invited for the work as mentioned in Appendix 'A' to this NOTICE INVITING TENDER (NIT).

2. The work is estimated to cost as indicated in aforesaid Appendix 'A'. This estimate, however, is not a guarantee and is merely given as a rough guide and if the work costs more or less, a tenderer/bidder will have no claim on that account.

3. The work is to be completed within the period as indicated in aforesaid Appendix 'A' in accordance with the phasing, if any, indicated in the tender from the date of handing over site, which will be on or about two weeks after acceptance of Performance Security by the Accepting Officer.

4. Contractors whose names are on the MES approved list and within whose financial category the estimated amount would fall and un-enlisted contractors may submit tender/bid subject to other criteria mentioned in Appendix A. However, in case of term contracts, enlisted contractors of Class SS to E may submit tender. Not more than one tender shall be submitted/uploaded by one contractor/ firm. Under no circumstances will a father and his son(s) or other close relations who have business dealing with one another be allowed to tender/bid for the same contract as separate competitors. Two firms shall be deemed to have business dealing if any of the partners/proprietor/director is common among both of them. A breach of this condition will render the tenders/bids of both the parties liable for rejection.

5. The Office of Chief Engineer (AF) Prayagraj Zone will be the Accepting Officer here-in-after referred to as such for the purpose of this contract.

6. Not more than one tender/bid shall be submitted/uploaded by one bidder firm. Under no circumstances will a father and his son(s) or other close relations who have business dealing with one another be allowed to tender/bid for the same tender as separate competitors. A breach of this condition will render the tenders/bids of both the parties liable for rejection.

7. The Technical Bid and Financial Bid (Cover-1 and Cover-2) shall be uploaded by the contractor on or before the date & time mentioned in NIT. A scanned copy of DD with enlistment details and other documents as specified in Appendix A shall be uploaded as Cover-1 (Technical bid) of the tender on e-tendering portal. DD is refundable in case the contractor is not considered eligible in technical evaluation of Cover 1 resulting in non-opening of Cover1. The applicant contractor shall bear the cost of bank charges for procuring and encashing the DD including revalidation of DDs and shall not have any claim from Government whatsoever on this account.

8. Tender form and conditions of contract and other necessary documents shall be available on website defproc.gov.in for download and shall form part of contract agreement in case the tender/bid is accepted.

9. In case of MES enlisted contractor who has not executed the Standing Security Bond and un enlisted contractor, the Cover-I shall be accompanied by Earnest Money for the amount mentioned in Appendix 'A' in the form of deposit at call receipt in favour of concerned CCE/GE/GE (I)/AGE (I) (see Appendix 'A') by a Scheduled Bank or in received treasury Challan the amount being credited to the revenue deposit of the concerned CCE/GE/GE(I)/AGE(I) (see Appendix 'A'). The CCE/GE/GE (I)/AGE (I) will return the Earnest Money, wherever applicable, to all unsuccessful tenderers/bidders by endorsing an authority on the deposit at call receipt for it's refund, on receipt of intimation from the Accepting Officer to do that.

10. In case of successful contractor i.e., the lowest contractor having submitted EMD, he shall have the option of converting the EMD instrument into part of the Performance Security to be deposited by him within 28 days from the receipt of intimation of acceptance of tender from Accepting Officer.

11. Sample of materials and stores to be supplied by the contractor will also be available for inspection by the bidder at the office of concerned GE/GE (I)/AGE (I)/Project Manager during working hours. The bidder is advised to visit the site of work by making prior appointment with GE/GE (I)/AGE (I)/CCE/Project Manager, who is the Executing Agency of the work (see Appendix 'A'). The bidder shall be deemed to have full knowledge of all relevant documents, samples, site etc whether he has inspected them or not.

NOTICE INVITING TENDER (NIT) (Contd…/-)

12. Any bid which proposes any alteration to any of the conditions laid down or proposes any other new condition whatsoever, is liable to be rejected.

13. The uploading of bid by a bidder implies that bidder has read this notice and the conditions of contract and has made himself aware of the scope and specification of work to be done and of the conditions and rates at which stores (as applicable) etc will be issued to him and local conditions and other factors having bearing on the execution of the work.

14. The bidder must be in possession of a copy of the MES Schedule (SSR) (Part-I & Part-II of latest edition) including amendments and errata thereto.

15. Accepting Officer does not bind himself to accept the lowest or any tender/bid or to give any reason for not doing so.

16. The Accepting Officer reserves the right to accept a tender submitted by a Public Undertaking/Small & Medium Enterprises (SMEs), giving a price preference/purchase preference over other tender(s)/bids which may be lower, as are admissible under the Government Policy. No claim for any compensation or otherwise shall be admissible for such tenderer/bidder whose tender/bid is rejected.

17. The Notice Inviting Tender (NIT) including Appendix 'A' and Annexures thereto, if any, shall form part of the contract agreement.

AAD (Contracts)
For Accepting Officer

(Signature of Contractor)

APPENDIX 'A' TO NOTICE INVITING TENDER (NIT)

1.	Name of work	PROVN OF RCC SHELTER ALONGWITH ASSOCIATED WORK SERVICES AT AF STN BAREILLY
2.	Estimated Cost	**Rs 4672.33 Lakhs (**At par Market)
3.	Period of completion	**30 Months**
4.	Cost of tender documents	Rs. **3000.00** in the form of DD / Bankers cheque from any Nationalized / scheduled Bank in favour of **GE (AF) Bareilly** and payable at **Bareiily**
5.	Website / portal address	www.defproc.gov.in
6.	Type of Contract	The tender shall be based on drawing and specifications (IAFW – 2159 and GCC (IAFW-2249) with Sch 'A'(list of items of work) to be priced by MES. The Contractor is required to quote the lump sum amounts for parts of schedule 'A' and quote rates against items of other parts of schedule 'A'/BOQ.
7.	Timeline Details	
	(a) Bid submission start date	**Refer critical dates on the website.**
	(b) Bid submission end date	
	(c) Date of Bid opening (Cover-I)	
8.	Eligibility Criteria	
	(A) For MES enlisted Contractors	Contractor shall satisfy the following: - (a) Enlistment in Class 'SS' (b) Category a (i) as per MES enlistment rule. (c) Contractor shall not carry adverse remarks in work load return or any similar report circulated by the competent Engineering authority. (d) They should submit requisite documents listed here- in-after. (e) Pre- qualification criteria as in Note No. 8(C) below. (f) Contractor will not be allowed to execute the work by subletting or through power of attorney holder on his behalf to a third party / another firm except sons / daughter /. proprietor / partner / director and firms own employees, director, project manager. This shall be subject to certain conditions which will be prescribed in NIT.
	(B) For other contractors	The firms not enlisted with MES shall satisfy the following: - (i) Contractor shall meet the enlistment criteria of 'SS' Class & a(i) category, contractor with regard to satisfactorily completion of requisite value works with Central/ State Government/ Central/ State PSUs/ AWHO/ AFNHB/ CGEWHO/DGMAP, annual turnover, bank solvency, working capital and other requirements given in para 1.4 & 1.5 of MES Manual of contracts 2020. (ii) Not carrying adverse remarks in work load report (WLR) or any other similar report circulated by any competent authority, if already working in MES. (iii) Not suspended/ debarred/ black listed (either permanently or temporarily) from participating in any bid or for business dealings by any central/ state Government Department or any ventral / State Government PSU or any autonomous body under Central/ State Government or any local body as on the bid submission end date. (iv) Details of work completed and under progress in MES be submitted in following format.

SN	CA No. & Name of Work	Value of CA	Date of Commencement	Date of Completion	Extended date of Completion

(v) Unenlisted Contractors who have secured two works in MES should get themselves registered in appropriate designated class with any registering authority, else the firm will not be eligible for participation in the tender unless until the firm is enlisted with the MES

	(C) For contractors MES Enlisted & Other Contractors	**Technical pre-qualification criteria (PQC): -** (a) Average annual financial turnover during the last three years, ending 31st March of the previous financial year, should be at least **30.00 %** of the estimated cost. (b) Firm should be well experienced having successfully completed similar works of Runway rigid pavement during last 7 years (last date of bid submission) and meeting one of the following criteria: - **(i)** Satisfactorily completed one Rigid Runway Pavement or national highway/Expressway with min six lane costing not less than Rs 8.00 Crore (At par with present cost) (c) **The similar work means -** Rigid Runway Pavement or national highway/Expressway with min six lane. (d) **For the purpose of Value of completed works -** The value of previously completed works will be enhanced by 7.50 % per year to bring them at par with present cost. (e) Bidder shall upload scanned copies & submit hard copies (duly signed) of following: - (i) The List of Completed Similar works of cost mentioned above in a self-explanatory tabular form giving details of Name of Accepting Officers their address/ Telephone/ Fax/ Email, date of acceptance of contract, date of commencement & original/ extended & actual completion date etc. (ii) Acceptance letter, work order, completion certificate & performance reports from concerned Dept/ Govt Agency (which will be considered only after due scrutiny & vetting). **(f) FOR CONTRACTORS OTHER THAN MES ENLISTED –** shall submit (i) Latest working Capital Certificate from the Nationalized /Scheduled Bank issued within last 6 months (Issued on or after 01 Jan 2024) for the amount not less than Rs. 75 Lakh in prescribed format. (ii) Latest Solvency Certificate from the Nationalized /Scheduled Bank issued within last 6 months (Issued on or after 01 Jan 2024) for the amount not less than Rs. 08 crore in prescribed format. (iii) Minimum reserve in the name of company (Total of Immovable/fixed & Movable assets) shall not be less than Rs. 75 Lakhs supported with Notarized affidavit & valuation reports (80 % of total asset shall be in the shape of Immovable asset). (iv) Unenlisted Bidder/contractor shall render Notarized affidavit affirming that they have not been banned by any Govt /Semi Govt / State Govt Dept/Agency /PSU & DSV for participating in tender. (v) Un-enlisted Bidder/ contractors must produce satisfactory performance report of qualifying Runway works at the time ofsubmission of bids. Certificate will be accepted only after duescrutiny and vetting. **(g)** Firm should be well experienced having successfully completed similar work of RE Wall of 50 m length and 10m high in last 7 years (last date of bid submission) with Central/ State Government/ Central/ State PSUs/ AWHO/ AFNHB/ CGEWHO/DGMAP or Firm should have MoU for execution of this particular work with a special service provider firm who has successfully completed similar work of RE Wall of 50m length and 10m high in last 7 years (last date of bid submission) with Central/ State Government/ Central/ State PSUs/ AWHO/ AFNHB/ CGEWHO/DGMAP **Note: -** (i) In a tender, the eligible firms can either bid as direct participant/ bidder or under MoU with MES enlisted /unenlisted contractors, but cannot bid simultaneously for the same tender as direct participant/ bidder as also under MoU with MES enlisted/ unenlisted contractors. (ii) MoU be permitted between one eligible firm and one MES enlisted / unenlisted contractor, bidding for the same tender (iii) In case any violation of condition at (i) and (ii) above is noticed, all such bids shall be treated as invalid.

APPENDIX 'A' TO NOTICE INVITING TENDER (NIT) (Contd…/-)

9.	Tender issuing and Accepting Officer	Chief Engineer (AF) Prayagraj Zone Bamrauli, Prayagraj (UP) – 211012 Civ-Tele: 0532-2975213
10.	Executing agency	GE (AF) Bareilly
11.	Earnest Money	₹ 15,00,000.00 in favour of GE (AF) Bareilly in the form of Deposit at call receipt, FDR not acceptable.

NOTES:

1. After opening of Cover 1, if the number of MES enlisted contractors of eligible class as well as un-enlisted contractors, if any, fulfilling the other eligibility criteria given in NIT are less than 7(Seven), applications in respect of contractors one class or two classes (in case of remote and difficult areas to be decided as per list circulated by CE command/ADG) below the eligible class shall also be considered subject to fulfilment of other eligibility criteria given in the NIT. Therefore, MES contractors one class below (two classes below in case of remote and difficult areas) may also bid for this tender. However, contractors of one/two classes below the eligible class shall not be considered in case their present residual work in hand is more than five times their present tendering limit. Such bidders shall upload in their Cover-1 bid details of works in hand showing names of work, names of Accepting Officers, Contract amounts, dates of commencement and completion (stipulated) and progress as on bid submission date. These details shall be verified by the Tender Issuing Authority from concerned formations in case bids of such contractors are considered for evaluation.

2. In case after opening of Cover 1, the number of MES enlisted contractors of eligible class as well as un-enlisted contractors, if any, fulfilling the other eligibility criteria given in NIT, are 7 (Seven) or more, applications of only those one class below the eligible class bidders shall be considered, who have previously completed similar works satisfactorily and are meeting the criteria of upgradation in respect of past experience of completed works (individual work experience and/or average annual turnover as applicable) and financial soundness (solvency/financial soundness and working capital) as per details given in Manual on Contracts. Therefore, such contractors shall upload the requisite information/documents in the Cover-1.

3. Un-enlisted contractor shall be considered provided he meets the criteria. Foreign firms shall not be eligible for this tender. However Indian Firms having foreign national / Indian nationals staying abroad / Indian national having taken foreign citizenship, as director(s) shall be considered subject to security clearance from the concerned authorities.

4. Contractors enlisted with MES will upload following documents in Cover 1 for checking eligibility:
(a) Application for tender on Firm's letterhead.
(b) Enlistment letter issued by the Registering Authority duly renewed for the cycle period in vogue.
(c) Scanned copy of DD /Bankers Cheque toward cost of tender and EMD instrument in case SSD bond is not signed at the time of registration.
(d) Any other document required as described in this Appendix.

5. Contractors not enlisted with MES will be required to upload following documents in Cover 1 for checking eligibility.
(a) Application for tender on Firm's letterhead.
(b) Scanned copy of DD/ Bankers cheque toward cost of tender and Earnest Money Deposit (EMD) instrument.
(c) Copy of Police Verification Report/Police Clearance Certificate/ Character Certificate from the Police Authority of the area where the registered office of the firm is located/notarized copy of valid passport of Proprietor/each Partner/each Director.
(d) All documents required for enlistment in MES for the class mentioned in Para 8(b) above as per Para 1.5 of Section 1of MES Manual on Contracts 2020.
(e) Details of works being executed in MES, if any.
(f) Any other document required as described in this Appendix.

6. Tenders not accompanied by scanned copies of requisite DD/Bankers Cheque towards cost of tender and earnest money (as applicable) in Cover 1 shall not be considered for validation of 'T' bid and their Financial Bids will not be opened.

7. Contractors should ensure that their original physical DDs and Earnest Money Deposit (EMD) instruments (as applicable) reach the office of Accepting Officer within 10 days of bid submission end date failing which following action shall be taken.

(a) In case of tenders from an enlisted contractor of MES, where scanned copies of requisite DD/ Bankers Cheque towards cost of tender have been uploaded in Cover 1 but physical copies are not received within the stipulated period, their financial bids (Cover 2) will be opened. However non-submission of physical copies of cost of tender shall be considered as willful negligence of the tenderer with ulterior motives and such tenderer shall be banned from bidding for a period of six months commencing from the date of opening of Financial Bid (Cover 2).

(b) In case of tenders from un-enlisted contractor, where scanned copies of requisite DD/Bankers Cheque towards cost of tender have been uploaded in Cover 1 but physical copies are not received within the stipulated period, their financial bids (Cover 2) will not be opened. Name of such contractors along with complete address shall be circulated for not opening of their bids for a period of six months commencing from the date of opening of financial bid (Cover 2).

(c) In case of tenders from enlisted and un-enlisted contractors, where scanned copies of instruments for Earnest Money Deposit (as applicable) have been uploaded in Cover 1 but the same are not received in physical form within stipulated period, such tenders shall not qualify for opening of financial bid (Cover 2).

8. Contractor will not be allowed to execute the work by subletting or through power of attorney to a third party/another firm on his behalf. However, a contractor can execute the work through power of attorney to sons/daughters/ spouse of Proprietor/Partner/Director and firm's own employees, director, project manager provided they are not having a separate enlisted firm in MES in their name as Proprietor/Partner/Director.

9. After opening of Cover 1 and during its technical evaluation, in case any deficiency is noticed in the documents required to be uploaded by the tenderers as per NIT, a communication in the form of e-mail/SMS/Speed Post etc shall be sent to the contractor to rectify the deficiency within a period of seven days from date of communication failing which their financial bid (Cover 2) shall not be opened and contractor shall not have any claim on the same.

10. Invitation for e-tender does not constitute any guarantee for validation of Technical bid and subsequent opening of financial bid of any applicant/bidder merely by virtue of enclosing DD. Accepting Officer reserves the right to reject the Technical bid and not to open the financial bid of any applicant/bidder. Technical bid validation shall be decided by the Accepting Officer based on eligibility of the firm as per criteria given in this Appendix. Tenderer/bidder will be informed regarding non-validation of his Technical bid assigning reasons therefore through tender evaluation report which shall be uploaded on the website. Such tenderer, if desires, may appeal to the next higher Engineer Authority (NHEA) viz Chief Engineer Central Command, Lucknow on email id ceengrll-mes@nic.in with copy to the Accepting Officer on email before the scheduled date of opening of Cover 2. NHEA shall decide the matter within a period of seven working days from the date of receipt of appeal. The decision of the NHEA shall be final and binding. The tenderer/bidder shall not be entitled for any compensation whatsoever for rejection of his bid.

11. In case an un-enlisted contractor is already executing works in MES, he shall not be considered eligible for the subject tender if the total value of such works is more than twice the tendering limit of the MES Class of contractor for which it is eligible. For this purpose, details of the works being executed by such a contractor shall be uploaded in the Cover -1 of the bid and shall be checked/verified by the Accepting Officer.

12. In case the BOQ is revised through the corrigendum and the bidder has failed to quote on revised BOQ (ie he has quoted on pre revised BOQ), such bid shall be treated as wilful negligence by the bidder and his quotation shall be considered non-bonafide. In such cases the lowest tender shall be determined from amongst the valid/bonafide bids only. Accepting Officer may decide whether to re-tender or consider the lowest bonafide tender for acceptance.

APPENDIX 'A' TO NOTICE INVITING TENDER (NIT) (Contd…/-)

13. Revoking the offer or revising the rates upward or offering voluntary reduction by the lowest tenderer after opening of Cover 2 shall be considered as a wilful default. For this default a penalty of an amount equal to Earnest Money shall be levied. In case of an unenlisted tenderer, Earnest Money deposited by him shall be forfeited. In case of MES enlisted tenderer having deposited the Standing Security Bond, an amount equal to the earnest money stipulated in the NIT, shall be notified to the tenderer for depositing through MRO and consideration of such tenderer in tender evaluation for future works shall remain suspended till the aforementioned amount is deposited in the Government Treasury. No other disciplinary/administrative action shall be taken against such tenderers. In such a situation, the next lowest offer shall not be considered for acceptance. Instead, retendering shall be resorted to in a transparent and fair manner and the defaulting tenderer and his related firm if any, shall not be eligible for this tender in second call or subsequent calls.

14. Tender to related firms shall not be issued simultaneously. Firms shall be termed as related if Proprietor/one or more Partners/Directors are common. Decision of Accepting Officer on issue/deny the tender to any one of the related firms shall be final and binding.

15. **INTEGRITY PACT (IP):** Integrity Pact (IP) in cover 1(Technical Bid) duly signing all the pages of Pact by the authorized signatory of the firm in token of their agreement to bound the firm by its provisions. Bidder who fails to upload signed copy of the Integrity Pact (IP) in cover 1 (Technical Bid), their T bid (cover 1) shall not be considered for validation and consequently their finance bid (Cover 2) will not be opened.

AAD (Contracts)
For Accepting Officer

(Signature of Contractor)

INTEGRITY PACT

General

1. Whereas the President of India, represented by Chief Engineer (AF)Prayagraj hereinafter referred to as Principal / Owner and the first part, has floated the Tender (NIT No **CE (AF)/ALD/BLY/28 OF 2024-25**) and intends to award, under laid down organizational procedure, contract for **PROVN OF RCC SHELTER ALONGWITH ASSOCIATED WORK SERVICES AT AF STN BAREILLY** hereinafter referred to as works / Services and M/s_____ represented by, _____(which term unless expressly indicated by the contract, shall be deemed to include its successors and its assignees), hereinafter referred to as the Bidder / Contractor and the second part is willing to carryout the works / services.

2. Whereas the Bidder is a Proprietorship Concern / Partnership Firm / Limited Liability Partnership Firm / Private Limited Company / Limited Company / Joint Venture constituted in accordance with the relevant law in the matter and the Principal / Owner is Chief Engineer (AF)Prayagraj performing its functions on behalf of the President of India.

Objectives

3. Now, therefore, the Principal / Owner and the Bidder agree to enter into this pre-contract agreement, referred to as **INTEGRITY PACT (IP)**, to avoid all forms of corruption by following a system that is fair, transparent and free from any influence / prejudiced dealings prior to, during and subsequent to the conclusion of the contract to be entered into with a view to:

3.1 Enabling the Principal / Owner to get the desired works / services at a competitive price in conformity with the defined specifications of the Services by avoiding high cost and the distortionary impact of corruption on public procurement.

3.2 Enabling Bidders to abstain from bribing or any corrupt practice in order to secure the contract by providing assurance to them that their competitors will also refrain from bribing and other corrupt practices and the Principal / Owner will commit to prevent corruption, in any form, by their officials by following transparent procedures.

Commitments of the Principal / Owner

4. The Principal / Owner commits itself to the following:

4.1 The Principal / Owner undertakes that, no official of the Principal / Owner, connected directly or indirectly with the contract will demand, take a promise for or accept, directly or through intermediaries, any bribe, consideration, gift, reward, favour or any material or immaterial benefit or any other advantage from the Bidder, either for themselves or for any person, organisation or third party related to the contract, in exchange for an advantage in the bidding process, bid evaluation, contracting or implementation process related to the Contract.

4.2 The Principal / Owner will, during the pre-contract stage, treat all Bidders alike and will provide to all Bidders the same information and will not provide any such information to any particular Bidder which could afford an advantage to that particular Bidder in comparison to other Bidders.

4.3 All the officials of the Principal / Owner will report to the appropriate Government office any attempted or completed breach(s) of the above commitments as well as any substantial suspicion of such a breach.

5. In case of any such preceding misconduct on the part of such official(s) is reported by the Bidder to the Principal / Owner with wilful and verifiable facts and the same is prima facie found to be correct by the Principal / Owner, necessary disciplinary proceedings, or any other action as deemed fit, including criminal proceedings may be initiated by the Principal / Owner and such a person shall be debarred from further dealing related to the tender / contract process. In such a case while an Inquiry is being conducted by the Principal / Owner, the tender process/ proceedings under the contract would not be stalled.

Commitments of Bidders

6. The Bidder commits himself to take all measures necessary to prevent corrupt practices, unfair means and illegal activities during any stage of his bid or during any pre-contract or post-contract stage in order to secure the contract or in furtherance to secure it and in particular commits himself to the following:

6.1 Bidder will not offer, directly or through intermediaries, any bribe, gift, consideration, reward, favour any material or non-material benefit or other advantage, commission, fee, brokerage or inducement to any official of the Principal/Owner, connected directly or indirectly with the bidding process, or to any person, organisation or third party related to the contract in exchange for any advantage in the bidding, evaluation, contracting and implementation of the Contract.

6.2 The Bidder further undertakes that he has not given, offered or promised to give, directly or indirectly any bribe, gift, consideration, reward, favour any material or non-material benefits or other advantage, commission, fees, brokerage or inducement to any official of the Principal/ Owner or otherwise in procuring the Contract or forbearing to do or having done any act in relation to the obtaining or execution of the contract or any other Contract with the Government for showing or forbearing to show favour or disfavour to any person in relation to the Contract or any other Contract with the Government.

6.3 The Bidder will not collude with other parties interested in the contract to impair the transparency, fairness and progress of the bidding process, bid evaluation, contracting and implementation of the contract.

6.4 The Bidder will not accept any advantage in exchange for any corrupt practice, unfair means and illegal activities.

6.5 The Bidder would not enter into conditional contract with any Agent(s), broker(s) or any other intermediaries wherein payment is made or penalty is levied, directly or indirectly, on success or failure of the award of the contract.

6.6 The Bidder commits to refrain from giving any complaint directly or through any other manner without supporting it with full and verifiable facts. Complaint will be processed as per Guidelines for Handling of Complaints in vogue. In case the complaint is found to be vexatious, frivolous or malicious in nature, it would be constructed as a violation of Integrity Pact.

7. **Previous Transgression**

7.1 The Bidder declares that no previous transgression occurred in the last three years immediately before signing of this Integrity Pact with any other company in respect of any corrupt practices envisaged hereunder or with any Public Sector Enterprises in India or any Government Department in India.

7.2 If the Bidder makes incorrect statement on this subject, Bidder can be disqualified from tender process or the contract and if already awarded, same can be terminated for such reason.

8. **Company Code of Conduct**

8.1 Bidders are advised to have a company code of conduct (clearly rejecting the use of bribes and other unethical behaviour) and a compliance program for the implementation of the code of conduct throughout the country.

INTEGRITY PACT (Contd…/-)

9. **Sanction for Violation**

9.1 Any breach of the aforesaid provisions by the Bidder or any one employed by him or acting on his behalf (whether with or without the knowledge of the Bidder) or the commission of any offence by the Bidder or any employed by him or acting on his behalf, as defined in Chapter IX of the Indian Penal Code, 1860 or the Prevention of Corruption Act 1988 or any other act enacted for the prevention of corruption shall entitle the Principal / Owner to take all or any one of the following actions, wherever required:

(i) Technical bid of the Bidder will not be opened. Bidder will not be entitled to or given any compensation. However, the proceedings with the other Bidder(s) would continue.

(ii) Financial bid of the Bidder will not be opened. Bidder will not be entitled to or given any compensation. However, the proceedings with the other Bidder(s) would continue.

(iii) The Earnest Money Deposit shall stand forfeited either fully or partially, as decided by the Principal / Owner, in case contract is not awarded to the Bidder and the Principal / Owner shall not be required to assign any reason therefor. For enlisted contractors an amount less than or equal to Earnest Money Deposit as decided by the Principal / Owner shall be deducted from any amount held with the Department / any payment due.

(iv) To immediately cancel the contract, if already concluded / awarded without any compensation to the Bidder.

(v) To encash the Performance Security furnished by the Bidder.

(vi) To cancel all or any other Contract(s) with the Bidder.

(vii) To temporarily suspend or temporarily debar / permanently debar the bidder as per the extant policy.

(viii) amount can be recovered from any outstanding payment due to the Bidder from the Principal / Owner in connection with any other contract for any other works / services.

(ix) If the Bidder or any employee of the Bidder or any person acting on behalf of the Bidder, either directly or indirectly, is closely related to any of the officers of the Principal / Owner, or alternatively if any close relative of an officer of the Principal / Owner has financial interest/stake in the Bidder's firm, the same shall be disclosed by the Bidder at the time of submission of tender. Any failure to disclose the interest involved shall entitle the Principal / Owner to debar the Bidder from the bid process or rescind the contract without payment of any compensation to the Bidder. The term 'close relative' for this purpose would mean spouse whether residing with the Government servant or not, but does not include a spouse separated from the Government servant by a decree or order of a competent Court, son or daughter or step son or step daughter and wholly dependent upon Government servant, but does not include a child or step child who is no longer in any way dependent upon the Government servant or of whose custody the Government servant has been deprived of by or under any law, any other person related, whether by blood or marriage, to the Govt. servant or to the Government servant's wife or husband and wholly dependent upon Government servant.

(x) The Bidder shall not lend to or borrow any money from or enter into any monetary dealings or transactions, directly or indirectly, with any employee of the Principal / Owner and if he does so, the Principal / Owner shall be entitled forthwith to cancel the contract and all other contracts with the Bidder.

9.2 The decision of the Principal / Owner to the effect that a breach of the provisions of this Integrity Pact has been committed by the Bidder shall be final and binding on the Bidder. However, the Bidder can approach the Independent External Monitor(s) (IEMs) appointed for the purposes of this Pact.

INTEGRITY PACT (Contd…/-)

10. **Independent External Monitors (IEMs)**

10.1 MoD has appointed the following Independent External Monitors for this pact in consultation with the Central Vigilance Commission:

Sl No	Name of IEM	e-mail id
1.	**Shri Narayan Murthy Ganapathy, IFoS (Retd)**	**gana_narayan@yahoo.com**
2.	**Shri Lalatendu Mohanti, IPS (Retd) Flat No. 9367 (GF & FF), Pocket 9, Sector-C, Vasant Kunj, New Delhi-110070**	**L.mohanti@gmail.com**

10.2 Details of Nodal officer nominated by E-in-C's Branch are as follows:

Name: Shri Bhavesh Gupta, Director (Contracts), E-in-C's Branch, Kashmir House, New Delhi
e-mail id: dircont1einc-mes@nic.in
Mobile No: 011-23019220, 9868145022
Postal Address: Room No 172(A)
Dt of Contract Management
Engineer-in-Chief's Branch
Kashmir House, Rajaji Marg
New Delhi-110011

10.3 In case of any complaint with regard to violation of Integrity pact, either party can approach IEMs with copy to the Nodal Officer and the other party. If any such complaint from bidder is received by the Principal / Owner, the Principal / Owner shall refer the complaint to the Independent External Monitors for their recommendations / Inquiry report.

10.4 If the IEMs need to peruse the relevant records of the Principal / Owner and / or of the Bidder / Contractor in connection with the complaint sent to them, the Principal / Owner and / or the Bidder / Contractor shall make arrangement for such perusal of records by the IEMs as demanded by them including unrestricted and unconditional access to the project documentation and minutes of meeting. If records / documents of Sub Contractor(s) are also required to be perused by the IEMs, the Bidder shall make arrangement for such perusal of records by the IEMs, as demanded by them. IEMs are under obligation to treat the information and documents of the Principal / Owner and Bidder / Contractor / Sub-Contractors with confidentiality.

10.5 The task of the IEMs, is to review independently and objectively, any complaint received with regard to violation of Integrity Pact and offer recommendations or carry out inquiry as deemed fit. The IEMs are not subject to any instructions by the representatives of the parties and shall perform their functions neutrally and independently. The report of inquiry, if any, made by the IEMs shall be submitted to either of the following for a final and appropriate decision in the matter keeping in view the provision of this Pact:

(a) Engineer-in-Chief in normal cases

(b) CVO(MES & BRO) / DMA / Department of Defence (DoD) in cases involving vigilance angle

11. Examination of Books of Accounts In case of any allegation of violation of any provisions of this Integrity Pact or payment of commission, the Principal / Owner or its agencies shall be entitled to examine the Books of Account of the Bidder and the Bidder shall provide necessary information of the relevant financial documents in English and shall extend all possible help for the purpose of such examination.

12. Law and Place of Jurisdiction This Pact is subject to Indian Law. The place of performance and jurisdiction is the seat of the Principal / Owner.

13. **Other Legal Actions**

The actions stipulated in this Integrity Pact are without prejudice to any other legal action that may follow in accordance with the provisions of the extant law in force relating to any civil or criminal proceedings.

INTEGRITY PACT (Contd…/-)

14. **Signing of Integrity Pact on behalf of Bidder**

(a) Proprietorship Concern - The Integrity Pact must be signed by the proprietor or by an authorised signatory holding power of attorney signed by the proprietor.

(b) Partnership firm - The Integrity Pact must be signed by all Partners or by one or more Partner holding power of attorney signed by all the Partners.

(c) Limited Liability Partnership firm - The Integrity Pact must be signed by all Partners or by one or more Partner holding power of attorney signed by all Partners.

(d) Private Limited / Limited Company - The Integrity Pact must be signed by representative duly authorised by Board resolution.

(e) Joint Venture- The Integrity Pact must be signed by all Partners and members to Joint Venture or by one or more Partner holding power of attorney signed by all Partners and all members to the Joint Venture.

15. **Validity**

15.1 The validity of this Integrity Pact shall be from date of its signing. It expires for the Contractor after the final payment under the contract has been made or till the continuation of Defect Liability Period, whichever is later and for all other bidders, till the Contract has been awarded.

15.2 Should one or several provisions of this Pact turn out to be invalid, the remainder of this Pact remains valid. In this case, the parties will strive to come to an agreement to their original intentions.

INTEGRITY PACT

To

_____

_____

_____

Tender ID No_____ for the work **PROVN OF RCC SHELTER ALONGWITH ASSOCIATED WORK SERVICES AT AF STN BAREILLY**

Dear Sir,

It is hereby declared that MES is committed to follow the principle of transparency, equity and competitiveness in public procurement.

The subject Notice Inviting Tender (NIT) is an invitation to offer made on the condition that the Bidder will sign the Integrity Pact, which is an integral part of tender/bid documents, filing which the tender/bidder will stand disqualified from the tendering process and the bid of the bidder be summarily rejected.

This declaration shall form part and parcel of the Integrity Pact and signing of the same shall be deemed as acceptance and signing of the Integrity Pact on behalf of MES.

Yours faithfully
Chief Engineer (AF) Prayagraj

INTEGRITY PACT

To

Chief Engineer (AF) Prayagraj

Sub - Tender ID No_____

Submission of Tender for the work of **PROVN OF RCC SHELTER ALONGWITH ASSOCIATED WORK SERVICES AT AF STN BAREILLY**

Dear Sir,

I/We acknowledge that MES is committed to follow the principles thereof as enumerated in the Integrity Pact enclosed with the tender/bid document.

I/We agree that the Notice Inviting Tender (NIT) is an invitation to offer made on the condition that I/We will sign the Integrity Pact, which is an integral part of tender document, failing which I/We will stand disqualified from the tendering process. I/We acknowledge that THE MAKING OF THE BID SHALL BE REGARDED AS AN UNCONDITIONAL AND ABSOLUTE ACCEPTANCE of the conditions of the NIT.

I/We confirm acceptance and compliance with the Integrity Pact in letter and spirit and further agree that execution of the said Integrity Pact shall be separate and distinct from the main contract, which will come into existence when tender/bid is finally accepted by MES. I/We acknowledge and accept the validity of the Integrity Pact, which shall be in line with Para 15 of the enclosed Integrity Pact.

I/We acknowledge that in the event of my/our failure to sign and accept the Integrity Pact, while submitting the tender/bid, MES shall have unqualified, absolute and unfettered right to disqualify the tender/bidder and reject the tender/bid in accordance with terms and conditions of the tender/bid.

Yours faithfully

(Duly authorised signatory of the Bidder)

IN LIEU OF IAFW 2159 (REVISED 1955)

(To be used in conjunction with General Conditions of contracts IAFW 2249, 1989 Print)

MILITARY ENGINEER SERVICES

Headquarters
Chief Engineer (AF) Prayagraj
Military Engineer Services
Bamrauli Prayagraj (UP)
Pin -211012

No: 958130/ /E8 **Dec 2024**

PROVN OF RCC SHELTER ALONGWITH ASSOCIATED WORK SERVICES AT AF STN BAREILLY

1.

M/s_____of_____is/arehere by authorized to tender for the above work. The tender is to be uploaded in the "defproc.gov.in" site before 1800 hours on _____

2. Any correspondence concerning this tender should be addressed quoting the reference as given to HQ Chief Engineer (AF) Prayagraj, 211012 and marked as **"CA NO: CE (AF)/ALD/BLY/28 OF 2024-25".**

3. "THE PRESIDENT OF INDIA DOES NOT BIND HIMSELF TO ACCEPT THE LOWEST OR ANY TENDER".

Signature of Contractor

AAD (Contracts)
For Accepting Officer

SCHEDULE 'A' NOTES

LIST OF WORKS AND PRICES

NOTES:

1. **GENERAL:**

1.1. **SCOPE OF WORK:** The scope of work included in various parts of Schedule 'A' is as under:

1.	Schedule 'A' Part-I	Building works (pre-priced)
2.	Schedule 'A' Part-II	Internal Water Supply (Pre-priced)
3.	Schedule 'A' Part–III	Internal Electrification (Pre-priced)
4.	Schedule 'A' Part-IV	External Electrification (Pre-priced)
5.	Schedule 'A' Part-V	Sewage Disposal (Pre-priced)
6.	Schedule 'A' Part-VI	Area Drainage (Pre-priced)
7.	Schedule 'A' Part-VII	Static Water Tank (Pre-priced)
8.	Schedule 'A' Part-VIII	Runway/taxi track/pavement (non-load bearing surface) (Pre-priced)
9.	Schedule 'A' Part-IX	Runway/taxi track/pavement (load bearing surface) (Pre-priced)
10.	Schedule 'A' Part-X	Lightening protection (Pre-priced)
11.	Schedule 'A' Part-XI	External water supply (Pre-priced)
12.	Schedule 'A' Part-XII	Road/path/culvert (Pre-priced)
13.	Schedule 'A' Part-XIII	Demolition (Pre-priced)
14.	Schedule 'A' Part-XIV	AC and Refrigeration (Pre-priced)
15.	Schedule 'A' Part-XV	Barricading (Pre-priced)
16.	Schedule 'A' Part-XVI	Rain water harvesting (Pre-priced)
17.	Schedule 'A' Part-XVII	Site clearance and earthwork (Pre-priced)
18.	Schedule 'A' Part-XVIII	Miscellaneous Items of B/R and E/M (**To be quoted by the tenderer in BOQ)**

2. **PERIOD OF COMPLETION:** Period of completion for the whole work is **30 (Thirty) Months.**

2.1 The Entire work of subject tender is grouped into two phases, i.e., Phase-I & II. The scope of work in each Phase with period of completion shall be as under:

PHASE-I: Phase-I constitute the following scope of work and shall be completed in **06 (Six) Months** from the date of commencement of work as per Work Order No 1 issued by the GE:

(i) Surveying of proposed area using total station survey Equipment.
(ii) Submission of survey report, proposed profiles for various locations as mentioned above, submission of initial levels, proposed longitudinal & transverse profile for various locations.
(iii) Mobilization and installation of T & P specified in tender documents.
(iv) Setting up of site office, erection of plants, and making approaches to site.
(v) Setting of site lab with full equipment as per Particular Specifications
(vi) Submission of project management report on MS project /Primavera showing allactivities, men and materials requirement on daily basis.
(vii) Approval of design mix for all layers as specified in tender documents.
(viii) Barricading of the required length for phase II & III (length of barricading at any point of time approx 3600 RM at work place material storing yard etc all as specified.
(ix) Deployment of Site Engineers & technical staff as specified with suitable reliever as specified in tender documents.
(x) Submission of design mix for PQC, DLC/WMM & RCC
(xi) Trial bay construction of all layers i.e stabilised soil layers, GSB, WMM, DLCand PQC, all as specified in tender documents
(xii) Procurement of 15% quantity of non-perishable material required for executionof all the items of BOQ /Schedule 'A'
(xiii) Demolition of existing bituminous macadam/PCC/WBM/GSB as catered in BOQ.

SCHEDULE 'A' NOTES (Cont'd. /-)

PHASE-II: Remaining complete work not covered under Phase-I shall be completed within **30 (Thirty) months** from the date of commencement of work as per Work Order No 1 issued by the GE.

Note: -

(i) Work order for Phase – I and Phase -II shall be placed concurrently. Both the phase will commence together.

(ii) Completion certificate shall be issued for each phase separately.

(iii) The whole work shall be deemed to be completed only after completion of Phase-II

(iv) Single Final bill shall be submitted for the entire Contract.

(v) Compensation for delay in completion of work: -

(i) Phase-I: - Rs 10,000/- per day, subjected to max 1% of CA Amount.

(ii) Phase-II: - Compensation for delay in completion of Phase-II work will be levied as per condition 50 of IAFW-2249 and amount for calculating compensation for delay of the phase as per contract rates for the work as catered for particular phase.

2.2 Accepting Officer shall be informed in advance by the GE before issue of completion of Phase – I work. Board of Officers/Rep of CE will inspect the site of Phase-I work and trial bays for DLC, PQC and all layers of pavement as specified in the BOQ and drawings.

2.3 Entire work executed under this contract shall be under defects liability period of 03 (Three) years from the certified date of completion. Contractor is fully liable to replace/rectify all defective work intimated to him during defects liability period. In case contractor fails to rectify/replace the defective work, GE shall get the replacement /rectification as necessary and recover the cost in full from the performance security deposit lodged by the contractor. If the contractor consider that any additional treatment is required over and above that specified here in before, the same shall be carried out by the contractor without any extra cost to the Government. During the defects liability period if any portion of pavement /paved shoulders sinks or damaged due to stagnation of water, the contractor shall rectify or replace defective work/material at his own expenses, when called upon in writing to do so by the GE.

2.4 Works covered under above mentioned phases shall be completed within 25 (Twenty-Five) months from the date of commencement as ordered by GE in the Work Order and in accordance with Project management Schedule agreed to between the GE and contractor in terms of condition 11 of IAFW-2249 (General Conditions of Contacts) forming part of this contract. The PRIMAVERA / MS PROJECT software shall be used for management of work. The route map through which materials / trucks and Contractor's men are to be moved (both entry & exit) shall be as instructed by the GE. Plant and material dumping area shall be on the location as instructed by GE and it shall be strictly adhered to. Any damage to existing road to be made good by the contractor at his own cost and no claim on this account will be entertained.

2.5 Mix design for PQC, DLC, soil stabilisation with cement content & RCC from IIT Delhi/IIT Kanpur/IIT Roorkee/ IIT Kharagpur (as specified).

2.6 Contractor shall submit detailed MS Project Schedule with Work Break down Structure (WBS) for each activity duly linked with manpower and T&P resources. This detailed schedule shall be submitted by the Contractor within 4 weeks of placing the Work Order. No extension shall be granted on account of any slippage in Project Schedule due to fault of contractor.

2.7 The contractor, if he so desires, will be allowed to work beyond normal working hours, except on Sundays and Gazetted holidays with prior written permission of GE. The contractor may also be permitted to work on Sunday and/or on Gazetted holidays if so desired by him/them with prior written permission of GE.

2.8 It is made clear that Contractor's quoted rates are deemed to include for cost of material for testing, all field apparatus required for sampling and testing as per MES Schedule / IS codes / MORTH specifications and manpower incidental to such testing will be provided along with necessary transport arrangement to and fro to the approved testing agency or laboratory by the contractor during the construction phase of the work and defects liability period. Field laboratory checked by SEMT Pune with all the required apparatus as listed in Particular Specifications specified here-in-after along with trained man power for conducting tests and requisite furniture, computer with connected equipment and trained staff shall be established by the contractor at site of work at his cost for carrying out field test at stipulated frequency. Any test which is required to be carried out as per MES Schedule / IS code / MORTH specifications, but cannot be performed at field laboratory established by the contractor, all such tests shall be carried out

by the Contractor at IITs/NITs /HQ E-in-C's approved /HQ CE (AF) Allahabad approved lab / NABL accredited Lab approved by HQ E-in- C's Branch New Delhi /SEMT Pune and cost of such tests shall be borne by the contractor and their quoted rates shall be deemed to include for the same. All the tests shall be carried out by qualified lab technicians employed by the Contractor in the presence of Engineer-in- charge or his representative. In addition to above, minimum required test as per frequency shown on Appx "B" to Special Condition shall be got done from third party viz. IIT/NIT/ HQ E- in-C's

approved /HQ CE (AF) Allahabad approved lab / NABL accredited Lab approved by HQ E-in-C's Branch New Delhi /SEMT Pune. Cost of materials, sampling, transporting & testing charges shall be borne by Contractor. The quoted rates shall be deemed to be included for this provision and nothing extra will be paid by Government in this regard. The quoted rates shall include rates for confirmatory test by third party of the soil design parameters mainly CBR and K value. Confirmatory soil investigation is to be carried out by the contractor before starting of cement soil stabilization.

2.9 Contractor shall deploy at least minimum machinery / T&P (excluding reserves) on the work as per Particular Specifications specified here-in-after. List of equipment to be installed in site lab for quality control is given in Particular Specifications. The contractor should note that list of machinery/equipment given in Particular Specifications is minimum requirement. The contractor may have to bring / use additional machinery / equipment other than specified in Particular Specifications for smooth functioning and timely completion of the work with respect to Project Schedule without any extra cost to the department.

2.10 PREPARATORY WORKS: -

2.10.1 The following items of works shall be considered as Preparatory Works under the subject Contract. These provisions shall be provided by the Contractor as specified here in below at his own expenses and the quoted rates shall have deemed to include for these provisions and nothing extra will be paid by Government in this regard.

(a) Temporary approach for Contractor's men and materials:

Contractor shall provide temporary approach as shown in Site Plan. Contractor is at liberty to use any specification for this provision. The contractor shall bring this stretch of approach roads to its original status by using excavated earth which shall be properly compacted and rolled to the satisfaction of Engineer-in-Charge before work is completed and certified so by GE.

(b) The materials i.e. aggregate for all types of pavements/concrete shall be stacked on paved surfaces in neat trapezoidal stack duly separated by suitable bulks. Platforms shall be 100 mm thick dry lean concrete having cement content not less than 140 kg/Cum and compacted with smooth wheeled rollers, using 40mm graded stone and shall be dismantled after completion of work.

2.10.2 During defect liability period the runway will be in operational mode and same will be made available in spells based on availability of site within 20 days after intimating to the contractor by the department. Repair methodology shall be as per relevant TI and if the same is not covered in TI relevant IRC clause shall be followed. For other than runway works relevant IS /MES Schedule shall be followed.

2.10.3 Where consolidation or rolling is specified in the description of items in Schedule 'A', the "Unit" of measurement shall be deemed to be for the compacted/consolidated quantities of work done. The quantities given in the Schedule 'A' against these items are of solid contents, i.e., after compaction/consolidation and payment shall be made accordingly.

2.10.4 In the absence of any definite provision in the specifications contained herein, reference made to :-

(i) In respect of pavement related items, TI – 09/2018 including up to date amendments for sub grade, GSB, WMM and TI – 10/2011 including up to date amendments for DLC and PQC, latest MORTH (Ministry of Shipping, Road, Transportation and Highways) / IS codes shall be followed. For design Mix of DLC IRC – SP- 49/2014 and design Mix of PQC IRC – 44/2017 should be followed.

(ii) In respect of other than pavement related items relevant IS codes, MES Schedule shall be followed.

(iii) Wherever these are silent, construction and completion of works shall conform to sound engineering practice, and in case of any dispute arising out of the interpretation of the above, the decision of Accepting Officer shall be final and binding on the Contractor.

2.10.5 All the Code of Practice, Standards and Specifications applicable shall be the latest editions with up to date correction slip and revisions etc.

2.10.6 Please note that no foreign exchange or import license facility will be arranged by the Department for any material, plants, equipments and tools required for execution of subject work.

2.10.7 The locations of Concrete Batching Plant(s) etc shall be as directed by GE. The location for these plants may be changed within the limits of layout plan and the Contractor shall have no claims whatsoever on this account.

2.10.8 **Levelling:** The initial levels shall be recorded at a grid interval of 2mx2m with the help of total station and it should be located after 30m and then levels at same interval and same places shall be taken after completion of each layer of work viz compacted and rolled soil base after demolition of existing WMM/GSB, refilled stabilized soil, GSB, WMM, DLC & PQC etc). All the levels so taken shall be fed into the computer on MS Excel sheet and longitudinal section and cross section will be got drawn by the contractor and signed jointly by the contractor and the GE. The first such L and cross section showing existing levels and proposed top level of excavated soil base shall be got approved from GE. Contractor's quoted rates and sum shall be deemed to be included for these provisions and nothing extra shall be payable on this account

2.10.9 PROJECT MANAGEMENT

2.10.9.1 The details in the Project Schedule Network will include: -

(a) All activities in detail duly supported with work break down structure (WBS) for each activity including assigned T&P, labour force and duration with facility to follow progress.

(b) Material Procurement planning and utilization (qty and time of availability), details to ensure actual availability prior to the commencement of the activity/task.

(c) Plant deployment details (activity wise).

(d) Activity Vs Labour employment details.

2.10.9.2 Detailed Dated Project Time Schedule will be jointly prepared by contractor and Garrison Engineer using MS Project/Primavera within Four weeks of date of commencement given in WO 1 of the contract and shall be kept on record of Garrison Engineer for further reference.

2.10.9.3 The Project Time Schedule will be updated daily with all necessary details and the "Work Done Report" will be signed by the contractor and included as part of the "Works Diary" by the JE and checked by the Engineer-in-Charge. A weekly review of the work progress will be done between the Engineer-in-Charge and the contractor to monitor the progress made during the week vis-à-vis total progress. The weekly report will also include forecast of resources to include plant, stores, labour, etc for the next week. A detailed monthly report will be prepared and submitted to the GE by the contractor and review done jointly to examine requirement of increasing resources to ensure completion within the laid down time period. The detailed monthly progress report shall be presented and discussed by Project Manager of Contractor with Accepting Officer/ Director (Contracts)/ Officers detailed by Accepting Officer at Chief Engineer (AF) Allahabad office every month initially for three months and then after every two months' interval from the date of start of work and if work is lagging behind then either Proprietor/ Director of the firm or one senior management member shall be present in the meeting.

SCHEDULE 'A' NOTES (Cont'd. /-)

2.11 **OBSERVATIONS OF SECURITY RESTRICTIONS: -**

2.11.1 The contractor, his agents, work people etc shall strictly adhere to the security instructions as laid down in the Contract Conditions. No claim whatsoever for observance of restrictions imposed at site shall be admissible at any stage of work.

2.12 **CONTRACTOR'S SUPERVISION**

2.12.1 Site of work is an IAF aircraft flying station. Contractor shall have to plan his resources, manpower and work keeping in view the aircraft flying schedule of AF Station.

2.12.2 All the minimum supervisory staff mentioned in clause 2.10.9.1 shall attend the site on all working days. Maximum two leave in a month per one appointment holder will be acceptable without any substitute arrangement. To monitor the attendance, the contractor is required to install biometric based attendance system with the facility auto generate attendance report in a required format. GE shall have full powers to put the contractor on notice on account of default either for non-employment of Engineers (s)/ Technical staff or absence of Engineer (s)/ Technical staff from the site and levy penalty of Rs 500/- per day per vacancy up to 30 days period. Thereafter, GE shall have the option to either suspend the work or employ Engineer (s) at Contractors cost and recover the amount from the contractor's due.

2.12.3 Contractor has to provide basic model of mobile phone without multi-media device to all their staff for effective communication between plant site, storage yard, site, lab MES and user. Their permission for use of mobile phone at site shall be obtained from Air Force authority through GE.

2.12.4 The equipments maintained at Site Lab and T&P deployed shall be got calibrated by the approved agency in every 6 (Six) months or earlier by the contractor so as to maintain preciseness / accuracy and efficiency.

2.12.5 On completion of the entire work, the contractor shall submit record drawing on tracing paper along with three prints showing layout plan of executed works / services drawn to the suitable scale mentioning actual routes / alignment of various electric cable, pipe lines giving sizes, position of fittings/ fixtures etc as executed together with any information which the Engineer-in-Charge may require, duly signed by the him and approved by Garrison Engineer before issue of completion certificate to the contractor.

2.12.6 Contractor shall provide photographic record in the shape of album size not less than A4 size in triplicate for each important stage in 6 different locations as directed by GE. After completion, the contractor shall also submit framed photo of complete project in triplicate size not less than 900x600mm. Contractor is not permitted to take the camera to restricted site. The soft copies of all photo's shall be provided by GE for printing and photo framing. The cost to this effect shall be deemed to be included in rates quoted by contractor for the entire work.

3. **NOTE FOR LUMPSUM SCHEDULE 'A' PART –I (BUILDING WORKS): -**

3.1 The lump sum considered towards Sch A Part I shall be deemed to include for all items of works complete as specified/referred in Particular Specifications and all minor details though not specifically shown on drawings/TD drawings or specified in Particular Specifications.

3.2 Cutting and forming chases in masonry, concrete work etc. including cutting and or leaving holes/recesses, sinking, etc. where required and as directed by the Engineer-in-Charge shall be done as far as possible while the work is in progress. The cost of materials and labour for cutting and/or forming chases, cutting or leaving holes/recesses, sinking and making good in cement mortar 1:3 for filling up to 20mm and in PCC 1:2:4 (12.5mm graded stone aggregate) for filling more than 20mm shall deemed to be included in the cost considered by the contractor in its lump sum.

3.3 In all the above and other similar cases, the details indicated elsewhere in the drawings as approved or TD drawings that are similar or near similar to the missed-out items of work shall be followed. In the absence of any other similar or near similar details, minimum essential requirement for completion of the work from structural and utility point of view shall be deemed to be included in the lump sum /rates quoted by contractor.

3.4 Lump Sum amount considered by the Contractor for the Building Works under Sch A Part-I shall be deemed to include for the following whether or not these are covered in the drawing:-

SCHEDULE 'A' NOTES (Cont'd. /-)

3.4.1 **For Item No. 01 (Blast Pen) of Sch 'A' Part -I: -**

(a) **SITE CLEARANCE AND SURFACE EXCAVATION**: Before setting out the building and commencing the construction, preparatory work such as removal of grass, vegetation, jungle clearance etc, surface excavation to a depth not exceeding 30 cm and averaging 15 cm, in any type of soil for the entire area occupied by the building and structure including plinth protection ramps, steps etc. shall be carried out by the contractor.

(b) **SURFACE DRESSING**: Surface dressing around the entire buildings and structure to a width of 3 Metres beyond the external edge of plinth protection / steps / ramp or external wall as applicable shall be carried out by the contractor as per clause 3.6 and 3.10 of MES Schedule Part-I [2009]. Area around outer edge of building / structure shall be dressed to slope away from the buildings as directed by Engineer –in-Charge.

(c) Any other work specifically mentioned in notes to other parts of Schedule 'A' and particular specification.

(d) Roof treatment and surface finishes as specified in drawings. Contractor has to give the Guarantee for Water Proofing Treatment over RCC Roof Slabs, sunken slab etc. for TEN YEARS.

(e) The measures to be adopted for prevention of leakage/seepage and dampness to building in terms of Drawing.

(f) Numbering of building all as directed by GE/Engineer-in-Charge.

(g) Earth filling in traverse including provision of weep holes, sump, hume pipe.

(h) Flooring including excavation as specified in drawings.

(i) All other items, which are not covered in notes but shown on drawings shall be deemed to be included in unit rate inserted against Schedule 'A' Part I.

3.4.2 **For Item No. 02 (Deflector wall) of Sch 'A' Part -I: -**

(a) **SITE CLEARANCE AND SURFACE EXCAVATION**: Before setting out the building and commencing the construction, preparatory work such as removal of grass, vegetation, jungle clearance etc, surface excavation to a depth not exceeding 30 cm and averaging 15 cm, in any type of soil for the entire area occupied by the building and structure including plinth protection ramps, steps etc. shall be carried out by the contractor.

(b) **SURFACE DRESSING**: Surface dressing around the entire buildings and structure to a width of 3 Metres beyond the external edge of plinth protection / steps / ramp or external wall as applicable shall be carried out by the contractor as per clause 3.6 and 3.10 of MES Schedule Part-I [2009]. Area around outer edge of building / structure shall be dressed to slope away from the buildings as directed by Engineer –in-Charge.

(c) Any other work specifically mentioned in notes to other parts of Schedule 'A' and particular specification.

(d) The measures to be adopted for prevention of leakage/seepage and dampness to building in terms of Drawing.

(e) Numbering of building all as directed by GE/Engineer-in-Charge.

(f) Earth filling in traverse including provision of weep holes.

(g) All other items, which are not covered in notes but shown on drawings shall be deemed to be included in unit rate inserted against Schedule 'A' Part I.

3.4.3 **For Item No. 03 (ORP Annxe with Garage) of Sch 'A' Part -I: -**

(a) **SITE CLEARANCE AND SURFACE EXCAVATION**: Before setting out the building and commencing the construction, preparatory work such as removal of grass, vegetation, jungle clearance etc, surface excavation to a depth not exceeding 30 cm and averaging 15 cm, in any type of soil for the entire area occupied by the building and structure including plinth protection ramps, steps etc. shall be carried out by the contractor.

(b) **SURFACE DRESSING**: Surface dressing around the entire buildings and structure to a width of 3 Metres beyond the external edge of plinth protection / steps / ramp or external wall as applicable shall be carried out by the contractor as per clause 3.6 and 3.10 of MES Schedule Part-I [2009]. Area around outer edge of building / structure shall be dressed to slope away from the buildings as directed by Engineer –in-Charge.

SCHEDULE 'A' NOTES (Cont'd. /-)

(c) Sanitary fittings, appliances, soil, waste /vent pipes upto first manhole or upto 3m beyond outer edge of plinth protection with connected items of excavation, earthwork, bedding, haunching of pipes, gully and nahani traps with chambers, AC fittings, CI fittings specials, CI pipe 150 mm dia from septic tank to soak well 5.0 meter length etc., complete plumbing work all as per normal Engineering practice, shown on drawings and as specified.

(d) Fan hooks fixtures and fittings, niches for boxes to house MCB's /DB's /main switch boxes etc.

(e) Flushing Cistern, WC /EWC shall be as per standard pattern, Urinal, Ready made PVC Connecting Pipes to Wash Hand Basin, etc., Wash Hand Basin with bottle trap including Mirror.

(f) Nahani trap with stainless steel grating, gully trap with grating, soil, waste and vent pipe, HDPE drain pipe upto and including first manhole and gully trap where bath/WC/toilet are occurring.

(g) Any other work specifically mentioned in notes to other parts of Schedule 'A' and particular specification.

(h) Concealing of conduit and conduit fittings for wiring in roof slab up to the junction of roof and wall.

(j) Cutting chases for concealed conduit wiring and GI (Water supply) tubing where tiles are in involved and making good with cement mortar (1:3).

(k) ANTI-TERMITE TREATMENT : Pre-construction anti-termite treatment shall be carried out to all the buildings. Pre-construction Anti-Termite Treatment shall be carried out as per IS.Contractor has to give the Guarantee for Anti-Termite Treatment for TEN YEARS.

(l) Damp Proof Course (DPC) as specified.

(m) The measures to be adopted for prevention of leakage/seepage and dampness to building in terms of Drawing.

(n) Built in furniture as indicated on drawings and as specified in particular specifications. In case specification of any built in item of furniture shown on main drawings is not given in particular specifications the same shall be as directed by GE.

(o) Numbering of building all as directed by GE/Engineer-in-Charge.

(p) All other items, which are not covered in notes but shown on drawings shall be deemed to be included in unit rate inserted against Schedule 'A' Part I.

(q) Water proofing roof treatment shall be provided to the buildings. Contractor has to give the Guarantee for Water Proofing Treatment over RCC Roof Slabs, sunken slab etc. for TEN YEARS.

(r) PCC Plinth Protection, PCC Plinth Protection with PCC Surface Drain as indicated/shown in approved drawings.

(s) RCC lintels band shall be provided as per provisions of relevant IS.

(t) RCC lintels beam/continuous lintel beam shall be provided over all doors, windows or any openings if required as per site condition and as well as structural purpose.

(u) HDPE water tank including staging all as specified and shown in drawings

(v) All other items, which are not covered in notes but shown on drawings shall be deemed to be included in unit rate inserted against Schedule 'A' Part I.

(w) Brick work as in filler wall of RCC frame work shall be Fly Ash Brick masonry. Fly ash bricks shall be used which is locally available best quality as approved by GE.

3.4.4 In addition to list of TD drawings attached with tender documents, if any other drawing(s) are referred to in these drawings or in the tender documents, but not listed in the list of TD drawings, these drawing shall also form part of the tender documents. The contractor shall be deemed to have taken the details etc, shown in these missing drawing into consideration before quoting the lump sum amount. Nothing extra shall be paid on this account.

4. The method of measurements to be adopted in the Contract shall be as laid down in MES Standard Schedule of Rates (SSR) Part-I & II, **IS-1200 (latest)** and other relevant BIS standard. The General Rules, Special Conditions, preambles and other provisions of these documents shall be deemed to apply to the work under this Contract unless otherwise mentioned in the tender documents.

SCHEDULE 'A' NOTES (Cont'd. /-)

5. Unless otherwise specified, rates tendered by the contractor shall include for all incidental and additional labour to cover all the work enumerated and described under the heading "description of items and preambles to similar items in Standard Schedule of Rates Part II 2020 as applicable. It is an express condition of this tender that the rates quoted by the contractor in tender for various items of work shall be deemed to include for full and entire completion of the items of work in accordance with the provisions of the tender. The Government will not entertain any claim, whatsoever; on account of inaccuracies/ misunderstanding, if any, in the aforesaid rates.

6. Description of items given in Schedule 'A' is in brief. These are deemed to be amplified and read in conjunction with special conditions, particular specification, specification of materials and workmanship and conditions given in the relevant section of SSR 2009 Part-I (specification) and Part-II 2020 (Rates) as applicable and contract drawings referred to against the items of Schedule 'A' and mentioned in list of drawings.

7. The work under this contract shall be executed in Restricted area where working hours and routes are restricted by the authorities controlling the area for security purpose. This being vital factor regarding loss of man-hours etc. shall be considered by the contractor while quoting his rates for this work. Govt shall not entertain any claims whatsoever on this account at subsequent date.

8. Specification in MES Schedule part I (2009) and preamble to items given in MES Schedule part-II rates (2020) under respective trade section shall be applicable. If any provision in the description of items of Schedule 'A' and particular specifications is at variance with the provisions given in specification in MES Schedule part-I (2009) and preambles to MES Schedule part-II rates (2020), provision in particular specifications shall take precedence there over.

9. Probable distributions of various items are indicated on drawing these are tentative and may be varied where necessary at the discretion of GE. The contractor shall not be entitled for any claim on account of such varied quantities, alignment or change in quantities required. No price adjustment shall be done on account of final approved layout within the site plan area.

10. The rates quoted by the contractor shall be deemed to include for all minor extra and constructional details which may not have been specifically mentioned in Schedule 'A' or given in particular specification but are essential for the execution of work and services in a workman like manner, sound construction and established engineering practice. The details of such items not specifically shown on drawings/specified shall be furnished during execution of the work. In case of difference of opinion between the contractor and the GE as to whether or not certain items of work constitute minor extras/minor constructional details which are deemed to have been included in the contractor's rates, the decision of the Accepting Officer shall be final, conclusive and binding.

11. The contractor shall calculate his own details and then insert unit rates in the BOQ.

12. The quantities shown in BOQ are approximate and are inserted as guidance only. These shall, however not be varied beyond the limits laid down in condition 7 of IAFW-2249, General conditions of Contracts forming part of this contracts. The quantities against similar items under various activities have been clubbed and shall be executed under various activities (locations) as detailed in Schedule 'A'.

13. Unit rate quoted by the contractor against various items of Schedule 'A' shall, interalia, be deemed to include for any minor details of construction which are obviously and fairly intended and which may not have been specifically included in the tender documents but which are essential for the execution and completion of works. In case of difference of opinion between the contractor and the GE as to whether or not, any item of work constitutes 'Minor details of construction', the decision of Accepting officer shall be final, conclusive and binding.

SCHEDULE 'A' NOTES (Cont'd. /-)

14. Method of measurement of all works shall be as described as in MES schedule, unless otherwise specified in these documents.

15. Unless otherwise specified, unit rate against each item of work shall be deemed to be inclusive of materials and labour, all plants, equipments and tools including testing for acceptance criteria complete.

16. The locations (s) of concrete batching plant(s) etc shall be as shown on site plan and as directed by the GE. The location for these plants may be changed within the limits of layout plan and the Contractor shall have no extra claims on this account.

17. DEFECTS LIABILITY PERIOD:

17.1 The defect liability period in this case is 3 (THREE) years as mentioned in condition 46 of IAFW-2249 & shall be commenced from the date of completion of the entire work under the Contract.

17.2 Accepting Officer shall convened /order Board of Officers (BOO) with representative of contractor, tentatively four months before expiry of defects liability period (DLP) for inspection of work to check & verify defects if any arose due to unsound, imperfect or unskilled workmanship, due to normal wear & tear, due to the Users negligence or any other reasons. Accepting officer shall take final call on defects noticed by the BOO.

18. **INITIAL MOBILISATION**

(i) Mobilization of serviceable T & P, procurement of minimum 15% quantity of material required for all items of Schedule 'A' (except sealant compound & cement being perishable items). The contractor shall also make arrangements for uninterrupted supply for balance materials required for entire completion of work, to carry out survey, getting Mix Design for PQC/RCC/DLC & WMM from the designated Lab including approval of mix design from the CWE and setting up site lab as specified here-in-after. This activity must be completed within 45 days from the commencement of the work.

(ii) Minimum T & P to be mobilized for the work as given in particular specification.

(iii) Contractor shall arrange T&P other than listed above which are required during execution of work without any extra cost to the Govt so as to ensure timely completion of the work. The contractor shall also make suitable arrangement to ensure uninterrupted execution of work in case of failure of any of above referred T&P. Contractor shall make his own arrangement for safety and security of T & P and material brought at site. Mix design from approved institution(s) as specified shall be submitted within 45 days of commencement of work to GE for taking approval from the CWE.

(iv) The materials and T & P collected shall be entered in the MB as "**NOT TO BE ABSTRACTED**" and shall be duly signed by both the parties

19. Prior to commencement of work, proper and accurate survey of existing surface of the area shall be jointly carried out by MES executives and contractor's representative to take and record existing levels after surface dressing under Sch 'A'. The initial levels shall be recorded at a minimum (2x2) Mtr or below grid interval as specified in Particular Specification with the help of suitable software and instruments like Total Station. All the levels so taken shall be fed into the computer on MS Excel sheet and L- section and cross section will be got drawn by the contractor and signed jointly by the contractor and the GE which shall form basis for fixing up longitudinal and cross profiles and calculation of volumetric quantities. Contractor's quoted rate shall be deemed to include for these provisions and nothing extra shall be payable on these accounts.

20. Unless otherwise indicated in the description of the items, rate quoted by the tenderer shall deemed to include for materials and labour complete and shall also deem to include for corresponding specifications given in particular specifications and MES Schedule Part I-2009 and Part II- 2020.

SCHEDULE 'A' NOTES (Cont'd. /-)

21. Unit rates quoted by the tenderer shall interalia be inclusive of all taxes levied by Central Govt/State Govt/Local Bodies and other levies etc including recovery of income tax, GST on works contract and labour welfare cess which is deductible at source. No claim of whatsoever nature shall be admitted by the Govt on this account.

22. The tenderers are advised to visit the site of work and see the condition of site where the work is to be carried out. To see the samples of various items and material to be incorporated in this work, contractor shall visit the sample room along with AGE / GE before quoting the rates. Please note that any claim whatsoever on account of misunderstanding or non-visit of site/ sample room shall not be entertained.

23. The contract sum shall be arrived at after deducting amount of schedule of credit from the total quoted amount in BOQ.

24. All precautions shall be taken by the contractor in demolition/dismantling of items. Any damage to existing structure/fittings during course of demolition/dismantling shall be made good by the contractor at his own cost and without any extra cost to the Govt.

25. The location of batch mix plants, material storage area, contractor's site office and lab and routes to follow to reach the plant and site shall be as approved by the Station authorities and as directed by the GE. Contractor shall not be allowed to follow any other route. Changes may, however, be made in the route on any later date. Nothing extra shall be paid to the contractor in this regard. The location of plant site from the construction site will be 2 to 4 Km along the approved route.

26. A duly equipped site lab of appropriate size shall be established by the contractor during mobilization phase for conducting tests of site level. The site lab & site office shall always be maintained in a neat & clean condition and the size & quality shall be got approved from GE.

27. The equipment and T&P to be deployed by the contractor shall be in good working condition and Vintage of T&P should be as per table , mentioned here –in- after.

28. TRIAL BAY

28.1 Before commencement of the work, a trial bay of PQC 36 mtr long and 7.5 mtr x 280 mm thick shall be constructed over DLC trial bay of size 36 mtr x 7.5 mtr x 150 mm thick, over WMM trial bay of size 36 mtr x 7.5 mtr x 150 mm thick in two layers of 75 mm each over GSb trial bay of size 36 mtr x 7.5 mtr x 200 mm in single layer over stabilised soil formation trial bay of size 36 mtr x 7.5 mtr x 500 mm thick in four layers not exceeding 150mm each as per the specifications and approved mix designs mentioned for run way pavement. The offsets shall be left uncovered till all required inspections done, test conducted and final clearance given by GE. All offsets in length should be one side. For the purpose of approach ramp, these offsets shall not be disturbed. For this purpose, the other side of trial bay should be utilised and shall be laid by contractor at his own expense at suitable place or as directed by GE. The method of mixing, spreading and compaction shall be done by the contractor as per CA provisions. At least three cores shall be cut and determined the properties/strength which is required as per CA. The construction of trial length will be carried out repeatedly till requirement achieved for satisfactory construction of runway. No additional /extra amount will be paid to the contractor for this work and its cost is deemed to be included in quoted rates/sum by contractor for the work. Testing and acceptance of trial bays for cement concrete pavements will be governed by condition contained in Para 11 of IRC-15-2011 and section 400-600 of "MORTH specifications of roads and bridge works (Fifth Edition)" and references thereof.

28.2 Onus of the delay and cost for relaying & recasting of Trial bays till achievement of required strength & properties as per CA lies with the contractor.

28.3 Specific attention is invited towards methodology for provisioning of top PQC layer (to required width) for new rigid pavement which include sub-base/base layers like excavation, GSB, WMM layer, all these layers are to be constructed 0.75 m extra on either side of final PQC along with main Runway to facilitate movement of pavers and to create working space. Extra width of GSB, WMM layer and DLC layers will be measured and paid in the respective items of BOQ.

SCHEDULE 'A' NOTES (Cont'd. /-)

29. PRESERVATION OF AGGREGATES/ CUBE & CORE SAMPLES.

29.1 Samples of the (i) fine aggregates, coarse aggregates used for the construction of GSB/WMM/DLC/ PQC and RCC work/ items (ii) Concrete cubes & core cutting samples of relevant layers duly marked with chainage & location by properly recording them in the Register shall be preserved by the GE till expiry of Defects Liability Period.

29.2 Record of above-mentioned preserved samples shall be handed over while proceeding on posting by the GE to the successor as part of the Handing over notes.

30. Garrison Engineer will stop execution of work at any stage under the following situation:

30.1 If any of the Tools/ Plants & Machinery deployed for the execution of work found in defective or unserviceable / breakdown condition.

30.2 Any Laboratory equipment & apparatus is found faulty, in defective condition or not calibrated from the NABL accredited laboratory.

30.3 Inclement weather condition at work site.

30.4 Decision of the GE to stop or permit execution of work under above mentioned situation/circumstances shall be final & binding on the contractor.

38. Material discarded at construction joints on commencement of next day work shall not be paid/ considered for payment. Finished work shall be only measured for payment.

39. **CONSTRUCTION OF SITE LAB / OFFICE / WATER AND CURING TANKS / PLATFORMS FOR STACKING OF AGGREGATES:**

Site Laboratory, site office, tank for curing of precast area drain cover slabs & other articles, water storage tank & platforms for stacking of aggregates shall be constructed all as mentioned in the particular specification. Cost of construction of site lab/office/tanks & platform shall be included in quoted rates & nothing extra is payable on this account.

40. **DEMOLITION /DISMANTLING**

40.1 The demolition/dismantling shall be done carefully to the extent required and shall made good as per existing surfaces/finishes without any extra cost.

40.2 All precaution shall be taken by contractor in demolition/dismantling of items. Any damage to the existing structure/fittings/cable & piping etc during course of demolition/ dismantling shall be made good by the contractor at his own cost and without any extra cost to the Govt. All serviceable materials shall be deposited by the Contractor to store yard of GE. Dismantled/ Demolished materials from the site shall be removed only after written permission of the GE. Rates quoted for dismantling/demolition shall include separation of serviceable materials from unserviceable, disposal of unserviceable materials, debris and rubbish at location as directed by Engineer– in- Charge.

40.3 The materials obtained from demolition/dismantling unless specifically catered shall not be used in the work under this contract. These materials shall be removed from the site of work. Proper inventory showing materials handed over to the contractor under schedule of credit shall be held on the record by the GE.

41. **Repair of existing roads used by contractor for plying of vehicle /machineries for execution of work. :-**

41.1 Contractor will be allowed to use existing road inside AF Station for transportation of men /material & machineries. These roads are not meant for frequent heavy traffic. Hence these roads which get damaged shall be repaired by the contractor regularly as & when needed and directed by the Garrison Engineer without any extra cost. Finally, roads used from plant site to construction site are to be perfectly repaired by providing suitable base coarse and wearing coarse to the entire satisfaction of GE. The contractor shall carry out the repairs and renewing of these roads without any extra cost. Quoted rates are inclusive of cost towards repair of road. Nothing extra will be paid by the Dept on this account.

41.2 Area /Portion of damaged road will be jointly surveyed and recorded time to time and after completion of the work by the (i) GE (ii) Rep of the AF station authorities / Users and (iii) Rep of contractor to assess quantum & required scope of repair. Decision of the GE on the damages & requirement of road repair during execution of work & finally after completion of work regarding scope /quantum & method for repair to be followed shall be final & binding on the contractor.

SCHEDULE 'A' NOTES (Cont'd. /-)

41.3 If contractor fails to repair the damaged road then GE shall get it repaired through other Agency & cost of repair will be recovered from the payment due to the contractor.

42 Any Drawing mentioned in reference on various drawings or anywhere on any drawing forming part of the tender document or anywhere in the tender documents but inadvertently not included in the list of drawings, shall be deemed to form part of the tender & shall be followed/referred during execution of respective items (s).

43 Integrity Pact (IP) enclosed hereafter is an integral part of the contract and both parties are bound by its provision.

44 **VISIT TO SITE:** The bidders are advised to visit the work site by taking prior appointment with GE before uploading of bid and BOQ. The bidders shall have no claim whatsoever on this account at a later stage whether he has actually inspected the work site or not.

45 **LAYOUT:** Layout of Building/Structures and Works Services indicated in the SITE PLAN are Tentative and no Adjustment in Price shall be done on account of Final Approved Layout.

46 **SAFETY MEASURES, PRECAUTION'S, RISKS ETC**

46.1.1 The Work shall be carried out with utmost care to ensure that no damage to existing/adjoining work is done failing which the damage, if any done shall be rectified by the contractor to match with the existing/ adjoining work to the entire satisfaction of Engineer-in-Charge under contractor's own arrangement and at his own expenses.

46.1.2 Suitable tools, plants, equipment's, mechanism, etc., as considered necessary shall be adopted during execution of the work. The Contractor shall take all precautions, safety measures, etc., to avoid any damage, mis-happening, accident, etc., to the workmen engaged by him to carryout the work. The lump sum amount quoted by Contractor shall be deemed to have included the element of adopting safety measures, precautions and also the risks, etc. involved in work, except Accepted Risks as covered under the General Conditions of Contract, and nothing EXTRA shall be admissible on this account.

47. **THIRD PARTY TECHNICAL INSPECTION (TPTI)**

The Contractor shall provide all assistance and full access to site to carry out inspection and perform tests at site, to provide samples for testing in outside laboratories and to show site records and their records as asked for by the inspecting teams. Findings of such inspection shall be notified to the contractor and contractor shall be bound to take remedial measures to the satisfaction of GE. In the case of Technical Audit /Third Party technical inspection by an independent agency/individual/firm/institute finding deficiency or quality lapse in construction, consequent upon which there is a recovery from the contractor, recovery shall be made with orders of the GE whose decision shall be final. Third party technical inspection (TPTI) of the work shall be assigned to group of professors of NITs/IITs/Govt Engineering Colleges and CVC approved Institutions by the Department. Frequency of this (TPTI) shall be at least once in a quarter. However, depending on the exigency and importance of a particular stage the GE may call (TPTI) team at any time. The expenditure incurred on such special (TPTI) Call by the GE shall be reimbursed as per actual payment made to the institution carrying TPTI. Testing of materials as suggested by the TPTI shall be bourn by the contractor and shall be deemed to be included in Lump Sum.

48. Draft Star Rate shall be made based on market enquiry through telephonic enquiry/quotations/email/rate lists/internet based sources, material & labour constants available in various civil engineering books and record available in respect of Star Rates approved in the past for similar items of work etc. Contractor may also assist GEs office in preparation of draft Star rate. The Draft Star Rate shall be purely a draft rate and shall not be used for claiming final payment during execution of work. However GE shall allow part payment to the tune of 80% during execution to avoid any financial hardship to contractor. After completion of the item of work involving Star Rate, contractor shall submit the vouchers/literature of product/test certificates (as applicable, decision of GE being final in case of any disagreement) for finalisation of Star Rate. The Star Rate shall be technically checked by DCWE (C)/Director (C) depending upon the financial effect & approved by competent authority within a period of one month from submission of the relevant documents by contractor as mentioned above. The Star Rate as approved by competent authority after technical check by DCWE(C)/Director(C) depending upon the financial effect shall be referred as "the rate decided by GE" under Para 62(G) of IAFW-2249.

(Signature of Contractor)

AAD (Contracts)
For Accepting Officer

BOQ/ Schedule 34 to 95

SCHEDULE 'B'
ISSUE OF MATERIALS TO THE CONTRACTOR (SEE CONDITION 10 OF IAFW 2249)

Ser No	Particulars	Rates at which materials will be issued to the contractor		Place of issue by name	Remarks
		Unit	Rate in Rs		
1.	2	3	4	5	6
--NIL—					

SCHEDULE 'C'
ISSUE OF TOOLS AND PLANTS (OTHER THAN TRANSPORT) WHICH WILL BE HIRED TO THE CONTRACTOR(SEE CONDITION 15, 34 AND 35 OF IAFW-2249)

Srl No	**Quantity**	Particulars	Details of MES crew supplied	Hire charges per unit per working day (Rs)	Stand by charges per unit per off day (Rs)	Place of issue by name	Remarks
1	2	3	4	5	6	7	8
--NIL—							

SCHEDULE 'D'
TRANSPORT TO BE HIRED TO THE CONTRACTOR(SEE CONDITION 16 & 35 OF IAFW-2249)

Srl No	Quantity	Particulars	Rates per unit per working day	Place of issue by name	Remarks
1	2	3	4	5	6
--NIL—					

Signature of Contractor

AAD(Contracts)
for Chief Engineer (AF)

TENDER

To
The President of India

Having examined and perused the following documents:

1. NIT & Instruction to Bidder

2. Particular specifications

3. Special Conditions of the Contract

4. Schedule "A" (BOQ) Notes & BOQ

5. Drawings detailed in the list of drawings.

6. Schedule 'A', 'B', 'C' and 'D' attached herewith.

7. MES Standard Schedule of rates 2009 (Part-I) specifications together with amendment No. 1 to 3 and Part-II 2020 (rates) (hereinafter referred to as the MES Schedule) together with amendment **No 1 to 122** and including Section30 for "Runway & Pavement" containing item Nos. 30001 to 30053 as applicable.

8. General Conditions of Contracts (IAFW-2249) (1989 Print) together with Amendment **Nos.1 to 49** and Errata 1 to 20.

9. WATER: Condition 31 of IAFW-2249: General conditions of contracts. Water will **not be** supplied by MES.

Should this tender be accepted, I/We agree:

(a) that the sum of **Rs 15,00,000/- (Rupees Fifteen Lakh only) forwarded as Earnest Money shall either be retained by the Government on account of the Performance Security or shall be repaid by the Government on receipt of the full amount of Performance Security within the time specified in Condition 19 of IAFW-2249.

(b) To execute all the works referred to in the said documents upon the terms and condition contained or referred to therein and as detailed in General Summary here-in-after and to carry out such deviations as may be ordered vide condition 7 of IAFW-2249 upto a maximum of 10 % (ten percent) and further agree to refer all disputes (in case of disagreement with the decision of competent engineer authority) to Dispute Resolution Board (DRB) as per condition 71 of General Conditions of Contract (IAFW 2249) and further in case of non-acceptance of the decision of DRB, in part/full, to the arbitration by an Arbitral Tribunal as required by Condition 70 to be appointed by the ENGINEER-IN-CHIEF or in his absence the Officer Officiating as ENGINEER-IN-CHIEF or DIRECTOR GENERAL OF WORKS if specifically delegated in writing by the ENGINEER-IN-CHIEF ARMY HEADQUARTERS NEW DELHI whose decision shall be final, conclusive and binding.

** Delete whichever is not applicable

Brought forward TENDERED SUM Rs. _____Rupees

_____

_____

Signature_____ Name (inblock/capital letters) _____in the capacity of _____ duly authorized to sign the tender for and behalf of _____ (in block capital).

Witness_____ Date _____

Address _____ Address: _____

_____ _____

Email Id:_____

Telephone No: _____

ACCEPTANCE

_____alterations have been made in these documents and as evidence that these alterations were made before the execution of the contract agreement; they have been initiated by the contractor and Shri _____. The said officer is hereby authorized to sign and initial on my behalf the documents forming part of this contract.

The above tender was accepted by me on behalf of the president of India for the CONTRACT SUM of Rs _____ (Rupees _____ _____) on the _____ day of _____ signed dated this _____ day of _____

Chief Engineer (AF)
Prayagraj
Accepting Officer
(For and Behalf of the President of India)

GENERAL CONDITIONS OF CONTRACTS (IAFW-2249(1989) PRINT)
FOR
LUM SUMP CONTRACTS IAFW –2159
AND
MEASUREMENT CONTRACTS IAFW -1779 & 1779 A

1. A copy of the General Conditions of Contracts) (IAFW-2249-1989 Print) with Errata 1 to 20 and **Amendments No 1 to 49** has been supplied to me/us and is in my/our possession. I/we have read and understood the provisions contained in the aforesaid General Conditions of Contracts before submission of this tender and I/We agree that I/We shall abide by the terms and conditions therefore, as modified, if any, elsewhere in these tender documents.

2. It is hereby further agreed and declared by me/us that the General Conditions of Contracts (IAFW-2249-1989 Print) including condition 71 and 70 thereof pertaining to settlement of disputes first through Dispute Resolution Board and thereafter by arbitration, containing 34 pages(including this page) with Errata 1 to 20 and Amendment No 1 to 49 shall form part of these tender documents.

3. It is also agreed by me/us that in case of any discrepancy in the interpretation of the Contents between ENGLISH and HINDI VERSION, ENGLISH VERSION shall take precedence over HINDI VERSION.

AAD (Contracts)
For Accepting Officer

Signature of Contractor

SCHEDULE OF MINIMUM WAGES

1. It is hereby agreed by me/us* that the “Schedule of Minimum Fair Wages” (SMFW) as published vide Government of India which specifies the minimum rates of wages for various categories of workmen as applicable on the last due date of receipt of this tender shall form part of the tender documents and is in my/our* possession. I / We* have read and understood the provision contained in the aforesaid Schedule of Minimum Fair Wages before submission of tender.

2. My/Our signature hereunder amounts to my/our having read and understood the provisions contained therein and I/we agree that I/we shall abide by the same and that aforesaid documents form part of this tender.

3. It is also agreed by me/us that although latest notification, as available with the Accepting Officer, has been formed part of the contract, in case the Government has further revised the wages, the latest revised wages for labour shall only be applicable.

4. Schedule of Minimum Fair Wages is not enclosed along with tender but the contractor is deemed to have full knowledge regarding the minimum wages payable to labourers as legally effective on the last date of bid submission end date and his tendered rates shall be deemed to have been based on the same.

5. The minimum wage legally effective referred to above is the minimum wages notified in Gazette / governed by any local regulations or by Central Government, whichever is higher.

Notes:

(a)The documents mentioned above can also be seen in the office of the **Chief Engineer (AF) Prayagraj– 211012** or in any other MES office during working hours.

(b)In case of difference in interpretation due to wordings of English and Hindi versions, the English version will prevail as per Article 348 of Constitution of India.

AAD (Contracts) Signature of Contractor
For Accepting Officer

SPECIAL CONDITION

1. GENERAL : -

1.1 The Special Conditions of Contract which include various deviations from General Conditions of Contracts and certain other Conditions particular to this Contract shall be read in conjunction with the General Conditions of Contract, particular Specifications, Conditions particular to various works & specific scope of each work as specified together with particular Specifications, Drawings etc.

1.2 The term 'General Specifications' referred to hereinbefore, as well as referred to in General Conditions of Contracts (IAFW-2249) shall mean the specifications contained in the MES schedule.

1.3 General Rules, Specifications, Special Conditions and all preambles in the MES Schedule shall be deemed to apply to the works under this contract. In case of any discrepancy, the provisions in these tender documents shall take precedence over the aforesaid provisions in the MES Schedule.

1.4 It shall be deemed that the contractor has satisfied himself as to the nature and location of the work, general and local conditions and particularly those pertaining to transport including restriction of movement of traffic/ vehicles etc., handling, availability and storage of materials, availability of labour, weather conditions at site and general ground/sub soil conditions and the contractor has to quote his rates accordingly and nothing extra shall be payable on any reason whatsoever.

1.5 MES will bear no responsibility for the lack of such knowledge and also the consequences thereof to the contractor. The information and site data shown in the drawings and mentioned in the tender documents are furnished for general information and guidance only. The Engineer-in-Charge in no case shall be held responsible for the accuracy thereof or/and deductions, interpretations or conclusions drawn there from by the contractor and no claim shall be entertained whatsoever on this account, if the site conditions/information is different or otherwise incorrect. It will be presumed that the contractor has satisfied himself for all possible contingencies, situations, bottlenecks and acts of coordination which may be required between the different agencies.

2. INSPECTION OF SITE BY THE TENDERER: -

2.1 The tenderers shall contact concerned GE for the purpose of inspections of site(s) and examination of relevant documents other than those sent herewith, who will extend reasonable facilities for the purpose. The tenderers shall also make themselves familiar with working conditions, accessibility to site (s), availability of materials and other cogent conditions which may affect the entire completion of work under this contract.

2.2 Once the tenderers submit their tender/bid, the tenderers shall be deemed to have visited the site (s) and made themselves familiar with the working conditions etc. irrespective of whether they actually inspect the site or not. No extra payment consequent to any mistake or misunderstanding or otherwise on this account will be allowed.

3 EMPLOYMENTS OF PERSONNEL: -

3.1 The contractor shall employ only Indian National as his representative, servants and workmen and verify their antecedents and loyalty before employing them for work. He shall ensure that no person of doubtful antecedents and nationality is in any way associated with the works. If for reasons of technical collaboration or other consideration the employment of foreign national is unavoidable, the contractor shall furnish the particulars to this effect to the Accepting Officer at the time of submission of the tender. As proof that the contractor has employed only Indian Nationals he shall render a certificate to the GE within one month from the date of acceptance of the tender to this effect. In case the GE desires, contactor will have the police verification done of the personnel employed by him. Contractor's attention is also drawn to condition 25 of IAFW-2249 in this connection.

3.2 The GE shall have full powers and without giving any reason, to require the contractor immediately to cease to employ in connection with this contract, any Agent or servant or employee whose continued employment is, in his opinion, undesirable. The contactor shall not Be allowed any compensation on this account.

SPECIAL CONDITION (CONT'D…../-)

3.3 The contractor's attention is also drawn to condition 25 of IAFW-2249 in this connection.

3.4 **Contractor's Supervision : -**

3.4.1 Contractor shall employ Engineering staff to supervise the execution of work/contract as mentioned in amendment No. 27 of IAFW-2249.

4. **CONDITIONS OF WORKING: -**

4.1 The entire work under this contract lies in "**RESTRICTED AREA**". The restrictions for entry to work site and conditions of working in restricted area shall be as under:

(a) **Entry and Exit: -**

The contractor/his agents/representatives/workmen etc. and his materials, carts, trucks or other means of transport etc will be allowed to enter through and leave from only such gate or gates and at such times as the GE or authorities in charge of the restricted area may at their sole discretion permit to be used. Contractor's authorized representative is required to be present at the places of entry and exit for the purpose of identifying his carts, trucks etc to the person in charge of the security of the restricted area.

(b) **Identity of Workmen: -**

(i) Every workman shall be in possession of an identity card. The identity card shall be issued after a thorough investigation of antecedents of the labourers by the contractor and attested by Officer-in- charge of the unit concerned in accordance with the standing rules and regulations of the unit. Contractor shall be responsible for the conduct of his workmen, agents or representatives.

(ii) **Identity Card or Passes: -**

The contractor, his agents and representatives are required individually to be in possession of an identity card or pass duly issued by station authorities **or** GE as per latest instruction applicable at station. The identity card or pass will be examined by the security staff at the time of entry into or exit from the restricted area and also any time or number of times inside restricted area.

(c) **Search: -**

Thorough search of all persons and transport shall be carried out at each gate and for as many times the gate is used for entry or exit and may also be carried out any number of times at the site within the restricted area.

(d) **Working Hours: -**

The units controlling restricted areas usually work during six days in a week and remain closed on the 7th day. The working hours available to the contractor's labour and staff are however appreciably reduced because of the time of entry and exit during working hours. The exact working hours, working days and number of working days observed for these restricted area(s) where works are to be carried out shall be deemed to have been ascertained by the contractor before submitting his tender. The tenderer's attention is invited to the fact that the number of working hours for a unit are prescribed in regulations and that these cannot be increased by the Garrison Engineer or authorities controlling the restricted area. The definition of "working days" as given under Condition 1(t) of IAFW-2249 does not apply in cases where the works are carried out in restricted areas.

(e) **Working on Holidays: -**

The contractor shall not carry out any work on gazette holidays, weekly holidays and other non-working days except when he is specially authorized in writing to do so by the Garrison Engineer. The GE may at his sole discretion declare any day as holiday or non-working day without assigning any reasons for such declaration.

SPECIAL CONDITION (CONT'D…../-)

(f) **Fire Precautions: -**
The contractor, his agents, representatives, workmen etc shall strictly observe the order pertaining to fire precautions prevailing within the restricted area. Motor transport vehicles, if allowed by the authorities to enter the restricted area, must be fitted with a fire extinguisher in working condition.

(g)Workers/supervisor/contractor and their agents are not allowed to carry smart mobile phone having features like camera, net connection and related apps to the worksite. Smart phones are to be deposited with security personnel's of the unit/station. If any worker/supervisor/contractor and their agent found with smart phones at work site then said phones will be seized and destroyed by the concerned unit/station security authorities.

(h) **FEMALE SEARCHER: -**

If the contractor desires to employ female labour on works to be carried out inside the area of a Factory, Depot, Park, etc and a Female Searcher is not borne on the authorized strength of the Factory, Depot, Park etc. at the time of submission of the tender, he shall be deemed to have allowed in his tender for pay and allowances etc. for a Female Searcher (Group D) calculated for the period female labour is employed by him inside that area. If more than one contractor employs female labour during any month and Female Searcher (s) has/have to be employed in addition to the authorized strength of the Factory, Depot, Park etc the salary and allowances paid to the additional Female Searcher (s) shall be distributed on an equitable basis among the contractors employing female labour taking into consideration the values and periods of completion of their contracts. The GE's decision in regard to the amount recoverable on this account from any contractor shall be final and binding.

(j) **ACCESS TO RESTRICTED AREA AFTER COMPLETION: -**

After the works are completed and surplus stores etc. removed, the contractor, his agents, representatives or workmen etc., will not be allowed any access to the restricted area except for attending to any rectification of defects pointed out to him by the GE.

5. **MINIMUM WAGES PAYABLE**:

(a) Refer condition 58 of IAFW-2249. Govt. of India Min of Labour (Deptt of labour and Employment) vide notification dated have fixed minimum rates of wages as up to date from time to time for various categories of employees under the Minimum Wages Act 1948. The contractor shall not pay wages lower than minimum wages including allowances as fixed under said notification as updated till the date of receipt of tender. Where in any area the minimum rates of wages are also fixed by state Government for employees in employment in the construction/maintenance of buildings/roads/runways, the labour wages as fixed vide above said updated notification of Ministry of labour or respective state Government whichever is higher in respect of these areas shall be deemed minimum rates of wages payable under this contract and the contractor shall not pay lower than this minimum rates of wages.

(b) Contractor's attention is also drawn, amongst other things to the "explanations" to the schedule of minimum wages referred to above.

(c) The fair wages referred to in condition 58 of IAFW-2249 will be deemed to be the same as the minimum wages, referred to above as up to date from time to time.

(d) Schedule of minimum wages has not been enclosed along with tender documents. However, the contractor shall be deemed to have verified the minimum fair wages payable as on the bid submission end date.

(e) The contractor shall have no claim whatsoever, if on account of local factor and/or regulations, he is required to pay the wages in excess of minimum wages as described above during the execution of work.

(f) Wages/salary of employed labourers and supervisor/deployed staff shall be paid by the contractor through Aadhar Linked bank account of individual labour and deployed staff. Relevant details of payment made by the contractor through bank shall be produced for verification of principal employer (Garrison Engineer).

SPECIAL CONDITION (CONT'D…../-)

6. WATER : -

6.1 Refer condition 31 of General Condition of Contract IAFW-2249, clause 1.13 of MES Schedule, Part-I. Water will **not be** supplied by the MES. The tenderers are advised to visit the site of works to ascertain availability of water from civil sources or from nearby natural sources outside MoD land. The contractor shall be allowed, if he so desires, to install hand pumps, tube wells at site of work at places as approved by Engineer-in-Charge and nothing shall be charged from the contractor. The contractor shall remove the hand pumps, tube wells as and when asked to do so by Engineer-in-Charge/GE and in any case on completion of the work and before issue of completion certificate, unless GE desires that these hand pumps, tube wells be left in position and the contractor agrees to do so without claiming cost thereof from department. No compensation whatsoever shall be admissible to the contractor, if he is required to remove the pumps(s), tube wells before completion of work. Use of water from such sources shall be permitted only if found potable after testing and fit for use in the work. The water from such sources shall be got tested by the contractor from a laboratory approved by the GE, who shall, after satisfying himself, permit the contractor to use the water from such sources. Testing charges shall be borne by the contractor. GE shall also carry out the independent testing of water.

6.2 The contractor, if he so desires, will be allowed to augment the departmental water supply by boring wells at his own cost at site(s) approved by the GE.On completion of the work the contractor shall remove pipes etc. installed and make good the site as directed by the Engineer-in-Charge without any extra cost to Govt. The contractor shall be given no rebate for augmentation of water supply and charges for water shall be recovered at the rate mentioned hereinbefore.

7. **CO-ORDINATION WITH OTHER AGENCIES: -**

The contractor shall permit free access and afford normal facilities and usual convenience to other agencies or Departmental workmen to carry out connected works or services under separate arrangements. The contractor will not be allowed any extra payment on this account and no compensation shall be admissible to the contractor on this account.

8. **ELECTRIC SUPPLY**:

(a) In case the contractor desires to buy electricity from the MES, he shall be charged for electric energy consumed in the following rates:

(i) At Rs 11.60 per unit for the light.
(ii) At Rs 11.60 per unit for power.

(b) Electric supply required for the work shall be made available by the MES at the point(s) marked on the layout plan. The main switch and energy meter to register the electric energy supplied shall be provided and installed by the MES. The contractor shall provide all necessary cable, fittings etc. from the main switches in order to ensure a proper and suitable supply of electricity for the execution of work. The contractor shall also provide necessary earthing keeping in view the security requirements. MES does not guarantee for the continuity of supply and no compensation what so ever shall be allowed for supply becoming intermittent or for breakdown in the system.

(c) The actual position of supply point(s) for electric energy will be ascertained by the contractor form the GE before submitting his tender. Irrespective of whether he actually inspects or not, he shall be deemed to have inspected and seen the supply point (s) in consultation with GE before submitting his tender. No claim, what so ever, shall be entertained on this account.

(d) GE or his representative shall be free to inspect all the power consuming devices or any electric lines provided by the contractor. Any device or electric lines provided by the contractor, which is not to the satisfaction of the GE, shall be discontinued from the supply if so desired by him.

SPECIAL CONDITION (CONT'D…../-)

9. **ADOPTION OF TIME AND PROGRESS USING CRITICAL PATH METHOD**

(a) The Time and Progress Chart to be prepared as per condition 11 of General Conditions of Contracts (IAFW-2249) shall consist of detailed network analysis and a time schedule. The critical path network will be drawn jointly by the GE and the contractor soon after acceptance of the tender. The time scheduling of the activities will be done by the contractor so as to finish the work within the stipulated time. On completion, a firm calendar date schedule will be prepared and submitted by the contractor to the GE who will approve it after due scrutiny. Four copies of the schedule will be submitted within two weeks from the date of handing over the site.

(b) During the currency of the work the contractor is expected to adhere to the time schedule and this adherence will be a part of the contractor's performance under the contract. During the execution of the work the contractor is expected to participate in the reviews and updating of the network undertaken by the GE. These reviews may be undertaken at the discretion of the GE either as a periodic appraisal measure or when the quantum of work ordered on the contractor is substantially changed through deviation orders or amendments. Any revision of the schedule as a result of the review will be submitted by the contractor to the GE within a week who will approve it after due scrutiny. The contractor will adhere to the revised schedule thereafter. In case of a contractor disagreeing with the revised schedule, the same will be referred to the Accepting Officer whose decision will be final, conclusive and binding. GE's approval to the revised schedule resulting in a completion date beyond the stipulated date of completion shall not automatically amount to a grant of extension of time. Extension of time shall be considered and decided by the appropriate authority mentioned in condition 11 of IAFW-2249 and separately regulated.

(c) The contractor shall mobilize and employ sufficient resources to achieve the detailed schedule within the broad framework of the accepted methods of working and safety.

(d) No additional payment will be made to the contractor for any multiple shift work or other incentive methods contemplated by him in his work schedule even though the time schedule is approved by the Department.

10. **Materials and Samples: -**

(a) Refer to Condition of 10 of IAFW-2249.

(b) The materials and articles, which have been specified from certain makers/manufacturers, shall be of makes/manufactures as specified. If the manufacturers specified in tender documents make both ISI marked and conforming to ISI, the materials/articles shall be ISI marked.

(c) The materials and articles, which have not been specified in tender documents by makes/manufacturers, shall be as under:
(i) If ISI marked materials are being manufactured, the same shall be ISI marked. For a list of ISI marked manufacturers refer to the website of BIS iewww.bis.org.in.
(ii) If ISI marked materials are not being manufactured, the same shall be conforming to IS Specifications.

(d) Materials of local origin shall be as specified and conforming to samples kept in GE's office. The tenderer is advised to inspect samples of materials which are displayed in the office of GE before submitting his tender. The tenderer shall be deemed to have inspected the samples and satisfy himself as to the nature and quality of materials he is required to incorporate in the work irrespective of whether he has actually inspected them or not.

(e) The contractor shall not procure materials and articles unless the samples are first got approved by the GE.

11. **TESTING OF MATERIAL: -**

11.1 The materials to be incorporated in the work shall be tested at a frequency as laid down in Indian standard/IRC Standard/ Standard as specified in the contract. The materials listed in **Annexure 'A'** shall be tested as per the frequency indicated therein.

SPECIAL CONDITION (CONT'D…../-)

11.2 If facility for testing of building materials for any particular test in not available in the site/ Zonal Laboratory/Command Test Lab (CTL), the same will be got tested in National Test House/SEMT WING/ NIT/ Engineering collage at discretion of Garrison Engineer, all expenses for testing shall be borne by the contractor.

11.3 Level of testing shown in legend as A, B & C are defined as under:

11.3.1 **LEVEL 'A': SITE LAB: -**

(i) "Site Lab" means own site lab established by contractor at the work site for such tests. This lab shall house all the facilities including T & P, machinery, equipment, manpower etc, required for conducting tests. A list of suggested essential equipments required for site lab is enclosed as per **Annexure 'B'**. Competent technical representative as approved by the GE shall be employed by the contractorto man the laboratory. This lab shall be operative for the entire duration of the contract till its completion. Tests shall be carried out in the presence of Engineer-in- Charge to be nominated by GE or any other departmental official to be nominated by the GE. Random check of compliance of frequency of testing shall be done by GE.Setting up site laboratory is mandatory for all works costing Rs.1 crore (Rupees one crore) and above.

(ii) Record shall be maintained at work site. These test result shall be signed bycontractor or his authorized representative and aforesaid departmental official.

(iii) Within 15 days of placement of work order No.1, site lab shall be established and facts reported by the contractor to GE in writing who will verify the fact and satisfy himself of the facilities provided. Thereafter, GE shall issue a certificate to this effect in writing listing outequipment particulars etc, of each material test. Only after issue of this certificate by GE, the tests shall be carried out and materials so approved shall be incorporated in the work.

(iv) Manpower, material and infrastructure like electricity, water etc, required for conducting these tests shall be provided by the contractor.

(v) Tenderer is deemed to cater for above provisions in his quoted lump sum.

(vi) Remedial measures, if any, required to achieve/obtain desired results for each tests shall be taken promptly by contractor. Lump sum is deemed to be included for this eventuality and nothing extra shall be payable to the contractor. No extension of time shall be admissible on this account.

(vii) No recovery shall be made for the tests carried out in Site Lab.

(viii) However, rates per test given are applicable for recovery in case of unavoidable circumstances where some tests and laid down could not be done and in the opinion of the GE non performances of tests do not affect quality control. However, in case GE in his opinion considers that contractor is purposely not adhering to laid down frequencies of tests, he shall reserve the right to get it tested in Zonal lab or any other lab as deemed fit and make penal recovery form RAR which shall be double the rate of testing charges indicated or testing charges actually paid to lab whichever is higher. GE's decision, in this regard, shall be final and binding.

11.3.2 **LEVEL 'B': MES LAB/ZONAL LAB/COMMAND TEST LAB (CTL): -**

(i) Zonal lab/Command Test Lab means any lab of MES department.

(ii) The tests shall be conducted as per frequencies laid down for these tests in these labs for which contractor shall provide all requisite facilities like samples, cubes, material etc, transportation to these labs for testing purpose. It will be contractor's responsibility to adhere to the laid down frequency of testing. Test results shall be sent by lab to the GE whose copies can be made by contractor at his own expense. Testing charges for the tests so conducted shall be recovered at the rates indicated from the running payments. The contractor's quoted lumpsum is deemed to be included for above provisions.

(iii) Provision of para 11.3.1(viii) of level 'A' shall be applicable to level 'B' also.

11.3.3 **Level 'C': NABL Accredited Lab (National Test House / SEMT Wing Pune / Govt Engg Colleges) : -**

SPECIAL CONDITION (CONT'D…../-)

(i) Level 'C' lab stands for NABL Accredited Lab which may be National Test House / SEMT Wing Pune located in CME Pune/ Govt Engg Colleges where such facilities exist.

(ii) Test provision contained in para 11.3.2(ii) of level 'B' above shall be applicable here except that contractor shall make necessary arrangement for transportation etc, to hand over the samples to these labs. Test results shall be forwarded to GE by these labs directly. The testing charges payable to these labs for conducting these tests shall be borne by the contractor and his quoted lump sum is deemed to include this provision.

11.4 In case the contractor has not set up the site laboratory and the tests are carried out in Zonal or any other laboratory, the recovery shall be made at the applicable rates indicated in **Annexure 'A'**.

11.5 In case non availability of testing facilities in MES lab. The tests shall be conducted in any outside Govt. approved labs. Testing charges of materials/cubes carried out in approved laboratory shall be as per actual and shall be directly borne by the contractor.

12. **LABOUR REGULATION AND ABOLITION ACT:-**

Contract labour (Regulation and Abolition) Act 1970 is applicable to MES contractors. Rates quoted by the contractor shall be deemed to take into account the cost/extra expenditure etc required to comply with the provisions contained in the said Act and the rules framed regarding labour from time to time by Central/State Govt.

13. **SECURITY OF CLASSIFIED DOCUMENTS: -**

Contractor's special attention is drawn to Conditions 2A and 3 of General Conditions of Contracts (IAFW-2249). The Contractor shall not communicate any classified information regarding works either to sub-contractors or others without prior approval of the Engineer-in-Charge. The Contractor shall also not make copies of the design/drawings and other documents furnished to him in respect of works and he shall return all documents on completion of the works or on earlier determination of the Contract. The Contractor shall along with final bill attach a receipt from the Engineer-in-Charge in respect of his having returned the classified documents as per Condition 3 of General Conditions of Contracts (IAFW-2249).

14. **OFFICIAL SECRET ACT: -**

The contractor shall be bound by the Indian Official secret Act 1923, particularly section 5 thereof, all as per condition 2A of IAFW-2249.

15. **STORAGE OF MATERIALS: -**

The contractor shall not be provided any storage accommodation at the site by MES. The contractor shall make his own arrangement as per condition 10 of IAFW-2249 and as directed by the Engineer-in-Charge.

16. **DAMAGE TO EXISTING STRUCTURES: -**

Any damage to the existing structures, any existing road etc., during the execution of work shall be made good by the contractor at his own expense and the site work left clean and tidy on completion Rectification or reinstatement, replacement, making good and touching up etc. shall be carried out to conform to the standard of materials and workmanship originally provided in the work and finished work shall be matched with existing works in all respects to the entire satisfaction of GE. In case of any dispute on this account, the matter shall be referred to the CWE whose decision in writing shall be final and binding.

SPECIAL CONDITION (CONT'D…../-)

17. **QUARRIES AND ROYALTIES: -**

Quarries are not available on land which is in the charge of MES authorities, Condition 14 of IAFW-2249 shall be treated as not applicable.

18. **APPROACHES: -**

The contractors shall make arrangements for and provide at his own cost all temporary approaches, if required to the site(s), after obtaining approval in writing of the GE to the layout of such approaches.

19. **ACTION WHERE NO SPECIFICATIONS EXIST: -**

(a) In case any class of work for which there is no such specifications in the tender documents including SSR, such work shall be carried out in accordance with Indian Standard Specifications and where Indian Standard Specifications do not cover the same, the work shall be carried out as per standard engineering practice subject to the approval of the GE.

(b) No claim what-so-ever will be tenable for any misunderstanding or for not having ascertaining requisite information /knowledge to muster the resources required in satisfactory completion of the entire work to requirements as per contract agreement.

20. **METHOD OF MEASUREMENT: -**

Measurement of all measurable work executed under this contract shall be taken in accordance with the rules of measurements as laid down in SSR 2009 Part-I and SSR 2020 Part-II pertaining to various items of work as applicable, unless otherwise indicated in the respective sections of schedule "A". In case the method of measurements for a particular item of work is not given in the SSR, a reference will be made to the relevant provision contained in the latest edition of IS-1200.

21. **SECURITY AGAINST LOSS OR DAMAGE: -**

(a) Contractor's plants/equipments at site :

The contractor shall furnish to the Engineer-in-Charge every morning a distribution return of his plants/equipments on the site of the work, stating the following particulars:

i. Particulars of plants/equipments, its make, manufacturer No, Model No if any, Registration No if any, capacity year of manufacture and year of purchase etc.

ii. Total No (Qty) on site of work.

iii. Location and indication of No (Qty) at each location of the site of work.

iv. Purchase value on the date of purchase.For the purpose of this condition, plant/equipment shall include vehicles i.e. trucks and lorries but neither the workman's tools nor any manually operated tools/equipment. The Engineer-in-Charge shall record the particulars supplied by the contractor in the works dairy and send monthly return to the GE for record in his office.

(b) Loss on account of Enemy Action :

If as a result of enemy action the contractor suffers any loss or damage, the Govt shall reimburse to the contractor such loss or damage to the extent and in the manner provided here-in-after:

i. The loss suffered by him on account of any damage or destruction of his plant / equipment (as defined in special condition (a) above), material or any part or parts thereof the amount of loss assessed by the Accepting Officer of the contract on this account shall be final, binding and conclusive.

ii. The compensation paid by him under any law for the time being in force, to any workmen employed by him for any injury caused to workmen or workman's legal successor for loss of the workmen's life.

iii. Payment of compensation for loss or damage to any work of part of work carried out. The amount of compensation shall be determined in accordance with condition 48 of General Conditions of Contracts (IAFW-2249).

SPECIAL CONDITION (CONT'D…../-)

iv. No reimbursement shall be made nor shall any compensation be payable under the above provision unless the contractor had taken Air Defence precautions ordered by the GE/or concerned, or in the absence of such order, reasonable precautions taken by the contractor. No reimbursement shall be made nor shall any compensation be payable for any plants/equipment or materials not lying at site of work at the time enemy action.

22. **FORCE MAJEURE: -**

Should any force majeure circumstances arise, each of the contracting party shall be executed for the no fulfillment or for the delayed fulfillment of any of its contractual obligations, if the affected part within 15 days of its occurrences informs the other party in writing.

Force majeure shall mean fires, floods, natural calamities or other acts such as war, turmoil, strikes (as no limited to be establishment of the spelled), sabotage, explosions, quarantine restrictions beyond the control of either party.

It is understood and agreed between the parties hereto that the rights and obligations of the parties shall be deemed to be in suspension during the continuance of the force majeure event as aforesaid and the said rights and obligations shall automatically revive upon the cessation of intervening force majeure event. The period within which the rights & obligations of the parties shall be in suspension due to the force majeure event shall not be considered as a delay with respect to the period of delivery and /or acceptance of delivery under the contract or otherwise the detriment of either party.

Notwithstanding the provisions of the immediately foregoing clauses it is further understood and agreed between the parties hereto that in the event of any force majeure persisting for an uninterrupted period exceeding 6 (Six) months, either party hereto reserves the right to terminate this contact upon giving prior written notice of 30 (Thirty) days to the other party of the intention to terminate without any liability other than reimbursement on the terms provided in this agreement of the goods received.

23. **VALIDITY OF TENDER: -**

The tender shall remain open for acceptance for a period of **90 days** from the next day subsequent to the bid submission end date.

24. **SAMPLE OF MATERIALS/ARTICLES: -**

(a) Materials/Articles provided by the contractor for in-corporation in the works shall unless otherwise specified in the particular specifications, comply with the requirements of the relevant Indian standards (IS) of the year of publication/ edition specified in the MES Schedule Part-I and shall have IS certification marking.

(b) Sample of all materials shall be got approved by the contractor from GE before incorporation in the work.

(c) In case of material/ articles for which Indian standards are not finalized/ issued and therefore not include in the MES schedule contractor shall produce the samples and get them approved by the GE in writing before incorporation in the works.

(d) The contractor is advised to inspect the samples of local building materials, e.g bricks, stone aggregate, sand etc and other materials maintained by the CWE/GE before submitting his tender. The contractor will be deemed to have inspected those samples and have full knowledge thereof, whether he inspects them or not. The materials to be provided by the contractor shall conform to or be superior to the sample.

(e) The selection and use of sources for materials shall be subject to the compliance of these specifications read in conjunction with MES schedule and relevant Indian Standard Specifications.

25. **RECORD OF CONSUMPTION OF CEMENT: -**

(a) For the purpose of keeping record of cement issued and consumed in works one properly bound register serially numbered and all pages initialed against the numbering by the Engineer-in-Charge and in the form approved by Engineer-in-Charge showing daily receipt quantity used in the work, balance in hand, shall be maintained.

SPECIAL CONDITION (CONT'D…../-)

(b) The register shall be kept at site in the safe custody of the contractor's representative during the progress of the work and shall be produced for verification of the Inspecting officer on demand.

(c) The entries in the aforesaid register will be signed daily by the contractor or his authorized representative and the JE for supervision and once in a week by Engineer-in-charge.

(d) On completion of the work, the contractor shall deposit the cement register with Engineer-in-Charge for record.

26. **WATCH/LIGHTING: -**

The contractor shall, at his own cost, take all possible precautions to ensure safety of life and property by providing necessary fencing, barrier, light, watchmen etc during the progress of work and as directed by the Engineer-in-Charge.

27. **MAKING GOOD: -**

The contractor shall leave or form holes of concrete (Plain or RCC), brick work, stone masonry and in any other situation as required for the work or as directed by Engineer-in-charge and make good all the holes in the same mortar and mix as specified for that portion of work.

28. **CLEANING DOWN: -**

(a) Refer Condition 49 of IAFW-2249. The contractor shall clean all floors, remove cement, lime or paint drops, clean joinery, glass panels etc, touch up all paint work and carry out all necessary items of work in connection therewith and have the whole premises clean and tidy to the entire satisfaction of Engineer-in-Charge before handing over the items/works. No extra payment shall be admissible to the contractor for his operation.

(b) The contractor shall remove from the site all unused stores and huts and the like belonging to the contractor provided for the execution of work under this contract and the site of works shall be cleared off all rubbish and waste materials by the contractor and the site of work delivered in clean and tidy manner to the satisfaction of the Engineer-in-charge on or before the date of completion.

29. **CONSTRUCTION LABOUR WELFARE CESS/TAX: -**

Consequent upon promulgation of ordinance by the President of India, the contractor is liable to pay element for construction labour welfare CESS/Tax. The rates quoted by the contractor shall be deemed to be inclusive of this CESS/TAX.

30. **PRICING OF DEVIATIONS: -**
The principle enumerated in condition 62 of IAFW-2249 shall apply to this contract.

31. **Reimbursement / Refund on Variation in Taxes Directly Related to Contract Value: -**

(a) The rates quoted by the Contractor shall be deemed to be inclusive of all taxes (including GST on materials, GST on Works Contracts, Labour welfare cess/tax etc.), duties, Royalties, Octroi & other levies payable under the respective statutes.

No reimbursement/refund for variation in rates of taxes, duties, Royalties, Octroi & other levies and/or imposition/abolition of any new/existing taxes, duties, Royalties, Octroi & other levies shall be made except as provided in sub para (b)(i) here-in-below.

(b) (i) The taxes which are levied by Government at certain percentage rates of Contract Sum/Amount shall be termed as "taxes directly related to Contract value" such as GST on materials, GST on Works Contracts, Labour welfare cess/tax and the like but excluding Income tax. The tendered rates shall be deemed to be inclusive of all "taxes directly related to Contract value" with existing percentage rates prevailing on the bid submission end date. Any increase in percentage rates of "taxes directly related to Contract value' with reference to prevailing rates on the bid submission end date shall be reimbursed to the Contractor and any decrease in percentage rates of "taxes directly related to Contract value" with reference to prevailing rates on the bid submission end date shall be refunded by the Contractor to the Govt/deducted by the Government from any payments due to the Contractor. Similarly, imposition of any new "taxes directly related to Contract value" after the bid submission end date shall be reimbursed to the contractor and abolition of any "taxes directly related to Contract value" prevailing on the bid submission end date

shall be refunded by the Contractor to the Govt/deducted by the Government from the payments due to the Contractor.

(ii) The Contractor shall within a reasonable time of his becoming aware of variation in percentage rates and/or imposition of any further "taxes directly related to Contract value" give written notice thereof to the GE stating that the same is given pursuant to this Special Condition, together with all information relating thereto which he may be in a position to supply. The Contractors shall also submit documentary proof/information as the GE may require.

(iii) The Contractor shall, for the purpose of this condition, keep such books of account and other documents as are necessary and shall allow inspection of the same by a duly authorized representative of Government, and shall further, at the request of the GE furnish, verified in such a manner as the GE may require any documents so kept and such other information as the GE may require.

(iv) Reimbursement for increase in percentage rates/imposition of "taxes directly related to contract value" shall be made only if Contractor necessarily and properly pays additional "taxes directly related to contract value" to the Government, without getting the same adjusted against any other tax liability or without getting the same refunded from the concerned Government Authority and submits documentary proof for the same as the GE may require.

32. **Goods & Services Tax (GST) & Labour Welfare Tax** shall be deducted by the disbursing officer at the prevailing rates from gross amount payable to the contractor. Contractors quoted rates shall include taxes to be paid/ deducted at source by the disbursing officer. Rate of recovery of taxes shall be all as per Govt notification /SRO as applicable on last date of receipt of tender. Nothing extra is payable on this account.

32.1 Rate of Tax to be paid/recovered at the source under Goods & Services Tax (GST) shall be the percentage on gross value of contract as prescribed by the Central Board of Excise & Custom authorities.

32.2 Contractor shall get registered with the Central Board of Excise & Custom for allotment of ARIN / ID / GSTIN. Contractor shall file monthly/ quarterly returns for GST as per instructions in the matter from the concerned authorities.

33. **ENFORCEMENT OF EMPLOYEES PROVIDENT FUND AND MISCELLANEOUS PROVISIONS (EPF & MP) ACT 1952: -**

(a) Contractor shall get registered with employees' provident fund organization. PF organization will allot **Provident Fund Code Number** to the contracting firm. Scanned copy of letter of allotment of Provident Fund Code number, (showing allotted code number) issued by the EPF organization shall be uploaded by the contractor in cover 1 (T bid) along with scanned copy of DD towards tender cost and application letter. Those bidders not registered with PF organization and not upload above scanned copy of letter of allotment of PF code number in cover 1 (T bid), they shall be disqualified during evaluation of T bid and consequently their financial bid (cover2) shall not be opened.

(b) All workers/tradesmen deployed by the contractor under subject contract shall be enrolled as members of Provident Fund and shall be given Universal Account Number (UAN).

(c) While submitting payment under RAR and final bill contractor shall render relevant certificate duly supported with challans to Engineer-in-Charge/Garrison Engineer giving confirmation that all workers employed directly/indirectly by him under subject contract are enrolled for EPF and contribution toward EPF credited till date into their respective accounts (UAN).

(d) Failure by the contractor of compliance of EPF and MP Act shall be an offence and attract penalty from the competent authorities in the matter.

34. **QUALIFIED TRADESMEN**: -

In compliance with Condition 26 of IAFW-2249 (General Conditions of Contracts), the contractor shall employ skilled /semi-skilled tradesmen who are qualified and possessing certificate in particular trade from Industrial Training Institute (ITI)/National Institute Of Construction Management and Research (NICMAR)/National Academy of Construction (NAC) Hyderabad and similar reputed and recognized Institutes by State/Central Government, to execute the works of their respective trades. The number of such qualified tradesmen shall not be less than 25% of total

skilled/semi-skilled tradesmen required in each trade. The contractor shall submit a list of such tradesmen along with requisite certificates to GE for verification and approval. Notwithstanding the approval of such tradesmen by GE, if the tradesmen are found to have inadequate skill to execute the work of their trades leading to unsatisfactory workmanship, the contractor shall remove such tradesmen within a week after written notice to this effect by the GE and shall engage other qualified tradesmen after prior approval of GE. GE's decision whether a particular tradesman possesses requisite qualification, skill and expertise commensurate with the nature of work, shall be final and binding. No compensation whatsoever on this account shall be admissible.

35. **BLASTING**: - Blasting in any form is prohibited.

36. **SITE DOCUMENTS: -**

In addition to the site documents which are normally maintained for works, the following additional documents (as applicable for this work) shall be maintained at site and signed by GE / his authorized representative and the Contractor:

i. Sieve analysis for coarse / fine aggregate
ii. Water absorption test for aggregates of different sizes/ grading.
iii. Stage passing register
iv. Daily works diary register
v. Register of materials daily arrived at site
vi. Materials approved register
vii. Cement consumption register
viii. Register for clay, silt and impurities in aggregate test.

37. **Defect Liability Period: -**

Refer amendment No. 49 to the GCC (IAFW-2249) defect liability period for the subject contract shall be **Thirty -Six (36) calendar months** from the date of handing over of the work by the contractor to the Government/MES.

38. **Location of Buildings and Works: -**

There may be some changes in location/siting of buildings shown in the site (layout) plan(s) to suit local conditions and/or departmental requirements. The contractor shall have no claim what-so-ever consequent to such changes in the location/siting of works.

39. SPECIAL CONDITION FOR DISPUTE RESOLUTION BOARD (DRB): -

39.1 During execution of the works or after completion or after determination / cancellation/ termination of the contract, all disputes between the parties to contract arising out of the contract (except those for which decision of Accepting Officer or any other officer (CWE and/or GE) is expressed to be final and binding), including any disagreement by either party with any action, inaction, opinion, instruction, certificate or valuation by the Accepting Officer or his nominee, the matter in dispute shall, in the first place be referred to the Dispute Resolution Board (DRB). In case of disagreement with the decision of such DRB, any party may invoke arbitration as per contract conditions.

39.2 The constitution of the DRB shall be a three member body as under :-

(i) **Chairman:** Jt DG (C) of the concerned Command Chief Engineer/ADG. Where Jt DG (C) is not posted in the Command/ADG office, any other Chief Engineer/ Brig level Officer posted in CE Command/ADG shall be nominated by Command CE at his sole discretion.219

(ii) **Member 1** : Col/Director level officers of command/ADG office or any other Zonal CE to be nominated by Command CE/ADG.

(iii) **Member 2** : Col/Director level officers of command/ADG office or any other Zonal CE to be nominated by Command CE/ADG.

SPECIAL CONDITION (CONT'D…../-)

39.3 The name of chairman and members shall be notified by the Accepting Officer within one month of the date of acceptance of contract.

39.4 Once the DRB is constituted the members and chairman shall disclose in writing their neutrality and impartiality about any personal interest in the work.

39.5 The dispute shall be referred to the chairman of the DRB by the concerned party after giving notice to the other party for invoking of this clause.

39.6 The DRB shall decide the dispute in accordance with the terms of the contract, principle of natural justice, equity and fair play.

39.7 The DRB may fix oral hearing at a place, date and time as decided by the Chairman.

39.8 The requisite administrative support to the DRB shall be provided by the Accepting Officer.

39.9 All the contract documents pertaining to the case shall be provided by the Accepting Officer for reference by the DRB.

39.10 DRB shall give its decision on the disputes within three months of notice from any party invoking the DRB clause. This period can be extended by one month with the consent of the parties.

39.11 All the decisions given by the DRB shall be by majority and such decisions shall be communicated in writing by the Chairman to the parties.

39.12 If the decision of the DRB is not to the satisfaction of either party or if the DRB fails to give decision within the laid down time, either party shall indicate his reservations to the other party within 30 days of such decision and to refer that dispute for arbitration within the provisions of Condition 70 of IAFW-2249 General Conditions of Contract.

39.13 It shall be mandatory for the party invoking arbitration on any particular dispute to have first exhausted the remedy provided under the DRB clause for that particular dispute.

39.14 The mandate of the DRB shall terminate on completion of one year from the date of completion/ determination/cancellation/termination of the contract.

39.15 If any member or Chairman of the DRB is unable to function due to any reason whatsoever, or he resigns his appointment, Chief Engineer Command/ADG as the case may be, shall fill the vacancy so caused within 15 days of the occurrence of such vacancy.

39.16 Any dispute referred to the DRB and having been decided by the DRB and not objected to by either party within 30 days shall attain finality and shall not be referable to arbitration.220

39.17 Accepting Officer shall ensure implementation of the decisions of the DRB which attain finality, ie except those which are objected to by him or by contractor within 30 days as per Para 12 above. The amount so decided shall be processed as per AIP and the authority be mentioned as 'decision of DRB'.

39.18 Findings and decision of DRB shall be admissible as evidence, to the extent permissible as per law, in the subsequent arbitration and/or litigation.

39.19 DRB Chairman/members shall not, in any case, be liable to be called as witness or to produce any evidence in any Arbitration or departmental proceedings of any kind.

39.20 During execution of work the disputes may be referred to the DRB as per the requirement of each party after having exhausted the decision making process provided in the contract. In case of completion of work or after determination/cancellation/termination of the contract all the disputes including payment/ nonpayment/delay in final bill shall be simultaneously referred to the DRB within six months of completion/determination/ cancellation/termination of the contract.

40. **ADVANCES ON ACCOUNT AGAINST MATERIALS: -**

40.1 Refer Condition 64 of IAFW-2249 Advances on Account. Add the following in continuation of Para 8. 1.

SPECIAL CONDITION (CONT'D…../-)

40.2 Provided further, the Contractor may be paid advance on account to the full value of the under mentioned materials only, brought on the site, on his furnishing Guarantee Bond (s) from a Schedule Bank for the amount of the retention money which should otherwise be recoverable from him under contract:

(a) Factory made door shutters
(b) Factory made steel windows/Vents.
(c) Sanitary fittings.
(d) Iron Mongery
(e) Electrical Equipments/fittings/fixtures
(f) Water supply pipes and pipe fittings/fixtures
(g) Steel
(h) Cement
(j) SGSW Pipes and Reinforced cement concrete pipes
(n) Any non-perishable materials which do not lose its identity, at the discretion of GE

40.3 The Bank Guarantee Bond (s) shall be executed for a period and on a form as directed by the Accepting Officer. The Contractor shall further arrange to extend the period of Guarantee Bond(s) if and when necessary, as directed by the Accepting Officer or shall furnish fresh Guarantee Bond(s) of similar value in lieu".

40.4 **PROPRIETORY MATERIAL/ ARTICLES**.

(i) The proprietary materials & articles such as cement , steel ,bitumen, HDPE sheet , polyester synthetic fibre , road marking paint, spray curing compound , joint filler board, joint sealing compound & primer , electrical cable & jointing kits , taxiway edge light, RCC hume pipe , Transformer housing boxes, PVC / HDPE pipes and fittings etc when brought to site shall be inspected and approved by the GE.

(ii) The quantity of proprietary articles brought to site shall be recorded in measurement book (IAFW-2261) and signed by the contractor and the Engineer-in-Charge as a check to ensure that required quantity has been brought to site for incorporation in the works.

(iii) The proprietary materials shall be stored as directed. Those already recorded in the measurement book shall be marked suitably for identification.

(iv) The contractor, shall on demand, produce to the GE, original receipt vouchers/ invoices in respect of the supplies, to ensure that the contractor has actually brought the articles of required quality and quantity from the authorised agent/manufacturer/ supplier and also to find out the rates thereof.

(v) Purchase voucher of following proprietary and branded materials from manufacturer/dealer shall be produced by the contractor to Engineer-in-Charge. Defaced copies of voucher shall be held on record by the Engineer-in-Charge. Details of vouchers of proprietary material shall be entered in MB.
(a) HDPE film
(b) Polyester synthetic fibre
(c) Road marking paint
(d) Spray curing compound
(e) Joint filler board
(f) Joint sealing compound & primer
(g) Electrical cable & jointing kits
(h) Taxiway edge light
(i) RCC hume pipe
(j) Transformer housing boxes
(k) PVC / HDPE pipes and fittings
(l) Reinforcement bars
(m)Cement
(n) Geo fabric textiles

The Vouchers/ Invoices shall clearly indicate the IS No & specific alternative to which the material conform in case of various alternatives given in IS.

41. RECORD OF MATERIALS AND PURCHASE VOUCHERS: -

SPECIAL CONDITION (CONT'D…../-)

41.1 The quantity of materials such as cement, steel, paints, water proofing compound, chemicals for anti-termite treatment and the like, as directed by Engineer-in-Charge (the quantity of which cannot be checked after incorporation in the works), shall be recorded in Measurement Books and signed by the contractor and the Engineer-in-Charge as a check to ensure that the required quantity has been brought to site for incorporation in the work.

41.2 Materials brought to site shall be stored as directed by the Engineer-in-Charge and those already recorded in Measurement Book shall be suitably marked for identification.

41.3 Contractor shall produce vouchers/invoices from the manufacturers and/or their authorized agents for the full quantity of the following materials as applicable as a pre-requisite before submitting claims for payment for advance on account of the work done and/or materials collected in accordance with Condition 64 of General Conditions of Contracts (IAFW-2249):
(a) Materials for which payment is claimed as material lying at site in RAR.
(b) Cement and Steel
(c) Ceiling fan, Exhaust fan and wall mounted air circulator fan
(d) LED fittings
(e) LT Panel board with accessories
(f) RCC hume pipe
(h) Electrical, water supply and firefighting fittings / fixtures where names of manufacturers / brands are specified or approved
(j) Cable / wires
(k) Transformer, APFC panel
(l) Galvalume sheet
(m) Flame proof fittings

41.4 The contractor shall, on demand, produce to the GE, original receipted vouchers/invoices in respect of the materials other than as stated in above condition. Vouchers/invoices so produced and verified shall be stamped by Engineer-in-Charge indicating contract number. The contractor shall ensure that the materials are brought to site, in original sealed containers/packing, bearing manufacturer 's marking except in the case of the requirement of materials(s) being less than smallest packing.

41.5 The vouchers/invoices will clearly indicate the contract number and the IS No., specific alternative to which the material conforms in case of various alternative in IS.

42. **OUTPUT OF ROAD ROLLERS: -**

42.1 A log book for each road roller shall be maintained by him for recording hours of working of the road roller. Entries in the log book shall be signed by the contractor or his authorised representative and the Engineer-in-Charge.

42.2 To ensure proper consolidation, roller must work for at least the number of days assessed on the basis of output given hereunder: -

Srl No.	Description	Output per Day of 8 Hour Work(Square Metres)
1.	Consolidation of formation surfaces/subgrade	1850
2.	Consolidation of stone soling 23cm thick with 8 to 10 Tonne Roller	518
3.	-Ditto—15cm thick –Ditto---	800
4.	Consolidation of water bound macadam (stone metal) 11cm spread thickness including spreading and consolidation with binding materials	248
5.	--Ditto-7.5 Cm—Ditto--	372
6.	Consolidation of single coat surface dressing	774
7.	--Ditto—two coat surface dressing	558
8.	Consolidation of 2.50 cm thick premix carpet including seal coat.	600
9.	Consolidation of 2 cm thick premix carpet including seal coat	744
10.	Consolidation of bituminous mixture 2 parts of broken stone metal and one part of sand and bitumen, consolidated thickness 4 cm.	372

SPECIAL CONDITION (CONT'D…../-)

43. REIMBURSEMENT/REFUND ON VARIATION IN PRICES: -

Refer Condition 63 of the General Conditions of Contracts (IAFW-2249/Re imbursement / Refund on Variation of Prices. The Condition 63 of General Conditions of Contract shall be deemed to be modified to the extent mentioned hereinafter. Increase or decrease in prices of Cement, Steel and other materials shall be adjusted on the basis stipulated hereinafter irrespective of the actual variation in prices (to the contractor):

(a) Cement: The cement cost component for the contract as a whole shall be taken as Kc% of the value of works executed under the contact. Accordingly, value/cost of cement consumed in the work as well as that lying at site for which reimbursement/refund is applicable shall be:

$$VMC = \frac{(Kc \times Vg)}{100} + Vc$$

Variation in prices of cement shall be worked out by applying the following formula:

-

$$EMC = (VMC2 - VMC1) \times \frac{(C1-Co)}{Co}$$

EMC = Variation in prices of cement to be adjusted.

Kc = Constant representing the percentage cost of Cement as compared to the total value of work under the contract as a whole. The value of Kc for this work shall be **26 (Twenty-Six)** only.

Vg = Amount of work done priced at contract rates up to for the last date of the period of reckoning excluding amount payable to the contractor towards items on star rate and PC sum.

Vc = Cost of all cement lying at site for incorporation in the work excluding cement issued under Schedule 'B' and excluding cement brought and paid or payable to contractor under Price Cost Sum and/or Star Rate (s).

C1 = Wholesale Price Index for Cement (Base Year 2011-12 (100)) published by Economic Adviser to the Government of India as on the date of commencement of the period of reckoning. In case the original contract period is extended under condition 11 of General Conditions of Contracts IAFW-2249), the Price Index as applicable on the date of commencement of the last period of reckoning before the original completion date (s) (phase wise except where phasing has been done only for sample quarter/block) shall only be applicable during the extended period. If phasing has been done for only sample quarter/sample block, the price index as applicable on the date of commencement of the last reckoning period before the original completion date of the project as a whole shall only be applicable during the extended period.

Co= As per C1 but the index as on the last due date of Bid Submission. In case of tenders for specialist works made in two parts viz "T" and "Q" Co shall be as on date of Opening of "Q" Bid.

Note — The type of cement wholesale Price index Shall be considered for Co & C1, shall be specified in the tender based on type of cement to be used in the work and after verifying the type of cement for which Wholesale Price Index is published by Economic Advisor to the Government of India.

VMc2 = Amount of cement up to the last date of the period of reckoning for which variation is adjustable as worked out as per formula for VMC.

VMc1 = As per VMc2 but as on date of immediately preceding period of reckoning.

(b) Steel: The Steel cost component for the contract as a whole shall be taken as Ks% of the value of works executed under the contract. Accordingly, value/cost of steel consumed in the work as well as that lying at site for which reimbursement/refund is applicable shall be

$$VMs = \frac{(Ks \times Vg)}{100} + Vs$$

Variation in prices of steel shall be worked out by applying the following formula: -

$$EMS = (VMs2 - VMS1) \times (S1-So)/ So$$

SPECIAL CONDITION (CONT'D…../-)

EMS = Variation in prices of steel to be adjusted.

Ks = Constant representing the percentage cost of Steel as compared to the total value of work under the contract as a whole. The value of Ks for this work shall be **30 (Thirty)** only.
Vs = Cost of all steel lying at site for incorporation in the work excluding steel issued under Schedule 'B' and excluding steel brought and paid or payable to contractor under Prime Cost Sum and/or Star Rate(s).
S1 = Wholesale Price Index for Mild Steel (Long Products) (Base Year 2011-12 (100)) published by Economic Advisor to the Government of India as or the date of commencement of the period of reckoning. In case the original contract period is extended under Condition 11 of General Conditions of Contracts (IAFW-2249), the Price Index as applicable on the date of commencement of the last period of reckoning before the original completion date(s) (phase wise except where phasing has been done only for sample quarter /block) shall only be applicable during the extended period. If phasing has been done for only sample quarter/sample block, the price index as applicable on the date of commencement of the last reckoning period before the original completion date of the period as a whole shall only be applicable during the extended period.

S0= As per S1 but the Index as on the last due date of Bid Submission. In case of tenders for specialist works made in two parts viz "T" and "Q", S0 shall be as on date of opening of "Q" Bid.

VMs2 = Amount of steel up to the last date of the period of reckoning for which variation is adjustable as worked out as per formula for VMS.

VMs1 = As per VMS2 but as on date of immediate preceding period of reckoning.

(c) Other materials (except Cement & Steel): The material cost component except cement & steel for the contract as a whole shall be taken as KoM % of the value of works executed under the contract. Accordingly, value/cost of other materials consumed in the work as well as that lying at site for which reimbursement/refund is applicable shall be:

$$VM = \frac{(KoM \times Vg)}{100} + (VoM - VB)$$

Variation in prices of materials (except cement & steel) shall be worked out by applying the following formula: -

$$EM = (VM2 — VM1) \times (W1-Wo) / Wo$$

EM = Variation in prices of material to be adjusted except cement & steel.

KoM = Constant representing the percentage cost of other material except cement & steel as compared to the total value of work under the contract as a whole. The value of K0M for this work shall be **4.00 (Four Point Zero)** only.

Vg = Gross value of work done at contract rates upto the last date of period of reckoning excluding amount payable to the contractor towards items on Star Rate and PC sum.

VoM = Value of all materials (except cement & steel) lying at site for incorporation in the work including materials (except cement & steel) issued under schedule 'B' and including materials (except cement & steel) brought and paid or payable to contractor under Prime Cost Sum and/or Star Rate (s).

VB = Value of all materials (out of Vg and VoM) (except cement & steel) issued under schedule 'B' plus value of all materials (except cement & steel) brought and paid or payable to contractor under Prime Cost Sum and/or Star Rate (s).

W1 = Wholesale Price Index for All Commodities (Base Year 2011-12(100)) published by Economic Advisor to the Government of India, as on the date of commencement of the period of reckoning. In case the original contract period is extended under Condition 11 of General

SPECIAL CONDITION (CONT'D…../-)

Conditions of Contracts (IAFW-2249), the price index as applicable on the date of commencement of the last period of reckoning before the original completion date (s) (phase wise exceptwhere phasing has been done only for sample quarter/block) shall only be applicable during the extended period. If phasing has been done for only sample quarter/sample block, the price index as applicable on the date of commencement of the last reckoning period before the original completion date of the period as a whole shall only be applicable during the extended period.

Wo = As for W1 but the index as on the last due date of bid submission.

VM2 = Value of material (except cement & steel) up to the last date of the period of reckoning for which price variation is adjustable as worked out as per formula for VM.
VM1 = -do- but as on date of immediate preceding period of reckoning.

Note:-

1. No adjustment, whatsoever, due to variation in prices of materials on account of coming into force of any fresh law or statutory rule or order as provided in Condition 63 of IAFW-2249 or otherwise than provided in this condition shall be made.

2. No adjustment in prices shall be made for any work done with materials brought at site after the original date of completion of the work as mentioned in work order No 01 under contract except as contemplated under definition of C1, S1 and W1 hereinbefore.

3. Periodicity of working out the escalation on account of variation in prices will be three months. The last calculation shall however be done for the value of work at contract rates and materials lying at site for incorporation in the work as on date of completion or extension thereof as mentioned in Note 2 above. Valuation of RARs is to be timed in such a manner that relevant date required for quarterly calculation under this condition is available from RARs. In case on these dates no RAR is preferred by the contractor, dummy RAR would be prepared & shall be kept on record duly technically checked and audited. Amount payable relevant to work done and materials collected in quarter will be worked out after firm whole sale price indices for the relevant quarter are available. Once the amount adjustable for any quarter is worked out, the same shall be adjusted as and alongwith advance on account payment in the subsequent RAR (s).

4. Any dispute arising out of interpretation or application of this Special Condition shall be referred to the Accepting Officer whose decision shall be final and binding.

5. For the purpose of calculation of retention money, liquidated damages, GST on works contracts, deduction of income tax at source and recovery of water charges (in case of un metered supply) the value of contracts as revised by the above price variation will be taken into account.

6. In cases, where value of VM2 – VM1 works out to minus on account of higher 117utilization of schedule 'B' stores (i.e value of Sch B stores under contract as a whole is higher than KoM value) and the reimbursement on account of variation in prices of materials (except Cement & Steel) works out to be negative in spite of the Wholesale Price Index for All Commodities published by Economic Adviser to Government of India going up from Wo, reimbursement on account of variation in prices of materials shall be treated as "Nil".

(d) Labour :- The Labour component for the work under the contract as whole shall be taken as KL of the value of the work executed under the contract . Variartion in labour wages shall be worked out by applying the following formula:

$$EL = \frac{KL}{100} \times VG1 \times \frac{(L1-L0)}{L0}$$

Wherein,

SPECIAL CONDITION (Contd…/-)

EL = Variation in wages of labour reimbursement to be made to the contractor or refund to bemade by the Contractor.

KL = Constant representing the percentage cost of labour element as compared to the total value of work under the contract as a whole. The Value of KL for the work shall be **20 (Twenty**).

VG1 = Gross Value of work done at contract rate during the period of reckoning less value of work paid or payable to the contractor based on actual cost (eg Star Rate (s) , workexecuted under prime cost sum etc) during the period of reckoning .

L1= Minimum wage in rupees of an unskilled adult male mazdoor as fixed under any law, Statutory rule or order as on date of commencement of the period of reckoning.

L0= As for L1 but the minimum wage in rupees of an unskilled adult male mazdoor as on the bid submission end date. If labour wage on the bid submission end date is increased afterward with retrospective effect, value of L0 shall be fixed keeping in view the following aspects: -

(a) If the increase /decrease in wages of labour are made known to the public by any means of media before the bid submission end date but the same is officially notified thereafter giving retrospective effect, the value of “L0” shall be as per notification though made subsequently.
(b) If a net –wage comprises a fixed basic wage and the living allowances revised from time to time based on Consumer Price Index (CPI) and increased CPI is made known to the public by any means before the bid submission end date , the “L0” will be revised wage corresponding to revised CPI , though the formal notification for the net wage (Considering the revised living allowances corresponding to revised CPI) is made subsequent to bid submission end date.
(c) In case the labour enforcement officer makes the announcement before the bid submission end date but Gazette Notification is made subsequently making wages applicable with retrospective effect, the value of “L0” shall be as per Gazette notification subsequently made.
(d) If the increase/decrease in wages of labour is notified /announced subsequently to the bid submission end date with retrospective effect without making the same publicly known by anymeans of publicity /media prior to the bid submission end date, then the value of “L0” shall be as per wage known at the time of bid submission end date.

Notes:

1. The Contractor shall within a reasonable time of his becoming aware of any alteration to the payment of wages of labour consequent on fixation of minimum wages under any law , statutory rule or order , give written notice thereof to the GE stating that the same is given pursuant to this Special Condition together with all information relating thereto which he may be in a position to supply .

2. Irrespective of the variation in minimum wages for any category of labour , for the purpose of adjustment under this Special Condition , the variation in minimum wage fixed under any law , statutory rule or order for an unskilled adult mazdoor , if any, shall only form the basis.

3. Periodicity of working out variation in wages of labour will be three months commencing from the bid submission end date . The last adjustment for variation in wages of labour shall however , be done for the period upto the date of completion or extended date of completion .Valuation of price adjustment due to increase/decrease in minimum wages under any law
,statutory rule or order for the purpose of making reimbursement /refund in RARs, will be timed in such manner that relevant data required for quarterly calculation under this Special Condition is available from the RARs. The first price adjustment in respect of variation in wages of labour will be worked out for the relevant quarter during which alteration to the wages of labour took place. For implementing this provision, the period of reckoning in such quarters will have to be divided in two periods ie. the first period upto the RAR payable immediately after the date of variation and the other upto the end of quarter. Value of L1 atthe beginning of the other period shall be the altered wage. If there is more than one change in wages in a quarter, there will be more than two periods of reckoning on

SPECIAL CONDITION (Contd…/-)

a similar basis. Amount payable relevant to the work done for any quarter will be worked out after the minimum wage of an unskilled adult male mazdoor as fixed under any law, statutory rule or order for the relevant quarter is available. Once the amount adjustable for any quarter is worked out, the same shall be adjusted in subsequent RAR as "advance on account" adjustments along with adjustment for 'Cement', 'Reinforcement' and 'Other material' .

4. No adjustment in prices shall be made for any work done after the due date of completion or extended date of completion on account of extension of time granted under Condition 11 ofIAFW-2249 (whichever is later) for the work under the contract.

5. No adjustment, whatsoever due to variation in wages of labour on account of coming into force of any fresh law or statutory rule or order as provided in Condition 63 of IAFW-2249 or otherwise, then provided in this Special Condition shall be made.

6. Any dispute arising out of interpretation or application of this Special Condition Shall be referred to the Accepting Officer whose decision shall be final and binding.

7. For the purpose of calculation of retention money, liquidated damages, GST on works Contract, deduction of income tax at source and recovery of water charges (in case of unmetered supply) the value of contract as revised by the above price variation will be taken into account.

43. **STANDARD OF WORKMANSHIP: -**

To determine the acceptable standard of workmanship and of fittings, wiring etc, the GE shall order the contractor to execute certain typical portion of work (different trades) and services sufficiently in advance of other work in one portion of the building. These shall be executed and completed under the close supervision of the Engineer-in-charge and shall be got approved from GE. On approval by the GE of such items, these items shall signed and/or suitably identified by the GE and labelled as guiding samples. The record of such inspection and passing of each stage of these samples shall be recorded by GE under his dated signatures. Work on such sample/portion of the building shall be progressed well ahead (minimum two stages ahead) of other portions of the building. Approved finishes/workmanship shall be followed in the work as a whole.

Signature of Contractor

AAD (Contracts)
for Chief Engineer (AF)

SPECIAL CONDITION (Contd…/-)

Annexure 'A' TO SPECIAL CONDITION

MATERIALS AND THEIR TESTS (OTHER THAN STEEL / CEMENT)

Sl No	Material	Tests	Method of testing	Frequency of tests			Level of test	Rate per test Rs.	Remarks
1	2	3	4	5			6	7	8
1.	Fly ash Brick	1)Compressive strength 2)Water Absorption 3) Efflorescence	IS-3495 (Part-II) --Do– ---Do… (Part-I)	As per IS-5454 as given under:			A A A	180/- 150/- 180/-	Checks for visual and Dimensional characteristics shall also be carried out as per IS:5454
				Lot Size.	sample Size	permissible Nos of defective bricks			
				1001 to 10000	5	0			
				10001 to 35000	10	0			
				35001 to 50000	15	1			
2.	Coarse Aggregate	1) Sieve Analysis	IS:2386(Part-I)	One test for every 15 cum of aggregates or part there of brought to site.			A	120/-	
		2) Flakiness Index	---do--				A	90/-	
		3)Estimation of deleterious materials	---do–	One test for every 100 cum of aggregate or part thereof.			A	120/-	
		4)Organic impurities	---do--	One test per source of supply			C	120/-	
		5) Moisture content	IS:2386(Part-II)	Regularly as Reqd.			A	120/-	
		6) Specific gravity	--do--	One test for each source of supply.			B	120/-	

SPECIAL CONDITION (Contd…/-)

MATERIALS AND THEIR TESTS (OTHER THAN STEEL / CEMENT)

Srl No	Material	Tests	Method of testing	Frequency of tests	Level of test	Rate per test Rs.	Remarks
1	2	3	4	5	6	7	8
3.	Fine aggregate	1) Sieve Analysis	(IS:2386Pt-I)	One test for every 15 cum of FA or part thereof when brought to site.	A	660/-	
		2) Test for clay, silt and impurities.	--do–	--do--	A	500/-	
		3) Specific gravity	--do. But (Part-II)	One for each source of supply	B	330/-	
		4)Test for organic impurities	--do--	One test for each source of supply.	C	275/-	
		5)Moisture content	-do-	Regularly as required subject to 2 tests/ per day when being used.	A	330/-	
4.	Cement	1) Setting time	IS-4031-63 Reaffirmed 1980	Once for each consignment or as and when Reqd.	B	500/-	
		2) Soundness	--do--	--do--	C	550/-	
		3)Compressive strength	--do--	--do--	B	550/-	
		4) Fineness	--do--	--do--	C	275/-	

SPECIAL CONDITION (Contd…/-)

MATERIALS AND THEIR TESTS (OTHER THAN STEEL / CEMENT)

Srl No	Material	Tests	Method of testing	Frequency of tests		Level of test	Rate per test Rs.	Remarks
1	2	3	4	5		6	7	8
5.	Structural concrete	1) Slump test or compacting factor time or VEE-BEE test	IS-1199	The Min frequency of sampling of concrete of each grade shall be as under:		A	300/-	(1) Random sample shall be carried out to cover all mix units.
		2) Compressive strength	IS-516	Qty of Conc. in the work(M^3) 1 – 5 6 -15 16-30 31-50 51 and above	No of samples 1 2 3 4 4 + 1 for each additional 50 Cum or part thereof	A	900/- per sample	(2) Refer IS-456-2000 Clause. 15.2 for frequency of sampling.
6.	Water for construction purpose	1)Test for acidity 2) Test for alkalinity 3) Test for solid content	IS:456 & 3015 -do- -do-	Once at the stage of approval of source of water		B B C	500/- 500/- 500/-	Also refer clause 4.3 of IS :456 and its subsequent sub clauses regarding suitability of water.
7.	Cement flooring tiles/Terrazzo tiles/Paver Block	1) Water absorption 2) Wet Transverse strength 3) Resistance to wear	IS:1237-1980(Appx 'D') -do-(Appx "E") -do- (Appx 'F')	6 Tiles out of 18		B	330/-	Samples: 18 Tiles from each source of supply selected at Random.
				-do-		B	330/-	
				-do-		C	540/-	

SPECIAL CONDITION (Contd…/-)

MATERIALS AND THEIR TESTS (OTHER THAN STEEL / CEMENT)

Srl No	Material	Tests	Method of testing	Frequency of tests	Level of test	Rate per test Rs.	Remarks
1	2	3	4	5	6	7	8
8.	Wood/Timber	1) Specific gravity and weight	IS:1708-1960	Minimum 3 sample from a lot of 4 Cum or 250 pieces of seasoned timber	B	120/-	
		2) Moisture content	-do-	-do-	A	120/-	
9.	Timber paneled and Glazed door/window Shutters (including Factory made shutters)	(a) Dimensions Sizes workmanship And finish	IS-1003 (Part-I)	Frequency of sampling from each lot shall be as under: Lot size / Sample size 26 to 50 / 05 51 to 100 / 08 101 to 150 / 13 151 to 300 / 20 301 to 500 / 32 501 to 1000 / 50 1000 & above / 80	A	180/-	
		(b) Strength test 1. Slamming 2. Impact indentation 3. Shock resistance 4.Edge loading	 IS: 1303 -do- -do- -do-	From each lot 5% of the factory made shutters shall be manufacturer tested for strength tests.			

SPECIAL CONDITION (Contd…/-)

MATERIALS AND THEIR TESTS (OTHER THAN STEEL / CEMENT)

Srl No	Material	Tests	Method of testing	Frequency of tests	Level of test	Rate per test Rs.	Remarks
1	2	3	4	5	6	7	8
10.	Plywood (IS:303)	Moisture content	IS:1734 (Part-I)	Six test pieces cut from each of the board selected as per table I shall be subjected to test	C	240	Sampling shall be as per IS :7835-1975 table-2
11.	Wood particle board (Medium density IS: 3097)	(a) Density	IS:2360 (Part III)	Three test specimens from each sample (size 150 mm x 75 mm)	A	60	Sampling shall be as per IS :3487 table-with moisture meter
		(b) Moisture content	-do-	-do-	A & B	60	
		(c) Water absorption	-do- (Part16)	-do- (size 300mm x 300 mm)	A	60	
		(d) Swelling due to surface absorption	-do- (Part17)	-do- (size 125 mm x 100 mm)	A	60	
		(e) Swelling water	-do-	-do- (size 200 mm x 100 mm)	A	60	
		(f) Modulus of rupture	-do- (Part 4)	Three test specimens as per IS: 2380	B	90	
		(g) Screw withdrawal strength	-do- (Part 4)	Three test specimens as per IS: 2385	C	120	
12.	Mangalore pattern roofing tiles	(a) Water absorption	IS: 654 (Appx A)	6 tiles out of 32	B	180	Sample: 32 tiles from each consignment of 3000 tiles or part thereof. These tiles shall be checked for dimension.
		(b) Breaking load	-do- (Appx C)		B	120	

Signature of Contractor

AAD (Contracts)
For Accepting Officer

SPECIAL CONDITION (Contd…/-)

Annexure 'B'

LIST OF SUGGESTED ESSENTIAL EQUIPMENTS IN SITE LABORATORY

1.	Cube Mould (150+/-0.2mm)	12 Nos
2.	Tamping Bar (16 mm dia, 600 long)	01 No
3.	Balance 12 Kg (LC 1 gm)	01 No
4.	Balance 220 gm (LC.001 gm) digital	01 No
5.	Weights	01 No
6.	Compression Testing machines with three gauge (capacity-2000 KN)	01 No
7.	Slump Test Apparatus	01 No
8.	Standard Test Sieve (80 to 4.75 mm) Square hole, perforated plate.	01 Set
9.	Standard Test Sieve (3.35 mm to 75 micron) fine mesh, wire cloth.	01 Set
10.	Soft brush & Camel hair brush	01 Set
11.	Lid & Pan	01 Set
12.	Hot air oven (Thermostatically controlled)	01 No
13.	Thickness Gauge	01 No
14.	. Measuring Cylinders (graduated)	01 No
15.	Steel Tape (LC 1 mm). Steel scale (Lc.5mm)	01 No
16.	Plywood sheet (2 No) 3 mm thick	02 Nos
17.	Dish (180 mm, 180 mm, 40 mm) of glass or porcelain or glazed stoneware.	01 Set
18.	Distilled Water	01 Set
19.	Sieve shaker coarse Aggregate Test Sieve	01 Set
20.	Sieve shaker for fine Aggregate Test Sieve	01 Set
21.	Thermometer for Hot bitumen	01 No
22.	Thermometer for Recording Day Temp	01 No
23.	Humidity recorder meter	01 No
24.	Cubes moulds 70.6mm x 70.6mm x 70.6mm	04Nos
25.	Cubes moulds 50mm X 50mm x 50 mm 4 Nos	04Nos
26.	Standard sand	50 Kgs
27.	Gauging trowel (Confirming to IS-10086)	02Nos
28.	Vicat apparatus	01 Set
29.	Le chaterlier's test apparatus	01 Set

Signature of Contractor

AAD (Contracts)
For Accepting Officer

PARTICULAR SPECIFICATIONS

1.0 GENERAL: -

1.1 Work under this contract shall be carried out in accordance with Schedule 'A', Special Conditions, Particular Specifications, drawings including notes thereon (unless specified otherwise) and general rules and specifications given in MES SSR Part-I(2009) as well as general rules, Special Conditions and preambles to the various rates given in MES SSR Part II– (2020) (MES SSR Part I & Part II hereinafter called MES Schedule) and these shall be read in conjunction with each other.

1.2 The term 'General Specification' referred to hereinbefore as well as referred to in IAFW-2249 (General Conditions of Contracts) shall mean the specifications contained in the MES Schedule Part I & IS.

1.3 General Rules, Specifications, Special Conditions, method of measurements, preambles in the MES Schedule shall be deemed to be applicable to the work under this contract, unless specifically mentioned otherwise in these documents.

1.4 The term "as specified", wherever appears in tender documents and drawings, relates to relevant particular specifications and in its absence general specifications.

1.5 Particular specifications in this section given hereinafter shall be generally applicable to all works covered under Schedule "A". The particular specifications are brief and are only to particularize, amend and emphasize the specifications given in MES Schedule, which are not repeated.

1.6 Where specifications/provisions given in these particular specifications are at variance with the provisions/specifications given in MES Schedule, specifications /provisions given in these particular specifications, hereinafter, shall be followed.

1.7 Where specifications for any item of work are not given in MES Schedule or in these particular specifications, specifications as given in relevant Indian Standard or Code of practice shall be followed.

1.8 Reference to any drawings which is mentioned on the drawing forming part of the tender but not specifically mentioned in the list of drawings shall be deemed to be forming part of the tender. The tenderer shall see such drawings/details in the office of Accepting Officer/concerned CWE/GE before quoting his tender.

1.9 The contractor shall not take cognizance of Note(s) appearing on drawings regarding the bearing capacity of soil taken into consideration while designing the foundation, as it is for departmental purpose only.

2.0 LAYOUT: -

2.1 In laying out buildings centre line dimensions mentioned in the drawings or derived there from shall be strictly followed.

3.0 EXCAVATION AND EARTH WORK: -

3.1 **PREPARATORY WORK**: -

3.1.1 The existing ground levels shall be recorded jointly at an interval not more than 3.0 metres grid on graph sheet. Level sheet shall be prepared clearly indicating proposed building work, external services such as roads, culverts, sewage disposal, area drainage and information such as proposed plinth level of building, invert level of man holes, drains, culverts etc required for proper execution of the work as per Engineering norms, within 15 days from the date of commencement of work (prior to execution of work) duly signed by GE and contractor. Level sheet shall be kept in record by GE in duplicate before execution of work.

PARTICULAR SPECIFICATION (CONT'D…../-)

3.2 **SITE CLEARANCE AND SURFACE EXCAVATION**: -

3.2.1 Before setting out the building and commencing the construction, preparatory work such as removal of grass, vegetation, jungle clearance etc, surface excavation to a depth not exceeding 30 cm and averaging 15 cm, in any type of soil for the entire area occupied by the building and structure including plinth protection ramps, steps etc. shall be carried out by the contractor.

3.3 **SURFACE DRESSING: -**

3.3.1 Surface dressing around the entire buildings and structure to a width of 3 Metres beyond the external edge of plinth protection / steps / ramp or external wall as applicable shall be carried out by the contractor as per clause 3.6 and 3.10 of MES Schedule Part-I [2009]. Area around outer edge of building / structure shall be dressed to slope away from the buildings as directed by Engineer –in-Charge.

3.4 **EXCAVATION: General**

3.4.1 Any depth in excess of required depth in excavation shall be made good by the contractor at his own cost with the concrete of same proportion of foundation base concrete.

3.4.2 The lump sum price quoted by the Contractor for Schedule "A" Part-I shall be deemed to include for all excavation and earthwork in any type of soil.

(i) Where soft/loose soil, hard/dense soil, soft/disintegrated rock and hard rock are mixed, the measurement for the total quantity shall be made by taking levels of the ground as directed by GE and the total quantity of excavation shall be computed from these levels. The soft/disintegrated rock and the hard rock excavated shall then be stacked separately and measured in stacks. The quantity of soft/disintegrated rock and hard rock so measured shall be reduced by 50% to allow for voids. These reduced quantities of soft/ disintegrated rock and hard rock shall only be admissible for payment of deviation under item of excavation of soft/disintegrated rock and hard rock respectively. From the total qty of the mixture the quantity of soft disintegrated rock and hard rock excavated thus arrived and shall be deducted to work out the quantity of soil i.e. soft/loose/hard dense soil excavated.

(ii) Where only soft /disintegrated rock and hard rock are mixed, the measurement for the total quantity shall be made by taking levels of the ground as directed by GE and the total quantity of the excavation shall be computed from these levels. The hard rock excavated shall then be stacked and measured in stacks. The quantity of hard rock so measured shall be reduced by 50% to allow for voids. This reduced quantity of hard rock shall only be admissible for payment of deviation under item of excavation of hard rock. From the total quantity of the mixture, the quantity of hard rock excavated thus arrived at, shall be deducted to work out the quantity of soft/disintegrated rock excavated.

(iii) All excavation shall be measured in successive stages of 1.5 meter depth starting from the commencing level. In case of excavation in rocks, if the depth of successive stage slightly exceeds 1.5 meter the levels shall be taken at that stage but excavated rock though obtained from this deeper depth, shall be measured in the previous successive stage only.

(iv) Hard rock obtained from excavation shall become property of the contractor for which he will be charged @ Rs.1000/- per cum and the same shall be deducted from payment due to the contractor. The method of measurement shall be as laid down in clause 3.1 on page 18 & 19 of SSR-2020 Part -II. The measurement shall be properly recorded in MB and signed by the Engineer - in-Charge and contractor.

(v) In case of any dispute with regards to the classification of various strata and their depth, the decision of GE will be final and binding.

3.5 **FILLING UNDER FLOORS**: -

PARTICULAR SPECIFICATION (CONT'D…../-)

(i) Approved soil obtained from excavation [other than those obtained from surface excavation and surface dressing] shall be used for filling in foundation up to ground level of developed area and under floor. No charges shall be levied for the use of soil obtained from excavation for filling. Nothing extra is payable on this account. Filling shall be spread, levelled, watered and well rammed in layers not exceeding 25 cm thick.

(ii) Any additional earth required for purpose of filling shall be arranged by the Contractor from outside MD land at no extra cost to the Department. Expansive or other unsuitable soil obtained from excavation shall not be used in filling. The decision of Engineer-in-Charge as to whether the soil obtained from excavation is suitable or not for filling, either partly or fully, shall be final and binding. If the quantity of suitable soil obtained from excavation falls short of the filling required, the contractor shall bring the requisite quantity of earth from the places approved by the Engineer-in-Charge, without any extra cost to the Government.

3.6 DEWATERING: -

3.6.1 No extra payment over the lump sum amount quoted for buildings and services shall be admissible for dewatering, if water is met with or accumulated in the foundations or any other excavations due to any cause, whatsoever, and for excavation in mud, bailing and pumping of water if required, shall be done as described in Para 3.17 of MES Schedule Part-I.

3.7 TIMBERING: -

(i) The contractor's lump sum shall not include the cost of timbering, if any, required to uphold the sides of excavation. If any timbering as specifically ordered to be provided in writing by the Garrison Engineer, consequent upon necessity arising as per site conditions, the same will be paid for under a deviation order.

(ii) The thickness of filling shown in drawings is after consolidation.

3.8 DISPOSAL OF SURPLUS EXCAVATED EARTH: -

3.8.1 All surplus spoil obtained from excavation for building work (Schedule 'A' Part-I) shall be disposed off to a distance not exceeding 250 m at the location(s) as directed by Engineer-in-Charge, deposited, spread and levelled as specified in MES Schedule. Quoted lumpsum for building works include this aspect.

3.9 MOORUM FILLING: -

3.9.1 Moorum shall be as specified in clause No 3.21 and shall be spread and leveled as specified in clause No 3.21.4 of SSR Part I 2009.

3.10 PRECONSTRUCTION ANTI- TERMITE TREATMENT IN FOUNDATION AND GROUND FLOORS.: -

3.10.1 The pre-construction anti termite treatment shall be carried out in buildings as catered under SI Item No. 1 of Schedule 'A Part I with the proportions and with specifications as specified in clauses 3.26.1 of MES Schedule Part-I of 2009 except that no mound treatment shall be provided. Wherever 5 litres or 7.5 litres is mentioned in these clauses of MES Schedule Part-I it shall mean 5 litres or 7.5 litres of emulsion of concentration as indicated in MES Schedule. Anti-termite treatment shall be carried out with emulsion of the chemical "chemical Chlorpyrifos 20% EC (IS-8944) or Imidacloprid 30.5% SC (IS-16131)" as per IS 6313 parts I and II- 2001. Con-centration by weight percent (Active Ingredients) shall be 1% in both cases of chemicals.

3.10.2 The work of anti-termite treatment shall be got done from a specialist pest controller and exterminator agency and shall be a member of Indian Pest Control Association, and holding license as per Clause 13 of the Insecticides Act, 1968. The specialist agency may be any one from the

specified firms. Prior approval of GE will be obtained before ordering the work on specialist agency.

3.10.3 Anti-termite treatment shall be carried out with emulsion of Chloropyriphos 20%EC purchased directly from the manufacturer or his authorized agent. Original purchase vouchers shall be produced before the GE for verification and defacement against this contract. Chemical banned by Government for use in anti-termite treatment shall not be used. In the event of deviations, the rates given in MES Schedule Part- II 2020 for the treatment shall be applicable.

3.10.4 The Lump sum quoted for the building mentioned in clause 5.8.1 here- in- before in above shall include the following pre-construction Anti-termite-treatment:

(i) Treatment to Masonry & column foundations, trenches/pits/ basements and treatment of RCC foundations and basements and back fill using chemical all as specified in Para 3.26.6 of MES Schedules Part I.

(ii) Treatment to top surface of plinth filling under floor and plinth protection all as specified in para 3.26.7 of MES Schedule Part I.

(iii) Treatment at junction of walls and the floor all as specified in para 3.26.8 of MES Schedule Part I.

(iv) Treatment along external perimeter of building all as specified in para 3.26.9 of MES Schedule Part I.

Chemicals brought to site in sealed container bearing ISI certification marks shall only be permitted to be used in the work

3.10.5 Chemicals shall be stored carefully at site. Seals of containers shall be broken only in the presence of Engineer-in-charge. Empty containers shall be removed off the site promptly. On any particular day, if the contents of the full container could not be used in the work, the container shall be sealed at the end of the day in the presence of Engineer-in-Charge.

3.10.6 Total quantity of chemical required for the work as per specification shall be worked out in order to ensure that full quantity is brought to site and used in the work. For this purpose, entries shall be made in measurement book (Not for payment/Not to be abstracted) indicating the brand, name, batch number, date of manufacturing, date of expiry, quantity brought etc, and signed by Engineer-in-Charge as well as the representative of the agency executing the work and contractor's representative.

3.10.7 GUARANTEE FOR ANTI-TERMITE TREATMENT: -

(i) The contractor shall stand a guarantee of **10 (TEN) years** to the Government for the work against recurrence of termite infestation (viz. effectiveness of the treatment carried out). For giving guarantee, if any additional treatment over and above that is specified is required, the same shall be carried out by the contractor without any extra cost to the Government. The contractor shall furnish a written guarantee for the effectiveness of the treatment for a period of TEN YEARS in favour of Garrison Engineer.

(ii) The amount of **security deposit @ 10% of ATT work at the contract rate** to be carried out for the building included in Schedule 'A' part I against guarantee period for Anti-termite treatment shall be retained by the Government from the contractor's dues. This amount shall only be released after successful expiry of the guarantee period. The contractor may however, furnish a fixed deposit receipt in lieu from a Schedule bank, pledged in favour of Garrison Engineer for the period of Guarantee in which case the amount if any, deducted from the dues/final bill shall be refunded.

(iii) Should the GE at any time during construction or reconstruction or prior to the expiry of the guarantee period, finds that the buildings have been infested with termites,

the contractor shall on demand in writing from the GE specifying the building(s) complained of, not-with-standing that the same may have been inadvertently passed, certified and paid for, undertake to carry out such treatment as may be necessary forthwith to render the building(s) free from termite infestation at his own expense till expiry of the guarantee period. In the event of his failure to do so within the specified period to be specified by the GE in his demand aforesaid, the GE may undertake such treatment at the risk and expense in all respects of the contractor. The liability of the contractor under this condition shall not extend beyond the period of ten years from the date of completion, unless the GE had previously given notice to the contractor to rectify the defects.

(iv) The contractor shall provide a plate/board of adequate dimension but not less than 90cm x 45 cm x 2 cm thick in cement mortar (1:4) on one side of each building and indicated thereon the Contract No, Name of contractor and Name of the agency executing the anti-termite and water proofing treatment and the date of expiry of the guarantee period.

3.11 HARD CORE: -

3.11.1 Refer Clause 3.27 of MES Schedule Part- I. Hard core shall be of broken stone of gauge not exceeding 63 mm, well graded to provide dense and compact sub base. Unless otherwise indicated or shown in drawings/specifications, the thickness of hard core shall be 100 mm consolidated. Hard core filling where indicated on drawings shall be spread, levelled in layers not exceeding 15 cm thick well rammed, watered and consolidated. Moorum may be used to fill interstices in hard core. Thickness wherever shown on drawings/schedule of finishes shall be treated as consolidated thickness.

4.0 CONCRETE: -

4.1 MATERIALS: -

4.1.1 CEMENT: -

Refer clause 4.3 on Page No. 51 of MES Schedule Part -I.

4.1.2 PROCUREMENT AND TESTING OF CEMENT: -

4.1.2.1 All cement required for completion of this contract shall be ordinary Portland cement 43 grade conforming to IS: 8112 or Portland Pozzolana Cement conforming to IS: 1489 and each bag shall be ISI certification mark and manufacture date.

4.1.2.2 However, the OPC and PPC cement both will not be used in one building. Before commencement of building the contractor will intimate type of cement to be used and get the same approved by GE for each building separately. Thereafter only one type of cement will be used in one building.

4.1.2.3 When the Contractor opts for using PPC in the work, GE can allow incorporation of PPC subject to following:

(i) GE is required to ensure that PPC meets the strength criteria of 43 Grade OPC as laid down in IS-1812.

(ii) The minimum period before striking formwork given in clause 11.3.1 of IS-456 is to be suitably modified at sites by the GE.

(iii) The contractor shall give an undertaking that he shall not claim anything extra whatsoever on account of extra time for stripping form work etc while using PPC in lieu of OPC.

(iv) In cold climate regions where temperature is lower than 15˚C and important structures like Over Head Reservoirs, Under Ground Sumps and Buildings with spans 10m or more only OPC shall be used.

(v) Mixing of OPC and PPC shall not be allowed in work except for plaster and mortar.

4.1.2.4 All cement required for completion of this contract shall be procured by the contractor, under his own arrangement directly from the manufacturers. The cement shall be procured from the manufacturers mentioned at **Appendix A**.

4.1.2.5 The contractor shall furnish the particulars of the manufacturer of cement along with the date of manufacture to the Garrison Engineer for every lot of cement separately. The cement so brought shall be fresh and in no case older than 90 days from the date of manufacture. The GE shall verify the document in support of the purchases of cement. Before placing order for supply of cement by the contractor, he shall obtain written approval from the GE regarding name of manufacturer, quantity of cement etc. Cement shall be procured for minimum requirement of one month and not exceeding the requirements of the same for more than two months at a time. The cement shall be consumed in the work within three months after receipt. Cement shall conform to the requirement of IS Specification and each bag of cement shall bear relevant ISI mark. The weight of each consignment shall be verified by the GE and recorded. The content of cement shall be checked at random to verify the actual weight of cement per bag. However, the content of cement per bag shall be 50 Kg only subject to tolerance given in Clause 9.2.1.1 and Annexure 'B' of IS -8112 and Clause 10.2.1.1 and Annexure 'B' of IS -1489 for OPC and PPC respectively.

4.1.2.6 <u>TESTING</u>: -

(i) The manufacturer is required to carry out inspections and testing of cement in accordance with the relevant BIS provisions. The contractor shall submit the Manufacturer's Test Certificate in original along with the test sheet giving the result of each physical test as applicable and chemical composition of the cement or authenticated copy thereof duly signed by the manufacturer, with each consignment. The Engineer-in-Charge shall record these details in the cement Acceptance/Rejection Register as per Appx B.

(ii) The GE may organise independent testing as per IS: 3635 (method of sampling hydraulic cement) and IS: 4001-1995 (Method of Physical test for hydraulic cement) and IS: 4030-1985 (Methods of chemical analysis of hydraulic cement) of random sample of cement drawn from each lot of the consignment from the National Test House, SEMT Wing, CME, Regional Research Laboratories, Govt Engineering collage, IS approved laboratories and Zonal lab etc.

(iii) The contractor shall make available required quantity of cement and other facilities for testing of cement by GE as specified here-in-before and shall bear the cost of cement and testing thereto irrespective of whether sample passes or fails. GE shall make payment to the testing house lab and direct the contractor to deposit the amount in Govt. treasury in favour of GE concerned and submit the treasury challan to the GE within 10 days of the letter issued by the GE for payment of cost of testing of cement. If contractor fails to do so, the testing charges shall be recovered from the dues of the contractor.

(iv) The required quantity of samples of the cement shall be taken in two polyethylene bags and sealed by the Engineer-in-Charge in the presence of representative of the contractor. The seal shall be authenticated both by the Engineer-in-Charge and representative of the contractor. Out of two samples, one sample shall be sent for testing and other retained by GE in his office. Record of these samples shall be kept by Engineer-in-Charge in the register and copy of the same shall be sent to GE/CWE/CE by him for record.

(v) Cement brought by the contractor shall be allowed to be incorporated in the work after satisfactory test results of sample sent for testing. GE shall inform the contractor, about the test results and permission to incorporate in the work, in writing. GE shall ensure that word "pass" is embossed on both sides of the cement bags of lot, which has been found as per specification as per test result. The cost of embossing as

indicated herein above shall be borne by contractor.

(vi) In case the sample got tested by the GE from the institutions/ laboratories, mentioned here in before, fails, the concerned lot of cement shall stand rejected. The GE shall emboss a mark 'X' on both sides of each bag of the rejected lot. The GE will inform the contractor in writing about the test result and direct him to remove the rejected cement from the site of works within two days of the order by the Engineer-in-charge. Contractor shall have no claim whatsoever on account of rejection and removal of cement.

(vii) It shall be the responsibility of the contractor to arrange cement well in advance, to facilitate it's testing before use in the work. The contractor shall submit the program of procurement of cement to the GE before the date of commencement of the work indicating date of procurement and quantity.

4.1.2.7 STORING:-

(i) Cement shall be stored in covered godown over dry platform at least 20 cm high in such a manner as to prevent deterioration due to moisture or intrusion of foreign matter. In case of store room, the stack should be at least 20cm away from floors and 60 cm from walls. The stacking of cement shall be done as specified in relevant IS. The storage accounting and preservation of cement supplied by the contractor shall be done as per standard Engineering practice till the same is incorporated in the work and the cost of the same shall be deemed to be included in the unit rate/amount quoted by the tenderer. The Engineer-in-charge (EIC) shall inspect once a day to verify that cement lying at site is stored, accounted, preserved and maintained as per the norms. The cement shall be stored so as to differentiate each tested and untested consignment separately with distinct identification. If the GE is not satisfied with the storage/preservation of cement, he may order for any test (s) of cement as applicable for that consignment to ensure its conformity to the quality mentioned in the in the manufacturer's test certificate. The contractor shall bear the cost of necessary testing (s) in this regard and no claim whatsoever shall be entertained.

(ii) Stacking of cement shall be done as per relevant IS and as under:

a) Each cement consignment shall be stacked separately and removal shall be made on the basis of First in First out.

b) Adequate top cover will be provided.

c) Stacks in no case shall be higher than 12 bags. The maximum width of each stack shall be 3.00 m. If the stack is more than 7 or 8 bags high, the bags shall be arranged in header and stretcher fashion, i.e. alternatively lengthwise and crosswise so as to pile together and avoid topping over.

d) Adequate space shall be kept between two stacks.

(iii) Cement godown shall be provided with two locks on each door. The key of one lock at each door shall remain with EIC or his representative and that of the other lock with the contractor's authorized agent at site of works so that cement is removed from the godown only according to daily requirement with the knowledge of both the parties. During the period of storage, if any cement bag (s) is found to be in damaged condition due to whatsoever reason, the same shall be removed from the cement godown on written orders of the GE and suitable replacement for the cement bag (s) so removed shall be made and no claim whatsoever shall be admissible on this account.

(iv) In case more than one type of cement is used in the work ie Ordinary Portland cement or Portland Pozzolana Cement, both type of cement shall be stored separately as directed by the Engineer-in-Charge to avoid mixing of these type of cement. Separate record shall be maintained including the location/items where these type of cement are used.

PARTICULAR SPECIFICATION (CONT'D…../-)

(v) Cement shall be removed from the store only according to daily requirement with the knowledge of both the parties and the EIC and the contractor shall record daily consumption of cement in cement consumption register, which shall be signed. Cement constants given in Appendix 'A' to E-in-C's branch letter No. 19280/E8 dated 03 May 1976 shall from the basis of consumption of cement for various items of works unless specifically indicated otherwise.

(vi) In case the consumption of cement as per cement consumption register is found to be more than the estimated quantity of cement due to whatsoever reason, the contractor shall not have any claim whatsoever for such excess consumption of cement.

4.1.2.8 SCHEDULE OF SUPPLY: -

The contractor shall procure the cement timely as required in accordance with CPM chart agreed between GE and the contractor. The contractor will forfeit his right to demand extension of time if the supply of cement got delayed due to his failure in placing order in time to the manufacturer.

4.1.2.9 MEASUREMENTS AND PAYMENT OF CEMENT: -

(i) The entire quantity of cement shall also be suitably recorded in the measurement book for record purposes as Not to be abstracted before incorporation in the work and shall be signed by the EIC and the contractor.

(ii) The payment shall only be allowed after production of original purchase voucher, certified copies of test certificates from manufacturer for each consignment and results of testing carried out in laboratory on receipt of cement (7 days compressive test) are found satisfactory after testing at site as per condition 64 of IAFW-2249. Rate of payment given in SSR shall be applicable for cement irrespective of type of grade of cement specified for use in the work.

4.1.2.10 DOCUMENTATION: -

(i) The contractor shall submit original vouchers, test certificates and test sheets from the manufacturer/producers for the total quantity of cement supplied under each consignment to be incorporated in the work. The GE along with the relevant documents before sample is taken for the testing shall inspect all consignments received at the work site. The original vouchers and test certificates (original/authenticated) shall be defaced by the Engineer-in-Charge and photocopy of the same shall be kept on record in the office of the GE duly authenticated and with cross reference to the control number recorded in the cement acceptance/rejection register. The cement acceptance /rejection register shall be signed by (JE Civil), Engineer-in-Charge, GE and the Contractor. The Accepting Officer may order a Board of Officers for verification of connected documents produced by the contractor. The entire quantity of cement shall also be recorded in the measurement book (not to be abstracted) for record purpose before incorporation in the work and shall be signed by the Engineer-in-Charge and the contractor.

(ii) Record of cement brought by the contractor shall be maintained on the format as per **Appx B**.

4.2 Lime shall be conforming to approved samples kept in GE's office.

PARTICULAR SPECIFICATION (CONT'D…../-)

4.3 AGGREGATES: -

4.3.1 FINE AGGREGATE: -

Fine aggregate (sand) for concrete work shall conform to materials specifications and grading within the limits of grading Zone I to III as specified in Clause 4.4.1 to 4.4.6 and 4.4.7(2) of MES Schedule Part-I and shall conform to approved sample kept in the office of GE. Mixing of sand for obtaining specified grading from two different sources shall not be permitted. Source of fine aggregate shall be as per **Appx C.**

4.3.2 COARSE AGGREGATE: -

(i) Coarse aggregate (stone aggregate) for all cement concrete work such as PCC/RCC shall be of approved quality all as specified in clause 4.4.1 to 4.4.7 (1) of MES Schedule Part-I. Mixture of two types of stones shall not be permitted.

(ii) Coarse aggregate (stone aggregate) 20 mm and below shall be machine crushed and above 20mm may be hand broken.

(iii) Grading of coarse aggregate (stone aggregate) unless specified otherwise in the specifications hereinafter in various locations/work i.e. PCC/RCC shall be as specified in MES Schedule Part I

(iv) Stone for lean concrete in foundation shall be sand stone conforming to IS: 383 from approved source. Unless otherwise specified in Schedule 'A' stone for all RCC/ PCC/PQC/WMM work shall be crushed Granite/Trap/ Basalt or any other igneous rocks stone from approved source irrespective of any distance from site of work.

(v) Source of fine aggregate shall be as per **Appx C.**

4.3.3 WATER: -

4.3.3.1 Water to be used in this work shall be all specified in IS: 456 of 2000 and in clause 4.9 of SSR Part I (specification). When the water is not issued by the department, the water shall be got tested in Govt. approved laboratory and test certificate shall be submitted to GE. The water shall be used in the work only after receipt of satisfactory test results as per IS.

4.3.4 TYPE/GRADE OF CEMENT CONCRETE MIX: -

4.3.4.1 Type/grade of concrete mix with nominal size of coarse aggregate required for works in various situations unless otherwise specifically mentioned elsewhere or shown on drawings or notes thereon and structural notes, etc shall be as under:

Sl No	**Situation**	**Mix of concrete**	**Nominal size of graded stone aggregate**
1	**2**	**3**	**4**
a.	All RCC works in beams, columns including footings, roof slabs including RCC bands, plinth beam, RCC walls, RCC lintels, chajjas, shelves except water retaining structure etc.	Design Mix controlled concrete of Grade M-25	20 mm
b.	Lean concrete, filling in sunken floor, Sub base for flooring/ platform etc.	1:4:8, Type D-2	40 mm
c.	PCC in foundation of walls of buildings lean concrete under plinth/grid/toe beam, in gaps be-	1:4:8, Type D-2	40 mm

PARTICULAR SPECIFICATION (CONT'D…../-)

	tween plinth beam/Column footing and foundation of brick steps under column footings,.		
d.	PCC filling in frames/chowkhat	1:3:6, Type C-0	12.5 mm
e.	PCC floor topping, DPC, shelves etc having thickness less than 40mm.	1:2:4, Type B-0	12.5mm
f.	PCC floor topping 40 mm thick and above but not exc. 100 mm thick.	1:2:4, Type B-1	20mm
g.	All PCC works for which no mix is specified else where	1:2:4	20 mm for works above 40 mm thick and 12.5 mm for work below 40 mm thick.
h.	PCC coping, padding under bearing of RCC beams/lintels/slabs, splash stones PCC blocks for fixing clamps, bed blocks. Cills, plugging for scaffolding holes etc.	1:3:6, Type C-1	20 mm

4.4 BATCHING, MIXING, DEPOSITING AND RAMING: -

4.4.1 Controlled concrete materials shall be batched by weight only. Combined batching with digital weighing system and mixing plant with auto cut off and computer printout facility shall be used for concreting. The capacity of mini batching plant provided at site shall be adequate enough to execute the work as per the CPM, or otherwise the contractor shall provide single batching plant of higher capacity or more number of batching plants of adequate capacities as required at site, as asked by the GE. No claim what so ever arising out on this account is admissible. The decision of GE in this regard shall be final and binding. The plant/plants shall have the digital system of adding specified quantity of water into concrete mix as per the design mix requirement.

4.4.2 Water shall be measured either by volume in calibrated tanks or weighed. All measuring equipment's shall be kept in a clean serviceable condition and their accuracy checked periodically.

4.4.3 Provisions as in clause 4.11.3.2 to 4.11.3.5 of MES Schedule Part-I shall be followed. All batching of concrete and accuracy of batching shall be as per Clause 10.2 of IS-456: 2000

4.4.4 The mixing shall be done for at least 2 minutes and until a uniform colour and consistency is achieved.

4.4.5 Quantity of concrete mixed in any one batch shall not exceed the rated capacity of the mixer. The whole of the mixed batch shall be removed before mate-rials for fresh batch enter the drum. Concrete mix as approved shall not be modified by addition of water or otherwise in order to facilitate handling for any other purpose. On ceasing of work and other stoppage exceeding 20 minutes, the mixer and other plants used for handling wet mix shall be thoroughly washed with clean water. Pickup and throw over blades in the drum of the mixer which are worn down 20mm or more in depth shall be replaced with new blades.

4.4.6 All cement concrete, both plain and reinforced shall be mixed in mechanical mixer as specified in para 4.11.5 and 4.11.5.1 of MES Schedule Part-I. However for small quantity of concreting (other than RCC works) i.e., the quantity of concrete required being less than one batch of mix, the contractor may after obtaining written approval of Engineer in charge which shall be exceptional, adopt hand mix subject to addition of 10% extra cement without price adjustment where hand mixing permitted, it shall be carried out on a concrete platform and care shall be taken to ensure that mixing is

continued until the concrete is uniform in colour and consistency.

4.4.7 All cement concrete both plain and reinforced concrete shall be deposited and compacted all as specified in Clause 4.11.10 and 4.11.11 of MES Schedule Part-I. However, RCC work in columns, foundation, beams walls, chajjas and slabs etc., shall be compacted using mechanical vibrator, compaction of lean concrete shall be carried out by ramming and consolidated by tamping and rodding as specified. In the event of break down of mechanical mixer and vibrator, the contractor must have arrangements for standby mechanical mixer and vibrator.

4.5 DESIGN MIX CONCRETE (CONTROLLED CONCRETE): -

4.5.1 Grade of design mix concrete shall be as specified here-in-before and shall be as referred to in IS-456-2000 and as specified hereinafter. Design mix concrete may also be referred to as controlled concrete. Mix design shall be done as per IS-10262 of 1982 (Recommended guidelines for design mix concrete) and as described in SP-23 of 1982 (an IS publication).

4.5.2 The requirement of cement per cubic meter of controlled concrete of grade M25 shall be as per IS-456 of 2000. The actual requirement of cement for the controlled concrete shall be ascertained by the tests as specified here-in-after. The design mix for M25 concrete shall be carried out for **MODERATE** environment conditions and good quality control. The tenderer shall ascertain the quantity of cement required and quote the lump sum accordingly. No claim whatsoever arising on account of quantity of actual cement incorporate in the work on account of design mix is admissible.

4.5.3 (a) Contractor shall use liquid admixtures (Super plasticizers) to achieve the workability and to reduce the water content in design mix. Admixtures shall confirm to IS 9103: 1999 shall be from approved manufacturers as given below:

(i) FOSROC Chemicals (India) Ltd.
(ii) Roff Construction Chemicals Pvt. Ltd.
(iii) STP Specialty Chemicals Ltd.
(iv) CICO Technologies Ltd

(b) Para 5.5 of IS 456: 2000 be also referred for quality of admixtures.

(c) For maximum dose of admixtures, please refer para 10.3.3 of IS 456: 2000.

(d) Various tests as specified in IS 9103: 1999 shall be carried out for each batch of Admixtures at contractor's cost.

(e) Contractor shall submit original purchase voucher and test certificate of manufacturer for complete quantity of admixtures used in the work before claiming payment for the same.

(f) Complete quantity of admixtures including name of manufacturer, its brand name, date of manufacturing, date of expiry, voucher No. and details of test certificates shall be entered in MB as **"Not to be abstracted**" duly signed by JE, Engineer-in-Charge, GE and representative of contractor before making payment in RAR."

4.5.4 As soon as possible after receiving the work order to commence the work, the contractor shall submit samples of the materials required for preparing design mix concrete viz cement, coarse aggregate, fine aggregate and admixtures for approval of GE and intimate the place out of the following where they propose to carryout the design mix and preliminary tests of RCC M-25 grade concrete: -

(i) **Any Indian Institute of Technology (IIT)**
(ii) **Any National Institute of Technology (NIT)**
(iii) **Any Govt. Engineering College**
(iv) **Command Test Lab (CTL), Lucknow**
(v) **National Test House**

PARTICULAR SPECIFICATION (CONT'D…../-)

4.5.5 The cement used in the work shall be as specified here in before. Coarse aggregate shall be crushed stone aggregate. The gradation shall be followed as per clause 4.2 table 2 of IS: 383 to obtain maximum density.

4.5.6 After the samples of all the materials are approved by GE in writing sufficient quantities of these materials shall be forwarded to the institutions as listed in clause No. 6.5.6 here-in-before, by GE at contractor's expense for carrying out design mixes. If design mix carried out from Command test Lab (CTL) Lucknow as decided by GE. A sum of **Rs. 10000/- (Rupees Ten thousand only)** on account of carrying out design mix shall be recovered from contractor's dues.

4.5.7 PRELIMINARY TESTS: -

(a) Preliminary tests are tests conducted on the trial mixes of concrete produced in the laboratory with the object of.

(i) Designing concrete mixes before the actual concreting operation starts.
(ii) Determining the adjustments required in the design, when there is change in the materials used during execution of work.
(iii) Verifying the strength of concrete mix at 28 days

(b) The preliminary tests shall consist of 3 separate sets of tests covering possible variation of gradation of aggregates and each set of test using a minimum 7 cubes of size 150 mm x 150 mm x 150 mm and one slump test. Three cubes shall be tested at 7 days to get indication of minimum strength of 28days. Other 03 cubes shall be tested at 28 days and 01 cube shall be preserved for Government use for subsequent testing. The compressive strength tests of cubes shall be performed as per IS-516. Casting of cubes and testing of these cubes shall be carried out in the presence of contractor's representative, GE/ GE`s representative. It will be contractor's responsibility to ensure that design mix is carried out at the earliest. Contractor shall ensure that design mix calculations, supporting trail mix (03 Nos) details and test results of trial mixes along with recommended trial mix are submitted to GE at the earliest for his further action. Based on test results, the GE shall approve the design mix in writing. Copy of approved design mix shall be submitted to Accepting Officer within 10 days of approval by GE. The testing charges for the design mix and the tests conducted shall be borne by the contractor.

4.5.8 WORK TESTS: -

4.5.8.1 The work tests shall be carried out at site Lab/Command Test Lab (CTL), Lucknow/NABL accredited lab.

4.5.8.2 Work test shall be conducted as per clause 15 of IS-456 of 2000. At the commencement of the concreting, samples of concrete shall be taken on each day as specified in clause 15 of IS-456 of 2000 and specimens made at the work site out of the concrete being used in the works, for the purpose of testing compressive strength.

4.5.8.3 From each of these samples, 7 test cubes of size 150 x 150 x 150 mm shall be taken to test 3 specimens at 7 days and 3 specimens at 28 days in MES laboratory. CA No., date of casting and location where concrete is being used shall be marked on each concrete cube. One test cube of preliminary and work test shall be preserved duly marking the date of casting and CA No. for verification / subsequent testing, if required. The cube shall be preserved by the GE / Engineer-in-Charge until the defects liability period of the work is over.

4.5.8.4 The testing charges for the work tests conducted in the Command Test Lab (CTL) shall be at the rates mentioned in **Annexure 'A'** to Special Conditions and the same shall be effected from the payments due to the contractor in RAR / Final bill whichever is earlier. The cost of materials, labour and transport shall be borne by the contractor. The lump sum quoted shall include the cost of testing the concrete cubes both for design mix /volumetric mix.

4.5.8.5 In the event of contractor setting up the laboratory at site as specified here -in-

before in Special Conditions, the contractor shall carry out cube testing in site lab in presence of Engineer-in-Charge. However, random testing up to 5 percent of total tests to check the compressive strength of cube shall be carried out in Command Test Lab (CTL), Lucknow for which testing charges shall be recovered from the contractor at the rate mentioned in **Annexure 'A'.** Contractor shall include this aspect in his lump sum while quoting his rates.

4.5.8.6 The Engineer-in-Charge shall maintain the record for all the test carried out in Site lab / Command Test Lab (CTL)/ NABL accredited lab separately. The cost of testing including material, labour etc., incurred shall be borne by the contractor and the lump sum quoted shall be deemed to include this aspect.

4.5.9 MIXING: -

4.5.9.1 The mix design and also execution of work shall be carried out by weigh batching. The quantum of cement for execution of work by weigh batching shall be as per mix design.

4.5.9.2 It shall be ensured that the grading characteristics as adopted in the mix design are followed throughout. Wherever the type and/or batch of cement/aggregate is changed, a fresh mix design shall be carried out. Nothing extra is payable on this account.

4.5.9.3 The contractor during the progress of work shall not change the mix design without the prior approval of the GE.

4.5.9.4 Engineer-in-Charge shall maintain a record of actual consumption of cement in proper register (other than the cement register mentioned in special conditions) and initial the entry for every day of quantity of materials used by contractor. The register shall be got checked and signed by GE/AGE(I). In case the consumption of cement as per cement consumption register is found to be more than the estimated quantity of cement due to what so ever reason, the contractor shall not have any claim, whatsoever for such excess consumption of cement.

4.5.10 WATER CEMENT RATIO: -

4.5.10.1 It is most important to maintain the water cement ratio constant and to it's correct value. To this effect determination of moisture content in both fine and coarse aggregate should be made as frequently as possible. The frequency for a given job shall be determined by the Engineer-in-Charge. According to weather conditions the amount of water to be added shall be adjusted to compensate any variations in the aggregate, IS-2386 method of test for aggregate, for concrete, Part II specific gravity, density, voids, absorption and bulking of aggregates due to variation in their moisture contents shall apply. The maximum quantity of water to be added shall be determined by mix design to be carried out as specified herein before.

4.5.10.2 Workability of concrete shall be checked at frequent intervals. The slump test or where facilities exist the compacting factor test in accordance with IS-1199 may be adopted for this purpose.

4.5.10.3 The slump for M-25 shall be between 25 to 75 mm and for M30 grades concrete (except for piles) 50mm to 100mm for medium degree workability as given in clause 7 of IS-456-2000.

4.5.10.4 Curing shall be carried out all as specified in MES Schedule Part I.

4.5.10.5 Acceptance Criteria: As per Clause 16 of IS-456: 2000.

4.6 NOMINAL MIX/VOLUMETRIC MIX: -

4.6.1 Wherever nominal mix concrete is specified, it shall be as per IS: 456. The periodicity of mix design and other tests on materials/mix shall be as decided by the GE.

4.6.2 MIXING :-

4.6.2.1 All concrete shall be mixed in mechanical concrete mixer. Where only small quantity

PARTICULAR SPECIFICATION (CONT'D…../-)

of plain cement concrete is involved, hand mixing may be adopted if approved by the Garrison Engineer. The contractor should arrange to wash out and clean the mixing drum on completion of work and or on stoppage of work if stoppage is for more than 20 minutes.

4.7 READY MIX CONCRETE: -

4.7.1 The contractor may use Ready Mix Concrete (RMC) from the approved RMC plant in lieu of Design mix concrete without any extra cost to the department.

4.7.2 RMC shall conform to the requirements of the following Indian Standards:

i. IS-4926 - Ready Mixed Concrete – Code of practice
ii. IS-9103 – Concrete Admixtures – Specifications
iii. IS-8112 – OPC Grade 43 – Specifications
iv. IS-1489- PPC - Specification
v. IS-456- Plain and RCC - Code of practice

4.7.3 SELECTION OF RMC MANUFACTURER: -

4.7.3.1 Immediately on commencement of the work, the contractor shall intimate the name of locally available manufacturers for manufacture and supply of RMC for approval of **CWE** subject to requirement of cement per cubic meter of concrete of grade M25 shall be as per IS-456 of 2000.Even though the firm for manufacturing and supply of RMC to site of work as approved by the CWE, the responsibility to maintain quality and grade of concrete fully rests with the contractor. It is the responsibility of the contractor to make payments to the RMC supplier independently and the department is not responsible for any disputes between contractor and RMC manufacturer for non-payment or delayed payment or on account of any other reasons. The contractor may alternatively establish Ready Mixed Concrete (RMC) plant of required capacity at site without any extra cost to Government.

4.7.4 MATERIALS: -

[a]. **Cement**: Cement shall conform to specifications as specified herein before.
[b]. **Fine Aggregates**: Fine aggregates shall be as specified hereinbefore.
[c]. **Coarse Aggregate**: The coarse aggregates shall be as specified herein before.
[d]. **Water**: Water shall comply with the requirements as per IS-456 and IS-3025.
[e]. **Admixtures (Liquid Type Only)**:
(i) Admixture shall be retarding super plasticizing type and shall conform to IS-9103 and of approved manufacturers as given in clause No. 6.4.12:
(ii) Admixture shall not exceed 1.5% of cement contents by volume in any case.

4.7.5 Mix design shall be got carried out by the contractor from the approved RMC manufacturer incorporating materials complying with the requirements given hereinbefore. The mix design shall be properly bound in booklet form and submitted in triplicate for approval by the Garrison Engineer. The RMC incorporated in the work shall be in accordance with the approved mix design. The CWE/GE/Engineer-in-Charge/JE (Civil) who are connected with administration and execution and other operations connected with the execution of this work shall have access to inspect /check the quality of materials used for manufacturer of RMC in RMC manufacturer's yard as well as the quality/grade of RMC supplied by the

manufacturer. The contractor shall make all arrangements for the aforesaid inspections and checks as required.

4.7.6 Contractor shall obtain a certificate from RMC manufacturer for the RMC supplied for each day to the effect that materials used for manufacturing of RMC complies strictly as per mix design requirements and the materials incorporated are conforming to the specifications given herein before. In addition, the contractor shall collect samples of materials for each days concreting in the presence of Engineer-in-Charge which shall be tested in approved lab as specified here-in-before to ensure that materials used are as per requirement as specified. Such tests for each material shall be made as per relevant BIS requirements and shall be entered in register of test results. Register shall be signed by the contractor, GE, Engineer-in-Charge & JE.

4.7.7 A register shall be maintained by the contractor duly signed by the Engineer-in-Charge showing the following details of RMC in addition to the information given on delivery ticket for each delivery of concrete (Refer Clause 9.4 and Annexure-G of IS-4926): -

(i) Time of mixing of each batch.
(ii) No. of batches in each delivery.
(iii) Location where used in the work and reference to cube test register.

4.7.8 For the RMC delivered at site and incorporated in work, sample for cube test shall be taken as per requirements of IS-4926 and as specified hereinafter.

4.7.9 If the condition of RMC delivered at site is not acceptable to the Engineer-in-Charge it shall be taken back and removed from site by the contractor at his own cost. The decision of the GE with regard to non-acceptability of RMC shall be final and binding. No claim of contractor, what so ever, shall be admissible on this account. Some of the conditions under which RMC can be rejected are given below:

i. Initial setting due to delay in transit. RMC shall be delivered at site of work within 2½ hours of mixing of first batch of concrete.

ii. Segregation of aggregate due to excessive rotation of mixer during transit.

4.7.10 <u>ACCEPTANCE OF CONCRETE</u>: -

4.7.10.1 Acceptance criteria for the RMC shall be as per IS-456. In case the RMC supplied and incorporated fails to meet the strength requirements as per IS-456, work done shall be rejected by GE and contractor shall demolish the rejected work and re-do the same with-out extra payments so as to produce the work complying with the strength requirements as per IS-456. The contractor will have no claim whatsoever on this account.

4.7.11 <u>DESIGN, MANUFACTURE, TRANSPORTATION, PLACEMENT & TESTING: -</u>

4.7.11.1 The design mix shall be carried out as per the durability condition stipulated in the contract. Concrete mix information shall be supplied by the Contractor to the RMC manufacturer on the format as per Annexure 'D' of IS: 4926, which shall form the basis of mix design.

4.7.11.2 RMC supplier will ensure that the concrete is transported in truck mixers conforming to IS-5892 to the point of placing as rapidly as possible by methods that will maintain the required workability and will prevent segregation, loss of any constituents or ingress of foreign matter or water.

PARTICULAR SPECIFICATION (CONT'D…../-)

4.7.11.3 RMC shall be used in the work only after design mix has been approved by GE in writing.

4.7.11.4 Contractor should plan their work in such a way so as to full load of concrete is discharged within 30 minutes of arrival at site and placed immediately. Re-handling should be avoided as far as practicable.

4.7.11.5 The concrete shall be discharged from the truck mixer within 2 hours of the time of loading at the plant.

4.7.11.6 Conveying equipment for concrete shall be water tight, well maintained and thoroughly cleaned before commencement of concrete mixing. Concrete shall be transported by transit mixers.

4.7.11.7 Concrete shall not be dropped from a height, thrown or otherwise treated so that segregation, undesirable finish, or defective structural quality results.

4.7.11.8 No extra water shall be added to the concrete mix after it has left the batching plant. The contractor shall take adequate precautions to protect concrete in transit from the effects of the weather.

4.7.11.9 Pumping operation whenever commences shall proceed continuously so as to prevent 'Cold' joints between placed sections. Concrete less than 6 cubic metres may be deposited manually. Concrete for columns may be deposited manually.

4.7.11.10 The delivery line of the pump shall be 100mm dia or greater and pump shall be capable of pumping concrete containing 20mm nominal size aggregate.

4.7.11.11 The pump shall have receiving hopper and pumping chamber shall be capable of pumping at least 15 Cum of concrete per hour against horizontal delivery head of at least 90 m and/or a vertical delivery head of 20m.

4.7.11.12 Pumping lines shall be of approved metallic type laid to avoid bends. The joints in pumping lines shall be sealed tight to prevent leakages.

4.7.11.13 All equipment, pump chamber, hoppers, lines etc. shall be kept clean at all times. Any build-ups in the lines of materials from previous operations shall be cleaned out prior to pumping.

4.7.11.14 In the event of breakdown in the equipment causing delay not exceeding 20 minutes, the time within which concrete cannot replaced, the following procedure shall be adopted:

"With the approval of Engineer-in-Charge, the concrete already placed shall have the 'Wet Edge' and vibrated into mass. Where atmospheric temperature exceeds 30°C, the receiving hopper and lines shall be cleaned out and concrete contained therein discarded and immediately removed from the site. The concrete shall be discarded if initial setting of the concrete has begun in the hopper or discharge lines. All lines shall be cleaned free of concrete prior to resumption of pumping after each breakdown. Concrete in the lines shall be pumped at approximately 8 minutes' intervals to ensure the concrete in the line is live, whenever delivery of concrete in the pump is delayed. This pumping interval shall be reduced to 5 minutes during extra hot weather conditions. Delivery lines where exposed to hot sun, shall be protected by covering with gunny bags, wet hessian or other approved means"

4.7.11.15 Due to mechanical malfunctioning, if concreting is required to be

PARTICULAR SPECIFICATION (CONT'D…../-)

stopped, necessary precautionary measures shall be taken by the contractor. Cost of any additional work caused due to these stoppages shall be contractor's responsibility.

4.7.11.16 No concreting shall be commenced until formwork and reinforcement and other preparatory work required are completed, inspected and approved by the Engineer-in-Charge/GE.

4.7.11.17 The contractor shall take adequate precautions and strengthening measures to strengthen the shuttering as required to withstand the pressure that will be created due to pumping of concrete.

4.7.11.18 Slump of concrete shall be as per IS-456 and as specified. The workability shall be within the following limits on the specified value as appropriate:

i. Slump : + 25mm or + 1/3of the specified value whichever is less.

ii. Compaction Factor : + 0.03, where the specified value is 0.90 or greater + 0.04, where the specified value is less than 0.90 but more than 0.80 and + 0.05, where the specified value is 0.80 or less.

4.7.11.19 Slump test shall be carried out at site by the contractor in the presence of Engineer-in-Charge/JE. The concrete shall be placed in position within the designed initial setting time. At the end of initial setting time, the unused concrete shall be rejected.

4.7.11.20 The contractor shall obtain from RMC manufacturer computer printout of the data sheet of every batch of concrete and submit to GE. The same shall be signed by the Contractor, Engineer-in-Charge & JE.

4.7.11.21 The minimum cement content shall be as per IS: 456(Durability Criteria).

4.7.12 CONSOLIDATION OF CONCRETE: -

4.7.12.1 Consolidation shall be done by mechanical vibrators, plate type for slab and needle type for other locations.

4.7.13 SAMPLING AND TESTING OF READY MIX CONCRETE: -

4.7.13.1 Allow at least the first ⅓ Cu.m. of concrete to be discharged from the truck mixer prior to taking any samples. Take required number of samples from the remainder of the load avoiding sampling the last cubic meter of concrete. Thoroughly re-mix this composite sample either on a mixing tray or in the sampling bucket and proceed with the required testing.

4.7.13.2 In addition to the tests carried out by the RMC manufacturer at the plant site, sampling and testing of concrete shall be carried out at the site after delivery as per IS-456 by the department along with the representatives of the contractor at contractor's expense.

4.7.13.3 Samples from fresh concrete shall be taken as per IS-1199 and cubes shall be made, cured and tested in accordance with IS-516 for 7/28 days compressive strength. The samples shall be taken as follows:

Place of sample	Quantity of Concrete	No of Samples	Remarks
At RMC Plant	For every 6 Cum or part thereof	1	One sample will comprise of 6 test specimens. 3 specimens for testing and 3 for preservation.

PARTICULAR SPECIFICATION (CONT'D…../-)

At Site	For every 6 Cu.m. or part thereof	1	a) One sample will comprise of minimum 9 test specimens, 3 specimens each for 7 days and 28 days testing and 3 specimens for preservation. (b) One sample forslump / compaction factor for workability.

Notes:

(i) At least one sample shall be taken from each delivery.
(ii) The test specimens will be marked showing clearly the C. A. No., date of sample, location and name of building/pile reference where it has been taken from.

4.8 TRANSPORTING, DEPOSITING AND COMPACTING: -

4.8.1 Transporting depositing and compacting generally shall be carried out as specified in Clause 4.11.9, 4.11.10 and 4.11.11 of SSR Part I 2009

4.9 PROTECTION AND CURING OF CONCRETE: -

4.9.1 This shall be carried out in accordance with specifications, given in clause No. 4.11.13 and 4.11.14 of MES Schedule Part I 2009.

4.10 PRECAST CEMENT CONCRETE: -

4.10.1 Lintels (without chajja) with a span of less than 1.5 m clear and PCC bed blocks, copings and the like may be precast at the discretion of the contractor all as specified in Clause 4.20 of SSR Part I as applicable. All precast articles shall be set in CM (1:3) with joint to match.

4.11 STRIPPING TIME FOR FORM WORK: -

4.11.1 The contractor's attention is invited to the stipulation in Clause 4.11.6.3 of SSR Part I regarding stripping of form work. The periods stipulated therein are for concrete using ordinary Portland cement. In case PPC cement used in the work, stripping time of form work applicable as per relevant IS.

4.12 LEAVE/FORM HOLES AND CHASES: -

4.12.1 The contractor as the work proceeds should Leave/form holes/chases in concrete/masonry and RCC where and as directed by the Engineer-in-Charge and make good in cement and sand mortar (1:3) when ordered to do so.

4.13 FINISH TO CONCRETE SURFACES: -

4.13.1 (refer clause4.11.16,4.11.16.1, 4.11.16.2(a),(b).(c),(d),(e) and 4.11.16.3 of MES Schedule Part-I.)

(i) Exposed surfaces of RCC chajjas, cantilevers and the like (bottom, sides) and soffits and sides of beams, soffit of RCC roof slab/floor slab/waist slab of stairs, sides of isolated column, shelves, RCC parapets, fins/fascia's and the like RCC walls shall be plastered not less than 5mm thick with cement mortar (1:3) and finished fair and even without using extra cement after removal of form work.

(ii) Exposed surfaces of RCC columns, beams, lintels, bands etc. which are continuous with plastered surfaces of walls shall be plastered & finished in the same specifications as for adjoining wall surfaces.

(iii) Exposed surface of columns, beams, lintels, bands and similar items which are not covered under clause 9.1 (a) and 9.1(b) above shall be finished with a thin layer of 5 mm thick plaster in cement mortar (1:3) and finished fair and even without using extra cement after removal of form work.

PARTICULAR SPECIFICATION (CONT'D…../-)

(iv) If thickness of plaster in cement mortar (1:3) as specified above is required to be increased in excess of 5 mm to achieve fair and even surfaces, it shall be provided by the contractor without any extra cost to Government and the same shall be deemed to have been included in the lump sum amount quoted by the contractor against Schedule 'A' Part I.

(v) Surfaces of concrete which are required to be plastered as indicated above after removal of formwork shall be roughened with wire brush and hacked out closely.

4.14 FORM WORK (FOR RCC/PCC WORK): -

4.14.1 Refer to clause 4.11.6.1 to 4.11.6.5 and relevant clauses of 7.15 of MES Schedule Part I. Formwork shall be of steel with adjustable telescopic steel vertical props and other steel accessories confirming to IS 14687:1999 (Guide lines for false work for concrete structure). Release agents shall be applied so as to provide a thin uniform coating to the finished flat surface of shuttering before laying RCC for slabs. Deformed steel sheets shall not be permitted for use as formwork. However, for the purpose of pricing DOs, the rates given in MES Schedule Part II for fair finished surfaces for concrete using steel formwork shall be applicable.

4.14.2 Where steel shuttering is not feasible in locations such as soffits and sides of lintels, beams, chajjas and other locations as decided by GE, the plywood board of adequate strength may be used after written permission of GE.

4.14.3 Wooden Ballies/ Bamboo & planks shall not be allowed as formwork.

4.15 SAMPLING AND TESTING OF CONCRETE: -

4.15.1 Refer Clause No. 4.11.17 of MES Schedule Part I.

4.15.2 Tests will be carried out on 15 cm cubes all as per IS: 516. Minimum Six cubes per sample shall be supplied by the contractor for testing.

4.15.3 For relatively small and unimportant works even though testing may be waived off by the GE, the contractor shall be responsible to achieve the desired strength of concrete.

4.15.4 The contractor shall provide all necessary materials (including moulds etc) and labour for mixing and casting of cubes, transporting the cubes to the testing laboratory and bringing back in same and any other assistance that may be required for getting the samples tested in the laboratory without any extra cost. The charges for testing of cubes shall be borne by the contractor. The testing charges shall be as indicated in Appendix 'B' to Special Condition. In case the tests are concluded in site laboratory set up by contractor, the recovery for testing charges shall not be effected.

4.15.5 The concrete which is not upto the desire strength shall be rejected and the same shall be made good by the contractor without any extra cost to Government.

4.16 PCC COVING:

4.16.1 PCC coving with PCC (1:2:4) type B-0 (using 12.5 mm graded stone aggregate) mixed with integral water proofing compound shall be provided at junction of RCC chajja with wall/lintel /beam to the radius of 75 mm irrespective of whether shown or not shown on drawings. Coving shall be 15 cm high on wall and to the full length of chajja.

4.17 FILLETING:-

4.17.1 Provide triangular filleting to skirting / dado as directed by Engineer-in-Charge with CM (1:3) finished even and smooth without using extra cement.

PARTICULAR SPECIFICATION (CONT'D…../-)

4.18 CONCRETE PADDING: -

4.18.1 Padding under bearing of RCC lintels to make up the height of opening in wall wherever required shall be done with PCC (1:3:6) type C-1 PCC blocks shall not be cut to under sized dimension and used to make up the required height in such positions.

4.19 RCC LINTELS: -

4.19.1 The bearing of lintels shall be 15 cm or effective depth (Whichever is more), unless otherwise shown on drawings.

4.19.2 Lintels (without chajja) for opening not exceeding 1.5 m clear span may be precast at contractor's option. However, these shall be priced as cast-in-situ lintels in the event of deviation if any. All other lintels and bands shall be cast-in-situ.

4.20 DRIP COURSE/THROATING / WEATHERING: -

4.21 Following works shall be executed whether shown on drawings or not: -

(i) Provision of proper drip course and / or throating and weathering to all chajja, roof, cills, coping and the like.

(ii) All flat surfaces exposed to weather shall be finished with imperceptible slope for smooth run of rain water. (In cases no slope is mentioned in drawings or specified).

4.21 WINDOW CILL: -

4.21.1 Irrespective of whatever shown on drawing or otherwise specified provide 18-20 mm thick machine cut polished kota stone slab in one piece slopping outside with round edged nosing in full width of wall and 25 mm projected outside in all windows and ventilators over 10 mm thick bedding layer in CM 1:6 and jointed in cement slurry in cills of windows & ventilators. Each side bearing shall be 40 mm.

4.22 PLINTH PROTECTION: -

4.22.1 Plinth protection shall be of 75 mm thick PCC (1:2:4) type B-1 using 20 mm graded stone aggregate with surface finished even & fair using extra cement, laid over 75 mm thick [compacted thickness] hardcore of broken stone aggregate as specified here-in-before, over rammed earth/finished surface. Construction joints shall be provided at every 3 m Interval and at every change of direction as specified in SSR and shall be filled with sealing compound grade 'A'.

4.23 RCC JALI (Factory made): -

4.23.1 RCC Jali with mix 1:2:4 (using aggregate of size 6 mm and below) shall be factory made of approved design & pattern pre-cast with wire reinforcement as per drawings and jointed in cement mortar (1:3). Thickness where not shown on drawings shall be 50 mm. Faces of RCC jali shall have fair and even surface.

4.24 DAMP PROOF COURSE: -

4.24.1 Damp proof course shall be 40 mm thick in PCC (1:2:4) type B-0 using 12.5 mm graded stone aggregate mixed with integral water proofing compound conforming to IS: 2645 and covered with 100-micron thick polythene film. The percentage of water proofing compound shall be as per manufacturer's instructions. The same shall be taken as 3% (by weight of cement) while pricing of any deviation order in omit portion.

4.24.2 Damp proof course shall be laid to the full width of walls, including under door and other opening at plinth level. Damp proof course shall not be provided for portion below door for 100/ 115 mm thick walls, dwarf walls and at locations where RCC / PCC plinth band or RCC plinth beams are provided.

PARTICULAR SPECIFICATION (CONT'D…../-)

5.0 BRICK WORK: -

5.1 BRICK WORKWITH SUB CLASS 'B' BRICKS): -

5.1.1 Bricks shall be locally available best quality sub class 'B' bricks as approved by GE. The compressive strength of bricks shall not be less than 75 Kg/Sq cm. Water absorption of bricks shall not exceed 20% when tested in accordance with IS-3495 of 1992. Sampling and testing of bricks shall be carried out as per IS- 5454 (1978) and IS-3495 (1992). The specification shall be as per IS-1077-1992. The nominal size of bricks shall be 230x115x75 mm with permissible tolerance in dimensions of bricks shall be plus or minus 8%.

5.1.2 The thickness of brick masonry walls one brick (230 mm) thick and above upto three thickness shall be in multiples of half brick (115 mm) thick including joints. Centre line dimensions shall however be maintained.

5.1.3 Thickness of joints shall be such that for course and three joints taken consecutively shall be equal to four times of actual thickness plus 4 cm.

5.1.4 Provide 8 mm dia TMT bars 2 Nos. at every fourth course for 115 mm thick wall. TMT bars shall be measured and paid for separately under relevant item. The half brick wall shall be constructed over sub base/RCC slab.

5.1.5 Provide horizontal RCC lintel band at 2.10 metres height/lintel level over 115 mm thick brick wall/ half brick wall. RCC lintel band shall be measured and paid for separately under relevant item.

5.1.6 The rating of efflorescence of the bricks shall not be more than moderate.

5.1.7 Sand for mortar shall be as specified in clause 5.4 of MES Schedule Part-I.

5.1.8 Bricks may be hand or machine moulded and shall be made from suitable soil. The brick shall have smooth rectangular faces with sharp corner & slight round edges, and shall be well burnt, uniform in colour, free from cracks, flaws, nodules of lime and emit clear ringing sound when struck. Bricks be moulded with frog 1 to 2 cm deep on one of its flat faces but those made by an extrusion process may not be provided with frogs.

5.1.9 PHYSICAL CHARACTERISTICES: -

(i) **Compressive Strength:** The minimum average wet compressive strength shall not be less than 7.5N/mm2 when tested in accordance with IS-3495 (Part-I). The wet compressive strength of any individual brick shall not fail below the minimum average wet compressive strength of brick by not more than 20 percent.

(ii) **Drying Shrinkage**. The average drying shrinkage of the bricks shall not exceed 0.15 percent when tested in accordance with IS-4139.

(iii) **Efflorescence test**: Rating of Efflorescence shall not be more than moderate.

(iv) **Water absorption**: Water absorption of bricks shall not exceed 20% when tested in accordance with IS-3495 (Part-II).

(v) **Sand:**Sand for mortar shall be as specified in clause 5.4 of MES Schedule Pt-I.

(vi) The brick shall have smooth rectangular faces with sharp corner & slight round edges, and shall be well burnt, uniform in colour, free from cracks, flaws, nodules of lime and emit clear ringing sound when struck.

(vii) **Sampling from a stack :** When it is necessary to take a sample from a

PARTICULAR SPECIFICATION (CONT'D…../-)

stack, the stack shall be divided in to number of real or imaginary sections and the required number of real and imaginary sections and required number of bricks drawn from each section. Sampling from trucks shall be as per IS-55. Scale of sampling and criteria for conformity for visual and dimensional characteristics.

(viii) The samples shall be taken as below:-

Table-1

No of bricks in a lot	For characteristics specified for individual bricks		For dimensional characteristics for 20 bricks - No of bricks to be selected
	No of bricks to be selected	Permissible no of defective in the sample	
1	**2**	**3**	**4**
2001 - 10000	20	1	40
10000-35000	32	2	60
35001-50000	50	3	80

5.1.10 Visual characteristics: -

5.1.10.1 The bricks shall be selected and inspected for ascertaining their conformity to the requirements specified in subsequent Para. The number of bricks to be selected from a lot and shall be in accordance of Col1 & 2 of Table-1 for visual characteristics in all cases and dimensional characteristics if specified for individual bricks.

5.1.11 Dimensional characteristics: -

5.1.11 The number of bricks to be selected for inspecting the dimensions and tolerance shall be in accordance with col 1 & 4 of Table 1. These bricks will be divided into groups of 20 bricks thus formed will be tested as per IS : 1077 for all the dimensions and tolerance as given here-in- before. A lot shall be considered having found meeting the requirements of dimensions and tolerance if none of groups of bricks inspected fails to meet the specified requirements.

5.1.12 Scale of sampling and criteria for physical characteristics shall as per IS-5454.The Lot, which has been found satisfactory in respect of visual and dimensional requirements, shall be next tested for physical characteristics like compressive strength, water absorption, and efflorescence

5.1.13 The physical requirements of the bricks shall be tested as per IS:3495 (Part I to IV)

5.1.14 The bricks shall have physical characteristics as specified here under: -
[i] Water absorption: 20% Max.
[ii] Compressive strength: 75 Kg/cm2.
[ii] Efflorescence: Nil to Slight sign of efflorescence i.e. 0-10% of surface area of the brick should show a thin deposit of salt.

PARTICULAR SPECIFICATION (CONT'D…../-)

5.1.15 A lot shall be considered having satisfied the requirements of physical Characteristics if the condition stipulated here in are all satisfied.
(i) The average compressive strength shall satisfy the requirements specified in Clause 10.3.1 (a).
(ii) The acceptance criteria shall be as per IS.
(iii) Average water absorption shall be as specified in Para K(i).
(iv) The number of bricks failing to satisfy the requirements of the efflorescence specified in Para (K) should not be more than the permissible in IS 5454.

5.1.16 Handling and storage of bricks shall be as per Clause 5.6.9 of SSR Part I.

5.1.17 The general quality of brick shall be as per Clause 5.6.5 of SSR Part I.

5.1.18 Bricks shall be laid & bonded all as specified in clause 5.23 & 5.24 of SSR Part I.

5.2 BEARING PLASTER: -

5.2.1 Two layers of indigenous bituminous craft paper shall be provided to the full width of brick wall under bearing of lintels, beams, slabs resting on load bearing walls, PCC bed blocks and floor/roof band where casted monolithic with RCC slab on 20mm thick bearing plaster in CM (1:4) finished even and smooth and given one coat of white wash. Each layer of indigenous bituminous craft paper shall weigh not less than 1.20 Kg per 10 Sqm. This provision, however, shall not be made where RCC slabs rest on a levelling course of PCC and in RCC framed structure.

5.2.2 Size of PCC bed blocks for beam where not shown in the drawings shall be twice the width of beams in length, covering the entire thickness of wall and 200 mm thick (depending upon type of brick) below the RCC beams resting on masonry.

5.3 FLY ASH BRICK WORK: -

5.3.1 Brick work as in filler wall of RCC frame work shall be Fly Ash Brick masonry. In case load bearing wall/compound wall (not a RCC frame work shall be sub class 'B' brick masonry.Fly ash bricks shall be locally available best quality bricks as approved by GE.

5.3.2 Fly ash bricks (Solid) will be of size 400x200x200mm/400x200x100mm /400x200x150mm /230x110x 70mm/ 250mm x 115mm x 75mm as approved by GE with 28 days average wet compressive strength not less than 50 Kg/Sq.cm when tested as per procedure laid down in IS-3495(Part-I) and water absorption not more than 15% by mass when tested as per procedure laid down in IS-3495 (Part-II) after immersion in cold water for 24 hours. Average drying shrinkage of the blocks will not be more than 0.15% when tested as per IS-4139. However, the contractor will be allowed to use fly ash brick of smaller size as per site condition with prior approval of GE without any extra cost on plus side. The fly ash used in the process of making blocks should confirm to Grade II of IS-3812. The tolerance on dimension of the blocks will be taken as per clause 5.2 of IS-12894. The Fly ash brick/block shall conform to IS-12894.

5.3.3 200/300/400mm thick (as applicable), brick wall shall be built in CM 1:6 and 100mm thick brick wall shall be built in CM 1:4.

5.3.4 The entire workman ship shall conform to relevant clauses as applicable for brick work in section 5 of MES Schedule Part-I. The thickness of joint in Fly Ash brick work shall not be more than 6mm.

5.3.5 Provide horizontal RCC lintel band at 2.10 metres height/lintel level over 115mm thick brick wall/half brick wall and hidden beam as specified and shown in drawing.

PARTICULAR SPECIFICATION (CONT'D…../-)

5.3.6 Fly ash bricks should be adequately cured and stored for six weeks before incorporating in the masonry works in both load bearing and partition walls.

5.3.7 The record of above action shall be maintained meticulously at site of work.

6.0 BLANK

7.0 WOOD WORK, JOINERY, FALSE CEILING, STAINLESS STEEL RAILING ETC: -

7.1.1 Timber for all wooden joinery and wood work shall conform to specifications given in clause 7.3 of the MES Schedule Part I and defects shall be within the permissible limits as specified in Clause 7.4 and 7.5 of the MES Schedule Part I. Unless otherwise specified timber for all joinery and woodwork shall be 2nd class hard wood (Hollock/Sal/Benteak/Bijasal) and as per the approved sample kept in GE's office.

Note: - However, the contractor may provide Shisham wood in lieu of Hollock/Sal/ Benteak/Bijasal wood without any extra cost to Govt.

7.1.2 Timber shall be well seasoned and chemically treated under pressure (kiln dried). Factory made shutters shall also be 'kiln' seasoned only. The moisture content of timber shall not exceed the limits laid down vide clause 7.7 of the MES Schedule Part I.

7.1.3 Preservative anti-termite treatment shall be carried out to all woodwork and joinery. Contractor's quoted rates shall be deemed to cater for this provision. Factory made plywood, particle boards and factory made shutters are not to be further treated with any chemical against termites.

7.1.4 Wood work and joinery other than factory made, ply wood particle board and factory made shutters as described above shall be treated with copper Napthanate or any other chemical specified in IS: 401 of 1982 and shall be applied in any one of the manners specified in IS or as directed.

7.1.5 Timber surfaces hidden or not exposed to view and in contact with brick work/masonry /concrete shall be left clean sawn and applied with two coats of creosote, unless specified otherwise hereafter.

7.1.6 **WORKMANSHIP**: -

(a) Dimensions of various parts of joinery shall be as shown on drawing and shall supersede those stipulated in clause 8.20& 8.24 of MES Schedule Part I. For the purpose of pricing deviations, factory made shutters shall be classified as second class hard wood and priced accordingly at the applicable rates in MES Schedule Part-II and no adjustment in SSR rates shall be done for change in sizes of stiles and rails.

(b) Irrespective of what is shown on drawings, use of nails in the joinery is prohibited. Wood screw of appropriate size shall be used and shall be of "Nettle fold" brand or any other equivalent brand as approved by GE.

(c) All joinery work shall be of full dimensions as shown on drawings. Overall sizes of joinery as shown on drawings shall be maintained. Tolerance in sizes and thickness shall be all as specified in SSR Part-I.

(d) Plugging/wooden gutties to walls wherever shown on drawings or where required shall be done with wooden plugs made out of hard wood (Hollock/Bijasal/ SAL/Benteak/Shisham/Teak) as per clause 7.29 of MES Schedule Part I.

7.1.7 **PLY WOOD AND PARTICLE BOARD:-**

(a) Un-veneered particle board where indicated shall be flat pressed, BWP grade bonded with phenol formaldehyde synthetic resin adhesive conforming to IS: 3087

PARTICULAR SPECIFICATION (CONT'D…../-)

(Type I Part I) all as specified in para 12.13 of MES Schedule Part I.

(b) Veneered particle board where indicated shall be three layers flat pressed with commercial or teak veneer (both sides) as indicated on drawing/specified and shall be BWP grade bonded with phenol-formaldehyde synthetic resin adhesive as per IS-3097-Grade I, Type-I for commercial veneer and decorative veneer all as specified in para 12.14 of MES Schedule Part I.

(c) Plywood where indicated shall be BWP grade bonded with phenol formaldehyde synthetic resin adhesive conforming to IS: 303 for general purpose plywood and IS: 1328 for decorative face plywood. Facing shall be of teak veneer as shown on drawing/specified.

(d) The pre-laminated particle board in all situations shall be three layered ISI marked (IS: 12823 Grade I) with pre-lamination of approved shade on one side and matching white on other side and pre-laminated on both sides (for partitions) as shown on drawings. The lump sum quoted by the contractor shall be deemed to include the above provisions.

(e) Marine plywood shall be as per IS: 710-1976 (specifications for marine plywood) (Kit ply) and bonded with best quality phenol formaldehyde synthetic adhesive BWP type or hot pressed at high pressure and temperature.

(f) Thickness of particle board shall 18/19 mm and of plywood 6 mm, if not specifically specified on drawings.

7.2 **FRAME (CHOWKHAT) FOR DOORS EXEPT WC/BATHPRESSED STEEL FRAME**: -

7.2.1 Where pressed steel frames/chowkhats are shown on drgs these shall be machine pressed steel rebated made out of 1.25 mm thick MS sheets (factory made). Pressed steel frame shall comply to IS: 4351(1976). Hollow space of chowkhat frames shall be filled with PCC (1:3:6) Type C-0 using 12.5 mm graded stone aggregate as the work proceeds unless shown in the drawing. Refer Clause 10.27.1 of MES Schedule Part-I. For other specifications unless specified otherwise hereafter profile and section, size of frame and lugs/hold fasts shall be as shown on drawing and not as in IS. Back of chowkhat and MS hold fasts/lugs shall be given one coat of bituminous paint over primer before fixing in position. All exposed surfaces of chowkhats shall be given two coats of synthetic enamel paint over a coat of red oxide primer.

7.3 **FACTORY MADE SOLID PVC DOOR FRAMES WC/BATH: -**

7.3.1 Solid Single PVC Door Frame solid PVC Door Frame of the size 50 x 47mm made out of 5mm PVC sheet. The door profiles are to be reinforced with 19xl9mm M.S. Square tube. EPDM rubber gasket weather seal to be provided throughout the frame. The profile shall be mitre cut at two corners and joined by inserting 2nos. of 150mm long brackets of 15x15mm M.S. square tube inside the 19mm x 19mm M.S. square tube and suitably screwed all as per table of clause 8B.3.1 of SSR Part II 2020.

7.4 **WOODEN SHUTTER: -**

7.4.1 Panelled, partly panelled and wire gauge shutters for all wooden doors are shown on drawings shall be factory made as under:

(a) Stiles, rails, sash bars: These shall be of 35 mm thick. Other details all as shown on drgs. ASCU Vacuum pressure treated out of kiln seasoned hard wood second class of species bijasal/hollock/ben teak all as per IS: 1003.

NOTE: - In case factory made shutters with aforesaid species of timber are not available due to any reason, the contractor shall provide factory made shutters made from Shisham wood without any extra cost to Govt.

PARTICULAR SPECIFICATION (CONT'D…../-)

(b) Panel for panelled doors: -

(i) Unless otherwise shown on drawings, panel for panelled doors for Internal and external doors (OTHER THAN BATH AND WC) shall be of 9 mm thick 7 ply grade BWR.

7.5 **FOR BATH, TOILET AND WC FACTORY MADE SOLID PANEL PVC DOOR SHUTTER: -**

7.5.1 30 mm thick Factory Made Solid Panel PVC Door Shutter consisting of frame made out of M.S. tubes of 19mmx 19mm for stiles, & 15mm x 15mm for top & bottom rails. M.S. frame shall be covered with 5mm thick heat moulded PVC sheet 'C' channel having a 5mm thick PVC sheet strip of 20mm width stuck inside with solvent cement, forming stiles, and 5mm thick PVC sheets for top rail, lock rail & bottom rail on either side, and 10mm (5mm x 2) thick, 20mm wide cross PVC sheet as gap insert for top rail & bottom rail. Panelling of 5mm thick PVC sheet to be fitted in the M.S. frame welded / sealed to the stiles & rails with PVC sheet beading, and joined together with solvent cement adhesive etc. complete as per direction of Engineer-in-charge, manufacturer's instructions and drawing. The width of stiles & rails will be in proportion to the width of the door shutter as per table of clause 8B.2.1 of SSR Part II 2020. Contractor may use PVC doorframe but at no extra cost to Govt.

8.0 BLANK

9.0 BUILDER'S HARDWARE:-

9.1 **GENERAL: -**

(a) All articles of builder's hardware shall comply with the relevant IS specifications and the specifications given in the MES Schedule for the relevant item. Builders' hardware shall be provided all as specified in Schedule of fittings and as specified in relevant drawings and shall be ISI marked.

(b) In case the size of particular fittings is not shown in the drawings, it shall be of size as directed by the GE.

(c) Screws used for fixing items of Builders hardware shall be as specified in clause 9.2.6 of MES Schedule Part-I. Unless specified otherwise elsewhere, screws shall be dipped in adhesive before fixing on particle board.

(d) Irrespective of what is shown on drawings, all articles of builder's hardware (other than for steel windows, butt hinges and rising door springs with magnetic catches) shall be of Stainless steel.

(e) Butt hinges shall be of SS, IS marked.

(f) The contractor shall produce samples of each article of builder's hardware (fittings) which he proposes to use and get it approved. The sample shall be labelled as such, signed by the GE and the contractor. These shall be kept in the custody of the GE till the completion of work. In case rat-tail spring door closers are mentioned on drawings these shall be changed to helical door springs, 150 mm long of an approved make.

9.2 **ARTICLES: -**

(a) Butt hinges for doors shall be SS conforming to IS and all as specified in clause 9.7.2 of MES Schedule Part I. Butt hinges shall be run welded to machine pressed steel sheet frame/angle frame and screwed to timber shutters.

(b) Size of Stainless steel sliding bolts unless otherwise specified shall be 300 mm with 16 mm dia shoot.

(c) Stainless steel tower/ Barrel Bolts shall be with bolt shoot 10 mm dia up to size 125 mm and 12mmdia for size 150mm and above. For steel window, the dia of bolt

shall be 10 mm.

(d) All other articles of builder's hardware shall comply with the requirement of relevant IS and as specified in MES Schedule Part I.

(e) Stainless steel wire cloth/gauge for mosquito/fly proofing shall be as specified in clause 9.25.3 of MES Schedule Part I. It shall have 0.36 mm nominal dia of wire and 1.40 mm average with of aperture for mosquito/fly proofing and as approved by GE.

10. STEEL AND IRON WORK: -

10.1 GENERAL: -

Steel and ironwork in various situations shall be carried out as specified in MES Schedule Part-I, Section-10 and as shown in drawings. All steel required for the work under the contract shall be procured, supplied and incorporated in the works by the contractor under his own arrangement.

10.2 GRADES AND QUALITY: -

10.2.1 Steel supplied by the contractor shall conform to the following grades and quality: -

(a) **Steel for Concrete Reinforcement: -**

(i) High strength deformed steel bars produced by thermo mechanical treatment process (TMT steel bars of grades Fe-500 /Fe 500D/Fe550/Fe550D) meeting all other requirements of IS-1786.
Note : For Swimming pool, Diving pool, Balancing tank ,Hazardous tank, RCC over Head water tank , High strength corrosion resistant deformed steel bars (from primary producers as approved by E-in-C's Branch) produced by Thermo Mechanical Treatment process (TMT steel bars) of grade Fe-500 /500Dmeeting all other requirements of IS-1786,shall be used. Minimum elongation shall be 18%.

(ii) Mild steel bars shall conform to IS-432 (Part-I) and Grade-I.

(b) **Structural Steel (Refer Clause 10.4 of MES Schedule Part-I): -**

(i) Structural steel standard quality shall conform to IS-2062-2006 (6th revision) and Grade Fe-410-W (Gde E-250), quality A. The above mentioned steel shall be provided in the location mentioned in drawings and in clause 10.4.1 of MES Sch Part-I.

(ii) Structural Steel ordinary quality shall be used for structures not subjected to dynamic loading. This steel shall not be used where welding is used in fabrication and in earth quake zone where severe damage is expected and design of structure based on plastic theory. The steel shall conform to IS-2062-2006 (6th revision) and Fe-290 (Gde E-165). The above mentioned steel shall be provided in the location specified in drawings and in clause 10.4.2 of MES Sch Part I.

(iii) Structural steel ordinary quality shall be used in railing, gates, fencing, guard bars, grills, holdfasts, door and window frames etc, and shall conform to IS-2062-2006 (6th revision), Fe-290 (Gde E-165). The above mentioned steel shall be provided in the location specified in drawing and in clause 10.4.3 of MES Schedule Part I.

(c) **Galvanised Steel Sheets (Plain & Corrugated):**Galvanized steel sheets (Plain and Corrugated) shall conform to IS-277. Unless otherwise specified the sheet shall be of Galvanized Class-4. Grade of zinc coating shall be as per clause 10.29.1 of MES Schedule Part-I.

(d) **Hard drawn Steel Wire Fabric for Concrete Reinforcement:-**Fabric reinforcement shall conform to IS-1566.

(e) **Steel Tubes for Structural Purposes:** Steel tubes for structural purposes shall conform to IS-1161 and shall be of grade YST-240.

PARTICULAR SPECIFICATION (CONT'D…../-)

10.3 **PROCUREMENT AND TESTING OF STEEL :-**

(a) **REINFORCEMENT STEEL** : All steel for reinforcement to be used in the work shall be contractor's supply. The contractor shall procure steel for reinforcement (TMT bars) directly from the main producers of steel as per **Appx D**.

(b) **STRUCTURAL STEEL:**Grade of Structural steel shall be all as specified here-in-after. Structural steel to be used in the work by the contractor shall be procured directly from primary producers as per **Appx E.**

(c) **GALVANISED STEEL SHEETS & FABRIC REINFORCEMENT FOR CONCERETE :** These shall be procured directly from main producers/BIS marked manufacturers as approved here-in-before at the option of Contractor without any minus price adjustment.

(d) Reinforcement steel, structural steel and Galvanised steel sheets & fabric reinforcement for concrete may be permitted to be procured from authorized dealers of main producers in case the total requirement of steel is less than 5 tonnes and specifically approved by Accepting Officer in writing.

(e) Steel sections for railings, gates, fencing, guard bars, grills, steel chowkhat, holdfasts etc., which do not constitute structural members, can be procured from main producers/ secondary producers/ BIS marked manufacturers or their authorized dealers at the option of Contractor without any minus price adjustment. Tests will not be insisted upon for such steel sections.

(f) The documents in support of purchase of steel shall be furnished to GE for verification. The particulars of the manufacturers of steel shall be furnished by the contractor for every lot of steel separately as per the Performa given in Appendix F.

10.4 **TESTING OF STEEL**: -

(a) The contractor shall submit the manufacturer's Test certificate in original alongwith the test sheet giving their results of each mechanical test as applicable and the chemical composition of the steel or authenticated copy thereof duly signed by the manufacturer with each consignment. The Engineer-in-Charge shall record these details in a steel Acceptance Register after due verification.

(b) It will be mandatory for the GE to carry out simple field test and record the findings before the contractor is allowed to use steel in the work. CWE may also carry out random checks and record his remarks in the Steel Test Register. Such simple testing includes sand papering the cross section of the TMT bar and dipping it in chemical solution to give a clearly defined annular ring of tempered steel. Cost of all such facilities for test shall be borne by the contractor.

(c) The Accepting Officer may order a board of Officers for random check of steel and verification of connected documents. In case of such check, if it is found that steel brought on site/incorporated in work does not meet the requirements laid down in the contract, such steel/work in which such steel has been incorporated shall be rejected or devalued based on the defects. In this regard decision of the Accepting Officer shall be final and binding. Contractor shall have no claim whatsoever on account of such rejection.

(d) Independent testing of steel by the GE from random samples of steel drawn from various lots from National Test House, SEMT, CME, Regional Research Labs, Govt Engineering College, Zonal Lab or Govt approved Labs etc, as per the recommended frequency given here-in-after shall be optional at the discretion of the GE in case of procurement of steel from main producers and testing charges shall be borne in accordance with Condition 10 A of IAFW-2249 i.e. testing charges shall be borne by the Deptt, if the test results are found in order otherwise these shall be borne by the Contractor.

(e) Independent testing of steel by the GE from random samples of steel drawn from various lots from National Test House, SEMT, CME, Regional Research Labs, Govt Engineering College, Zonal Lab or Govt approved Labs etc as per the recommended frequency given here-in-after shall be mandatory in case of procurement of steel from secondary producers and testing charges shall be borne by the contractor irrespective of the outcome of test results.

PARTICULAR SPECIFICATION (CONT'D…../-)

(f) In both the cases at sub Para 10.3 (d) & (e) above, the Contractor at his cost shall provide all facilities required for the testing and cost of materials consumed in tests shall also be borne by the Contractor.

(g) The elongation of reinforcement steel shall not be less than 18%.

10.5 **FREQUENCY OF NORMAL MASS, TENSILE, BEND AND REBEND TESTS OF STEEL: -**

NOMINAL SIZE STEEL REINFORCEMENT	**QUANTITY**
(a) Bars size less than 10mm.	One sample (three specimens) for each test for every 25 tones or part thereof).
(b) Bars size 10mm to 16mm.	One sample (Three specimens) for each test for every 35 tones or part thereof)
(c) Bars size over 16mm.	One sample (Three specimens) for each test for every 45 tones or part thereof
STRUCTURAL STEEL	
(a) Tensile test including %age elongation	One test for every 25 tones of steel or part thereof)
(b) Bend test	One test for every 10 tones of steel or part thereof

Note: - Applicable for all Tests.For various tests, acceptance criteria, tolerance etc. refer to steel supply/Acceptance form given at **Appx F** and relevant BIS Codes.

10.6 **DOCUMENTATION: -**

(a) The contractor shall submit original purchase vouchers from the manufacturer for the total quantity of steel supplied under each consignment to be incorporated in the work. All consignments received at work site shall be inspected by the GE alongwith relevant documents before acceptance. The original vouchers and the test certificates shall be defaced by the Engineer-in-Charge and photocopy of the same shall be kept on the record in the office of GE duly authenticated and with cross reference of control number recorded in the steel Acceptance Register and will be signed by JE (Civil), Engineer-in-Charge, GE and the contractor duly endorsed CA No & Year.

(b) The entire quantity of all steel items shall be recorded in Measurement Book indicating size, quantities, voucher numbers, dates brought on site for record purposes as "not to be abstracted" and signed by Engineer-in-Charge and the contractor before incorporation in the work.

(c) Irrespective of what is shown in drgs or specified elsewhere, steel reinforcement to be used in this project shall be Thermo Mechanically Treated steel bars.

10.7 **STORAGE ACCEPTANCE/PRESERVATION OF STEEL**:-

10.7.1 The steel procured by the contractor shall be stored in the site of work as directed by Engineer-in-Charge/ GE neatly in separate stacks at least 15 cm above GL for various grades/quality/sizes/ consignments with distinct paint marks for identification. The steel so stacked shall be removed for incorporation in the work only in the presence of departmental representative. The quantity of steel of various sizes received at site and recommended for incorporation in the work shall be entered in a separate register and signed by the contractor and the Engineer-in-Charge daily.

10.7.2 Steel shall be stored in a manner so as to prevent distortion and corrosion till it is consumed in the work. Any section that has deteriorated and corroded or if, considered defective for any other reason, the same shall be removed from site by contractor at his cost.

10.7.3 The contractor will keep a separate stack of steel brought at site for inspection, away from the accepted stack of steel. In case, the consignment does not meet any of the requirements of the relevant IS codes, the steel will be rejected by the GE and it

PARTICULAR SPECIFICATION (CONT'D…../-)

will be removed from the site within 24 hours at the cost of the contractor.

10.7.4 The steel shall be procured and brought to the site one month before it is required to be incorporated in the work taking into account the time required for its inspection and acceptance as specified here-in-before.

10.7.5 Contractor shall be responsible for proper storage, preservation and maintenance of steel at site till it is consumed in the work.

10.8 **SAFETY OF STEEL:-**

10.8.1 It will be responsibility of contractor to make sure that all possible arrangement are made for safe custody of the steel. In case of any loss of steel, only contractor will be responsible and the loss will be made good by contractor without any delay or claim what so ever.

10.9 **SCHEDULE OF SUPPLY:-**

10.9.1 Contractor shall work out complete requirement of steel size wise and phasethe same as per the activities planned to be executed in terms of CPM networking.The contractor shall procure all the steel sections in accordance with this CPM chart. Schedule of supply of steel will be finalised by GE in consultation with contractor and same will be incorporated in CPM chart so that supply of steel is monitored in a way to avoid any delay in completion of the work. The schedule of supply of steel will be vetted by CWE from time to time.

10.10 **CONVERSION WEIGHT OF STEEL**: -

10.10.1 The weight of steel shall be calculated as per the conversion factors specified in the MES Schedule. For sections not listed in MES Schedule, ISI conversion table shall be followed or manufacturer's certificate if the weights are not available in MES Schedule /ISI tables.

10.10.2 Normal waste and off-cuts shall be stacked neatly which shall be the property of contractor. Contractor shall be allowed to remove such cut pieces after inspection and certification by the Engineer-in-Charge

10.10.3 Advance on account of payment made towards these cut pieces shall be adjusted from advance on account of payment immediately falling due and before removal of such cut pieces from site.

10.11 **PAYMENT IN RAR: -**

10.11.1 Payment of the steel brought by the contractor should only be released by the GE after taking action on points enumerated hereinbefore and after completing the documentation mentioned here-in-before in this regard.

10.11.2 Before procurement of steel, contract and structural drawing shall be read thoroughly and various grades/types of steel to be incorporated in the work shall be identified by contractor and got approved by the GE. Steel shall be procured sufficiently in advance as mentioned here-in-before.

10.12 **REINFORCEMENT STEEL**: -

10.12.1 Reinforcement of mild steel and deformed cold twisted bars shall be of TMT bars in lieu thereof without change of any spacing all as shown on drawings in accordance with the provisions of IS: 456 of 2000 without any extra cost to Govt.

10.12.2 Reinforcement shall be fabricated, placed in position all as shown on drawings and specified in Clause 10.17 to 10.22 of MES Schedule Part- I.

10.12.3 Laps and crossing shall be tied with mild steel wire (annealed) (soft drawn) of diameter not less than 0.90mm. Laps shall be staggered.

10.12.4 Irrespective of whether corner reinforcement is indicated or not in RCC plan showing reinforcement of slabs, corner reinforcement shall be provided as shown in TD (typical drawings).

PARTICULAR SPECIFICATION (CONT'D…../-)

10.12.5 The contractor shall be responsible for accurate fixing of reinforcement shown on drawings, and shall not pour any concrete until the reinforcement has been inspected in position and approved in writing by the GE. The contractor shall take necessary precaution to prevent any displacement of reinforcement bars during concreting.

10.13 **STRUCTURAL STEEL WORK** :-

10.13.1 Structural steel work shall be executed as specified in relevant clauses of section 10 of MES Schedule Part I.

Note: Steel for holdfasts, grills, guard bars, cooler brackets etc shall be ordinary quality. Balance structural steel shall be of standard quality. Test certificate for ordinary quality steel is not required.

10.13.2 **MS BLACK, BOLTS, NUTS AND WASHERS: -**

(i) All MS Black, Bolts and nuts shall conform to IS: 1353 part I to III of 1984 and round washer shall conform to IS: 5370, IS: 5372 or IS: 5374 where ever specification are not available, the American standard specifications to be followed.

(ii) Bolts, heads, nuts, shall be of such length as to project and clear thread beyond the nuts when fixed in position and these shall fit in the holes without any shake. The nut shall fit in the threaded ends of bolts properly.

(iii) The round washers shall be placed under the heads and nuts of permanent bolts. Maximum two washers for one nut and one for each bolt head shall be used. Bolts threads shall be out side limit of jointing members and un threaded portion of bolt shall not be out side the washer.

(iv) Bolts, Nuts & washers shall be thoroughly cleaned and dipped in double boiled linseed oil before used. The bolts shall be tightened starting from the enter of joints towards the edge.

10.14 **HOLD FASTS:-**Provision of hold fasts to Chowkats of doors/windows/ventilators and cupboards, etc all as per details shown on drawings and as specified. These shall be embedded with PCC (1:2:4) type B1 using 20mm graded aggregate in the wall/mix of concrete as per the columns.

10.15 **WIRE MESH:**Stainless steel wire cloth/gauge for mosquito/fly proofing shall be as specified in clause 9.25.3 of MES Schedule Part I. It shall have 0.36 mm nominal dia of wire and 1.40 mm average with of aperture for mosquito/fly proofing and as approved by GE.

10.16 **FAN HOOKS WITH BOXES:**Wherever fan hooks/fan points have been shown, MS boxes with fan hooks shall be provided as per detail shown on drawings. Exposed faces shall be given two coats of white paint over a coat of red oxide primer. However, fan hooks without boxes shall be provided in roof slab. Fan hook boxes shall be covered with 3mm thick plastic laminated sheet of required colour.

10.17 **MS GRILLS:** Provide fabricated MS Grills to windows where shown on drawings**.**MS grills shall be fabricated with MS bars/flat to required size all as per the details shown on drawings**.** MS Grills shall be fixed to chowkats with suitable GI screws all as directed**.** All the surfaces of grills shall be treated with paints as specified hereinafter for steel surfaces**.**

10.18 **MS GUARD BARS:** MS guard bars shall be provided to ventilators wherever shown on drawings. The surfaces of guard bars shall be treated with paints as specified hereinafter.

10.19 **BENDING OF BARS BY BAR BENDING MACHINE:**TMT bars used for reinforced concrete works shall only be bent using bar bending machine driven by motor powered by not less than 5 HP suitable for bars of dia. up to 32mm as approved by GE. Bending of bars shall in no case be allowed to be done manually. The number of bar bending machines to be provided at site shall be adequate enough to execute the

works at all sites simultaneously as per the CPM. No claim whatsoever arising out on this account shall be admissible.

10.20 **WELDING:**The welding work shall be executed all as specified in clause No. 10.15 & 10.16 of MES Schedule Part-I as applicable. All Welding shall be carried out by metal arc welding process conforming to the requirements of IS-1024. The welding electrodes shall conform to IS-814. Filler wire & flux shall conform to IS-3613.

11 ROOFING/ RCC SLABS: -

11.1 RCC floor/roof slab shall be laid to the required thickness all as shown on drawings.

11.2 RCC roof slabs shall have a finished slope of all as shown on drawing. However, where slope is not shown the same shall be finished to the slope of 1:40.

11.3 **WATER PROOFING TREATMENT OVER ROOF SLAB: -**

11.3.1 The Contractor shall carryout water proofing work over roof to ensure the entire roof water tightness during the guarantee period of **10 (Ten) years** and he will be responsible to keep the entire surface of roof water tight for a period of 10years from certified date of completion of work. The security Deposit towards the guarantee for water proofing treatment shall be retained from the contractor's bill against the guarantee for water proofing treatment shall be calculated **@ 5% of water proofing treatment at contract rates**. This shall be worked out by GE and intimated to the Contractor. The amount shall be released to the Contractor after the satisfactory completion of **TEN YEARS** guarantee period. During the guarantee period if any leakage is found out same shall be got rectified by the Contractor without any extra cost to the Government. In any case during the guarantee period the Contractor who executed the WPT works should inspect and examine the treatment once in every year and make good any defects observed. Defects liability period of two year as laid down under condition 46 of IAFW-2249 is not applicable for water proofing work covered under this contract and shall deemed to be amended to the extent mentioned above for water proofing treatment work. The Garrison Engineer may accept BankGuarantee Bonds from Schedule Bank or fixed deposit receipt (pledged in favour of GE) from any approved Bank for the above said amount for a period of 10 years in which case no further amount will be recovered from the final bill on this account.

11.3.2 Tenderer shall submit **10 years guarantee certificate** (as per performa enclosed herewith)duly undertaking and accepting the responsibility to carryout all required rectifications at his own risk and cost. In case of Contractor's failure to carryout the rectifications, the Government shall be at liberty to get all such rectifications carried out at his risk and cost from security amount. The tenderer is required to visit the buildings in which the water proofing treatment is to be done in order to ascertain the exact nature and extent of work involved. His quoted rate in the Schedule-'A' Part-I against the respective items shall be deemed to include for the following aspects for effective water proofing treatment and no extra claim on whatever account will be entertained from the Contractor.

11.4 **WATER PROOFING MEMBRANE: -**

11.4.1 APP modified water proofing membrane to be incorporated in the work for WPT shall be obtained from one of the manufacturers as mentioned in **Appendix 'G'** to PS.

11.4.2 **PHYSICAL PROPERTIES:-**
The physical properties of APP modified water proofing membrane thickness 4mm shall be as under:

(i)	Membrane weight	Not less than 3.60 Kg/Sqm
(ii)	Not woven polyester weight	Not less than 160gms/sqm
(iii)	Softening paint of membrane	Not less than 150°C
(iv)	Joint tensile strength	Longitudinal > 650kg/sqcm at 230°C Transverse > 450 kg/sqcm

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(v)	Tear strength (L/T)	Longitudinal > 300N/ sqcm, Transverse > 250N / sqcm
(vi)	Elongation of reinforcement	> 30%
(vii)	Impermeability under water pressure	> 3kg/cm^2

The water proofing membrane treatment shall be carried out by the approved applicator of the manufacturer as approved by the GE.

11.5 **CERTIFICATE BY SPECIALISED AGENCY: -**

11.5.1 A certificate by the specialized firm, who has executed the work of water proofing treatment under this contract will be rendered to the department stating that the said work has been executed by them and shall be signed jointly with the main building contractor. This certificate however does not absolve the main contractor from his responsibility in respect of specialist work as per contract condition. The main contractor shall be responsible to ensure that the roof slab shall be water tight/leak proof during guarantee period.

11.5.2 **APPLICATION AND WATER PROOFING WITH APP MEMBRANE OVER ROOF:-**

a) RCC roof slab shall be cleaned with water for roof treatment.

b) Cement bedding not less than in 10mm thick in CM 1:3 mixed with integral water proofing compound as per manufacturer's instructions shall be provided on RCC slabwith smooth and true surface and slope and as specified under relevant item of Schedule'A'.

c) Apply oil bound bituminous primer @ 0.30litre/sqm manufactured by the same agency as that of APP with a density of 0.87-0.89kg/litre at 25^{0}C on the prepared dryroof surface. After primer the entire roof surface shall be applied with hot blown bitumen grade 85/25 conforming to IS – 702 @ 1.2 Kg / Sqm, Provide and lay APP modified water proofing membrane reinforced with non woven polyester mat one layer weighing > 3.5kg/sqm on prepared surface over bitumen layer.

d) **For non accesssableroof :-** Laying 40 mm thick brick tiles over bedding layer 20 mm screed in CM 1:4 mixed with polymer based liquid water proofing additive @ 5% by weight of cement over APP membrane as specified above. Tiles joints shall be filled with CM 1:3 mixed with crude oil as per SSR- 2009 (Part-I).

e) **For accesssable roof /Terraces:** Laying 20 mm thick machine pressed factory-made PCC tiles as per IS 1237 – 1980 over bedding layer 20 mm screed in CM 1:4 mixed with polymer based liquid water proofing additive @ 5% by weight of cement over APP membrane as specified above. Tiles joints shall be filled with polymer-based cement proof joint filler as approved by GE.

11.5.3 **TESTING OF MEMBRANE: -**

The following tests shall be carried out in approved Government Laboratories and necessary test reports shall be submitted to the Garrison Engineer for the materials before incorporation in the work:

(a) Thickness of membrane
(b) Weight of membrane
€ Tensile strength (longitudinal/crosswise)
(d) Tear Resistance
€ Softening point
(f) Weight of Carrier (Polyester).

Apart from above tests Contractor shall also produce manufactures test certificate. Test certificate shall be produced with every consignment of material brought at site by the contractor.

11.5.4 **TEMPLATE (LABEL) :**Template (label) made of 15mm thick plaster in CM (1:3) of size 60x60cm showing the following particulars shall be provided on each bldg

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before the final completion of work without any extra cost to the Department:

(a) Name of Contractor
(b) CA Number
€ Date of completion
(d) Guarantee period

The above information shall be written with white paint with 5cm size letters and numbers.

11.5.5 The entire quantity of materials such as water proofing membrane bitumen primer, bitumen and bituminous aluminium paint etc required for the whole work shall be brought at site in sealed position and shall be entered in the MB as **'NOT TO BE ABSTRACTED**' showing the date, total quantity brought. Each seal shall be opened in front of Engineer-in-Charge only. Paid vouchers from authorized dealer shall be submitted to the Engineer-in-Charge immediately after every consignment and copy of such paid voucher shall be forwarded to Garrison Engineer by the Engineer-in-Charge duly endorsed.Guarantee to be executed by the contractor for removal of defects after completion in respect of water proofing works shall be as per **Appx H**.

12 FLOODING TEST TO RCC SLABS: -

12.1 On completion of casting of RCC slabs and sealing up pores the Contractor shall carryout flooding test of each RCC slab after 28 days of curing of RCC slab. Flooding test shall be carried out by forming earthen bunds over parts of the roof slab and at selected locations as directed by the GE. Water shall be ponded for minimum depth of 50 mm and shall be allowed to remain over slab for a period of 72 hours. During this period if any defect such as leakage/seepage etc is noticed it shall be rectified by the contractor by grouting to the satisfaction of the GE. After rectification of the defects more tests shall be carried out at the discretion and to the satisfaction of the GE.

12.2 **Testing** : The contractor will submit original purchase voucher along with the test certificate for each consignment/lot of material brought at site. Independent testing will also be carried out be GE for least one test sample consisting of three test e specimen at random from each lot. The polymeric membrane shall be got tested from the test house as mentioned here under :-

Frequency of test At least one test consisting of three sample specimen of polymeric membrane at random from each lot. The polymeric membrane shall be got tested from the test house as mentioned hereunder:-

(a) Chemical and metallurgical, Services T.S. No. 63, SIDCO Industrial Estate Ekkaduthangal, Chennai-600097 Fax No: 44-2312944.
(b) Central Institute of Plastic Engineering and Technology (Ministry of Chemical and fertilizer, Govt of India) Guindy, Chennai-600032, Fax No : 94-44-2344673
€ National Test House, Alipore, Kolkatta.
(d) Regional Testing Centre (Northern Region), NABL Accredited laboratory, Shaeed Capt. Gaur Marg, Okhla Industrial Estate, New Delhi-20

12.3 **Execution of work**: -

12.3.1 The work will be got executed through authorized applicator of the manufacture under strict supervision of Engineer-in-Charge.

13 FLOOR FINISHES: -

13.1 **GENERAL**:

15 Type of floor topping shall be provided to details as shown on drawings/ Schedule of finishes and as specified here in after. Flooring shall be laid as indicated in SSR Part –I.

15 Floor topping shall be extended over dwarf walls, doors and other openings.

15 ICement for floors shall be as described here in before.

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15 PCC sub floor may not be laid in panels/bays. For the purpose of deviation, the rates as given in SSR Part-II for flooring shall be applicable.

(e) Unless otherwise shown in drawings, the dividing line between the floors of different types, wherever they so meet between adjoining rooms, shall be determined on the basis of the finish visible when the doors are closed so that no dissimilarity appears and the finish shall accordingly be provided.

13.2 **FLOOR TOPPING**: -

(a) Cast in situ PCC floor topping shall be laid in panels as per clause 13.32.1 and 13.38 of SSR Part-I respectively. The pattern/layout of panels in rooms shall be as directed by GE.

15 Joints in floor topping shall not be staggered while providing glass strips.

15 IUnless otherwise shown on drawings/Schedule of finishes and/or specified hereinafter, PCC floor toppings shall be broom finished .

13.3 **TYPES OF FLOORS**: - Unless otherwise mentioned in drgs and in schedule of finishes floors shall be provided as specified here-in-after.

13.3.1 **CEMENT CONCRETE SUB FLOOR:** Unless otherwise shown on drawings, cement concrete sub floor i.e.PCC sub base laid below floor finish in ground floor shall be plain cement concrete and thickness shall be as indicated on Schedule of Finishes. PCC sub floor need not be laid in alternative bays and the same shall be worked upto an even surface. Cement concrete sub base wherever indicated to be provided over hard core laid over consolidated approved earth filling all as indicated in Schedule of finishes.

13.3.2 **PCC FLOOR:** Unless otherwise shown on drawings, PCC floors (except floors in sub base) shall be laid in bays adopting panel dimension of 1200x1200mm. The bays shall be so arranged that the end bays shall be of equal width not exceeding 1200mm. The floor shall be laid in alternate bays. Floor topping shall be finished even and smooth using extra cement. 3mm thick glass dividing strips shall be provided to form joint in all cement concrete floors. The glass dividing strips shall be 3mm less than the thickness of PCC floor finish. The glass dividing strip shall be inserted in the PCC while the concrete is still plastic and the strips shall be left in position. PCC floors upto 40mm thickness only shall be provided with glass strips. PCC floors to be provided with glass strip shall not be laid in alternative bays and this will not involve any price adjustment. The surfaces of PCC floor shall be finished with a steel trowel to even and smooth surface using extra cement. Floors finishes of Ramps shall be worked upto an even and smooth surface without using extra cement and making of expanded metal impressions while the concrete is green. Wherever indicated in schedule of finishes and on treads of steps floor shall be provided with chequered finish.

13.3.3 PCC Floor finish shall be provided at locations mentioned in Schedule of finishes and shall be laid over cement concrete sub floor to the thickness as indicated on drawings. PCC floor finish shall be in 1:2:4 type B1 (Using 20mm graded stone aggregate). The surface shall be cured efficiently by water ponding as directed by Engineer-in-Charge.

13.3.4 **NON-SKID CERAMIC TILES FLOORING: -**

(a) Unless otherwise shown on drawings, Joint less coloured tiles shall be of uniform texture. Water absorption shall be less than 2%.Tiles shall be made finished; of size 600mmx 600mm, 7-8 mm thick hydraulically pressed with breaking strength 350 to 400 Kg/Sqm and weighing not less than 14 Kg/Sqm. It shall be free from bubbles, distortion due to overheating. Size and shade of tiles shall be directed by GE.

(b) Unless otherwise shown on drawings, The ceramic tiles shall be set, jointed with cement slurry and pointed in neat coloured cement to match the colour of tiles. Tiles shall be procured by the contractor from any one of the make given in list of makes as approved by GE. The work shall be carried out all as specified in clause 13.40 of MES Sch Part I and /or as per manufacturer's instruction.

13.3.4.1 **NON SKID TILES FLOORING IN BATHROOM/WC/KITCHEN: -**

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Wherever shown on drawings, non-skid coloured tiles of size 300x300/400x400x 7-8 mm thick shall be of uniform texture. Water absorption shall be less than 2%. Tiles shall be made finished hydraulically pressed. It shall be free from bubbles, distortion due to overheating. Size and shade of tiles shall be as approved by GE.

13.3.5 **INTERLOCKING TILES FLOORING:**

Unless otherwise shown on drawings, Interlocking coloured tiles for flooring in porch shall be of the premium quality hard, sound, dense and homogenous in texture, free from defects and conforming to relevant IS. The shade & size shall be as approved by GE. Joints shall be pointed flush using polymer based cementitious tiling joint filler to match the shade of the tile. Tiles shall be set, jointed in neat cement slurry and pointed flush using polymer based cementitious tiling joint filler to match the shade of the tile.

13.3.6 **KOTA STONE FLOORING: -**

Unless otherwise shown on drawings, Kota stone flooring shall be provided as specified in para 13.47 of SSR (Part-1) 2009. Kota stone in flooring shall be 20 to 22 mm thick laid over 20mm thick screed bed in cement mortar (1:6), over 25 mm thick plain cement concrete 1:2:4, type B-0 (using 12.5 mm graded crushed stone aggregate) sub floor over 75mm thick sub base plain cement concrete (1:5:10) type E-2 if not specified in drawings (using 40mm graded crushed stone aggregate), over well rammed approved earth. Tiles in first and above floors shall be laid over 20 mm thick cement screed 1:6 over 25mm thick of PCC (1:2:4) type B0 (Using 12.5 mm graded stone aggregate) if not specified in drawings. Kota stone shall be of size 550x550mm except at borders and shall be hard, of uniform texture& without cracks flaws etcif not specified in drawings. Joints shall be filled in grey cement using pigment. Top surface shall be grinded and polished as per PS clause No. 13.47.4 of SSR Part-I. Kota stone slab shall be of selected quality, hard, sound, dense and homogenous in texture, free from cracks, decay, weathering and other flaws and shall be laid and finished all as described in clause 13.47 on page 13-14 of SSR Part-I. The floor shall be achieved mirror polished surface. Full length Kota stone in one piece shall be provided on all the steps, stairs irrespective of what ever shown on drawing or mentioned elsewhere in the tender documents**.**

13.3.7 **RAMP/APRON: -**

13.3.7 Unless otherwise shown on drawings, Ramp/Apron shall be 150 mm thick PCC 1:2:4, type B1 (Using 20 mm graded coarse aggregate) over 75 mm thick PCC 1:4:8, type D2 over 150 mm thick hard core over rammed & compacted earth if not specified in drawings. Top finish of Ramps/Aprons shall be worked up to an even and smooth surface without using extra cement and making of expanded metal impressions while the concrete is green. PCC topping shall be laid in alternate bays. Proper construction joints shall be formed as specified in SSR and filled with sealing compound. Sides of ramps shall be finished with 15 mm thick plaster in CM(1:4) in one coat finished even and smooth without using extra cement.

13.3.7A **VITRIFED TILES FLOORING AND SKIRTING: -**

Wherever shown on drawings, for room/passages/main room (Bed room, living room/drawing room) vitrified coloured tiles of size 600mm x600mm x 8-10 mm thick shall be of uniform texture double charged. Water absorption shall be less than 2%.Tiles shall be made finished hydraulically pressed. It shall be free from bubbles, distortion due to overheating. Size and shade of tiles shall be as approved by GE.

13.3.8 **SKIRTING: -**

13.3.8.1 **CEMENT SKIRTING:-**

Wherever shown on drawings cement skirting shall be 15mm thick consisting of 10mm thick rendering coat in CM(1:4) and 5mm thick setting coat in CM (1:3). Top edge of skirting / dado shall be neatly finished to straight line. Setting coat shall be finished smooth using extra cement. Height of skirting shall be 100mm.

PARTICULAR SPECIFICATION (CONT'D…../-)

13.3.8.2 **KOTA STONE SKIRTING AND ON STEPS: -**

Where ever shown on drawings, Kota stone skirting/dado and topping on steps shall be provided. Thickness of Kota stone shall be of 12 to 15mm pre machine cut, machine polished (granite finish). Kota stone in skirting/dado shall be laid/set over 10mm thick screed bed in cement mortar (1:3) and jointed with Gray cement slurry mixed with pigment to match the shade of the slab all as specified in MES schedule Part-I. Height of skirting shall be 100mm. Full length Kota stone in one piece shall be provided on all the steps, stairs irrespective of what ever shown on drawing or mentioned elsewhere in the tender documents.

13.3.8.3 **NON SKID CERAMIC TILE SKIRTING: -**

Unless otherwise shown on drawings, Non skied ceramic tile shall be minimum 7mm thick first quality and shall be provided all as shown in schedule of finishes. Non-skid ceramic tile shall be laid over 10mm thick cement and sand screed (1:3) all as specified in clause 13.4.2 of SSR Part I. The colour of tiles shall be approved by GE to match the floor tiles. Jointing and pointing of tiles shall be carried out with white cement and coloured pigment to match with the colour of tiles. The tiles at all external corners shall be raced and have mitred joints.

13.3.8.4 **GLAZED CERAMIC TILE DADO:-**

13.3.8.4.1 Unless otherwise shown on drawings, Glazed Ceramic tile shall be minimum 5.5mm thick 200 mm x 300 mm size glossy finish first quality and shall be provided if otherwise shown in schedule of finishes. Ceramic tile shall be laid over 10mm thick cement and sand screed (1:3) all as specified in clause 13.4.2 of SSR Part I. The colour of tiles shall be approved by GE to match the floor tiles. Jointing and pointing of tiles shall be carried out with white cement and coloured pigment to match with the colour of tiles. The tiles at all external corners shall be raced and have mitred joints.

13.3.8.4.2 Height of glazed ceramic tiles and dado shall be as indicated in schedule of finishes / drawings. Height of glazed ceramic tiles dado, where not indicated in schedule of finishes/drawings shall be 0.90 metre for WC, upto lintel height in case of Bathrooms and 1.50m for Toilet portion (excluding WC and Bath chambers). Also, all other places where Glazed tile Dado shown in the drawing/schedule of finishes but height of Dado is not mentioned, the same shall be 1500 mm.

13.3.8.4.3 **WALL TILES IN BATHROOM/WC/KITCHEN: -**

Wherever shown on drawings, coloured wall tiles of size 300x450x 6-7 mm thick shall be of uniform texture. Water absorption shall be less than 2%. Tiles shall be made finished hydraulically pressed. It shall be free from bubbles, distortion due to overheating. Size and shade of tiles shall be as approved by GE.

13.3.8.5 **MARBLE STONE SLABS/GRANITE STONE SLAB:-**

Unless otherwise shown on drawings, Provide mirror polished marble slabs/Granite stone slab for kitchen/pantry/Glass cupboard platforms etc. At locations shown in the drawing and as specified in schedule of finishes drawing. The size of the marble slab/Granite stone slab shall be in accordance with sizes shown in drawings as a single piece. The slabs shall be machine cut and shall be of uniform colour as approved by GE. The thickness of marble stone/Granite stone slab shall be 18 to 20mm thick. The marble slab/Granite stone slab shall be set and jointed in neat cement slurry @ 3 Kg/Sq.m. over screed bed as indicated otherwise If details of screed bed are missing then it shall be minimum 15 mm screed in CM 1:4 and pointed in white cement with pigments or coloured cement to match the slabs. The grinding & tin oxide polishing shall be done after laying of slabs to mirror polish.

13.3.8.6 **CHEQUERED CEMENT CONCRETE TILES** : Chequered cement concrete tiles shall be conform to IS-1237-2006 at all as specified in Sch 'A' and Clause No. 13.16 of MES Schedule Part-I. Chequered cement concrete tiles shall be laid as specified in Sch 'A' Tiles shall be laid straight or to the pattern and all as specified in Clause No. 13.39.1 and 13.39.2 of MES Schedule Part-I. Cutting and polishing shall be done by machine as specified in Clause No. 13.39.3 of MES Schedule Part-I. However hot mixture of turpentine and beeswax shall not be applied

PARTICULAR SPECIFICATION (CONT'D…../-)

14 PLASTERING AND POINTING :-

14.1 **GENERAL:**

a) All external plaster shall be carried out upto 150 mm below ground level exceptwhere steps, ramps, plinth protection are provided.

(b) Plaster and skirting/dado shall be returned in jambs, soffits of lintels and window cills etc, unless otherwise shown on drgs, height of skirting & dado shall be 100 mm & 2100 mm respectively.

€Thickness of plaster shall be the finished thickness exclusive of dubbing. Dubbing wherever required shall be executed without any extra cost.

All corners, angles, junctions and edges unless otherwise specified shall be truly vertical or horizontal as the case may be and shall be carefully finished.

€ Irrespective of what is specified and shown on drawing, 12mmx12mm continuous groove shall be provided in plaster throughout at the junction of wall & RCC slab. Trowel groove shall be provided at the junction of walls and RCC columns/RCC beams as directed. In other situations where it is not desirable to make groove, chicken wire mesh 30 cm wide shall be provided under the plaster.

14.2 **PLASTER TO INTERNAL SURFACES: -**

15 mm thick plaster in CM (1:6) finished even and smooth without using extra cement above skirting/dado in one coat.

14.3 **PLASTER TO EXTERNAL SURFACES**:-

15mm thick in CM (1:4) mixed with water proofing compound as per manufacturer's instructions. Plaster shall be finished even and smooth using extra cement. Top and sides of parapet on roof shall also be plastered in CM 1:4 except wall cladding or where any other finishes are shown on drawings.

14.4 **PLASTER ON RCC CHAJJAS** :-

Top surfaces of the RCC chajjas shall be plastered in CM (1:4) 15mm thick mixed with water proofing compound as per manufacturer's instructions.Irrespective of whether shown on drawings or not, the plaster shall be taken up to 200 mm height over adjacent vertical surface of wall in addition to external plaster for a length equal to length of chajja. Thickness of plaster at junction of wall and chajja shall be rounded to 50mm radius.

15 SURFACE FINISHES TO BUILDINGS:

15.1 **WHITE (LIME) WASHING:**

[a] Lime used for white washing shall be freshly burnt fat lime Class C) white in colour, conforming to IS-712. 3 coats of white wash shall be provided to walls all as specified in clause 15.12 of MES Schedule Part-I. Skirting and dado are not to be white washed. Unless otherwise shown in Schedule of finish drawing the ceiling are to be treated with 3 coats of white wash all as specified herein above.

[b] All exposed surfaces of parapets, sides of steps, risers, ramps, faces of columns, jallies etc and the like shall be treated with same finish as provided for the surrounding surfaces, plastered surfaces above false ceiling shall be given three coats of white wash unless otherwise shown in drawings.

15.2 **OIL BOUND DISTEMPER:**

Wherever indicated in Schedule of finishes drawing, provide Oil bound distemper two coats over a coat of alkali resistant primer over layer of 3 mm thick wall care putty. Distemper oil emulsion shall conform to IS-428. Preparation of surfaces application of primer and distemper shall be in accordance with clauses specified in MES Schedule Part-I. Oil bound Distemper shall be of any of the makes specified

PARTICULAR SPECIFICATION (CONT'D…../-)

hereinafter. Primer shall be ofsame makeof Oil bound distemper.

15.2A **DRY DISTEMPERING**

Dry Distemper: Two coats of Dry distemper over a coat of primer, on plastered surface preparatory to distempering as indicated on drawings and specified in relevant clauses of MES Schedule Part–I. Preparation of surfaces application of primer and distemper shall be in accordance with clauses specified in MES Schedule Part-I. Distemper shall be of any of the makes specified hereinafter. Primer shall be ofsame makeof Oil bound distemper.

15.2 B **PLASTIC EMULSION PAINT**

2 coats of plastic emulsion paint over one coat of wall primer over 3mm thick wall care putty as per approved colour scheme shall be provided on plastered surfaces as indicated on drawing and specified in relevant clauses of MES Schedule Part–I. Preparation of surfaces application of primer and paint shall be in accordance with clauses specified in MES Schedule Part-I.

15.3 **EXTERNAL WEATHER PROOF PAINT (EXTERIOR EMULSION):**
GENERAL:

Unless samples of all materials are approved, the contractor will not be allowed to commence the work. Brushing for painting etc., shall be got approved by Engineer-in-Charge based on manufacturer's recommendations. Noimprovised brushes or sub standard brushes shall be brought to site and used.

15.3.1 **MATERIALS:**

15.3.1.1 (a) Paint shall be weather proof Acrylic emulsion, exterior grade (100% acrylic) premium quality. Paint shall be procured from any of the makes listed hereinafter

(b) Shade of the paint shall be as approved by GE.

(c) Primer shall be water based acrylic suitable for exteriors as per manufacturer's instructions. Primer shall be of same make as of paint.

(d) The paint and primer shall be brought in manufacturer's sealed containers only by the contractor duly marked with batch number from the manufacturer.

(e) The contractor shall produce manufacturers test certificate along with purchase voucher in original for the paint and primer brought to site before claiming payment for the same Purchase voucher of paint and primer shall contain the complete description of material, batch No., net weight, test certification No., quantity in each package, No. of packages etc., The quantity of material brought at site indicating No. of packages, quantity in each package, batch No., purchase voucher number, test certification number, date of manufacturing, date of expiry etc., shall be entered in MB as ;Not to be Abstracted; and shall be signed by the JE, Engineer-in-Charge, GE and contractor.

15.3.1.2 Each container of paint and primer shall bear the following particulars:

(a) Manufacturer's Trade Mark (b) Reference to Indian Standard to which they comply
(c) Name of Product (d) Net Weight
(e) Date of Manufacturing (f) Batch No.
(g) Storage Requirement (h) Storage Life
(i) Date of Expiry

Each lot of paint and primer shall be checked by Engineer-in-Charge and approved by himafter verifying from invoices, package, batch No. and test certificate.
Materials shall beincorporated in the work only after written approval from Engineer-in-Charge.

15.3.2 **WORKMANSHIP:**

15.3.2.1 All brushes, tools, pots, kettles etc., used in carrying out the work shall be free from foreign matter and shall be thoroughly cleaned with hot water and solutions at the end of a day's work or before use for a different type of material. No finish shall be executed until a sample of the finish to the required colour and shade has been approved by the GE. Where more than one finish is indicated, each coat shall be approved by the GE before the subsequent coat is applied.

15.3.2.2 The colour shall be even shade over the whole surface, if it is patchy or otherwise bad, the work shall be redone by the contractor at his own expense.

PARTICULAR SPECIFICATION (CONT'D…../-)

15.3.3 **PREPARATION OF SURFACES:**-

15.3.3.1 The surface shall be thoroughly cleaned of loose particles, dust, dirt, efflorescence,
chalking, grease, mortar drops and other foreign matter.The surface shall be sand papered with Grade-I abrasive paper and dusted off to achieve an even and smooth surface free from all dust particles. The contractor will use electric blowers for this purpose. If surface so obtained is uneven, it shall be brought to a perfectly even surface by applying putty and allowing it to dry completely and then it shall be rubbed with the abrasive paper and dusted off and finally area cleaned by use of electric blower.

15.3.3.2 **PRIMING COAT:**

After preparing the surface as approved by GE, one coat of exterior water based
acrylic primer as approved by manufacturer, thinned with water in 1:1 ratio shall be applied with brush as per manufacturer's instructions and as directed by Engineer-in-Charge.

15.3.3.3 **APPLICATION OF WEATHER PROOF PAINT:**

(a) The acrylic emulsion weather proof paint shall be applied by brush or roller. No stainer or colorants shall be used. The paint shall be stirred well before use. The primer coat shall not be left without application of top coats for a long period time.

(b) Two coats of 100% Acrylic emulsion weatherproof paint thinned with 400 ml water per litre of paint shall be applied. The drying period between two coats shall be minimum 4 hours or as per Manufacturer's instructions. The shade shall be as approved by GE. The finish of Acrylic emulsion weather paint shall be smooth matt finish.

(c) The paint shall be as per Manufacturer's original colour as available or shade card. No mix of tint shall be made into original shade.

15.3.3.4 **GUARANTEE:**-

15.3.3.4.1 The work of applying primer and paint shall be got carried out under the supervision and guidance of accredited representative of the manufacturer. A certificate from their representative shall be obtained by the contractor to the effect that work of painting and primer has been carried out under their strict supervision and as per manufacturer's instructions. The same shall be submitted by the contractor to GE. The contractor shall also obtain a written guarantee for effectiveness of paint against fading out, peeling off, cracking, dust/algae accumulation etc., for a period of **5 (Five) years** from the certified date of completion of entire work from the manufacturer and submit the same to GE before completion of work.

15.3.3.4.2 Should the GE at any time during construction or reconstruction or prior to the expiry of the Guarantee period, finds defective performance of the paint, the contractor shall, on demand in writing from the GE specifying the location complained of notwithstanding that the same may have been inadvertently passed/certified and paid for, undertake to carry out such treatment as may be necessary forthwith to rectify the defects to the full satisfaction of GE and render complaint, free from any type of defects. In the event of his failure to do so, within the period as specified by the GE in his aforesaid demand, the GE may undertake such defective work through other agency at the risk and cost of the contractor in all respects. The liability of the contractor under this condition shall not extend beyond the period of **5 Years** from the certified date of completion, unless the GE had previously given notice to the contractor to rectify the defects. Defect Liability Period under Condition-46 of General Conditions of Contracts (IAFW-2249) shall be deemed to be amended to the extent mentioned abovefor Acrylic Emulsion Paint.

PARTICULAR SPECIFICATION (CONT'D…../-)

15.3.3.4.3 Guarantee amount shall be retained out of the contractor's Final Bill as Security Deposit against the guarantee for Weather proof paining for **5 Years @2.5% of weather proof painting work at contract rates**. If contractor fails to rectify the defects noticed in the treatment or found in the material the aforesaid amount so retained shall be utilized for rectification of defects and contractor shall have no claim whatsoever on this account. This amount shall be released after successful expiry of the 5 Years guarantee period from the certified date of completion of the entire work by GE, provided always that the contractor shall first have been paid the Final Bill and have rendered. No Demand Certificate (IAFA-451). Alternatively, the contractor may furnish Fixed Deposit Receipt inlieu of Security Deposit from a Schedule Bank in favour of GE for the above said Guarantee Period.

15.4 **SCAFFOLDING:-**

(a) The exterior painting work shall be carried out by using scaffolding. No zoola is permitted for the work under any circumstances. Suitable scaffolds shall be provided for workmen.

(b) Scaffolding or staging more than 3.5 Metres above the ground or floor, swung or suspended from any over head support or erected with stationary support shall have a guard rail properly attached, braced and otherwise secured at least 1 Metre high above the floor or platform of such scaffolding or staging and extending along the entire length of the outside and ends thereof with only such openings as may be necessary for the delivery of materials. Such scaffolding or staging shall be so fastened as to pervert it from swaying from the building or structure.

(c) Every opening in the floor of a building or in a working platform be provided with suitable means to prevent the fall of persons or materials by providing suitable fencing or railing whose minimum height shall be 1 Metre.

(d) Safe means of access shall be provided to all working platforms and other working places.

(e) The rates quoted are deemed to include the above provision of scaffolding and no extra will be payable to contractor on this account. The scaffoldings shall be removed only after obtaining clearance of Engineer-in-Charge/Garrison Engineer after considering the quality of the work undertaken on completion of the painting.

15.4.1 **SAFETY PRECAUTIONS: -**

(a) Contractor shall provide all safety precautions for the labour engaged for this work. All the labours shall be provided with safety belts, helmets, Nose screens etc and the contractor shall adhere to all safety precautions as per Labour Welfare Act.

(b) It is also advised, contractor to have insurance cover for his workmen working at heights against any eventuality from any reputed insurance agencies. Department will not be responsible for any untoward incident happening due to lack of safety precautions taken by contractor.

15.4.2 **STAGE PASSING: -**

15.4.2.1 The work shall be executed in a workman like manner and to the entire satisfaction of the GE. Contractor shall obtain the approval of GE stage wise as indicated below. The contractor shall give due notice in writing with sufficient time in advance to the Engineer-in-Charge and the GE, when each stage is about to be completed. The contractor shall start the work of subsequent stage only after obtaining written approval of the GE for previous stage. In default of such notice being received from the contractor, if he commences the work of subsequent stage without approval of the GE, then the GE shall have all the rights reserved to reject the work and ask the contractor to demolish the said portion executed. The contractor shall have no claim on this account or otherwise. The decision of the GE in this regard shall be final and binding.

15.4.2.2 **STAGES:**

(i) Extent of area to be painted.

(ii) Sample of painting work (Area to be decided by GE)

PARTICULAR SPECIFICATION (CONT'D…../-)

(iii) After preparation of surface and before applying priming coat.
(iv) After applying priming coat.
(v) After applying first coat of Acrylic emulsion paint.
(vi) After applying second coat of Acrylic emulsion paint.

15.5 **CEMENT BASE PAINT**: -

15.5.1 Whatever shown on drawings external Schedule of finishes of Sch `A` Part-I shall be applied two coats of cement base paint over a coat of alkali resistance primer after preparation of surfaces, all as specified in MES Schedule ABOVE Plinth Protection. To prevent algae and moss growth and efflorescence, silicon base water repellent compound may be added to mixture at the rate as recommended by the manufacturer. Cement based paint conforming to IS: 5410-1969 and shall be ready mix & of make and shade as approved by GE. After the surface is dry, primer shall be applied with brush followed by two consecutive finishing coats of paint/distemper.

15.5.2 **GLAZING: -**

15.5.2.1 Glazing shall be with plain sheet glass or frosted glass or tinted glass as indicated on drawings and as specified. Sheet glass shall be of ordinary glazing quality conforming to the requirement of IS: 1761. In case of bath & WC rooms the glass shall be frosted on one side. Glass shall be fixed to wooden windows and ventilators using teak wood beading with putty all as specified in MES Schedule Part-I.

15.5.2.2 Unless otherwise shown on drawings/specified elsewhere, the sheet glass in panes up to 0.5 Sqm (each) shall be 3mm thick and exceeding 0.5 Sqm (each) it shall be 4 mm thick plain sheet glass 'B' quality as specified in MES Schedule conforming to IS: 1761.

16 PAINTING: -

16.1 The synthetic enamel paint shall be of 1st quality as approved by GE.The contractor shall inform the GE, within three weeks of the acceptance of the tender, thebrand names of the manufacturer of paint proposed to be used in the works and submitsamples thereof and obtain prior written approval of the GE beforetheir use in the work.Paint for priming coat, under coat and finishing coat will be of the same manufacturer.Tint of paint, if not mentioned in drawings/schedule of finishes will be approved by the GE.Contractor shall execute painting under the guidance of the Engineer-in-Charge andmarked as such before commencement of painting work. Each coat of paint shall bepassed by the Engineer-in-Charge beforethe next coat is applied.If the undercoat of paint is not executed within six months after applying the priming coat ofpaint, the priming coat shall be redone by the contractor at no extra cost to theGovernment.Surfaces which become inaccessiblefor painting after execution shall be painted before execution.Surfaces which are specified to be treated with synthetic enamel paint in the schedule of finishes and drawings shall be prepared and painted with two coats of approved synthetic enamel paint over a coat of primer all as specified. Primer for wood surfaces shall be pink primer and for that of steel surfaces shall be zinc chrome primer. colour and shade for under coat and finishing coat shall be decided by Garrison Engineer.

16.2 **PAINTING TIMBER SURFACES/STEEL SURFACES:**

16.2.1 Where painting to timber/steel surfaces is indicated on drawings prepare surfaces and apply two coats of synthetic enamel paint over one coat of primer as specified hereinbefore. Preparation of surfaces for all locations except gantry girders shall be all as specified in MES Schedule Part-I. The steel surfaces of gantry girder shall be prepared by sand blasting. The prepared surface shall be got passed by the GE before applying paint, gantry girders shall be painted with primer and under coat before erection. Painting shall be carried out by spray painting.

16.2.2 **WORKMANSHIP: -**

16.2.2.1 All wood work required to be painted shall be smoothened, sized and knotted and then applied with priming coat. Stopping and filling(filler coat) shall be done after priming coat and surfaces rubbed down to a level and smooth surface and thereafter

PARTICULAR SPECIFICATION (CONT'D…../-)

under coat and finishing coat applied, all as specified in clause 17.6 of MES Schedule Part-I.

16.2.2.2 The steel surfaces which are required to be painted shall be given two coats of paint, priming coat and undercoat after fabrication but before assembly and erection and finishing coat after assembly and erection.

16.2.2.3 Unless otherwise indicated on drawings and/or specified in these particular specifications all surfaces of iron and steel work shall be prepared and treated with two coats (one undercoat and one finishing coat) of synthetic enamel paint over a coat of primer all as specified here-in-before. Reinforcement bars, tinned or galvanized iron surfaces and steelwork embedded in concrete/plaster shall however not to be painted.

16.2.2.4 Bottom of door shutters shall be given one coat of primer only.

16.2.2.5 Irrespective of what is indicated on drawings and specified elsewhere, finishing coat and undercoat shall be with the same paint.

16.2.3 **CLEANING:**
Before commencement of plastering/painting work all doors/windows/ventilators, Balconyrailings etc shall be covered properly for protecting from the drips of paint/primer while painting, to the entire satisfaction of Engineer-in-Charge. Covering shall be done with tarpaulin or ply wood. After completion, the surrounding area, glass panes, flooring shall be cleaned and all the paint marks on it shall be removed. Contractor's quoted rates shall be deemed included of the above said provision and nothing extra shall be admissible on this ground.

16.2.4 **TARRING:-**
Prepare and apply two coats of tar to the hold fasts, backs of wooden frame in contract with brick work/plaster, etc. Hold fast shall be given two coats of tar and sanded.

16.3 **MISCELLANEOUS**:

16.3.1 **FORMATS**: -Formats for numbering of buildings and display for anti-termite treatment shall be in 10mm thick setting coat in CM (1:4) over plastered surfaces of external walls, Size of letters and figures to be engraved shall be 50 mm x 100 mm (minimum) and painted black on white background of paint. Details for ATT and water proofing of roof shall be shown on the same format.

16.3.2 **TOWEL RAIL**:Towel rail shall be of brass chromium plated 18/19mm dia (wall thickness shall not less than 1 mm) with heavy weight bracket at ends and fixed with chromium plated steel screws on wooden gutties. Unless otherwise shown on drawings the size of towel rail shall be 600 mm long.

16.3.3 **STAINLESS STEEL RAILING**:
Stainless steel railing shall be provided as shown on drawings. The vertical pipes and hand rail shall be of stainless steel, standard quality of grade SS-304 polish finish of wall thickness 1.6mm and of size as shown on drawings. SS tubes shall be welded at the turnings and junctions. All the welded portions shall be completely grinded andbrought to bright and smooth finish. Anchor fastening bolts shall be provided with grouting so as to achieve full strength of railing. The final finish of the railing shall be bright and smooth.

16.3.4 **MILD STEEL RAILING:** MS Railing shall be fabricated to shape and design all as shown on drawings. MS square pipe in balusters shall be of 20 mm square hollow bar 16 gauge thick and 50 mm dia GI pipe medium grade shall be provided in hand rail. As shown in drawing. Exposed steel surfaces shall be applied two coats of synthetic enamel paint over a coat of red oxide primer.

16.3.5 **SOAP NICHES** :
These shall be provided where shown on drawings. Inside finish of the soapniche shall be same as that of dado. Where niche is not coming in dado, itshall have 5 to 7 mm thick ceramic tiles white. Where it is shown to be of vitreous china, it shall be got approved from GE before procuring.

PARTICULAR SPECIFICATION (CONT'D…../-)

16.3.6 **FAN HOOK AND BOX: -**

Fan hook shall be made out of 12mm dia MS round bars of shape and size complete with box all as shown on drawings. It shall be given a coat of steel primer before placing in position.

16.3.7 **OVERHEAD WATER TANKS** :

HDPE service overhead water tanks having triple layered wall shall be of capacityand placed at location as shown on drawings/as directed by GE depending uponload distribution on roof slab. It shall have lockable cover. In case tanks of capacity shown on drawings are not available, tank of next higher capacity shall be provided without any price adjustment. Tanks shall be placed on 75 mm (average) thick PCC (1:3:6) Type C-1 padding unless otherwise shown in drawings. For eachtank GI light grade 300mm long, over flow pipe of same dia as that of inlet pipe withPVC anti-mosquito cap and 25mm dia GI light grade wash out pipe extended300mm from outer edge of PCC padding shall be deemed to be included in the cost of respective building in addition to tank fittings such as connection, float valve,ball valve and stop cock for inlet, outlet and washout pipes.

16.3.8 **BRICK STEPS:**

Brick steps shall be provided all as shown on drawings. Brick work shall be in CM (1:4) sides and risers of steps shall be plastered CM(1:4), 15mm thick andtreads of steps shall have 40 mm thick PCC (1:2:4) type B-0 using 12.5 mm stoneaggregate finished even and smooth without using extra cement and having XPMimpression unless otherwise shown in drawings.

16.3.9 **RUNGS** :-

In blocks where stair is not provided or the stair is not upto roof slab, rungs shall be provided at one place so as to reach the roof for repairs etc, Rungs shall be of mild steel round bars of 20 mm dia and shall be fixed in RCC as the work proceeds and on brick wall embedded in PCC block (1:3:6) type C-1. Rungs shall be provided at spacing shown on drawing and shall be fixed staggered. Rungs shall be given two coats of black bituminous paint all over. Size of PCC block if not shown on drawings shall be 150 mm x 150 mm x 230 mm.

17 NITCHES BOXES FOR MAIN SWITCH/ELECTRIC METER/WATER METER :

17.1 Boxes shall be fabricated out of MS Sheet, MS angle, MS flat etc all as shown in drawings. MS work shall be painted as specified here-in-before.

17.2 Niches shall be provided as per details on drawings and be finished with 15mm thick plaster in CM (1:4)

17.3 Wood work in nitches shall be of first class hard wood Teak.

18 CURTAIN ROD :

18.1 Curtain rod where indicated in the drawings shall be provided with 18/19mm dia of steel chromium plated, pipe 16 gauge with Stainless steel brackets etc all as shown on the drawing.

18.2 **PELMET BOX (PB):**

Pelmet box shall be provided to details & situations where shown on drgs. These shall be made with pre-laminated particle board 18 mm thick, three layers flat pressed melamine face pre-laminated particle board with exterior grade ISI marked, IS: 3087 grade I and IS: 12823 Grade I type II melamine finish on one face and balancing coat on other side. 6 mm thick teak wood beading shall be provided on all open edges of particle board.

PARTICULAR SPECIFICATION (CONT'D…../-)

19.1 **PEGS :**

Aluminium anodized pegs commercial type (cast or extruded section) shall be provided on wall/ wood with cadmium plated screws.

19.2 **MIRROR:-**

Mirrors shall be 6.00 mm thick with square corners polished fixed to 6 mm thick plywood (kitply) backing and with aluminium channel frame fixed to walls with brass chromium plated screws to wall all as shown on drgs.

19.3 **OPENING FOR EXHAUST FAN** :

Opening for exhaust fan shall be provided wherever shown on drawing. Provn of exhaust fan and making connection shall be payable separately.

19.4 **STEEL DOORS**:

Mild steel doors shall be provided to details all as shown on drawings. These shall be given two coats of synthetic enamel paint over a coat of red oxide zinc chromate primer.

19.5 **GLAZING FOR STEEL WINDOW**:

(a) Unless otherwise shown on drawings/specified elsewhere, the sheet glass in pans up to 0.5 Sqm (each) shall be 3mm thick and exceeding 0.5 Sqm (each) it shall be 4 mm thick plain sheet glass 'B' quality as specified in MES Schedule conforming to IS: 1761.

(b) Pin headed figured glass shall be conforming to IS: 5437 for glazing. Where thickness of pin head glass is not shown on drawings it shall be 3mm thick.

(c) Glass will be fixed as specified in Clause 16.9.2 (for wood surrounds) and clause 16.10.1 (for steel surrounds) of SSR Part I. Putty shall be as specified in Clause 16.5 of MES Schedule Part I.

19.6 **RAIN WATER PIPES:**

(a) Irrespective of what is shown on drawings, rain water pipes shall be of HDPE medium density (ISI marked) 160 mm dia for hanger and 160 mm dia for other buildings. RWP fixed to the walls with MS holder bat clamps embedded in cement mortar (1:3) conforming to IS: 13512-1992 or in RCC as the case may be. These shall be provided all as specified in MES schedule Part I and given same finish as for wall. Pipes shall be jointed properly as per manufacturer's instructions or as specified.

(b) Grating shall be of PVC round type for PVC pipes and CI grating for CI pipes not less than weighing 1.50 kg each at the inlet to RW/down take pipes.

19.7 **DRIP COURSE:**

Drip course shall be provided under the projection of roof slab, projected balcony and chajjas etc, It shall be made in CM (1:4) and shall of size 40 mm x 20 mm thick (average).

19.8 **CONSTRUCTION JOINTS/CRUMPLE JOINTS:**

These shall be provided to details as shown on drgs Gap shall be filled with 25 mm thick preformed filler fiber board. Covering sheet shall be fixed with screws in wooden gutties /plugs to wall at 150 c/c.

19.9 **SPLASH STONE:**

Splashed stone shall be provided below the shoe of each rain water pipe as shown on drawings. In case details not shown on drawings, it shall be 500 x 750 x 75 mm thick self faced Shahabad/Dholpur stone slab set on 10mm thick CM (1:6) screed.

20 INTERNAL ELECTRIFICATION: -

PARTICULAR SPECIFICATION (CONT'D…../-)

20.1 **SCOPE: -**

The work covers Internal Electrification as described in Schedule 'A' under respective parts, Notes to Schedule 'A', as shown on drgs and as specified here in after.

20.2 **WORKMANSHIP: -**

(i) The work carried out shall be of the highest standard conforming to latest edition of Indian Electricity Act and rules framed there under as applicable and Para 19.2 of SSR 2009 (part I) read in conjunction with relevant IS codes for Internal electrification and code of practice for electric wiring installation as per IS: 2274 with amendments. All cable connection to the MCBs shall be by using copper lugs/thimbles only.

(ii) Refer condition 25 of General conditions of contracts (IAFW 2249). All electric work shall be carried out properly by skilled and licensed electrician under the supervision of qualified electrical supervisor. The contractor shall on demand by Engineer in Charge produce such evidence of qualifications of his workmen and supervisor during the currency of contract. This provision shall be complied with even if the contractor himself is qualified and enlisted for electrical work.

(iii) All the work shall be executed as per the code of practice and as listed in the IS. All electrical work shall be governed by the requirement as specified under the Indian electricity Act/Rules with latest amendments. The code of practice for electric wiring installation as per IS: 2274 with amendment No. 1 will be strictly followed.

(iv) Internal wiring shall conform to IS: 732 codes of practice for electrical wiring installation. Point wiring shall be with 1100 Volts grade cable with stranded copper conductor all as described in Schedule 'A' items: -

(a) Screws: All screws shall be of galvanised iron or steel chromium plated.
(b) Link clips: These shall be of aluminium conforming to IS: 2412
(c) Sunk cast iron/pressed steel terminal boxes. Terminal boxes shall be of suitable sizes required to mount switches, regulators and 3 pin sockets.
(d) Wiring shall be carried out all as described in Schedule 'A' and as specified in MES Schedule part I.

20.3 SAMPLES:-

20.3.1 The GE in writing before incorporation in the work shall be first approved samplesof all electrical fittings. Utility light fittings shall be of the best quality available and conform to the samples kept in GE's office duly approved by him.Samples of the following shall be supplied within 30 days of the date of issue of work order:-

(i) All types of cable.
(ii) Fittings, wires and accessories used for Internal Electrification.
(iii) The contractor shall ensure that the materials used in the works are identical with the approved samples. The materials shall be brought to the site by the contractor in the makers' original packing with the seal intact or with wrapper and shall not be installed unless approved by Engineer in Charge.

20.3.2 Height of switch boards, socket outlet, light fittings and fans etc shall be as shown in drawings or as directed by the GE.

20.3.3 All electrical fittings & wiring shall run clear of doors, windows & other openings.

20.3.4 Arrangement for lighting circuit shall be as decided by Engineer in Charge. Permissible number of points per circuit shall not be more than 8.

20.3.5 Loop in system of wiring shall be followed throughout the Installations.

20.3.6 No burrs or twisted joints shall be made at intermediate points in the through run of cable. If inevitable, such joint shall be made through proper cut out.

PARTICULAR SPECIFICATION (CONT'D…../-)

20.3.7 In any condition cables shall not come in contact with metallic pipe or non-earthed metal work.

20.3.8 Not more than two power socket out let points shall be wired on one circuit.

20.3.9 Diagonal run of wiring shall not be permitted.

20.3.10 Suitable inspection boxes shall be provided for periodical inspection or to facilitate removal of wiring. In concealed wiring these shall be mounted flush with the walls.

20.3.11 For conduit wiring colour coding and colour of insulation of wires shall be got approved before the wires/cables are procured.

20.3.12 Where conduits are concealed, these shall be held in chases with MS hooks except in RCC work where these can be secured to reinforcement with binding wire.

20.3.13 Cutting and forming chases in brick work/concrete where directed shall bedone while the building work is in progress prior to the commencement of plastering to brick work. Gaps/voids of all cut/formed chases including recesses formed for terminal boxes for housing power plugs, switches, sockets etc shall be filled flush with cement concrete (1:3:6) type C-1, using 20 mm graded stone aggregate or in cement mortar (1:3) mix after fixation of pipes/boxes etc prior to the plastering of walls at no extra cost.

20.3.14 Clause 19.101 to 19.111 and 19.114 of SSR Part I will be referred to for workmanship. Special attention is drawn to clause 19.111 of SSR Part I for method of passing conductors through walls and floors.

20.4 MATERIALS: -

20.4.1 All materials to be incorporated in the work shall be of the highest standard and unless specified otherwise, shall strictly comply with the relevant ISS or BSS for which no ISS has been published.

20.4.2 Unless otherwise specified point wiring shall be with 1100volt grade cable conforming to IS: 694 with stranded copper conductor all as described in different items of Schedule 'A'.

20.4.3 Wooden battens/blocks/Boards shall be of well seasoned hard wood wrought on all exposed surfaces and coated with shellac varnish. The finished minimum thickness of batten shall be 10 mm.

20.5 CONDUIT & CONDUIT FITTINGS :-

20.5.1 PVC conduits shall be of PVC medium grade of 25mm dia ISI marked. Conduit fittings shall be of un plasticized PVC conforming to IS: 3419.

20.5.2 **MS BOXES**: -

For housing the various fittings such as switches, sockets, fan regulators etc, shall be made of pressed steel sheet not less than 1.6 mm thick conforming to IS: 5133. The boxes shall be painted with two coats of black bituminous paint and shall be embedded in walls in cement mortar 1:3 and need not be fixed with plugs or screws. The minimum depth of these boxes shall be 60 mm and each box shall have an earth dolly. Corner of boxes shall have 3 mm thick MS sheet corners with screwed holes for fixing laminated sheet, 3 mm uniform thickness. Laminated sheet shall be fixed with brass screws & washer to MS box. It shall be of the color as decided by GE.

PARTICULAR SPECIFICATION (CONT'D…../-)

20.5.3 **PLASTIC LAMINATED SHEET TO COVER CAST IRON BOXES FOR POINTWIRING**:

Hylam / Wood lite / Anchor brand with the make embossed on the sheet.

Sl No.	Materials	Reference to Para of SSR (Part-I)
(a)	Cable cords and earthing leads	19.24
(b)	Cable for internal wiring for light power	19.25
(c)	Wooden battens, blocks and boards	19.28
(d)	Conduit and conduit fittings and accessories	19.29
(e)	Plug/Gutties	19.30
(f)	Screws and fastenings, link clips	19.31
(g)	Ceiling rose	19.32
(h)	Shades	19.33
(j)	Bulkhead fittings	19.34
(k)	Socket outlets	19.40
(l)	Lamp holders	19.41
(m)	Main switch	19.45
(n)	Switch boards	19.105
(o)	Distribution fuse board	19.44
(p)	Miniature circuit breaker	19.46

20.6 EQUIPMENTS AND MATERIALS: -

20.6.1 Piano type switches, lamp holder, main switches and socket outlets/switch fuse, cable for internal wiring, miniature circuit breaker etc shall be of the make as approved by GE. Flexible chord 23/.0076 shall conform to IS: 434. Steel Conduit shall be stove enameled black rigid steel conduit conforming to IS: 9539 (Part I).

20.6.2 The following tests shall be carried out to the entire satisfaction of the Engineer -in-Charge before the work is finally handed over by the contractor and completion certificate is issued: -

(a) Polarity test	Test results shall be lower then the minimum specified in IE rules/ISS.
(b) Insulation resistance test	
(c) Continuity test sectional and overall	
(d) Earth resistance test	This is not to exceed one ohm

20.6.3 Clause 19.146 of SSR (Part I) is also applicable.

20.7 LAYOUT PLAN

20.7.1 Locations of various light fittings/fans/socket outlet, main switches and DBS are tentatively shown on drawings.

20.7.2 The layout of equipment, fittings etc and alignment of cables shown on drawings are tentative. The exact position of these shall however, be approved by the GE in wiring before the contractor shall take the work in hand without any price adjustment.

PARTICULAR SPECIFICATION (CONT'D…../-)

20.8 TYPE OF ELECTRIC SUPPLY

20.8.1 Type of electric supply will be 4 wire, 50 cycles, 415/230 volts, AC.

20.9 EARTHING: -

20.9.1 The earthing shall be strictly in conformity with MES Schedule Part I electrical plate No.3 Earth plate shall be galvanised iron pipe shall light grade, concrete PCC type D-2 (1:4:8), wire mesh 0.63 mm and funnel of PGI sheet, PCC cover and frame shall be 10mm thickness: -

(a) All metal parts shall be provided with a continuous earth wire stranded PVC insulated copper conductor of size 1.00 Sqmm conforming to IS-732.

(b) Supply and fixing of switch fuse distribution boards shall include for the earth continuity conductor as specified above.

(c) No extra payment shall be made if the contractor is required to provide extra depth of pit and extra quantity of charcoal dust to achieve specified test result.

(d) Necessary tests as per SSR part I/Relevant IS shall also be conducted by the contractor without any extra cost and test results thereon shall be kept on record.

21 RECORD DRAWINGS:

21.1 The contractor shall submit 6 copies of the following drawings duly signed by him to the Engineer-in-Charge without any extra cost:-

(a) Internal wiring diagram to include: -
(i) Lights, switches, sockets, fans etc and circuit diagram as per symbols adopted in the extract drawings.
(ii) All outlets marked with circuit number to which they belong.
(iii) Typical section indicating mounting heights adopted.

(b) Drawing for each panel board shall be submitted by the contractor before execution for approval of the Engineer-in-Charge.

(c) One copy of the above drawings shall be submitted to the Engineer-in-Charge within 15 days of the placing of the work order for his approval and remaining with any alterations as directed by the Engineer-in-Charge within 15 days of the completion of work.

22 EXTERNAL ELECTRIFICATION: -

22.1 **GENERAL:-**

22.1.1 Specifications of materials and workmanship shall be all as described in MES Schedule (Part-1 Except where specifically mentioned otherwise in these documents. General rules, preambles to various sections, rates, special conditions, methods of measurements etc given in MES Schedule shall apply to this contract unless otherwise mentioned in these tender documents. The location and distribution of various items in the drawings are tentative. Alterations can be made by the GE as per site requirements and there shall be no price adjustment on this account. The lump sum quoted by the contractor against Schedule 'A' shall be deemed to include expenditure on account of all extras which are not specifically shown on drawings or given in particular specifications but are essential to the execution of work.

PARTICULAR SPECIFICATION (CONT'D…../-)

22.1.2 In case of difference of opinion in this respect between the contractor and GE/CWE, the decision of the Accepting Officer shall be final, conclusive and binding.

22.1.3 On completion of entire work the contractor shall submit in triplicate to the Engineer in Charge layout plan showing actual position of poles, overhead lines, routes, cable run and all other information that will be necessary for maintenance and operation.

22.2 **LAYOUT:-**

22.2.1 The layout of OH lines/UG cable shown on site plan is tentative. The electric lines and cables shall be laid by the route as actually directed by the Engineer in Charge. The contractor shall measure on the ground the actual length along with route and work out the exact requirement of electric lines and cable before placing order for the same.

22.3 **EXCAVATION AND EARTH WORK:-**

22.3.1 Excavation for laying of underground cables/poles/struts and fixing of stay, etc shall be restricted to dimensions shown in the drawings and as directed by the Engineer in Charge. The minimum depth and width of the trenches shall be as laid in paras 19.74, 19.74.1 and 19.76 of SSR Part I respectively. In case of excavation of poles/struts and stays, excavation in excess of required depth shall be made good by the contractor in cement concrete 1:4:8 type D-2 without any extra cost to the Govt.

22.3.2 The trenches for cables shall be excavated to levels/gradients as directed by the Engineer in Charge.

22.3.3 All excavated earth shall be deposited at least one m clear to the edge of the excavation. When roads have to be crossed, trench up to half the width of road shall be dug at a time and proper notice, sign boards and lights shall be displayed and watchman posted to prevent accident.

22.3.4 Refilling of earth in trenches shall be done in 25cm thick layers with excavated soil approved by the Engineer-in-Charge in a manner and to ensure greater compactness and solidity. Each layer shall be well rammed, surplus soil removed as provided in Schedule 'A'.

22.4 **CONCRETE, BRICKS, AGGREGATE SAND ETC:-**

22.4.1 Concrete work shall be carried out all as specified here-in-before. For other general specifications such as bricks, aggregates, sand etc refer relevant items in Schedule 'A' and Particular Specifications given here in before.

22.5 **CABLE PROTECTION:-**

22.5.1 Cable shall be protected in trenches by providing sand cushioning and cable covers all as described in Schedule 'A' items.

22.5.2 On poles/walls cable shall be protected by GI protection pipes all as specified. The pipes shall be fixed to pole by means of proper MS clamps.

22.5.3 Pipe for laying of cable and earth wire shall conform to IS: 1239.

22.5.4 Ends of pipe shall be properly sealed with wooden bush and bitumen compound

after drawing the cables through pipe.

22.6 **CABLE END BOX:-**

22.6.1 Cable end box shall be epoxy cold resin filled type suitable for relay voltage grade as described and jointed as per relevant IS/manufacturers instruction.

22.7 **CABLE GLANDS:-**

22.7.1 Cable glands shall be of brass chromium plated of appropriate size as per IE Rules.

22.8 **SUPPORTS, CLAMPS ETC:-**

22.8.1 The clamps shall be made out of MS flat iron of size not less than 50 x 6 mm. All nuts, bolts, washers used in fixing clamps shall be of galvanised mild steel and of size 19 mm dia.

22.8.2 While calculating the weight of steel, the weight of nuts/bolts/washers etc shall be included for payment and weight of paint will not be included for payment.

22.9 **EARTH WIRE:-**

22.9.1 Earth wire shall be of 4 mm diagalvanised iron and shall be stretched on poles with clamps or eye body hooks.

22.9.2 The earth wire shall be paid by weight either by calculating from table or actual, should these differ only lesser shall be paid. No extra allowance for sag and jointing shall be made while computing the weight of GI wire.

22.9.3 Earthing to metal supports with and accessories shall be done by 2.5 mm dia wire, connected to main continuous earth wire running on poles. Nothing extra shall be paid on this account.

22.9.4 The contractor shall furnish the test certificate from the manufacturers of the following equipments/instruments, material and accessories to GE, in accordance with the relevant ISS.

22.10 **LT CABLES:-**

22.10.1 LT cables shall be of 2/3/3½ or 4 core aluminium conductors, XLPE insulated Heavy duty and suitable for 1100 volts grade and conforming to IS: 7098 (Part-I). Cross section area of these cables shall be armoured unless specified otherwise in Schedule 'A'.

22.10.2 The Aluminium conductors used should be stranded, compacted and circular/shaped and the main insulation should be of cross linked polyethylene (XLPE) with inner sheathing PVC wrapped or PVC extruded and each core of the cable should have colour identification all as specified in IS.

22.10.3 Armour over the inner sheath should be either of strip or wire type and outer sheath should be of extruded PVC conforming to IS: 5831 of 1970.

22.11 **LAYING OF HT/LT CABLES:-**

22.11.1 The work of underground cables shall be carried out as specified in IS: 1255-1983 or latest edition and as described in section 19 (Electrical works section) sub clause 19.74 to 19.82.2 and 19.83 to 19.86.1 of MES Standard Schedule of Rates 2009 (Part–I specifications). The cables under road crossing, railway tracks etc (existing or proposed) shall be enclosed in GI pipes. The sizes of the pipes shall be as decided by

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the Engineer-in-Charge or as described in Schedule 'A'. Size of the pipe shall be such that the suitable space around the cable is available. The pipe shall be provided through out the width of roads, drains and extended for 1200mm on either side of the road. The joints of GI pipe shall be as specified in standard schedule of Rates 2009 (Part-I specifications). GI pipes shall be of light grade conforming to IS: 1239. The both ends of pipes shall be sealed with wooded plug/pegs and bitumen compound after drawing the cable through pipe.

22.11.2 Where the cables are to be laid in the trenches, 80 mm thick cushion of clean dry sand shall be provided in the trench before laying the cable. After the cable has been uncoiled and laid properly into the trenches, it shall be covered with dry sand to a depth of 150mm. This sand will gently punned down thus providing good bedding for the protective cable covers. The punning shall be done by hand and mechanical punners shall not be used. The depth of cable trenches shall be minimum 600 mm and 800 mm respectively for LT and HT cables and width of the trenching shall not be less than 450 mm.

22.11.3 The reinforced pre cast concrete cover or bricks for protection of cable for covering the cables shall be as specified herein before. The pre cast cover shall be laid flat (width of protection being equal to length of brick). After the layer of pre cast cover/brick is laid the remaining trenches will be filled with excavated earth all as described in Schedule 'A'.

22.11.4 Care should be taken during laying of cable to avoid sharp bending and twisting. Cable shall be unwound from the drum by lifting the drum on the center shaft supported both ends with suitable Jacks/stands. Under no circumstances the cable winding shall be lifted off a coil or drum lying flat at the flanges to avoid serious twist and damages. Further the minimum bending radius of 15 x diameter of the cable should be ensured.

22.11.5 The cable shall be snaked to provide a 4 m length of cable for further joints at all straight joints (both side of the joint i.e. 4 m on each side) of cable laid in trenches and a 3 m lengths at terminal cable boxes. The above additional length of cable specified for snaking shall also be measured and paid for under the respective items of cables in Schedule 'A'. Laying pre cast cover/brick protection and sand cushion at places where cables are snaked shall be done as directed by the GE.

22.11.6 Cable path indicators shall be provided at distance of every 100 meters and at cable joints and turning of all cables except for street light cables and 2 core cables. The cost of these indicators shall be deemed to be included in the unit rates quoted for the respective cables. Cable path indicators shall be of CI plate of minimum 150 mm dia and 4mm thick indicating cable size and voltage written on it. The CI path indicator shall be fixed on MS angle iron with minimum two numbers of nuts and bolts. The nuts shall be tucking welded with bolts after full tightening. The size of MS angle iron shall be 30 mm x 30 mm x 6 mm and the length of angle iron should be 900mm adequate to ensure minimum planting depth of 400 mm below ground and 300 mm above GL. The angle iron, at the location of planting should be splitted and bent to shape to provide better grip and should be embedded in PCC (1:3:6) type C-1 using 20 mm graded stone aggregate and of size 200 x 200 x 300 mm. MS angle iron shall be painted with one coat of red oxide primer throughout length and thereafter it will be painted with two coats of synthetic enamel paint and bituminous paint in the portion above ground and below ground respectively. The cable path indicators shall be erected at the time of refilling the trenches and the cable path indicator shall be painted as under:-

(i)For HT 11 KV cables - "BROWN"

(ii) For LT cables - "BLUE"

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22.11.7 While laying cables in ducts, care should be taken to ensure that the cables do not touch each other and minimum clear gap equivalent to diameter of the bigger cable should be provided. Further the cables should be fixed with suitable MS Flat iron (Minimum size 20 mm x 3 mm) clamps in the cable ducts to avoid jumbling. If necessary, cables may be fixed with clamps on the walls of the duct. Where cables are to be fixed by means of suitable clamps or bracket or on cable racks fixed along poles/steel structure, they shall be drawn through pipe as directed. Wherever the cables are embedded in the floor, it should be laid in the GI pipe all as directed. The cost of clamps etc shall be deemed to be included in the unit rate quoted for the respective cables.

22.11.8 Lay out of cables shown in the drawings is for guidance only. The contractor shall, however, verify the cable lengths actually required at site before procurement of the same.

22.11.9 Joints in cable shall not be at distance less than 300 meters. When large length of cable are to be laid, exceeding 500 m, which is a standard drum length, the joints will be done at interval of 500 m. Straight through joints shall be measured and paid separately under the respective items of Schedule 'A'.

22.11.10 The unit rates quoted for respective LT cables, shall be deemed to include the cost of cable glands, lugs/thimbles, preparation and making cable end joints including crimpling etc all as specified. The termination of LT cables in to Transformers/Switch gears/Feeder pillar-boxes/Distribution boxes shall in-variably be done through lugs or thimbles only to avoid loose connections. No termination of cables without suitable lugs/thimbles will be accepted. However, cable termination in case of HT cables shall be measured & paid separately under the respective items of Schedule 'A'.

22.11.11 The metal sheath, metal screen (if any) and armour of all the LT/HT cables should be effectively earthed at both the ends all as specified in IS: 1255-1983 and SSR part I.

22.11.12 The cables should be purchased directly from the manufacturers or their authorized dealers only. Contractor should also submit the test certificate from the manufacturers besides carrying out the field tests. All new cables should be tested for insulation resistance using 1000 V, 2.5 KV and 5.0 KV IR Tester for LT, 11 KV and 33 KV grade cables respectively. Cables should be tested for insulation resistance between all the phase to phase and from each phase to earth before decoiling the cable from the drum and also after laying the cable in the trenches. Only after satisfactory results are obtained, cable jointing and termination work should commence. The test results of completed cable installation should also be measured and entered into record book for comparison purposes during service life of the cable installations. Arrangement for carrying out all these tests shall be made by the contractor without any extra cost to Govt in the presence of Garrison Engineer/ Engineer-in-Charge. The record of all such tests shall also be signed by Engineer-in-charge and contractor and countersigned by Garrison Engineer. At the closing of the work on every day, cable ends should be covered with suitable rubber/plastic caps and sealed effectively by insulating tap/bitumen to avoid any ingress of moisture into the cable.

22.11.13 Unless otherwise specified in Schedule 'A' the cost of cutting through roads/drains for laying cable under roads/drains and making good shall be deemed to be included in the unit rate of cables laid in pipes.

22.11.14 Wherever HT/LT cable will be crossing or running parallel to existing telephone cables, suitable measures will be taken and distance will be observed as per relevant ISS/IE Rules on the subject.

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22.12 **CABLE END TERMINATION AND STRAIGHT THROUGH JOINT FOR 1100V GRADE XLPE CABLES:-**

22.12.1 Cable and termination Indoor/ outdoor type and straight through joints for LT cables shall be based on cast resin system. The casting resin should be polyurethane based having compatibility with the cable components. The cast resin should be suitable to provide adequate electrical insulation, mechanical protection and moisture imperviousness to the joint. The joint should be suitable for compression connection/ mechanical connector/soldering of cable conductors to meet the technical requirement as required at site. The termination kit should be designed for easy installation and have a crutch sealing arrangement for environmental protection on the core trifurcation.

22.12.2 All the jointing of cables in joint boxes etc and filling in of the compound shall be done strictly as per the instructions furnished by the manufacturers of the cables and joint boxes.

22.12.3 The joints should be suitable to withstand mechanical impact, heat shock tests, load cycle test and water penetration test (under pressure) without damage to the outer sheath.

22.12.4 No separate payment for providing above joints for LT cables shall be admissible. Rates quoted for cables shall be deemed to include for this provision.

22.13 **STANDARD OF QUALITY AND WORKMANSHIP:-**

22.13.1 The supply and installation of all electrical equipment and accessories shall strictly comply with provisions contained in the latest edition of Indian Standard Code of practice/specifications as applicable to above mentioned works, except where such regulation and rules are modified by these Specifications.

22.13.2 Refer condition 25 of General Conditions of Contracts IAFW-2249. All electrical work shall be carried out by properly skilled and electrical supervisors/ Engineers. The contractor shall, on demand by the Engineer-in-Charge produce such evidence of Qualifications of his workmen/ supervisors/ Engineers, either at commencement of work or at any time thereafter during the currency of the contract. The entire work shall be of high class with the best workmanship and to the entire satisfaction of GE/Engineer-in-Charge.

22.13.3 The Kiosk, transformer, cables, accessories, etc. incorporated therein shall be capable of giving satisfactory operation under the climatic conditions.

22.13.4 The design and manufacture of the equipment shall be of highest standard to ensure continuous and trouble free service.

22.14 **MOULDED CASE CIRCUIT BREAKERS (MCCBs):-**

22.14.1 Moulded case circuit breakers (MCCBs) shall be suitable for operational voltage of 415 Volt AC, 50 Hz, 3 phase, 4 wire system for a rated current and ultimate breaking capacity as mentioned in Schedule 'A' and it shall conform to IS: 13947 (Part 2) and IEC: 60947 (Part 2). All MCCBs shall be of four-pole version unless mentioned other wise in Schedule 'A'.

22.14.2 MCCBs shall be of Compact and elegant design suitable for reversible load and line terminations without affecting its performance. MCCBs shall be suitable for fixing flush on the panels and shall be provided with handle operating mechanism including Rotary Handle vari-depth type.

22.14.3 The insulating case and cover of MCCBs shall be made of high resistant and flame retardant thermosetting insulating materials. The switching mechanism shall be

quick make, quick break and trip free. The position of the operating knob/handle shall clearly indicate ON, OFF and TRIP position.

22.14.4 Each pole shall be provided with a pair of contacts, which shall open at a high speed over a large distance under short circuit faults. The special designed arc cut outs of insulating materials shall be provided to contain the arc by providing effective arc quenching device.

22.14.5 The tripping mechanism shall be thermal release or Electronic release or thermal magnetic release for protection for over load and short circuit as per the details given below: -

Capacity Of MCCBs	Ultimate breaking Capacity	Overload release	Short circuit release
1	2	3	4
Upto 125 Amps	16 KA	Adjustable Thermal release (range 70% to 100 % of rated current)	Fixed magnetic type suitable for 10.0 In
160 to 630 Ams	36 KA	Adjustable Thermal release (range 70% to 100 % of rated current)	Adjustable magnetic release (3.5 to 10 times of rated current)
800 and 1250 Amps	50 KA	Adjustable electronic release (over load Zone range 40% to 100 % of rated current)	Adjustable electronic release(range (5 to 10 times of rated current)

22.14.6 Under voltage trip, mechanical interlocks etc shall be provided as per standard practice and ISS. Under voltage trip shall be designed to operate when the control voltage drops below a tripping threshold i.e. 20% to 70% of rated voltage and shall be suitable for operation on 230V/415V AC.

22.14.7 The terminals shall be suitable for both copper and aluminium terminations as specified. The MCCBs shall be one of the makes mentioned herein after in these Particular Specifications.

22.15 **MCBs, DISTRIBUTION BOARDS AND PLUG & SOCKET DBs:-**

22.15.1 Distribution boards shall be factory made and conforming to IS: 8623 suitable for universal mounting copper bus bar, Neutral bar, and Earth bar, Standard DIN bar Rail and cable ties for cable management. Top and bottom shall have removable gland plates with knock outs. The copper bus bar shall be suitable for 100 amps unless specified other wise in Schedule 'A'.

22.15.2 SPN Distribution boards shall be suitable for provision for DP/SPN MCBs/ Isolators as incomer and SP outgoings all as specified and shall be flush mounting type. The degree of protection shall be IP-43 protection with acrylic door unless specified otherwise in Schedule "A".

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22.15.3 TPN distribution boards shall be suitable for FP/TPN MCBs/Isolator as in comer and TPN/SP outgoings all as specified and shall comply with IP-43 protection with metal door unless mentioned otherwise in Schedule 'A'.

22.15.4 All distribution boards shall be fixed flushed on the wall at locations as directed by Garrison Engineer.

22.15.5 MCBs: - All MCBs shall be conforming to IS: 8828 of 1996 and shall be ISI marked. These shall be suitable for 'C' curve and 10KA breaking capacity and shall be provided with box terminal on top and bottom both suitable for adopting cable size up to 35 Sqmm. The enclosure of MCBs shall be of 'Molded Self Extinguishing thermo set plastic and these shall be suitable for snap fixing on standard Din Rail. The current capacities and poles shall be all as mentioned in Schedule 'A'.

22.15.6 MCBs Isolators – All MCB type isolators shall conform to IS: 13947-3 and shall be suitable for impulse voltage of 6KV and short time with stand capacity of 1000 Amps for 0.3 seconds.

22.15.7 Plug and Socket DBs :- Plug and socket distribution Boards including SP/DP MCBs for protection of appliances like window type/split type Air conditioners and Geysers shall only be provided all as specified. Similarly distribution boards with MDS Legrands type P-17 Tempera plugs and sockets including TPN/FP MCBs shall be provided to make available 3 phase 415 volts, 50Hz AC electric supply in Technical Buildings. All these plug and socket DBs shall be of universal mounting type.

22.15.8 The makes of Distribution Boards, Plug and socket DBs and MCBs shall be all as specified here in after and the models/Type shall be as indicated in the respective Schedule.

23.0 **BLAST PEN AND LINK TAXI PAVEMENT**

23.1 **EXCAVATION & EARTH WORK**

23.1.1 Excavation shall be carried out all as specified in the respective Sch 'A' items. Rate quoted by the contractor shall be deemed to include for bailing, pumping & dewatering of foundation, trenches if water is met with or accumulated from any source other than that specified in Clause 3.1 of MES Schedule Part-I. No price adjustment shall be made on this account. Filling in trenches, in embankment etc shall be done in layers as specified here-in-after with approved earth. The soil obtained from excavation of foundation and excavation over areas if found not suitable shall not be used for filling. All surplus soil shall be disposed off to a distance as mentioned in the respective items of Sch 'A' and as directed by the Engineer-in-charge.

23.1.2 Excavation in trenches for laying of electric cable & construction of area drain for the approx length up to 350 Running Metre at the location earmarked by the GE shall be carried out manually to avoid damage to existing electrical and communication cable. Excavation by the JCB or other mechanical means shall not be allowed for said length. Stretch of around 350 RM shall be excavated carefully so that existing cable in the area will not be punctured /break. Any damage to the cable occurred during excavation shall be made good by replacing cable by the contractor without any extra cost to the Dept.

23.1.3 Contractor's representative and the Engineer-in-Charge shall jointly take the levels of existing ground at 2m x 2m grid intervals using Total station Survey Equipments as directed by the GE and enter the same in measurement book/level book.

23.2 **TEMPORARY BENCH MARK(S):**

23.2.1 For the transfer of levels, contractor shall construct temporary Bench Mark(s) of brick masonry duly plastered at the locations all as directed by the GE. Quoted rates are inclusive of cost of construction of temporary Bench Mark(s) & nothing extra is payable on this account.

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23.2.2 The drawings shall be prepared jointly showing block levels, plan, longitudinal sections, cross sections thereof & shall form part of the Measurement Book (MB). Authorised quantity of work to be executed for the purpose of payment shall be worked out from 'L' sections. Earth works/refilling other than authorised shall not be considered for payment. Excavation and earth filling shall commence only after the aforesaid 'L' and cross sections are drawn & signed by both the parties. Commutation of areas shall be done with "Trapezoidal Formula" and volume using "Prismoidal formula". Earth filling, in formations and sub grade of dispersal shall be carried out in layers exceeding 10 cm & not exceeding 20 cm thick (spread thickness). Each layer shall be sufficiently watered and consolidated with 8 to 10 tonne vibratory power roller. The grade should be brought to the proper levels, camber and gradient by filling all sunken and soft portions with earth so as to prepare the surface for receiving sub base.

23.2.5 **SETTING OUT**

After the site has been cleared, the limits of excavation shall be set true to lines, curves, slopes, grades and sections as shown on the drawings or as directed. The contractor shall provide all labour, survey instruments and materials such as strings, pegs, nails etc, required in connection with setting out of works and the establishment of benchmarks. The contractor shall be responsible for the maintenance of benchmarks and other marks of stakes for the period as directed by the GE.

23.2.6 **REUSE OF EXCAVATED SOIL**

Where the excavated soil is directed to be reused in the construction/ refilling in formation of sub-base & sub-grade, it shall be stacked properly for pulverization/mixing with cement etc.

23.2.7 **PREPARATION OF CUT FORMATION:**

The cut formation, which serves as a foundation base, shall be prepared to proper line and level as directed by GE. Any loose pockets shall be well compacted. Any unsuitable material encountered at the formation level shall be removed to a depth indicated by the GE and replaced with suitable material as approved and directed by GE and compacted in accordance with embankment and sub grade construction.

23.2.8 **FINISHING OPERATION**

Finishing operation shall include the work of properly shaping and dressing all excavated surfaces. The finished cut formation surface shall not vary from those calculated with reference to the profile as directed by the GE beyond the tolerance of 20mm to 25mm.

23.2.9 **METHODS, TOOLS AND EQUIPMENT**

Only such methods, tools and equipment as approved and directed by the GE shall be adopted /used in the work. Contractor shall demonstrate efficiency of type of equipment to be used before the commencement of particular work/activity.

23.2.10**MEASUREMENT FOR PAYMENT**

Excavation and earthwork shall be measured by taking cross sections/ grid at the intervals of 2 metre x 2 metre before commencement of excavation activity and subsequently after completion of connected activity to compute cubical content using Prizmoidal Formula & Simpson Rule. Where it is not feasible to compute volumes by this method because of erratic location of isolated deposits, the volumes shall be computed by other accepted methods as approved by GE. The contractor shall leave depth indicators during excavations of such shape and size and in such positions as directed so as to indicate the original ground level as accurately as possible. The contractor shall be responsible to keep these intact till the final measurements are taken.

24.0 **SOIL STABILISATION FOR SUB BASE**

24.1 The soil shall be stabilized with mixture of excavated earth and min 3% of cement by weight of earth (or as per design mix to achieve requirement) laid in layers not exceeding 20 cm in thickness (each layer) after mixing in the mechanical plants/rotellier, watering the mixture

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properly to enable the excavated earth and cement to react, banking the mixture in layers on the prepared sub-grade, rolling each layer with soil compactor/ tandem vibratory roller as directed by the Engineer-in-Charge under optimum moisture content to achieve density of at least 95% of the maximum modified standard proctor density at optimum moisture content, including levelling and fine dressing the top layer to the required grade, camber and slope etc complete as per Technical Specification and as directed by GE. (Payment to be made for compacted quantity only).

24.2 After proper compaction of refilled earth the "K" value and CBR value shall be ascertained /tested through agency approved by the CWE (AF) Gwalior to ensure that minimum "K" value of 5 kg/cm3 and soaked CBR value of 8% shall be achieved at the top of sub base layer (stabilized soil). The soil samples to be stabilized shall be tested in laboratory and proportion of cement shall be determined to yield soaked CBR value of 8% in terms of IS:2720 and sub grade reactions 'K' value of 5 kg/cm3 and unconfined Compressive Strength shall be tested in accordance with IS:9214 in the field.

24.3 Cost of testing of 'K' and 'CBR' value shall be borne by the Contractor.

24.4 **APPROVED EARTH FOR SOIL STABILIZATION**

24.4.1 Only approved earth shall be used for stabilization. Any vegetation, organic impurities shall be removed from the excavated earth. Samples of soil to be used for stabilization shall be tested in laboratory for the following properties:

(i) Sieve analysis
(ii) Liquid limit
(iii) Plastic limit
(iv) Modified proctor density and corresponding optimum moisture content.(v)Natural moisture content
(vi) Deleterious constituents.
(vii) Soaked CBR in Lab

Requirement for soil modification/subgrade improvement
Cement stabilized materials can be used for soil modification or improvement of subgrade soil. It is recommended from economic consideration that mix in-place methods of construction be used for subgrade improvement and only granular materials and silty cohesive materials be used. (The assumption being that more clayey materials would be more effectively stabilized with lime). The main requirements for cement modification or stabilization of subgrade soil are summarised in table mentioned table:-
Soil Characteristics for Cement Modified Soil/Improved Subgrade/Capping Layer

Properties	Specified Value
Liquid Limit (%)	<45
Plasticity Index	<20
Organic content (%)	<2
Total SO4 content	0.2 % Max
Minimum Laboratory CBR at specified density (%)	15
Minimum cement content (%)	2 (Ref note)
Degree of pulverisation (%)	>60
Temperature for mixing	More than 10ºC
Time for completing compaction	2 hrs Max

Note: -
In case better mechanical equipment for spreading of cement, for breaking clods and blending is used, the minimum percentage of cement for stabilization could be 0.5 percent. However extensive lab testing must be done to arrive at this minimum percentage. Sample at site of blended loose soil be collected and remoulded in lab to confirm that the desired CBR can be achieved.

24.5 **CEMENT**

24.5.1 Cement for soil stabilization shall be ordinary Portland cement 43 grade conforming to IS 8112-1989 from approved manufacture all as specified here-in-after.

24.6 **PROPORTION OF CEMENT MIX**

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24.6.1 In case the laboratory tests prescribe the cement proportion in excess of 3% of total weight catered in the BOQ item then the prescribed higher percentage of cement shall be used and excess cement so used shall only be paid.

24.7 **CONSTRUCTION OPERATION WEATHER LIMITATIONS**

24.7.1 Stabilization shall not be done when the air temperature in the station/ site is less than 10°C.

24.8 **DEGREE OF PULVERISATION**

24.8.1 For stabilization, the soil before addition of cement as stabilizer shall be pulverized mechanically using agricultural machineries such as disc, harrows motivators etc. The degree of pulverization shall be such that it passes the requirement of Table 400-7 of MORTH.

IS sieve 26.5mm 100% by weight passing IS sieve

IS sieve 5.6 mm 80% by weight passing IS sieve

24.9 **DETERMINATION OF DEGREE OF PULVERISATION**
Degree of Pulverisation shall be checked as per SSR Part-I clause 20.A.15.B.6.2.1.

24.10 **EQUIPMENT FOR CONSTRUCTION**

24.10.1 Stabilized soil sub-bases shall be constructed by central plant mixing of construction or as otherwise approved by the GE. Manual mixing shall not be permitted. The equipment used for central plant mixing construction shall be a soil stabilising plant or similar approved equipment capable of mixing the soil, cement and water to specified degree and of achieving the desired degree of mixing and uniformity of the stabilized material. If so desired by the Engineer, trial runs with the equipment shall be carried out to establish its suitability for work.

24.11 **CENTRAL PLANT MIXING METHOD OF CONSTRUCTION**

24.11.1 **SOIL**: Before deploying the equipment, the soil after it is made free of undesirable vegetation & other deleterious matter shall be pulverised till 100 percent passes through a 26.5 mm sieve and at least 80 percent of the soil, excluding any stone or gravel, passes through 5.6 mm sieve. It shall have enough stockpile material meeting the requirements for at least one day of base construction before operations begin.

24.12 **CEMENT:**

24.12.1 The cement (OPC-43 grade) shall be measure cement by weight and added uniformly into the mixture. The cement incorporated per ton of soil, shall be within ±5 percent of the prescribed weight. Mixing of cement shall be checks to compare the actual percentage of cement in the mixture with the required cement percentage as specified in the approved Mix Design on each of the first two tankers supplying cement to the plant. If these checks are within the specified tolerance then one cement check per day will be required.

24.13 **MIXING:**

24.13.1 A proportion of soil, cement, and water separately and accurately shall be measured before mixing. Charge all materials into the mixer together and begin mixing immediately until a homogeneous and uniform mixture is produced. If the final blend of materials is not homogeneously mixed or does not meet the moisture range specified in "Moisture Control," cease plant operations until corrections are made in the plant or to the materials.

24.14 **HAULING:**

24.14.1 Spread soil-cement material so that compaction can begin within 45 minutes after the soil, cement, and water have been charged into the mixer. The mixture shall be protected in transit by using a securely fastened waterproof cover large enough to extend down over the sides and the end of the bed of each haul vehicle.

24.15 **SPREADING:**

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24.15.1An approved MIXTURE SPREADER shall be used to obtain the specified thickness and spread the mixture to the full width of the area to be covered. It shall be ensure that trucks and other construction equipment, including motor graders, do not travel over the material until compaction equipment has made initial passes over the mixture. Ensure that less than 30 minutes elapse between the placement of cement- treated material in adjacent lanes at any location, unless longitudinal joints are specified.

24.16 **THICKNESS**: The earth layers to be stabilized with cement shall be of uniform thickness not exceeding 200 mm in each layer.

24.17 **COMPACTING AND FINISHING**

24.17.1 Test Section: Test section shall be constructed by the contractor of 50 m long and 12 m wide. Before constructing a test section a Construction Work Plan shall be submitted by the contractor to the Garrison Engineer for approval. The Construction Work Plan must indicate proposed equipment and compaction procedures. After Construction Work Plan is approved by the GE, Engineer-in-Charge will evaluate the Work Plan during test section construction. The Engineer-in-Charge will evaluate compaction, moisture, homogeneity of mixture, thickness of course, and laminations or compaction planes (scabbing). If the Engineer- in- Charge determines that the Work Plan is not satisfactory then revise the compaction procedure and augment or replace equipment as necessary so as to complete work according to the Specifications.

24.17.2 Time Limits: Compaction shall be commenced within 45 minutes from addition of water to the soil-cement mixture and to be completed within 2 hours. All operations from adding cement to finishing the surface shall be completed in four hours.

24.17.3 Moisture Control: During compaction, ensure a uniform moisture content of the mixture that is between 100 and 120 percent of the optimum moisture content. If the moisture content exceeds the tolerance at any time, cease operations immediately and make the adjustments necessary to bring the moisture content within tolerance.

24.18 **ROLLING AND COMPACTION**

24.18.1Immediately after spreading, grading and levelling of the mixed material, each layer shall be compacted with soil compactor and /or sheep footed roller/vibratory roller of minimum 80 to 100 KN static weight with plain drum or pad foot drum or heavy pneumatic tyre roller of minimum 200 to 300 KN weight and minimum tyre pressure 0.7 Mn/m2 or equivalent capacity roller as approved by the GE. If necessary rolling may be preceded by a few passes of light roller. Successive layers of mix shall not be placed until the lower layer has been thoroughly compacted to satisfy the density requirements, and all quality control tested as per requirement specified. Rolling shall commence at edges and to be progressed towards the centre, except at super elevated portions where it shall commence at the inner edge and progress towards outer edge. During rolling the surface shall be frequently checked for grade and cross fall (camber) and any irregularities noticed shall be corrected by loosening the material and removing/adding fresh material. Compaction shall continue until the density/achieved is at least 98 per cent of the maximum dry density for the material determined in accordance with IS: 2720 (Part-VIII). Utmost care shall be taken to see that the compaction of cement stabilized material shall be completed within two hours from its mixing or such shorter period as may be found necessary in dry weather. During rolling it shall be ensured that roller does not bear directly on hardened or partially hardened treated material previously laid other than what may be necessary for achieving the specified compaction at the joint. The final surface shall be well closed, free from movement under compaction planes, ridges, cracks or loose material. All loose or segregated or otherwise defective areas shall be made good to the full thickness of the layer and re-compacted. The field density of compacted layer shall be measured by "Sand replacement Method" as specified in IS: 2720 (Part- XXVIII).

24.19 **CONSTRUCTION JOINTS:**

24.19.1 Form a straight transverse joint at the end of each day's construction or when the work is interrupted so that the material cannot be compacted within the time limit specified in this Subsection. Create the straight transverse joint by cutting back into the completed work to form a

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true vertical face free of loose or shattered material. Joint shall be formed at least 600 mm from the point at which the strike-off plate of the spreader comes to rest at the end of the day's work or at the point of interruption. Longitudinal joint shall be formed as described above if the soil-cement mixture is placed over a large area where it is impractical to complete the full width during one day's work. Ensure that waste material is removed from the compacted base.

24.20 **CURING :** The surface shall be cured suitably for a minimum period of seven days all as per MORTH para. 403.3.

24.21 **SURFACE FINISH :** The finished surface shall be as per SSR Part-I, clause 20.A.15B.6.9.

24.22. **HORIZONTAL ALIGNMENT/ PROFILE:** Horizontal alignments shall be as per SSR Part-I clause No. 20.A.15B.6.9.1

24.23 **LONGITUDINAL ALIGNMENT/PROFILE** :Longitudinal profile shall be as per SSR Part-I, clause No. 20.A.15B.6.9.2.

24.24 **SURFACE REGULARITY OF SUBGRADE AND PAVEMENT COURSES**

24.24.1 The surface regularity of completed sub-grade and sub-base shall be as per SSR Part-I, clause no. 20.A.15B.6.9.3 and as per permitted tolerance table given hereinafter:-

TABLE 5.2
PERMITTED TOLERANCE OF SURFACE REGULARITY FOR SUBGRADE AND SUB-BASE

Sl No.	Type of Construction	Longitudinal profile with 3 meter straight edge					Cross-profile
		Maximum permissible undulation	Maximum number of undulations permitted in any 300 meters length exceeding				Maximum permissible variation from specified profile under camber template (mm)
			Mm				
			10	12	10	6	
1.	2.	3.	4.	5.	6.	7.	8.
1.	Earthen Sub-grade	24	30	-	-	-	15
2.	Granular Sub-base	15	-	30	-	-	12

24.25 **RECTIFICATION OF SUBGRADE IRREGULARITY**

24.25.1 When the surface regularity falls outside the specified tolerance, the Contractor shall be liable to rectify these in the manner described below and to the satisfaction of the Engineer-in-Charge.

24.26 **HIGH SURFACE**

24.6.1 Where surface is high, it shall be trimmed in such a way that the materials below are not disturbed due to this operation.

24.27 **LOW SURFACE**

24.27.1 For cement treated material, when the time elapsed between detection of irregularity and the time of mixing of the material is less than 2 hours, the surface shall be scarified to a depth of 50mm supplemented with freshly mixed materials as necessary and re-compacted to the relevant specification. When the time is more than 2 hours, the full depth of the layer shall be removed from the pavement and replaced with fresh material laid to Specification.

24.28 **QUALITY CONTROL TEST DURING CONSTRUCTION**

PARTICULAR SPECIFICATION (CONT'D…../-)

24.28.1 The material supplied and the works carried out by the Contractor shall conform to the specification. To ensuring requisite quality of construction, materials and works shall be subjected to quality control tests, as described hereinafter. The testing frequencies given below in Table are the desirable minimum frequencies and GE shall have the full authority to carry out additional tests as deemed necessary in his opinion to satisfy himself that the materials and works comply with the appropriate specifications. Where no specific testing procedure is mentioned, the tests shall be carried out as per the prevalent accepted engineering practice as directed by the GE/Engineer-in-Charge. Following quality control tests shall be carried out at frequencies noted against each:-

TABLE-5.3

(a) **EARTH**

TEST	TEST METHOD	FREQUENCY
Sieve Analysis (Gradation)	IS:2720 (Part-IV)	1 test per 250 Cum of good earth.
Sand Content	IS:2720 (Part-V)	2 tests per 3000 Cum of good earth.
Plasticity Test	IS:2720 (Part-V)	2 tests per 3000 Cum of good earth.
Density Test	IS:2720 (Part-VIII)	2 tests per 3000 Cum of good earth (Each soil type).
Deleterious Content Test	IS:2720 (Part-XXVII)	As and when required by Engineer-in-Charge.
Moisture Content	IS:2720 (Part-II)	1 test for every 250 Cum of good earth.

(b) **CEMENT STABILIZED SOIL SUB-BASE**

TEST	FREQUENCY (MINIMUM)
Quality of Cement	One test for each consignment subject to a minimum of one test per 5 tonnes.
Cement Content	Regularly through procedural checks.
Degree of pulverization	Periodically as considered necessary by the Engineer-in-Charge.
CBR (IS:2720 Part XVII) on soaked/ uncooked samples	One test per 3000 cum. Mean of 3 samples shall constitute one test result.
Moisture Content prior to compaction (IS:2720 Part-II)	One test per 250 Sqm.
Density of compacted layer (IS:2720 Part XXVIII)	One test per 500 Sqm.
Surface Accuracy	Regularly.

24.29 **MEASUREMENTS**

24.29.1 The initial levels shall be taken at a fixed interval/grid of 2 metre longitudinally as well as transversally. These levels shall be recorded in field books and plotted in plan before start of work both duly signed by the Engineer-in-Charge and the Contractor in token of acceptance. After completion of work, the final levels shall be taken at grid points, where previously the ground levels were taken, in the presence of Contractor or his authorized representative and his dated signatures obtained on the level book in token of his acceptance. These levels shall make the basis of payment. The quantity shall be computed in cubic metre from cross-sections after compaction. No deduction shall be made for voids. The quantity for payment shall be the theoretical quantity (based on formation levels) or actual quantity (based on actual levels) whichever is less.

25 **GRANULAR SUB- BASE (GSB)**

25.1 The material to be used for the work shall be crushed stone of 40 mm to 60 mm size from the source mentioned above, natural river sand, moorum, or combination thereof depending upon

grading approved by the CWE. The material shall be free from organic or other deleterious constituents and conforming to the grading given in the Table here-in-after.

25.2 **PHYSICAL REQUIREMENTS:**

The water absorption value of the coarse aggregate shall be determined as per IS: 2386 (Part-III); if this value is greater than 2 per cent, the soundness test shall be carried out on the material delivered to site as per IS: 383. For Grading-II and III materials, the CBR shall be determined at the density and moisture content likely to be developed in equilibrium conditions which shall be taken as being the density relating to a uniform air voids content of 5 per cent. Aggregate Impact value shall be < 40.

25.3 **TABLE: GRADING FOR CLOSE GRADED GRANULAR SUB BASE MATERIALS**

IS Sieve Designation	**Grading II Table 400.1 of MoRT&H.**
75.0 mm	-
53.0 mm	100
26.5 mm	70 – 100
9.50 mm	50-80
4.75 mm	40 – 65
2.36 mm	30-50
0.425 mm	10-15
0.075 mm	< 5
CBR Value (Min)	30

IS Sieve Designation	**Grading V Table 400.1 of MoRT&H.**
75.0 mm	100
53.0 mm	80-100
26.5 mm	55-90
9.50 mm	35-65
4.75 mm	25-50
2.36 mm	10-20
0.85 mm	2-10
0.425 mm	0-5
0.075 mm	-

Note:- Physical requirements of material of GSB will be as per Table 400-2 and QC test frequency will be as per Ser No 1 of Table 900-3 of MORTH-Specifications for road and bridge works (Fifth Edition).

25.4 **CONSTRUCTION OPERATIONS**

25.4.1 PREPARATION OF SUBGRADE:- Immediately prior to the laying of sub-base, the sub-grade already finished here-in-before as applicable shall be prepared by removing all vegetation and other extraneous matter, lightly sprinkled with water if necessary and rolled with two passes of 80 – 100 KN smooth wheeled roller.

25.4.2 **SPREADING AND COMPACTING**

25.4.2.1 The sub-base material of grading specified in the Contract shall be spread on the prepared sub-grade with the help of a MOTOR GRADER of adequate capacity, its blade hydraulic controls suitable for initial adjustment and for maintaining the required slope and grade during the operation or other means as approved by the Engineer-in- Charge.

25.4.2.2 When the sub-base material consists of combination of materials mixing shall be done mechanically by the mix-in-place method. The equipments used for mix-in-place construction shall be a ROTAVATOR or similar approved equipment capable of mixing the material to the desired degree and shall be carried out to establish its suitability for the work. Moisture content of the loose material shall be checked in accordance with IS: 2720 (Part 2) and suitably adjusted by sprinkling additional water from a truck mounted or TRAILER MOUNTED WATER TANK and suitable for applying water uniformly and at controller quantities to variable widths of surface or

other means approved by the Engineer-in- Charge, so that, at the time of compaction, it is from 1 per cent above to 2 percent below the optimum moisture content corresponding to IS: 2720 (Part 8). While adding water, due allowance shall be made for evaporation losses. After water has been added, the material shall be processed by mechanical or other approved means like disc harrows, motivators until the layer is uniformly wet. Immediately thereafter, rolling shall start, if the thickness of the compacted layer does not exceed 100 mm, a smooth wheeled roller of 80 to 120 KN weight may be used. For a compacted single layer up to 200 mm the compaction shall be done with the help of a vibratory roller of minimum 80 to 120 KN static weight with plain drum or pad foot drum or heavy pneumatic tyred roller of minimum 200 to 300 KN weight having a minimum tyre pressure of 0.7 MN/m2 or equivalent capacity roller capable of achieving the required compaction. Rolling shall commence at the lower edge and proceed towards the upper edge longitudinally for portions having unidirectional cross-fall and super elevation and shall commence at, the edges and progress towards the centre for portions having cross fall on both sides. Each pass of the roller shall uniformly overlap not less than one third of the track made in the preceding pass. During rolling the grade and cross-fall (chamber) shall be checked and any high spots or depressions, which become apparent, corrected by removing or adding fresh material. The speed of the roller shall not exceed 5 km per hour. Rolling shall be continued till the density achieved is at latest 98 per cent of the maximum dry density for the material determined as per IS: 2720 (Part 8). The surface of any layer of material on completion of compaction shall be well closed, free from movement under compaction equipment and from compaction planes, ridges, cracks or loose material. All loose, segregated or otherwise defective areas shall be made good to the full thickness of layer and re-compacted.

25.5 **STACKING OF AGGREGATES FOR GSB/WMM/DLC/PQC & RCC WORK.**

25.5.1 Contractor shall stack aggregates obtained from the approved source (mentioned here-in before) of the required grading and type as per design mix on a raised PCC platform. Measurements of stacks shall be recorded in the MB under the heading 'NOT TO BE ABSTRACTED '.

25.6 **GRADING/TESTING OF CONSTITUENTS MATERIAL FOR GSB/WMM/DLC/PQC & RCC WORK**

25.6.1 Test reports of all the constituents viz. coarse/ fine aggregate, crushed screening, water and cement etc approved for the construction of GSB/WMM/DLC/PQC & RCC work shall form part of their Design Mix.

25.6.2 If source/ make of any constituent is changed by the contractor then redesigning of sid course/ item is required. Onus of the delay due to redesigning which hamper progress of work lies with the contractor.

26. **WET MIXED MACADAM (WMM)**

26.1 Coarse aggregate shall be crushed stone as per sieve designation mentioned below. The coarse aggregate shall be hard and durable and shall be free from excess of flat, elongated, soft and disintegrated particles, dirt and other objectionable matter. Coarse aggregate shall be tested as per IS: 2386 (Part-IV) to check the physical requirements and strength . WMM shall be laid in layers as specified in Sch 'A'. Screening shall be of the same materials as of coarse aggregate.

26.2 Coarse aggregate shall conform to the following:

Srl No	TEST	TEST METHOD	REQUIREMENT
1	Los Angeles Abrasion	IS-2386 (Schedule 'A'-IV)	40% Max
	Value or aggregate impact value	IS-2386 (Schedule 'A'-IV) or IS 5640	30% Max
2.	Combined Flakiness and elongation	IS-2386 (Schedule 'A'-I)	35% Max

NOTE: To determine this combined grading & proportion the flaky stone from a representative sample should first be separated out. Flakiness index is weight of flaky stone metal divided by weight of stone sample. Only the elongated particles shall be separated out from the remaining (non-flaky) stone metal. Elongation index is weight of elongated particles divided by total non-flaky particles. The value of flakiness index and elongation index so found are added up. If the

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water absorption value of the coarse aggregate is greater than 2%, soundness test shall be carried out as per IS-2386 (Part-5).

26.3 GRADING REQUIREMENTS:

The aggregate shall conform to the grading 1 as per IRC: 109-1997 given below:-

IS sieve designation	Percent by weight passing the IS sieve
53.00 mm	100
45.00 mm	95-100
26.50 mm	-
22.40 mm	60-80
11.20 mm	40-60
4.75 mm	25-40
2.36 mm	15-30
600 micron	8-22
75 micron	0-5
11.20 mm	40-60

Material finer than 425 micron shall not have plasticity Index (PI) more than 6. The final gradation approved within these limits shall be well graded from course to fine and shall not vary from the low limit on one sieve to the high limit on the adjacent sieve or vice- versa. Coarse aggregate and screening specified above shall be stacked closely packed on level and hard surface after 'level' and shall be measured and recorded in MB before incorporation. A deduction of 1/13th from the stack measurements shall be made for unevenness of the ground.

26.4 Laboratory design mix of the WMM shall be approved by the CWE before commencement of WMM work / Trial bay laying.

26.5 **CONSTRUCTION OPERATION**

26.5.1 **PREPARATION OF BASE**

The surface of the sub grade/sub-base/base to receive the Wet Mix Macadam course shall be prepared to the specified lines and cross fall (camber) and made free of dust and other extraneous material. Any ruts or soft yielding places shall be corrected in an approved manner and rolled until firm surface is obtained if necessary by sprinkling water. Any sub- grade/sub-base/base surface irregularities where predominant, shall be made good by providing appropriate type of profile corrective course (levelling course).

26.5.2 **PROVISION OF LATERAL CONFINEMENT OF AGGREGATE**

(a) While constructing wet mix macadam, arrangements shall be made for the lateral confinement of wet mix. This shall be done by laying materials for adjoining shoulder along with that of wet mix macadam.

26.6 **PREPARATION OF MIX**

26.6.1 Wet mix macadam shall be prepared in a continuous MIXING PLANT OF 90 TO 120 TPH CAPACITY having provision for controlled addition of water and forced positive mixing arrangement like pug mill or pan type mix of concrete batching plant. Optimum moisture for mixing shall be determined in accordance with IS: 2720 (Part 8) after replacing the aggregate fraction retained on 22.4 mm sieve with material of 4.75 mm to 22.4 mm size. While adding water, due allowance shall be made for evaporation losses. However, at the time of compaction, water in the wet mix should not vary from the optimum value by more than limits as approved by the Engineer-in-Charge. The mixed material should be uniformly wet and no segregation should be permitted.

26.7 **SPREADING OF MIX**

(a) Immediately after mixing, the aggregates shall be spread uniformly and evenly upon the

prepared sub grade/sub-base/base in required quantities. In no case should these be dumped in

heaps directly on the area where these are to be laid nor shall their hauling over a partly completed stretch be permitted.

(b) The mix will be spread by a paver. For portions where mechanical means cannot be used, manual means as approved by the Engineer-in-Charge shall be used.

(c) The paver finisher for WMM shall be self propelled, having the following features:-

(i) Loading hoppers and suitable distribution mechanism.

(ii) The screed shall have tamping and vibrating arrangement for initial compaction the layer as it is spread without rutting or otherwise marring the surface profile.

(iii) The paver shall be equipped with necessary control mechanism so as to ensure that the finished surface is free from surface blemishes.

(iv) The surface of the aggregate shall be carefully checked with templates and all high or low spots remedied by removing or adding aggregate as may be required. The layer may be tested by depth blocks during construction. No segregation of larger and fine particles should be allowed. The aggregates as spread should be of uniform gradation with no pockets of fine materials.

26.8 **COMPACTION**

(a) After the mix has been laid to the required thickness, grade and cross fall/camber the same shall be uniformly compacted, to the full depth with suitable roller.

(b) In portions having unidirectional cross fall /super elevation, rolling shall commence from the lower edge and progress gradually towards the upper edge. Thereafter, roller should progress parallel to the centre line of the WMM, uniformly overlapping each of the preceding track by at least one third width until the entire surface has been rolled. Alternate trips of the roller shall be terminated in stops at least one metre away from any preceding stop.

(c) In portions in camber, rolling should begin at the edge with the roller running forward and backward until the edges have been firmly compacted. The roller shall then progress gradually towards the centre parallel to the centre line of the WMM uniformly overlapping each of the preceding track by at least one third width until the entire surface has been rolled.

(d) Any displacement occurring as a result of reversing of the direction of a roller or from any other cause shall be corrected at once as specified and/or removed and made good.

(e) Along forms, kerbs, walls or other places not accessible to the roller, the mixture shall be thoroughly compacted with mechanical tempers or a plate compactor. Skin patching of an area without scarifying the surface to permit proper bonding of the added material shall not be permitted.

(f) Rolling should not be done when the sub grade is soft or yielding or when it causes a wave like motion in the sub base/base course or sub grade. If irregularities develop during rolling which exceed 12 mm when tested with a 3 metre straight edge, the surface should be loosened and premixed material added or removed as required before rolling again so as to achieve a uniform surface conforming to the desired grade and cross fall. In no case should the use of unmixed material be permitted to make up the depressions.

(g) Rolling shall be continued till the density achieved is at least 98 percent of the maximum dry density or the material as determined by the method outlined in IS-220 (Part 8).

(h) Longitudinal joints and edges shall be constructed conforming to IRC: 109-1997

(j) After completion, the surface of any finished layer shall be well closed, free from movement under compaction equipment or any compaction planes sags, cracks and loose material. All loose, segregated or otherwise defective area shall be made good to the full thickness of the layer and re-compacted.

26.9 **SETTING AND DRYING**

26.9.1 After final compaction of wet mix macadam course, the surface shall be allowed to dry for 24 hours. Preferably no vehicular traffic of any kind should be allowed on the finished wet mix macadam surface till it has dried and the wearing course is laid.

26.10 **RECTIFICATION OF SURFACE IRREGULARITY**

26.10.1 Where the surface irregularity of the wet mix macadam course exceeds the permissible tolerances as laid down in IRC or where the course is otherwise defective due to sub grade soil getting mixed with the aggregate, the full thickness of the layer shall be scarified over the affected area, reshaped with added premixed material or removed and replaced with fresh premixed material as applicable and recomputed. The area treated in the aforesaid manner shall not be less than 5 m long and 2 m wide. In no case shall depressions be filled up with unmixed and ungraded material or fines equipment for determining the maximum dry density as per IS shall be provided by the contractor and cost of the same and conducting the test shall deemed to be included in the quoted rates.

27. **DRY LEAN CONCRETE**

27.1 The work shall consist of construction of dry lean concrete sub-base for cement concrete Pavement (PQC) in accordance with the requirements of these specifications and in conformity with the lines, grades and cross-sections shown on the drawings or as directed by the GE. The work shall include furnishing of all plant and equipment, materials and labour and performing all operations, in connection with the work, as approved by the GE.

27.2 **SOURCE OF MATERIALS:**

37.2.1 The contractor shall submit the sample of materials indicating source to the GE with relevant test data to be used in the lean concrete work sufficiently in advance and the approval of the GE for the same shall be obtained at least 15 days before the scheduled commencement of work of laying dry lean concrete. If the contractor later proposes to obtain the materials from a different source, he shall notify the GE for his approval at least 15 days before such materials are to be used. In no case materials shall be used at site without written approval of GE.

27.3 **CEMENT**

37.3.1 Cement shall be procured by the contractor as specified here-in-before. In case the sub grade is found to consist of soluble sulphates in a concentration more than 0.5 percent, cement used shall be sulphate resistant and shall conform to IS 6909.

27.4 **AGGREGATES**

Aggregates for lean concrete shall be natural material comply with IS: 383. The aggregates shall not be alkali reactive. The limits of deleterious materials shall not exceed the requirements set out in IS:

383. In case the Engineer-in-Charge considers that the aggregates are not free from dirt, the same may be washed and drained for at least 72 hours before batching as directed.

27.4 **COARSE AGGREGATE:**

Coarse aggregate shall consist of clean, hard, strong, dense, non-porous and durable pieces of crushed stone or crushed gravel and shall be avoid of pieces of disintegrated stone, soft, flaky, elongated, very angular or splintery pieces. The maximum size of the coarse aggregate shall be 26.5mm. Aggregate which has water absorption of more than 2% shall be tested for soundness in accordance with IS-23186 (Part-5). After 5 cycles of testing, the loss shall not be more than 12 percent if sodium sulphate solution is used and not more than 8 percent if magnesium sulphate solution is used. Los Angeles Abrasion value shall not exceed 35 percent. The combined flakiness and elongation index of the aggregate shall not be more than 35 percent.

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27.5 **FINE AGGREGATE:**

27.5.1 The fine aggregate shall consist of clean, natural sand and shall conform to IS: 383. Fine aggregate shall be free from soft particles, clay, shale, loam, cemented particles, mica, organic and other foreign matter. Aggregates which have water absorption of more than 3 % shall not be used. Deleterious substance should not exceed the requirements set out in table 600-2 of MORTH specifications (Fifth Revision). The gradation of DLC will conform to Table 600-1 of MORTH-Specifications for Road and Bridge Works(Fifth Edition).

27.6 **HANDLING OF AGGREGATES**

(i) Stockpiles shall be made immediately on receipt of aggregates. Aggregates shall be stacked separately according to the nominal sizes of the single sized coarse aggregates. For fine aggregates separate stocks should be made for different grading zones, if required.

(ii) Aggregates shall be stacked on a PCC stacking platform so as to exclude the possibility of soil or grass being mixed up. When stacks are in close proximity the stockpiles shall be separated by the baulk heads to prevent the aggregates from mixing together. Aggregated shall not be allowed to stack on ground.

(iii) Before batching, the aggregates shall have been stockpiled for at least 24 hours to allow for draining of water, if any. Contractor shall make adequate provisions for stock piling aggregates to an extent sufficient to meet the needs of the work taking into account the availability of supplies and rate of delivery etc and shall include cost element in quoted rates for necessary double handling and carting / transportation of materials from stock piles to the mixing plant etc.

27.7 **WATER:**

27.7.1 Water shall conform to the requirement stipulated in IS 456:2000 and as per Clause 4.9 of MES Schedule (Part-I) 2009. GE to ensure by testing suitability of water supplied by the contractor for use in DLC regularly so as to ensure water is free from oil, acid, vegetable matters, dissolved

salts or other substances harmful for casting & curing of concrete & meets requirements stipulated in IS 456-2000. Test details shall be held on record by the GE. If warranted & directed by the GE contractor has to use filtered water for concreting work by installing water filtration plant without any extra cost.

27.8 **STORAGE OF MATERIALS:**

27.8.1 All efforts must be made to store the materials in proper places so as to prevent their deterioration or contamination by foreign matter and to ensure their satisfactory quality and fitness for use in the work. The storage place must also permit easy inspection, removal and storage of materials. All such materials even though stored in approved go downs must be subjected to acceptance test immediately prior to their use.

27.9 **PROPORTIONING OF MATERIALS FOR THE MIX**

27.9.1 The mix shall be proportioned with a maximum aggregate cement ratio of 15:1. The water contents shall be adjusted to the optimum as specified here-in-after for facilitating compaction by rolling. The strength and density requirements of concrete shall be determined in accordance with specifications mentioned herein after by making trial mixes.

27.10 **MOISTURE CONTENT:**

27.10.1 The right amount of water for the lean concrete shall be decided so as to ensure full compaction under rolling and shall be assessed at the time of rolling the trial length. Too much water will cause the lean concrete to be heaving up before the wheels and picked up on the wheels of the roller and too little will lead to inadequate compaction, a low in situ strength and an open textured surface. The optimum water content shall be determined and demonstrated by rolling during trial length construction and the optimum moisture content and degree of compaction shall be got approved from the GE. While laying in the main work, the lean concrete

shall have a moisture content between the optimum and optimum + 2 percent, keeping in view the effectiveness of compaction achieved and to compensate for evaporation losses.

27.11 **CEMENT CONTENT**

27.11.1 The minimum cement content in the lean concrete shall not be less than 250 kg/Cum of concrete. If this minimum cement content is not sufficient to produce concrete of the specified strength, it shall be increased as necessary without additional cost/payment to the contractor.

27.12 **DESIGN MIX AND CONCRETE STRENGTH**

27.12.1 The design mix complying with the above clauses shall be got approved from the CWE through GE and demonstrated in the trial length construction. The DLC (M-15 grade design Mix) shall be tested all as specified in IS: 456(2000). Method of Testing and frequency of tests of DLC

shall be as per MoRTH clause 903 and acceptance of trial bay and regular work as per clauses contained in Section 600 of MoRTH-Specifications for road and bridge works (Fifth Edition).

27.13 **CONSTRUCTION**

27.13.1 **GENERAL**: The pace and programme of the lean concrete sub-base construction shall be matching suitably with the programme of construction of cement concrete over it. The sub-base shall be overlaid with cement concrete pavement only after 7 days after sub-base construction.

27.13.2 **BATCHING AND MIXING:** The batching plant shall be capable of proportioning the materials by weight each type of material being weighed separately. The cement from bulk stock shall be weighed separate from the aggregates. The capacity of batching and mixing plant shall be at least 25 percent higher than the proposed capacity for laying arrangements. The batching and mixing shall be carried out preferably in a forced action CENTRAL BATCHING AND MIXING PLANT having necessary automatic controls to ensure accurate proportioning and mixing. Other types of mixers shall be permitted subject to demonstration of their satisfactory performance during the trial length. The type and capacity of the plant shall be approved by GE before commencement of the trial length. The weighing balances shall be calibrated by weighing the aggregates, cement, water and admixtures physically either by weighing with large weighing machine or in a weigh bridge. The accuracy of weighing scales of the batching plant shall be within +2 percent in the case of aggregates and + 1 percent in the case of cement and water. The design features of batching plant should be such that the shifting operations of the plant will not take very long time when they are to be shifted from place to place with the progress of the work.

27.13.3 **TRANSPORTING.** Plant mixed lean concrete shall be discharged immediately from the mixer, transported directly to the point where it is to be laid and protected from the weather by covering the tippers/dumpers with tarpaulin during transit. The concrete shall be transported by tipping trucks, sufficient in number to ensure a continuous supply of material to feed the laying equipment to work at a uniform speed and in an uninterrupted manner. The lead of the batching plant to paving site shall be such that the travel time available from mixing to paving as specified.

27.13.4 **PLACING:** Dry lean concrete shall be laid/ placed with mechanical paver. The equipment shall be suitable of laying the material in one layer in an even manner without segregation, so that after compaction the total thickness is as specified. The paving machine shall have high amplitude tamping bars to give good initial compaction to the sub-base.

27.13.5 **COMPACTION**

(a) The compaction shall be carried out immediately after the material is laid and levelled. In order to ensure thorough compaction which is essential, rolling shall be continued on the full width till there is no further visible movement under the roller and the surface is closed. The minimum DRY DENSITY OBTAINED SHALL BE 97 PERCENT of that achieved during the trial length construction as specified here-in- after. The densities achieved at the edges i.e 0.5m from the edge shall not be less than 95 percent of that achieved during the trial construction. The spreading, compacting and finishing of the lean concrete shall be carried out as rapidly as possible and the operation shall be so arranged as to ensure that the time between the mixing of

the first batch of concrete in any transverse section of the layer and the final finishing of the same shall not exceed 90 minutes when the concrete temperature is above 25 and below 30 degree Celsius and 120 minutes if less than 25 degree Celsius. This period may be reviewed by the Engineer-in-Charge in the light of the results of the trial run but in no case shall it exceed 2 hours. Work shall not process when the temperature of concrete exceeds 30 degree Celsius. If necessary, chilled water or addition of ice may be resorted to for bringing down the temperature. It is desirable to stop concreting when the ambient temperature is above 35 degree Celsius. After compaction has been completed, roller shall not stand on the compacted surface for the duration of the curing period except during commencement of next day's work near the location where work was terminated the previous day. Double drum smooth wheeled vibratory roller of minimum 80 to 100 KN static weight is considered to be suitable for rolling dry lean concrete. In case any other roller is proposed, the same shall be got approved from the GE, after demonstrating its performance. The number of passes required to obtain maximum compaction depends on the thickness of the lean concrete, the compatibility of the mix, and the weight and type of roller etc, and the same as well as the total requirement of rollers for the job shall be determined during trial run by measuring the in situ density and the scale of the work to be undertaken.

(b) In addition to the number of passes required for compaction there shall be a preliminary pass without vibration to bed the lean concrete down and again a final pass without vibration to remove roller marks and to smoothen the surface. Special care and attention shall be exercised during compacting near joints, kerbs, channels, side forms and around gullies and manholes. In case adequate compaction is not achieved by the roller at these points, use of plate vibrator shall be made, if so directed by the Engineer-in-Charge.

(c) The final lean concrete surface on completion of compaction and immediately before overlaying shall be well closed, free from movement under roller and free from ridges, low spots, cracks, loose material, pot holes, ruts and other defects. The final surface shall be inspected immediately on completion and all loose, segregated or defective areas shall be corrected by using fresh lean concrete material laid and compacted as per specification. For repairing honeycombed surfaces, concrete with aggregates of size 10 mm and below shall be spread and compacted. It is necessary to check the level of the rolled surface for compliance. Any level/thickness deficiency should be corrected after applying concrete with aggregates of size 10 mm and below after roughening the surface. Similarily the surface regularity also should be checked with 3 m straight edge. The deficiency should be made up with concrete with aggregates of size 10 mm and below. Segregation of concrete in the dumpers shall be controlled by premixing each fraction of the aggregates, by moving the dumper back and forth while discharging the mix on it and other means. Even paving operation shall be such that the mix does not segregate.

27.13.6 **CURING:** As soon as the lean concrete is set, curing shall commence. Curing shall be done by covering the surface by gunny bags/Hessian cloth, which shall be kept moist for 7 days by sprinkling water.

27.13.7 **TRIAL MIXES**

(a) The mix shall be proportioned with a maximum aggregate cementatious material ratio of 15:1. The contractor shall make trial mixes of dry lean concrete with moisture contents like 5.0, 5.5, 6.0, 6.5 and 7.0 per cent using cement content specified and the specified aggregate grading but without violating the requirement of aggregate cement ratio specified in clause 5.31 here-in-before. Optimum moisture and density shall be established by preparing cubes with varying moisture constants. Compaction of the mix shall be done in three layers with vibratory hammer fitted with a square or rectangular foot. After establishing the optimum moisture, a set of six cubes shall be cast at that moisture for determination of compressive strength on the 3rd and the 7th day. Trial mixes shall be repeated if the strength is not satisfactory either by increasing cement content or using higher grade of cement. After the mix design is approved, the Contractor shall construct a trial section in accordance with clause 18.4.8 here-in-after.If during the construction of the trial length, the optimum moisture content determined as above is found to be unsatisfactory, the contractor may make suitable changes in the moisture contents to achieve a satisfactory mix. The cube specimens prepared with the changed moisture content should satisfy the strength requirement. Before production of the mix, natural moisture content of the aggregate should be

determined on a day-to-day basis so that the moisture content could be adjusted. The mix finally designed should neither stick to the rollers nor become too dry resulting in ravelling of surface.

27.13.8 **TRIAL LENGTH OF DLC**

(a) The trial length will be constructed as per clause 601.8 of "MORTH Specifications of Roads & Bridge works (Fifth Edition)" and references thereof. In order to determine and demonstrate the optimum moisture content which results in the maximum dry density of the mix compacted by the rolling equipment and the minimum cement content that is necessary to achieve the required strength, trial mixes shall be prepared as per clause 18.4.7 here-in-before. After construction of trial length, the in-situ density of the freshly laid materials shall be determined by sand replacement method with 20 cm dia density cone. Three density holes shall be made at location equally spaced along a diagonal that bisects the trial length; average of these densities shall be determined. These main density holes shall not be made in strip 50 cm from the edges. The average density obtained from the three samples collected shall be the reference density and is considered as 100 percent. The field density of regular work will be compared with these reference densities in accordance with clause 18.4.5.1. A few cores may be cut as per the instruction of the GE to check segregation or any deficiency.

(b) The hardened concrete shall be cut over 3 m width and reversed to inspect the bottom surface for any segregation taking place. The trial length shall be constructed after making necessary changes in the gradation of the mix to eliminate segregation of mix. The lower surface shall not have any honey combing and the aggregates shall not be held loosely at the edges. The trial length shall be outside the main works. The main work shall not start until the trial length has been approved by the GE. After approval has been given, the materials, mix proportion, moisture content, mixing, laying, compaction plant and construction procedure shall not be changed without the approval of GE. Cost of trial length shall be burned by the contractor.

27.13.9 **TRAFFIC**

(a) No heavy commercial vehicles like trucks and buses shall be permitted on the lean concrete sub-base after it's construction. Light vehicles if unavoidable may, however be allowed after 7 days of its construction with prior approval of GE. The unit rate quoted by the contractor for dry lean concrete sub-base shall include carrying out the required operation including full compensation for all labour, materials and equipment, mixing, transport, placing, compacting, finishing, curing, testing and incidentals to complete the work as per specifications, all royalties, fees, storage and rents where necessary and all leads and lifts.

(b) DLC layer will be laid / constructed in a manner that it is at least 600mm wider than the proposed width of PQC on both sides. This is provided so that PQC paver moves on dry lean concrete. Cost of extra width is deemed to be included in quoted rate. Net area excluding extra width will be measured and paid and nothing extra admissible on this account.

28. **PAVEMENT QUALITY CONCRETE (PQC)**

28.1 Pavement quality concrete shall be design mix concrete and shall have flexural strength of 45 Kg/cm2 (to be achieved in field), as per IS 456 -2000 and design mix with approved admixtures (if required necessary as per design mix without any price adjustment) conforming to IS-9103 shall be permitted to be used. The total amount of chloride content in the admixture mixed concrete shall satisfy the requirement of IS 456-2000. The design criteria for pavement quality concrete will be flexural strength of 45 Kg/Sqcm to be achieved at site.

28.2 The concrete mix design with or without admixture shall be carried out by the contractor through one of the following laboratories test houses which will be approved by CWE through GE.

(a) Any IIT

(b) SEMT Pune

(c) CRRI New Delhi

(d) Any reputed NIT/Govt Engineering college approved by CWE.

PARTICULAR SPECIFICATION (CONT'D…../-)

28.2 In case the laboratories are unable to carry out the requisite design / testing, the contractor shall have to get the same done from any other reputed recognized laboratory with prior approval of CWE through GE. The various ingredients for mix design, laboratory tests shall be sent to the lab/test houses through the Engineer-in-Charge and the samples of such ingredients sent shall be preserved at site till completion of work or change in Design Mix whichever is earlier, The samples shall be taken from the approved materials, which are proposed to be used in the work. The contractor shall submit the mix design report from approved laboratory for approval of CWE through GE within 30 days from the date of commencement of work as mentioned in work order No. 1 based on 07 days strength.

28.3 The contractor shall make cubes and test beams of trial mixes as per approved mix design at site laboratory for all grades of concrete in presence of the Garrison Engineer using same ingredients as adopted for design mix, prior to commencement of concreting and get them tested in presence of Engineer- in-Charge for 7 days and 28 days strength. For each design mix, a set of six cubes and six test beams shall be prepared from each of the three consecutive trial mixes. Three cubes and test beams from each set shall be tested at the age of 7 days and three cubes and test beams at the age of 28 days. The cubes and test beams shall be made, cured, transported and tested strictly in accordance with I S specification. The average strength of nine cubes and test beams at the age of 28 days shall exceed the specified target mean strength for which design mix has been approved. No concreting shall be done until the mix design is approved.

28.4 Tests shall be done at the site laboratory/ Zonal lab.

28.5 For each change of source or quality/characteristic properties of the ingredients during the currency / progress of work, from that approved & used in the concrete mix , a fresh mix design shall be got done by the contractor. Revised trial mix test shall be conducted and shall be submitted by the contractor. In case of any unforeseen eventuality, alternate mix design for different quarry to be got approved in advance and kept ready as reserve to avoid delay.

28.6 The cost of packaging, sealing, transportation, loading, unloading cost of samples and the testing charges for mix design in all cases shall be borne by the contractor.

28.7 The mix design shall be done considering the degree of quality control as "Very Good" (as mentioned in Table 1 of SSR (2009) Part-I) in all cases. Minimum cement content for required Concrete strength is 400 Kg/cum for pavement quality concrete. However in case quantity of cement used is more than 400 Kg/Cum and any admixture (super plasticizer) is recommended to be used as per design mix to achieve specified properties, then contractor shall use additional quantity of cement including admixture if any and nothing extra shall be payable to the contractor on this account.

28.8 **PRODUCTION OF CONCRETE**

28.8.1 The concrete shall be produced in central batching and mixing plant installed at site. The location for installation of batching and mixing plant and storage of aggregate shall be as shown in the drawing. The batching plant shall be fully-automatic type of standard make with printing facility. A fully Automatic batcher shall be charged by devices which, when activated by a single starter switch will automatically start the weighing operation of each material and stop automatically when the designated weight of each material has been reached. A batching plant essentially shall consist of the following components: -

(i) Storage bins for different sizes of aggregates and cement.

(ii) Batching equipment.

(iii) Mixers.

(iv) Control Panels

(v) Mechanical material feeding and elevating arrangements

PARTICULAR SPECIFICATION (CONT'D…../-)

28.8.2 The compartment of storage bins shall be approximately of equal size. The cement compartment shall be centrally located in the batching plant. It shall be water-tight and provided with necessary air vent, aeration fittings for proper flow of cement, emergency cement cut off gate. The aggregate and sand shall be charged by power operated centrally revolving chute. The entire plant from mixer floor upward shall be enclosed and insulated. The batch bins shall be constructed so as to be self cleansing during draw-down. The batch bins shall in general conform to the requirement of IS-4925.

28.8.3 The batching equipment shall be capable of determining and controlling the prescribed amounts of various constituent materials for concrete accurately i.e. water, cement, sand, individual size of coarse aggregates, etc. The mixer in the batching plant shall be so arranged that mixing action in the mixers can be observed from the operator's station. The mixer shall be equipped with a mechanically or electrically operated timing, signalling and metering device that will indicate and assure completion of the required mixing period. The mixer shall have all other components as specified in IS: 4925. The batching, mixing plant and other accessories shall always be available at site and should be removed only on completion of work after approval of Engineer-in-charge.

28.9 **WORK STRENGTH TEST**

28.9.1 **TEST SPECIMEN**

(a) Work strength test shall be conducted in accordance with IS : 516 on random sampling.

Each test shall be conducted on six specimens, three of which shall be tested at 7 days and remaining three at 28 days. If required additional samples shall be prepared as per direction of Garrison Engineer for testing samples cured by accelerated method as described in IS : 9103.

(b) FREQUENCY- The minimum frequency of sampling as per IRC / MORTH shall be followed.

28.9.2 **ACCEPTANCE CRITERIA**

(a) Acceptance of pavement quality concrete: Concrete shall only be accepted if it satisfies the following main conditions:-

(i) LCL of every lot (at least 30 samples) is not less than specified value
(ii) Co efficient of variation is not greater than 10%
(iii) Tolerance level factor is 1.5
(iv) There is no honey combing in the concrete

For determination of LCL at least 3 beams for every slab will be casted and tested for flexural strength as per IS:516 of 1959. After casting of minimum 30 samples have been cast for slab laid in similar conditions their results should be tabulated and LCL determined as follows:-

LCL = X – t.v

Where LCL = Lower control limit (minimum flexural strength)

X = Mean flexural strength from the samples tested t

t = Tolerance level factor

v = Standard deviation of the samples tested

LCL so determined should not be less than specified value i.e. 45 Kg/Sqcm. Along with the beams, cubes will also be cast and tested for compression as IS:516 of 1959. The compressive strength will be tabulated along with the corresponding flexural strength to establish correlation between flexural and compressive strength.

(b) **Critical examination of test data:-**

PARTICULAR SPECIFICATION (CONT'D…../-)

In case LCL of a lot is less, then the following procedure shall be adopted before core tests are undertaken:-

(i) Omit the slab having lowest average strength and revaluate the remaining test data of the samples.

(ii) If the revaluated data conforms to the above acceptance criteria, accept the lot less the slab omitted.

(iii) In case of unsatisfactory result, repeat the process by omitting the next lowest till all weak slabs are segregated for further testing by core cutting and the Part lot gets specified value.

(c) **Core Tests :-**

In case the concrete fails in flexure test i.e the LCL is less than specified for a particular lot, then concrete shall not be rejected unless it also fails in core test. In core test, at least two cores of the same will be cut per slab. The crushing strength of this core is then determined. The crushing strength should not be less than 0.8 times the corresponding crushing strength of 15 cm cubes.

The crushing strength determination will be as per IS: 516 of 1959. In case the H/D ratio of the core is between 1 and 2, then the crushing strength of the cube will be reduced. The correction will be carried out as per the formulae given below:-

f = 0.11n + 0.78

Where f = correction factor

n = H ratio

D

In case these concrete fails the flexure (LCL) test, but is found satisfactory is core test, it shall be accepted as the core test takes the precedence over the flexure test, However, in case the concrete fails both flexure as well as core test then it shall be rejected and replaced by the contractor without any extra cost to the Govt.

28.9.3 **MIX PROPORTION AND STRENGTH:** It shall be all as specified in PS clause No.10.9.9 here in after.

28.9.4 **PAVEMENT UNIFORMITY**

The finished cement concrete pavement surface shall be such that when tested with 3m long straight edge placed anywhere within the same or adjoining slab in any direction on the surface, there is no gap greater than 3 mm between the bottom of the surface of pavement. Surface irregularity exceeding 3mm but not exceeding 6 mm may be rectified by bump cutting or scrabbling or grinding using approved equipment. When required by the GE, areas which have been reduced in level by the above operations shall be retextured in an approved manner either by cutting grooves (5 mm deep) or roughening the surface by hacking the surface. If high areas in excess 6 mm or low areas in excess of 3 mm occur, if the contractor cannot rectify, the slab shall be demolished and reconstructed at the contractor's expense and in no case the area removed shall be less than the full width of the lane in which the irregularities occurs and full length of the slab. If deemed necessary by the GE, any section of the slabs, which deviates from the specified levels and tolerances, panel between construction/dummy/expansion joints shall be demolished and reconstructed at the contractor's expense.

28.9.5 **PRODUCTION OF PAVEMENT QUALITY CONCRETE :** Pavement quality concrete (design mix) shall be produced using fully automatic Central Batching & Mixing Plant installed at site.

29.10 **MATERIAL FOR PAVEMENT QUALITY CONCRETE**

29.10.1 **CEMENT :** Cement required shall be all as mentioned in the Particular Specifications"

29.10.2 **HANDLING OF AGGREGATE**

PARTICULAR SPECIFICATION (CONT'D…../-)

29.10.2.1 Stock piles shall be made immediately on supply of aggregate at the site of work. Aggregate of each nominal size shall be stacked separately. For fine aggregate, separate stacks shall be made.

29.10.2.2 Aggregate shall be stacked on a hard surface so as to exclude the possibility of soil or grass being got mixed up. When stacks are in close proximity, the stock piles shall be separated by bulk heads. Special care shall be taken to clean and wash the bottom layer of aggregate in contact with ground surface, before use. Before batching, the aggregate shall be stocked/ piled for at least 24 hours to allow for drainage of water, if any. Contractor shall make adequate provision for stock piling work taking into account the availability of supplies and rate of delivery, etc and he shall include in his tender for necessary double handling and transportation of materials from stock piles to mixing plant etc.

29.10.3 **GRADING OF AGGREGATE FOR FINAL MIX**

(a) The grading of single size aggregate shall be in accordance with IS-383 of 2016 and shall conform to the following grading:-

Sieve/size	40mm	20mm	16mm	12.5mm	10mm
80 mm	-	-	-	-	--
63 mm	100	-	-	-	-
40 mm	85-100	100	-	-	-
20 mm	0-20	85-100	100	-	-
16mm	-	-	85-100	100	-
12.5 mm	-	-	-	85-100	100
10 mm	0-5	0-20	0-30	0-45	85-100
4.75 mm	-	0-5	0-5	0-10	0-20
2.36mm	-	-	-	-	0-5

29.10.3.1 Aggregate shall be stored in single size in separate bins or piles on PCC stacking platform, which shall be self draining.

29.10.3.2 The fine aggregates shall be natural sand (river sand) and shall conform to IS-383.Only Natural River sand shall be used in this work. The fine aggregates shall have a sand equivalent value of not less than 50 when tested in accordance with the requirement of IS:2720 (Part 37).The grading of fine aggregate shall conform to Zone II of table IV of IS-383-2016 which is as shown below : -

IS SIEVE	**% PASSING BY WEIGHT**
10mm	100
4.75mm	90-100
2.36mm	75-100
1.18mm	55-90
600 Micron	35-59
300 Micron	8-30
150 Micron	0-10

29.10.3.3 The grading of coarse and fine aggregate to be used in the work shall be determined. The coarse aggregate and fine aggregate shall be mixed in suitable proportion so that the grading of the mix shall be in the range indicated in the following table for each size of IS Sieve.

IS SIEVE	**% PASSING BY WEIGHT**
31.5mm	100
26.5mm	85-95
19mm	68-88
9.5mm	45-65
4.75mm	30-55

PARTICULAR SPECIFICATION (CONT'D…../-)

600 Micron	8-30
150 Micron	5-15
75 Micron	0-5

29.10.3.4 The sieve analysis test of the mixed aggregate shall be carried out at random and results recorded to ensure that the grading lies within the specified percentage for each size as given above. The proportioning of coarse and fine aggregate require adjustment in case it does not satisfy this grading for any subsequent change in grading the contractor shall have no claim whatsoever on this account.

29.10.3.5 **WATER**

(a) Water shall conform to the requirement stipulated in IS 456:2000 and as per Clause 4.9 of MES Schedule (Part-I) 2009. GE to ensure by testing suitability of water supplied by the contractor for use in PQC regularly so as to ensure water is free from oil, acid, vegetable matters, dissolved salts or other substances harmful for casting & curing of concrete & meets requirements stipulated in IS 456-2000 . Test details shall be held on record by the GE. If warranted & directed by the GE contractor has to use filtered water for concreting work by installing water filtration plant without any extra cost.

29.11 **MIX PROPORTION AND STRENGTH**

29.11.1 The strength of pavement quality concrete shall be as specified here in after:

(a)	The strength to be achieved	Not less than 45 kg per Sq cm flexural strength for PQC
(b)	Aggregate/cement ratio by weight	As per Mix Design
(c)	Water/Cement ratio	As per Mix Design
(d)	Slump for concrete	Not more than 25mm + 10mm
(e)	Compacting factor test	0.78 + 0.02 for PQC
(f)	Degree of quality control	Very Good (PQC)
(g)	Exposure	Moderate
(h)	Maximum size of aggregate	31.5mm
(i)	Type of Aggregate	Crushed stone aggregate
(j)	Type of cement	As per clause here-in-before

Note: The design mix shall be carried out without adding synthetic fibres.

29.12 **TRIAL MIX**

29.12.1 At the time of tendering, the contractor after taking into account the type of aggregate, plant and method of laying concrete he intends to use/adopt, shall allow in his tender for aggregate/and water cement ratio which he considers will achieve the strength required specified.

29.12.2 As soon as possible after receiving the order to commence work, the contractor shall make trial mixes to satisfy the Engineer-in-Charge that the strength specified is obtained.

29.12.3 From approved design mix, nine numbers of preliminary test beams and 9 cubes shall be made and tested in accordance with IS-516 and IS-1199, for the flexural strength and compressive strength. For compaction in moulds for beams, the following procedure shall be adopted. The test specimens shall be made as soon as practicable after mixing and in such a way as to produce full compaction of the concrete with neither segregations nor excessive laitance. The concrete shall be filled in the mould in layers of 5 cm. While filling the scoop shall be moved around the top edge of the mould to distribute the concrete uniformly. Each layer of concrete shall be compacted fully by the use of the tapping bar/vibrator all as per IS-516. The number of strokes per layer required to produce the specified condition will vary according to the type of concrete, but in no case shall the concrete be subjected to less than 15 strokes per layer for 10 cm specimens. The strokes shall penetrate into the underlying layer and the bottom layer shall be rodded throughout its depth. Three of these beams shall be tested at seven days (i.e., on the 8th day) from the date of casting and another set of three beams at 28 days (i.e., 29th day) from the date of casting. The remaining three beams shall be preserved by the Garrison Engineer

for one year after certified date of completion of the work for any subsequent check. The test after 7 days (ie on the 8th day) is intended only to give early indication of the strength likely to be achieved. The strength achieved should be 70 to 75% of the 28 days strength specified hereinbefore. In case the strength achieved is less than 75% the mix shall be suitably re-designed.

29.12.4 The acceptance criteria for the design mix for RCC works shall be all as per IS-456: 2000 and as approved by the GE. Acceptance criteria for PQC shall be as per clause 10.9.2 of PS.

29.12.5 Whenever there is any change in the types of grading or material or change in quality of cement or method of laying etc, the tests as enumerated above shall again be conducted for ascertaining the suitable design mix for obtaining the specified flexural strength.

29.12.6 On the results of the above tests, the mix actually to be used shall be agreed to and approved by the Garrison Engineer. The approval of Garrison Engineer shall not relieve the contractor of his responsibility for obtaining the required minimum strength of the quality concrete in the works.

29.12.7 All tests recorded in support of mix design shall be maintained as Schedule 'A' of the records under the contract.

30. NON CIRCULAR SYNTHETIC FIBRE

30.1 Synthetic fibres with cut length 12mm+1 mm, tensile strength>48 Mpa, melting point>2400C, specific gravity 1.34-1.39 and effective diameter 20-40 micron shall be used which will be independently tested from Govt lab/SEMT/NITRA/IIT or Sri Ram Lab or any other lab specially approved by the Accepting Officer in writing.

30.1.1 **SCOPE:**The specifications mentioned below mainly prescribe , the physical , application functional requirements of synthetic polyester micro fibres for use in cement based matrix as secondary reinforcement in pavement construction works , especially for various air field facilities i.e. runways , dispersals , taxi tracks and hangar floors etc.

30.1.2 **Purpose:** The fibres selected for the purpose must be able to reduce concrete cracking due to plastic shrinkage, reduce permeability , enhance impact and abrasion resistance and resist impact of alkalis , chemicals , chlorides on concrete .

30.1.3 IS 16481 : 2016 gives out standard requirements for the synthetic fibres for use in concrete as secondary reinforcement for varied construction works such as roads, pavements, industrial floorings, residential and commercial buildings etc. However, use of synthetic microfibers in pavement quality concrete, used for construction of defence airfield pavement wherein fighter aircrafts (which generate very high temperatures while landing and take - off) use these pavements, needs special attention to ascertain specific requirements . Hence, the purpose of the Particular specifications is to give out specific requirements to be followed by executives for use of synthetic microfibers in pavement quality concrete (PQC) for construction of defence pavement facilities .

30.1.4 Requirements:- Following are the main requirements for synthetic fibres to be used as secondary reinforcement in PQC:-

(a) Physical Fibre Forms. Only virgin, synthetic, monofilament polyester fibres of cut length 12 mm to 18 mm, inert to concrete environment shall be used as secondary reinforcement in cement based matrix and no recycled material shall be permitted .

(b) Dispersal. Fibres shall be well dispersed in the cement based matrix so as to have uniform functional properties. Fibres shall be identified as per the confirmatory tests specified in IS 667.

(c) Type and Doses. Most synthetic fibre applications are at the low doses 0.06 percent to 0.3 percent by volume depending upon the type of fibre and as specified by the manufacturer.

(d) Certification by Manufacturer. The executives shall obtain a certificate from the fibre manufacturer along with a test report including that the fibre used is virgin fibre for every lot of fibre purchased. The iso - phthalic acid (IPA) content for polyester virgin fibre shall be zero when tested as per method prescribed in Annexure C of IS16481 : 2016.

PARTICULAR SPECIFICATION (CONT'D…../-)

(e) Polymer Content. Polymer content of the virgin polymer shall be minimum 97 percent and other additives such as oxidation and UV stabilizers shall not be more than 3 percent. This shall be validated by a certificate from the polymer supplier for each lot.
(f) Resistance to Alkalis . Synthetic fibres shall retain at least 90 percent of their original breaking strength when tested by the method prescribed in Annexure D of IS16481 : 2016.
(g) Resistance to Acids . Synthetic fibres shall retain at least 90 percent of their original breaking strength when tested by the method prescribed in Annexure ' E ' of IS16481 : 2016.
(h) Resistance to Ageing . The test specimen of synthetic fibres in the form of a sheet (refer G - 3 of Annex ' G ' in IS16481 : 2016) when subject to ageing at 70 ± 2 ° C for 168 hours by the procedure described in IS : 7016 (Part 8) shall retain at least 90 percent of their original tensile strength.
(j) Resistance to Ultra- Violet Light . The synthetic fibres when tested for resistance to ultra - violet light as specified in Annexure ' F ' of IS16481 : 2016, shall not have tensile strength less than 90 percent of the original value.
(k) Water Absorptive Capacity. The water absorption capacity of synthetic fiber when tested by prescribed in Annexure G of IS 16481:2016 shall be less than 1 percent.
(l) Physical and Application Requirements. The synthetic fibre used as secondary reinforcement in the PQC shall comply with the requirements as given in Tables 1 and 2 below :-

Table 1 Physical Requirements

Sr No	Characteristic	Desired parameters	Method of Test (Reference)
(i)	Fiber type	Polyester	
(ii)	Cut length	12-18 mm	IS: 10014 (Part 1)
(iii)	Physical Fibre Forms	Only virgin synthetic monofilament fibres	IS: 667
(iv)	Cross section	Noncircular	IS: 667
(v)	Minimum tensile strength (Mpa)	300	IS: 235
(vi)	Specific gravity	1.34 to1.39	Annexure 'H' of IS 16481:2016
(vii)	Minimum Secant modulus (10^3 Mpa)	6	IS: 235
(viii)	Minimum Ultimate elongation, percent	15	IS: 235

Table 2 Application Requirements

Sr No	Characteristic	Application for concrete	Method of Test (Reference)
(i)	Equivalent fibre diameter,(Micron)	15-40	Annexure 'L' of IS16481:2016
(ii)	Length, (mm)	12 to 18	IS: 10014 (Part 1)
(iii)	Aspect ratio	450 to 800	Annexure 'M' of IS16481:2016

(m) Functional Requirement To be as specified in table 3 at para 4 of IS 16481: 2016.

30.1.5 Packing and Marking To be as per guidelines specified at para 5 of IS 16481: 2016.

30.1.6 Sampling. To be as per guidelines specified at Para 6 of 1816481 : 2016. However the fibres from the bags or pouches selected from the lot shall be tested for various requirements as specified in Para 4 above .

30.1.7 No of Test Specimen and Criteria for Conformity. To be as per guidelines specified at Para 7 of IS16481: 2016 .

30.1.8 For any clarification on matters not explained in this policy, IS 16481: 2016 be referred.
30.1.9 Executive and supervisory staff will be fully responsible for ensuring use of correct synthetic polyester fiber meeting the laid down specifications, Proper check and record on the quality of the product supplied shall be maintained by the executives . Apart from manufacture's

PARTICULAR SPECIFICATION (CONT'D…../-)

certificate / test results , the engineer in - charge must get the material tested through an NABL approved laboratory as per the existing policies .

30.2 **TESTING OF NON CIRCULAR SYNTHETIC FIBRE**

30.2.1 The contractor shall submit manufacturer's test certificate lot wise in original for verification of GE/Engineer- in-Charge. Copies of defaced Test Certificates shall be kept by GE on record

30.3 Besides manufacturer's Test Certificate at least 3 sample from each lot (brought at site) selected randomly will be got tested independently by the GE from Govt lab/SEMT/NITRA/IIT or any other lab approved by the CWE in writing.

30.4 **FORM WORK:** PQC shall be laid using slip form paver as describe in list of plants & machinery.

30.5 **PLACING AND COMPACTION OF CONCRETE**

30.5.1 Concrete shall be transported without delay and incorporated in work as per MES SSR Part I clause 20.B.7.8 to 20.B.8.2 at the position of laying within 20 minutes from the time of discharge from the transit mixer. Concrete shall be deposited and spread to such depths that when compacted and finished, it shall conform to the grade and cross section required and to ensure the thickness required. Contractor is free to use dumper at his discretion.

60.5.2 Concrete shall be deposited in such a manner as to require as little handling as possible. Spreading, compacting and finishing shall be completed within a period not exceeding one hour from the time the mixing starts. In case of dry and hot weather, this time shall not exceed 45 minutes. If the air temperature exceeds 40 degrees Centigrade or 35 degrees Centigrade with relative humidity less than 25% and or wind velocity higher than 10 KM per hour, concreting precautions as required vide IRC - 61 or 76 shall be taken.

30.5.3 Concrete shall be spread paved, compacted and finished by approved mechanical plant. The contractor shall only use slip form paver to commensurate with overall period of completion of the work. Compaction of top layer shall be carried out mechanically by means of approved mechanical paver in accordance with the instructions of the manufacturer of paver. Needle vibrators shall be used all over the area for obtaining initial compaction of concrete. These should be of diameter not less than 40mm. If the vibrators are pneumatic, the pressure shall not be below 4 kg/Sq cm. If electrically operated, they shall have a minimum frequency of 3,500 to 4,500 vibrations per minute and amplitude under no load state (operation in air) measured in accordance with IS-2506, shall not be less than 1.5 mm and forward speed travel 60 cm per minute. The initial compaction shall be carried out by manual labour employing needle vibrator all as specified here in after.

30.5.4 There shall be at least three needle vibrators working for each area under concreting. Vibrating screed consisting of a steel section or timber section weighing not less than 15 Kg/m with a tapering edge of not less than 7cm width and having a vibrator mounted there may be used in addition to needle vibrators to obtain full compaction. The face of the timber tapering edge of the screed shall be lined with a mild steel plate rigidly fixed by means of counter sunk screws. The final compaction, final finish and levels shall however be obtained by rollers and vibrators mounted on concrete paver which will follow and run over initially compacted concrete mass.

30.5.5 Mounted vibrators, on mechanical paver and vibrating screed paver shall be of a frequency of not less than 3600 impulses per minute and an amplitude not less than 0.4 mm and be of sufficient capacity to agitate the concrete for a width of 25 cm on either side of its position. Where screed vibrators are used for compaction, a stand by unit shall always be kept ready for use, should the other one go out of order. At the discretion of the Engineer-in-Charge for compaction of edges and joint, vibrators may be supplemented by hand tamping and roding for securing satisfactory results. In no circumstances honey combing of concrete at joints or elsewhere shall be permitted.

30.5.6 Concreting shall be carried out in one operation between the expansion joints and construction joints without any break at the dummy joints.

60.5.7 Workmen shall not be allowed to walk on freshly laid concrete.

30.5.8 Minimum of one day shall be allowed to harden the poured concrete slab before adjacent slabs are laid. Otherwise, the vibration caused by the equipment used for the concreting on unhardened concrete may disturb the edges.

30.6 **JOINTS IN CONCRETE**

30.6.1 Joints in concrete pavements shall be provided as per IRC-15(2001) read in conjunction with IRC-57(2006). Joints shall be of the relevant type and of dimensions (width/depth) and spacing mentioned in the relevant BoQ items and shall be located in all cases as indicated in the drawing or as directed. Sealing compound shall be as mentioned in Schedule 'A' & specified in clause 11.8 here-in-after. Being specialized work the priming, filling the sealing compound to be carried out only through the original applicator of the manufacturer. Relevant details/letter & correspondence to that effect shall be held on record by the GE. All joints shall be sealed after minimum 28 days of placing of the concrete in the slabs.

30.6.2 **CUTTING JOINT:**

(a) Normally in summer when ambient temperature is more than 30°C initial cutting may be carried out after 4-8 hours of laying and in winter when ambient temperature is less than 30° C, initial cut may be done after 8-12 hours of laying. Subsequent widening of joint groove will be done after 14-16 days of casting concrete pavement. First cut in above time frame should be by using 4 mm circular diamond blade up to depth as shown on drawing across the width (For Dummy Joint).

30.6.3 **TRANSVERSE CONTRACTION JOINTS OR DUMMY JOINTS**

(a) These are the dummy groove type and the width of dummy joint shall be described in Schedule 'A' and shall be extended vertically from the surface of slab to a depth as detailed in Schedule 'A' /drawings. The joint shall be formed by sawing the concrete with diamond saw joint cutting machine of the approved design within 4-8 hours of laying when the ambient temperature is more than 30°C as in summer and within 8-12 hours of laying in winter when ambient temperature is less than 30°C and when concrete has sufficiently hardened. In all cases except where cutting is done with saw, the joint edge shall be bull nosed. Care shall be taken that the edges of the joints are not damaged. The edge will not stand proud of the concrete slab.

30.6.4 **CONSTRUCTION JOINTS**

(a) The construction joints shall be straight and vertical through the full thickness of the slab. The vertical edge of the concrete on the side of the joint shall be treated with a coat of epoxy tar before the adjacent bay is concreted. A groove of size as mentioned in Sch 'A' shall be cut at the top surface of the joint to receive the sealing compound. This groove will be formed in the same manner as that for a dummy joint by using diamond saw cutting machine.

(b) The construction operation shall be so planned that the location of the construction joint coincides with either a expansion joint or a dummy joint. Construction joints shall be provided at locations where concreting has to be suspended due to unforeseen reasons. If the joint is at the location of expansion joint, regular expansion joint shall suffice.

30.6.5 **EXPANSION JOINT**

(a) Expansion joint shall be provided both longitudinally and transverse direction at spacing as directed by Engineer-in-Charge.

(b) The transverse expansion joint shall be continuous from edge to edge of the pavements constructed at the same or different times.

(c) The expansion joint shall be straight and extend through the full thickness of the slab and shall be of the shape and dimension as shown on the drawings and specified. The slab edges

adjacent to the joint shall be formed truly vertical. The joints shall be filled as described in Sch 'A'. The groove to receive the sealing compound may be formed by keeping the depth of filler board 25 mm more than the thickness of slab at the time of casting of adjacent slab and tearing off the excess depth (including 25 mm protruding out portion and groove required) before joint filling. A cut in filler board should be made at appropriate height just before placing the same between two slabs to facilitate easy and accurate tearing off the same as described above. Care should be taken that the edges of the grooves are not damaged and that no bridging or plugging of the joint with concrete occurs, also that joint is of uniform width from top of slab to top of the joint filler.

(d) For longitudinal expansion joints, the same procedure as followed for transverse expansion joint shall be adopted.

30.7 **POLYURETHANE JOINT SEALING COMPOUND-SEALANT (SINGLE COMPONENT)**

(a) Polyurethane cold joint sealant shall be from the approved manufacturer listed in Appx 'A' to the Particular Specification & meet hydrolysis/water resistance tests as per EN141875-2003. Movement Accommodation Factor (MAF) of the sealant shall be minimum ± 30%. Sealant shall be flexible, resistant to age hardening, fuel and heat. Manufacturer shall offer minimum ten years (10) performance guarantee all as mentioned in SSR Part-I clause No 20.B.7.11.5. Use of Polysulphide joint sealant is not permitted in the work. Contractor shall note that joint sealant with primer shall be applied on ground by engaging original applicator of the approved manufacturer. The frequency of testing for hydrolysis / fuel/water resistance test and movement accommodation tests shall be done for every procured 5MT quantity and got tested from Indian Rubber Manufacturers Research Association Laboratory or any reputed testing laboratory possessing NABL accreditation. Test reports, test certificates, original purchase vouchers shall be submitted to the department/GE for the payment purpose.

(b) The following procedure shall be adopted while execution:-

(i) The joint shall be made dry and free of dirt and vacuum cleaned by a mechanical device as well as free of oil, vegetation and other debris.

(ii) Two coats of primer should be applied with a thin brush by the forward and backward moment at an interval of 30 of minutes before pouring the sealant (i.e. one side will be brushed twice forward and backward) with adequate quantity of primer as per manufacture's instruction.

(iii) The application temperature of the sealants should be strictly adhered to as per the manufacture's instruction.

(iv) The pneumatic pressure, while application should not be too high which causes high speed flow and results in over filling of joints. Too low pressure form small bubbles, which can be starting point of cohesive failures. Every air bubble that appeared on the surface should be immediately popped with a levelling tool.

(c) **PRIMER**:- Primer shall be procured from the same manufacturer of joint sealant procured as mentioned in "clause 24(b).5.1".

(d) **BACK-UP MATERIAL** Closed cell Polyethylence Rod or Sheet (Foam) is recommended as Back-up material to prevent three sided adhesion and to control sealant depth. In case joint depth is insufficient, Bond Breaker tape may be used as back-up. Back-up material should fit tightly into joint to prevent seepage of sealant through joint edges. Rod diameter or sheet (foam) width should be 20 to 25% more than joint width.

(e) **SEALANT AND PRIMER APPLICATION** - Sealant shall be applied by machine. The process of joint sealing shall be carried out strictly in accordance to steps and methods recommended by the manufacturers of sealing compound and primer which shall involve following steps:-

(a) Joint Inspection
(b) Joint preparation
(c) Masking tape application
(d) Insertion of Back-up material

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(e) Primer application
(f) Mixing of two part sealant
(g) Application of sealant
(h) Masking tape removal
(i) Final surface touch up
(j) Clean up

(f) **SOME MISCELLANEOUS ASPECTS**: The sealant are sensitive to temperature while in storage as well as during mixing. Temperatures for all phase of handling of the sealing compounds viz mixing placing and curing conditions must be in accordance with manufacturer's recommendations. All safety precautions during handling and application of these sealants as prescribed by the manufacturers shall also be strictly adhered to.

(g) **GUARANTEE FOR JOINT SEALING COMPOUND**

(i) The main contractor shall submit guarantee to the Govt for a period of Ten Years from the certified date of completion of the work for the effectiveness of the sealing compound in favour of Garrison Engineer in writing.

(ii) If the GE at any time during construction or reconstruction or prior to the expiry of the

Guarantee period, finds that the joints have cracked from the pavement or any type of adhesive, cohesive and material failure is noticed, the contractor shall, on demand in writing from the GE specifying the joints complied of, not withstanding that the same may have been inadvertently passed certified and paid for, undertake to carryout forthwith such treatment as may be necessary to render the paver from damage/crack at his own expense, till expiry of the guarantee period. In the event of his failure to do so, within the period to be specified by the GE in his demand aforesaid, the GE may undertake such treatment at the risk and expense in all respect, of the Contractor. The liability of the Contractor under this condition, however shall not extend beyond the period of TEN YEARs from the certified date of completion, unless the GE had previously given notice to the contractor to rectify the defects.

(iii) The amount of security deposit to be held back from the Contractor's bill against the guarantee for joint sealing compound shall be calculated on the amount of joint sealing compound at contract rates as per the scales given below enhanced by 1.25 times of the amount so calculated. This shall be worked out by GE and intimated to the Contractor. This amount shall be refunded to him after the expiry of the guarantee period. Alternatively, the Contractor may give a separate interest bearing security deposit to GE valid for 10 Years for this amount.

(h) **GUARANTEE OF JOINT SEALANT FROM THE MANUFACTURER**: The contractor shall obtain Guarantee for 10 years for joint sealant from the manufacturer and submit to the GE after completion. Any defect observed during this period notified by the GE in writing shall be made good by the contractor through the manufacturer or their authorized applicator to the entire satisfaction of GE within a reasonable period, failing which, the defects shall be got rectified at contractor's risk and cost. The defects liability period mentioned purpose of this condition. All the expenditure incurred in getting the defects rectified shall be borne by the contractor.

30.8 **FINISHING OF CONCRETE:**

30.8.1 **FLOATING**

(a) As soon as practicable after the concrete has been struck off and compacted, it shall be further smoothened and compacted by means of floating system provided with the paver finisher. If any manual correction is required it shall be carried out from a bridge between the rails generally part of these paving machines. A longitudinal float 1,200 mm long 75mm wide may be used.

30.8.2 **STRAIGHT EDGING**

30.8.2.1 After the longitudinal floating has been completed, the excess water has disappeared, but while the concrete is still plastic, the slab surface shall be tested for trueness with a 3 metre (10 Ft) straight edge swung from handles. The straight edge shall be held in successive positions

parallel to the pavement centre line in contact with the surface and the whole area gone over from one side of the slab to the other. Advance along the pavement be in successive stages of not more than one half length of the straight edge. Any depressions found shall be filled immediately with freshly mixed concrete, struck compacted and refinished. High areas shall be cut down and refinished. The straight edging and re-floating shall continue until the entire surface is found to be free from observable departure from the straight edge and the slab has the required grade and cambers.

30.8.2.2 The slab surfaces shall be retested for trueness before the concrete begins to set, with the 3 metre master straight edge and the wedge gauge (as per IRC 43/72). The straight edge shall be placed on the surface in successive positions, parallel to the centre line. Irregularities shall be measured with the help of the wedge more transversely at various points until it touches both the straight edge and the concrete surface. At any point tested, the concrete shall not show a departure from the true surface, greater than 3mm if at any place the said departure is greater than 3 mm, not more than 3 passes of the vibrating machine shall be allowed and the surface tested again in the specified manner, if the irregularity still exceeds the limit aforesaid, the concrete shall be removed to depth of 50mm.

30.8.2.3 The area of concrete to be removed shall be that represented by the length of the straight edge in the position of measurement across the full width of the slab. Where the point of measurement in default is less than4.5 mm from the transverse expansion joint, the whole area up to the joint shall be removed to the required depth. The concrete so removed shall not be reused in the work.

30.8.2.4 Fresh concrete shall be placed, compacted and finished in the manner already described in these specifications and shall again be subjected to test for accuracy of finish.

30.8.2.5 The forgoing procedure shall be adopted at each shifting of the straight edge and the whole shall be done over from one side of the slab to the other. The straight edge shall advance longitudinally in successive stages of not more than one half the length of the straight edge.

30.8.2.6 No extra payment shall be made for the removal of the rejected concrete and for laying fresh concrete. Although the concrete is to be removed immediately, following measurement of the irregularity and while it is still wet, this shall not mean any waiver from complying with the requirements of this clause if for any reasons the concrete to be removed has hardened.

30.8.2.7 After straight edging of the surface, it shall be finished by belting or brooming or by combination of both, in manner specified here-in-after.

30.9 **TINING**

30.9.1 Pavement surface shall be given tining surface texturing using tanning machine. The tining shall be done as per clause 602.9.11.1 of MORTH specification (Fifth Revision).

30.9.2 Tining shall be completed before concrete reaches such stage that surface is likely to be unduly roughened by the operation. Tining textured surface shall be free from porous or rough spots, irregularities, depression and all pockets caused by accidental disturbing of particles of coarse aggregate embodied near the surface. Depth of Tining shall be all as directed by the GE in consultation with the AF station authorities.

30.10 **EDGING**

After completion of tining of the pavement surface but before the initial setting of the concrete the edges of the slabs shall be carefully finished with a edging tool of 6 mm radius and the edges left smooth and to true to the alignment.

30.11 **HONEY COMBING**

30.11.1 As soon as the side forms are removed, any minor honey combed areas shall be filled with mortar composed of one part of cement to two parts of fine aggregate however these repairs

shall be carried out only after the clearance from the site executives. Major honey combed areas or segregated concrete or other defective work or areas damaged by rain or any other reasons whatsoever shall be considered as defective work and the entire affected slab shall be removed and replaced by the contractor at his own expense. The total area of honey combed patches each more than 2.5 Sqcm area shall not exceed 4 percent of the area of the slab thickness. This decision as to what constitute major honey combing shall rest entirely with the Garrison Engineer but in any case honey combing exceeding 300 sq cm in area at any one location shall be considered as major honey combing. GE's decision whether the concrete is defective or not shall be final and binding. Every slab shall bear an impression on top of slab not exceeding 3mm in depth comprising the number allotted to the slab and the date on which it was cast. This impression shall be formed by the contractor when the concrete is green so as to leave a permanent mark on setting.

30.12 **CURING AND PROTECTION OF CONCRETE**

30.12.1 Constructed slabs shall be marked/ numbered with reference to chainage and location by the contractor to give following details.

(i) Date of casting

(ii) Date of commencement of curing

(iii) Application of curing compound (date wise)

(iv) Date of completion of curing

Record of curing details mentioned above shall be preserved by the Engr-in-charge as site document.

30.12.2 Initial curing shall be done immediately after the surface texturing, the surface and

the sides of the slab shall be cured by the application of approved resign based aluminized reflective curing compound with the help of mechanical sprayer as per manufacturer's instruction and as directed by Engr-in-Charge. Curing compound shall contain sufficient aluminum flakes in a finely divided dispersion to produce a complete coverage of the sprayed surface with a metallic finish. The curing compound shall become stable and impervious to evaporation and shall be as approved by the GE. The curing compound shall have a water retention index of 90 percent in accordance with BS-7542 or ASTM-C-309-81, Type II/relevant IS/IRC code. After initial curing the concrete shall be kept under wet condition with Hessian cloth in position for concrete shall subsequently be cured by making earthen bunds (as approved by GE) of 8 to 10cm high at the edges as well as longitudinally and transversely at about one meter interval to form ponds to cover the pavement surface with water up to 28 days. After completion of 28 days period, the contractor shall clean and sweep the surface so that it is dirt free.

30.12.3 The workmen shall not be allowed to walk over the prepared clean surface either prior to or during the concreting operations. Walking over concrete being laid , compacted or finished shall also be prohibited. All operations on prepared cleaned surface including laying, compacting with needle vibrators, floating and straight edging shall be done from appropriate wooden bridges spanning the width of the slab being laid. Adequate number of such wooden bridge shall be provided.

30.12.4 The contractor shall appoint chowkidars at his own expense to prevent workmen, cattle etc., straying on to the pavement concrete for a minimum of three days after laying of concrete.

30.12.5 Concrete shall not be subjected to any load or weight of any plant until at least 28 days after laying. Concentrated loads or sharp objects like iron wheels of concrete mixers and any vehicular traffic including construction traffic shall not be allowed on the concrete surface for 28 days.

30.13 **CONCRETING DURING RAINS:** To prevent damage to previously laid concrete during monsoon or sudden rains the contractor shall have an adequate stock of tarpaulins or other water proof cloth which shall be spread over the concrete as directed. Any concrete damaged by rains

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shall be removed and replaced by the contractor at his own expense as directed by Engineer-in-Charge. Two temporary shelters of size 36m and required width for concreting (PQC) to be erected on rails as directed by the GE .

30.13.1 **CONCRETING DURING HOT WHEATHER :** If concreting for DLC/PQC/RCC during hot weather, when temp at work site more 320 C is permitted by the GE, then contractor shall arrange/deploy water chilling plant of required capacity without any extra cost to cool down/bring down temperature of water used for mixing & curing of concreting work to permissible limit as stipulated in the IS 456-2000.

30.13.2 :Concreting in hot weather without use of the chiller plant shall not be permitted.

30.13.3 CONCRETING DURING COLD WHEATHER Concreting during cold weather (when temp at the work site during execution falls below 10 degree Celsius) shall not be allowed.

30.14 **TESTING OF PAVEMENT QUALITY CONCRETE**

30.14.1 Testing of controlled concrete (pavement quality concrete) shall be done as per frequency specified. Expenses for samples, supply and transporting of samples etc., including assistance in testing shall be borne by the contractor. The following tests shall be carried out during the progress of work in addition, if any, to these specified here-in-after.

30.14.2 The contractor at his own expenses will establish a lab at site to conduct all relevant tests. The lab equipment shall be calibrated and certificate should be made available along with equipment.

30.15 **COMPACTING FACTOR TEST AS PER IS-1199**

30.15.1 A compacting factor test shall be taken at frequencies as specified here-in- after. Any batch from which compacting factor is being made shall not be transferred to the place of laying until the compacting factor test has been completed. Any batch which gives compacting factor that is in excess of that specified shall be rejected and removed from the site.

30.15.2 **FLEXURAL TEST**

(i) The concrete shall be sampled at the place of deposit. Sampling and testing shall be as per

IS: 1199 and IS: 516. The frequency of sampling shall be as specified in IS. One sample consists of 9 (nine) beams. All the 9 (nine) beams of the same sample shall be cast at the same time. Samples shall be made at well spaced intervals so as to represent entire slab from different batches. A slab shall mean the pavement bounded by expansion and construction joint.

(ii) For the purpose of subsequent identification, the test beam samples and the slab to which these pertain shall be cross referred and record of this maintained, duly signed by the Engineer-in-Charge and contractor.

(iii) Moulds for the test beams shall conform to IS-516 and shall be supplied by the contractor. Beams shall be prepared by the contractor as specified/as directed. The size of the mould for the test beam shall be 15cm x 15cm x 70 cm. The test beam samples shall be maintained and cured under conditions as specified in IS-516. The compaction will be done as specified, Flexural strength test results shall be signed both by the Engineer-in-Charge and the contractor.

(iv) The slab shall stand rejected if the average flexural strength of three specimen of any sample is less than specified.

30.15.3 **TEST**

30.15.3.1 During the currency of the contract, the contractor may demand conducting of core tests/sinusoidal tests of the concrete in respect of such of these slabs which stand rejected as well as such of these slabs which are proposed to be accepted subject to devaluation. Decision of the Garrison Engineer with regard to the type of test to be conducted shall be final and binding.

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30.15.3.2 Three cores each of 15 cm dia for every 30 cubic metres of concrete which is required to be verified shall be cut from each affected slab. If the depth of any core cut is more than the thickness of the concrete provided under the contract for the corresponding slab, the depth of core shall be suitably modified to correspond to the thickness of the concrete laid, by carefully cutting out the extra portion with a diamond edged saw so that in no case the height to diameter ratio of core shall exceed two. In case, when the depth of the concrete laid in a particular slab under the contract exceeds 20 cm, the core cut out of such a slab shall be modified as described here in before, so that the height/diameter ratio shall not exceed two. All cores for tests shall be cut and taken out in the present of the Garrison Engineer/Engineer-in- Charge and the contractor's representative. Testing of cores shall be done in a laboratory as approved by GE. Expenses for such testing shall be borne by the Contractor.

30.15.3.3 All cores for tests shall be cut and taken out in the present of the Garrison Engineer/Engineer-in- Charge and the contractor's representative. Testing of cores shall be done in a laboratory as approved by GE. Expenses for such testing shall be borne by the Contractor.

30.15.3.4 The compressive strength of the cores shall be determined as per IS-516. A correction factor as given here under according to the height/diameter ratio of core samples shall be applied to the average of three compressive strengths so determined:-

F = 0.11 n + 0.78 (Where F is correction factor)

n = H/D (ranging from 1 to 2)

30.15.3.5 The product of the correction factor and the compressive strength (average) shall be known as the corrected compressive strength (this being the equivalent strength of a cylinder having height/diameter ratio of two). The cube strength of concrete shall be determined by multiplying the corrected cylinder compressive strength by 5/4. This shall be further corrected for age factor, if applicable, as under. The cube strength thus determined shall be divided by the age factor as given in the table below, to obtain the equivalent cube strength at 28 days (ECS). The product of the correction factor and the compressive strength (average) shall be known as the corrected compressive strength (this being the equivalent strength of a cylinder having height/diameter ratio of two). The cube strength of concrete shall be determined by multiplying the corrected cylinder compressive strength by 5/4. This shall be further corrected for age factor, if applicable, as under. The cube strength thus determined shall be divided by the age factor as given in the table below, to obtain the equivalent cube strength at 28 days (ECS

Age of Slab in Month(s)	Age Factor
1	1.0
3	1.10
6	1.15
12 and over	1.20

If equivalent cube strength so calculated is K Kg/Sq cm or more, the concrete of the corresponding slab shall be accepted for strength requirement, irrespective of whether the concrete stood rejected/was proposed to be accepted subject to devaluation. It the equivalent cube strength of the concrete of a particular slab ranges between 0.9 K Kg/Sqcm to K Kg/Sqcm the concrete of the corresponding slab shall be considered for acceptance subject to devaluation as stipulated hereinafter

30.15.3.6 **ACCEPTANCEOF CONCRETE SUBJECT TO DEVALUATION:**

The decision of CWE whether concrete shall be accepted subject to devaluation or not shall be final, conclusive and binding. In case CWE decides not to accept concrete subject to devaluation, the slabs casted with such concrete shall be replaced and re-laid as per contract specification by the contractor at his own expense.

30.15.3.7 Acceptance of concrete (with regard to strength) subject to devaluation shall be regulated as under:

(a) Up to an extent of 5% (five percent) of the total quantity of the concrete provided under the contract be accepted subject to devaluation by allowing payment for such concrete at a reduced

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rate(as compared to the contract rate), the reduced rate being in direct proportion to the strength as given below .

The reduced rate payable shall be equal to :-

Contract rate X (AFS / 44)

OR

Contract rate X ECS (if core testing is resorted to.)

K

Where,

AFS- Average flexural strength pertaining to a slab in kg/sqcm of the sample.

ECS = Equivalent cube strength as defined.

(a) concrete in excess of 5% (five percent) of the quantity provided under the contract, but not exceeding 10% (Ten percent) may also be accepted, subject to devaluation but in respect of such concrete, the reduced rate payable to the contractor shall be :-

Contract rate X (2 x AFS-44) 44

OR

Contract rate X (2 x ECS-K) (if core test is resorted to)

K

Where,

K = Characteristic compressive strength of cubes obtained from cubes cast from approved design mix concrete (meeting flexural strength criteria of 45 Kg/Sqcm). K & ECS shall have same unit i.e. Kg/Sqcm.

(b) For calculating the quantity of concrete that may be accepted subject to devaluation as stipulated in preceding (tests) Clauses, the affected slabs in the order of casting (date wise) shall be taken into account. No sooner the limit of five percent has been reached, than acceptance of concrete of the remaining slabs, to an extent of a further five percent, subject to devaluations shall be governed by the stipulations given in preceding clauses (TESTS)

30.15.3.8 **REPLACEMENT OF REJECTED SLABS** Concrete in the slabs which stands rejected shall be replaced by the contractor. The concrete cast in replacement shall satisfy the strength requirement all as mentioned here in before. If due to operational or other reasons (the decision of the Garrison Engineer in this aspect shall be final, conclusive and binding) rejected slabs are not to be replaced, the cost of demolition and cost of replacement of such slabs including the affected joints shall be recovered from the contractor at the contract rates, irrespective of whether such slabs are subsequently actually replaced by Govt or not.

30.16 **COMMUNICATION FACILITIES**

Efficient communication facility for executives and contractor's representative with vehicular conveyance shall be made available by the contractor regarding site control, safety precautions, quality control and to ensure smooth execution of work.

30.17 **SEPARATION MEMBRANE:** A separation membrane shall be used between the concrete slab and the sub base and on exiting surface as directed by GE. HDPE sheet shall be 400 microns thick laid flat.

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30.18 **QUALITY CONTROL & DEFECTS RECTIFICATION**: Surface irregularities & cracks developed in the PQC pavement surface shall be rectified as per method & criteria given in the respective MoRTH Section Nos. 902.5, 602.14 & in the IRC: SP-83. Decision of commander works Engineer (CWE) regarding acceptance of work after rectification of defects or otherwise shall be final, conclusive & binding on the contractor.

30.19 **FWD & FRICTION TESTS**: Before issue of completion certificate FWD & Friction tests all be shall be carried out by the contractor through the authorized laboratories at their cost. Relevant details of FWD & Friction tests results shall be held on record duly defaced by the GE/Engineer-in-charge.

30.20 **LATERAL CONFINEMENTOF EACH LAYER** : Lateral confinement of each layer of the dispersal / manoeuvring area shall be ensured by subsequently constructing shoulder portion before laying of next layer .

30.21 **BATCHING AND MIXING** (MoRTH clause 602.9.2)Pavement concrete materials shall be batched by weight. Weigh batching shall be done with fully automatic computerized batch weighing ready mix concrete plants of capacity not less than 90 Cu m/hour. Polyester triangular synthetic fibres of approved make as secondary reinforcement shall be uniformly mixed through dosing attachment mentioned below at the rate of 0.25% of weight of cement (or at the rate as per manufacturer's instructions). The contractor shall install fully computerized batch mix plant of specified capacity at site to ensure that concrete is placed at desired location within 20 minutes of its pouring out from batching plant.

30.22 **PLACING AND COMPACTION OF CONCRETE** Placing and compaction shall be carried out as specified in Clause No. 20.B.7.8 to 20.B.8.2 of MES Schedule Part – I. The placing of concrete in front of PQC shall be preferably from side placer to avoid damage to DLC by concrete tipping trucks. Refer Para 602.9.4.1 and 602.9.4.2 of MORTH book for additional reference.

40.22 **WEATHER AND SEASONAL LIMITATIONS**

Concreting should be avoided during rainy season. Sufficient precautions shall be taken as per guidelines given in para 602.7.1 of MoRT&H book. Working limitation and precautions/use of chilling plant in hot weather shall be as per MORT&H para 602.7.2. THE TREATMENT OF RIGID PAVEMENT ON CULVERTS MORTH clause 602.9.10 may be referred for treatment to rigid pavement on culverts.

40.23 **CHEMICAL ADMIXTURES** : As per MoRTH book para 602.2.3 or as recommended in design mix, admixtures to be used in PQC shall be retarder to improve workability and setting time to facilitate work in temp above 35 degree centigrade temp and shall be non chloride based.

30.23 **AUTOMATIC DOSING ATTACHMENT FOR USAGE OF FIBRE IN CONCRETE PLANT:** Concrete batching plant shall have attachment of automatic dosing (FIB-A model) of synthetic Fibre. The attachment is manufactured by M/s Autosys Engg (P) Ltd, Chennai. It is portable and shall have PLC panel control with feeding capacity @ 0.6 kg-2.0 kg /cum.

30.24 **PAINTING OF DISPERSAL & MANUVERING**

30.24.1 **MATERIALS**

Painting shall be carried out all as specified in schedule 'A' with road marking paint of first quality water based from any of the following manufacturer:

(i) M/s. Asian Paints. (APCOTRAKIS BRANDS) or equivalent from the following:-
(ii) M/s. Jenson and Nicholson,
(iii) M/s. Shalimar Paints,
(iv) M/s. Berger Paints

The road marking paint shall be applied strictly as per manufacturer's instructions. The contractor shall submit samples of the paints proposed for the project along with results of tests conducted in approved laboratory, as directed by the Engineer-in-Charge, for approval. The manufacturer's recommendations shall also be submitted for the consideration and approval of the GE well in advance. The paint shall be brought to the site of work by the contractor in original sealed

containers. The material shall be brought in one lot in adequate quantity sufficient for the entire work. The material shall be kept in the joint custody of the contractor and the Engineer-in-Charge. The empty containers shall not be removed from the site of work, till the work has been completed and permission obtained from the GE. The paint for Runway marking paint shall be single component air drying water based acrylic paint confirming to IS 164.

TECHNICAL DATA Related to the ''APCOTRAK IS'' water based road marking paint are given here-in- below :

Colour	White and Golden yellow
Gloss	Matt
Volume Solids	Approx 35%
Recommended DFT/Coat	50 microns
Thoretical covering capacity	7.00 sq.mtr/lt @ 50 microns DFT
Drying time at 30 degree C	Surface dry – 20 minutes Hard dry – 4 hours
Over-coating interval at 30 degree C	Min : 45 minutes Max : unlimited, provided surface is dry and clean from all contamination
Shelf life (cool 7 dry place)	At least 12 month

30.24.2 **WORKMANSHIP:**

30.24.2.1 **PREPARATION OF SURFACE**: The surface shall be thoroughly cleaned and dusted. All dirt, scales, oil and grease shall be thoroughly removed before painting is started. Wire brushing and water washing shall be carried out for thorough removal of dust from the road surface. The surface shall be allowed to dry thoroughly before painting. The prepared surface shall be inspected and approved by the GE before painting is commenced. The weather shall not be foggy/ rainy, dusty or windy. Surface temperature must be at least 300C above dew point but not above 500C. The suitability of the weather will be determined by the Engineer-in-Charge.

30.24.2.2 **APPLICATION:** Before pouring into smaller containers for use, the paint shall be stirred thoroughly in its original container. The paint shall be continuously stirred in the smaller container while applying to runway surface so that its consistency is kept uniform. A time interval as recommended by the manufacturer or as directed by the Engineer-in- Charge shall elapse, between laying and finishing of the bituminous and cement concrete surfaces and the time of marking the pavement.

30.24.2.3 **MIXING** If settling is observed in the drum, loosen the settled material with the help of hand stirrer followed by power driven stirrer for quick homogenous mixing.

30.5 **APPLICATION BY AIR-SPRAY**

By Air Spray

Recommended thinner : Municipal/Potable water Volume of thinner : 20-40%

Nozzle Orifice : 1.5-3.00mm

The paint shall be applied evenly and smoothly by means of crossing and laying off. The crossing and laying off consists of covering the area over with paint, brushing the surface hard for the first time over and then brushing alternatively in opposite direction, two or three times and then finally brushing lightly in a direction at right angle to the same. In this process, no brush marks shall be left after the laying off is finished. The full process of crossing and laying off will constitute one coat. Final coat of painting may be provided with road marking paint machine at the discretion of the contractor without any extra cost to the department.

a Each coat shall be allowed to dry out thoroughly before the next coat is applied.

b Earlier applied coat shall be cleaned off dust before the next coat is applied.

PARTICULAR SPECIFICATION (CONT'D…../-)

c No left over paint shall be put back into the stock tins. When not in use, the containers shall be kept properly closed.
d No hair marks from the brush or clogging of paint puddles shall be left on work.

e The surface shall ordinarily not be painted until it has dried up completely. Trial patches of paint shall be laid at intervals to check if drying is satisfactory
f The runway marking shall be done as instructed by the Garrison Engineer.

g Safety Information

As a general safety measure, inhalation of paint mist and contract of liquid paint with skin & eyes, should be avoided. Forced ventilation shall be provided when applying paint in confined spaces or stagnant air. Even when ventilation is provided, respiratory, skin and eye protection are always recommended when spraying paint.

h The work of Road marking shall be all as per Sch Á' item and applied mechanically with all specification as per MORTH para 803.1 to 803.6.

31 SITE CONDITION:-

31.1 It is assumed that before tendering the Contractor would have visited the site and familiarized himself with all the local conditions and means of transportation and communications. No claim of whatsoever nature would be entertained at a later date on account of the Contractor's ignorance of the local conditions.

31.2 **WORKMANSHIP:-**

31.2.1 All the work shall be carried out in a workmanship like manner and as per the best practices of the trade.

31.3 **MATERIALS:-**

31.3.1 All materials to be supplied by the Contractor shall be new. All packed items shall arrive at site in original packing only. Any items found defective or damaged shall be replaced by the Contractor at his own expenses.

31.4 **INSTRUMENTS FOR MEASURMENT AND TESTING:-**

31.4.1 The Contractor shall provide, free of cost, all equipment, instruments, labour and all other allied assistance required by the Engineer-in-Charge or their representatives for measurement and testing of the works.

31.5 **COORDINATION WITH OTHER TRADES**:-

31.5.1 The Contractor shall be responsible for coordinating his work with works of other trades sufficiently ahead of time to avoid unnecessary hold ups. Hangers, sleeves, recesses etc. shall be left in time as the work proceeds.

31.6 **UP-KEEP OF THE SITE:-**

31.6.1 It shall be the responsibility of the Contractor to clear away, from time to time all debris and excess material generated by the activities of his workers.

31.7 **PROTECTION:-**

31.7.1 All work shall be adequately protected, to the satisfaction of the Engineer-in-Charge, so that the whole work is free from the damage throughout the period of construction up to

the time of handing over. Before handing over the work, the Contractor shall clean all elements of the complete installation, remove plasters, stickers, rust stains and all other foreign matter and leave every part in acceptable condition and ready for use to the satisfaction of the Engineer-in-Charge.

31.8 **TESTING OF CABLES:-**

31.8.1 Testing of HT/LT cables shall be carriedout as specified in Para 19.93.1 and 19.94 on page No. 19.22 & 19.23 of MES standard schedule of rates 1991 (Part specifications) duly modified by these particular specifications.

32 **BLANK**

33 **COMPLETIONDRAWINGS & PSMB:-**

33.1 The contractor shall submit, after the completion of the work, one set of originals and two sets of prints of the As-Fitted drawings / Completion drawings of all floors giving followings: -

(a) Run and size of conduits, inspection, and junction and pull boxes.
(b) Size and numbers of conductor in each conduit.
(c) Location of all detectors, manual call points, response indicators, hooters etc.
(d) Location and details of all zonal control panels and main control and indicating panels.
(e) A complete wiring diagram as installed and schematic drawings showing all connections in the complete fire detection and alarm system.
(f) Layout and particulars of all cables.

33.2 **PSMBs** : After completion of work and before payment of final bill, the Contractor shall prepare the PSMB (Periodical Services Measurement Book) for complete painting work for the buildings covered under Schedule 'A', The contractor shall provide all necessary details in his possession for preparation of PSMBs.

34 **GUARANTEE FOR LED LIGHT FITTINGS:-**

34.1 Contractor shall furnish a written guarantee to GE for a period of **70 Months** from the certified date of completion of the entire work for effective functioning and continuous rated lumen output of LED Light Fittings on Non-Judicial Stamp Paper of Rs. 100.00. If the LED light fittings become non functional or in case of any defect found in the light fitting or rated lumen is not produced during aforesaid Guarantee Period, the contractor shall carry out the necessary repairs/rectification'/replace the necessary spares including LED bulbs to the entire satisfaction of Garrison Engineer at his own cost in the following cases:

(a) Manufacturing Defects
(b) Failure due to mechanical and electrical impact
(c) Drop in Lumen (or Lux at 1m) below 90% of claimed values of Lumen (or Luxat 1m) of the LED luminaire

34.2 An amount calculated @ 2% of the value of LED Light Fittings at contract rates enhanced by 25% and rounded off to next hundred rupees (Subject to a minimum of Rs. 5,000.00) shall be retained out of the contractor's bill as Security Deposit against the Guarantee for effective functioning and continuous glowing of LED Light Fittings for **70 Months**. This Security Deposit shall be released after successful expiry of the **70 Months** guarantee period from the certified date of completion of the entire work by GE, provided always that the contractor shall first have been paid the Final Bill and have rendered 'No Demand Certificate (IAFA-451)'. Alternatively, the contractor may furnish Fixed Deposit Receipt in lieu of Security Deposit from a Schedule Bank in favour of GE for the above said Guarantee Period.

34.3 Should the GE at any time during the Guarantee period, finds defective performance of the LED Light Fittings, the contractor shall, on demand in writing from the GE specifying the location complained of notwithstanding that the same may have been inadvertently passed'/certified and paid for, undertake to carry out such repairs/

rectification/replacement of spares including LED bulbs as may be necessary forthwith to rectify the defects to the full satisfaction of GE and render complaint, free from any type of defects. In the event of his failure to do so, within the period as specified by the GE in his aforesaid demand, the GE may undertake such defective work through other agency at the risk and cost of the contractor in all respects. The liability of the contractor under this condition shall not extend beyond the period of **70 Months** from the certified date of completion, unless the GE had previously given notice to the contractor to rectify the defects. Condition-46 of General Conditions of Contracts (IAFW-2249) shall be deemed to be amended to the extent mentioned above.

35 SITE CLEARANCE:-

35.1 Refer Condition-49 of IAFW-2249, General Conditions of Contracts. The contractor shall remove from the site all unused stores and materials, tools and plants, equipment, scaffolding, temporary buildings, huts and like belonging to the contractor provided for the execution of the work under this contract and the site of work shall be cleared off rubbish and all waste materials by the contractor and deliver the site in neat, clean and tidy manner to the satisfaction of the Engineer-in-Charge on or before the date of completion. Nothing extra whatsoever shall be paid to the contractor for such clearance of site and the contract sum quoted shall deem to be including the same.

36-37 BLANK

38. Road, Path and culverts works:

Works pertaining Road, Path and culvers shall be carried out as described in items of sch 'A' here in before and here in below, shown on drawings and as directed.

38.1 **STACKING AND MEASUREMENTS OF MATERIALS:**

The contractor shall collect and stack, approved materials like aggregates, sand etc on levelled ground at places indicated by GE near the site of work in closely packed, stacks, stacks measurements of these materials shall be taken and recorded in measurement book and signed by the contractor and Engineer-in-Charge as per special condition 20.A.1.2 on serial page No 433 of MES Schedule 2004 part-II rates before commencement of hard standing/road work. It shall be ensured that the required quantity of materials have been incorporated in the work. However, these provisions do not absolve the contractor from his responsibility of providing more materials as necessary to complete the work. Any additional materials required for making good hollows and the depressions during rolling shall be supplied and provided by the contractor at his own expense in addition to the stacked materials mentioned here-in-before.

38.2 **ROLLING FORMATION SURFACES:**

(a)The formation surfaces shall be rolled to the required gradient and camber with power roller including sprinkling the surface with water as required.

(b)Where rolling is not feasible, prior permission of the GE shall be obtained in writing for hand ramming. The laying of the base course will commence only after the earth formation is approved by the GE.

38.3 **EARTHEN BERMS:**

Earthen Berms as described in respective items of Schedule 'A' shall be watered and well consolidated with 0.80 to 1.50 Tonne animal roller and finished to profile in line with finished surface of the road/path.

38.4 **DRAINAGE OF EARTH FORMATION:**

At all times, the formation surfaces shall be kept drained by the contractor. The contractor shall provide such temporary open drain as may be necessary to prevent accumulation of water from any cause whatsoever.

38.5 **ROAD , PATH AND CULVERT**

PARTICULAR SPECIFICATION (CONT'D…../-)

38.5.1 SOLING: Soling shall consist of broken stone laid and levelled in one layer, watered and rolled to required surface and camber all as per clause 20A.2 and 20A.2.1 of MES Schedule Part – I.

38.6 **WATER BOUND MACADAM**

(a)Stone aggregate, screenings and binding materials for water wound macadam shall be as specified in clause 20A.3 of MES Schedule Part-I and conform to the samples kept in GE's office and approved by the GE before incorporation in the work. Screenings shall be of grading 'A' as specified in clause 20A.3.2 of MES Schedule Part-I. The coarse aggregate shall be of grading No. 2 with size ranging from 63 mm to 40 mm as specified in sub clause 6 of clause 20A.3.1 of the MES Schedule Part-I.

(b)Water bound macadam shall consist of 75 mm thick (compacted thickness) broken stone aggregate consolidated in one layer. Spreading, rolling, applying screening and watering shall be as specified in clause 20.A.21 to 20.A.21.13 of MES Schedule Part-I. The rolling shall be done with power roller after the application of screenings and wet rolling as described in MES Schedule Part-I.

38.7 **Premix Carpet**

Binder shall be as specified in Schedule 'A' Quantity of stone chippings shall be as per para 20.A.25.8.

Preparation of surface and binder application shall be as specified for premix carpet with hot bitumen

Preparation, spreading, consolidating mix, surface finishing shall be same as that of premix carpet using hot bitumen shall not be heated but it shall be poured to wet the aggregate at atmospheric temperature at correct rate before spreading on the road surface. The rolling shall commence after spreading the mixture. The surface shall be protected by a suitable device such as barricading & posting of watchman for closing the traffic.

38.8 **Sand Seal Coat**

The binder shall be heated to the temperature appropriate to the grade of bitumen being used. Also dry sand shall be suitably heated to the required temperature before it is placed in a mechanical mixer. Mixing of binder with sand to the specified proportions shall be continued till the sand is thoroughly coated with the binder. As soon as sufficient length has been covered with premixed material, the surface shall be rolled with 8 to 10 tonne power roller. Rolling shall be continued till the premixed material completely seals the voids in the bituminous course and a smooth uniform surface is obtained.

38.9 **ROLLING FORMATION SURFACES IN CUTTING**: Rolling shall be carried out as per Sch `A' specified in clause 20.A.22 of SSR Part I.

38.10 **Wet Mix Macadam (WMM):-** WMM shall be provided as described and as per Schedule `A' and specified in clause 20.B.2.5 & 20.B.5 and its sub clauses of MES SSR 2009 Part-I and as directed by Engr-in-Charge.

38.11 **Dry Lean Concrete (DLC):-** DLC shall be provided as described in Schedule `A' and specified in relevant clauses of SSR Part-I 2009 and as directed. The maximum size of coarse aggregate shall not exceed 26.5mm.

Dummy Joints.

38.12 The size of joints shall be as specified in Schedule-`A' special in clause 20.B.7.11.2 of SSR Part-I and as shown on drawings /as directed. 32.

The joint shall be formed using mechanical equipment (diamond cutter) within 6 Hours of placing of concrete under moderate climatic conditions and when the concrete has sufficiently hardened. Cutting or sawing by a saw mounted at movable frame and driven mechanically will also be permitted as a method for making the joint. Care shall be taken that the edge of the joints are not damaged. The edge will not stand proud of the concrete slabs.

39. TECHNICAL SPECIFICATION FOR REINFORCED SOIL RETAINING WALL (RE)

PARTICULAR SPECIFICATION (CONT'D…../-)

A. DESCRIPTION

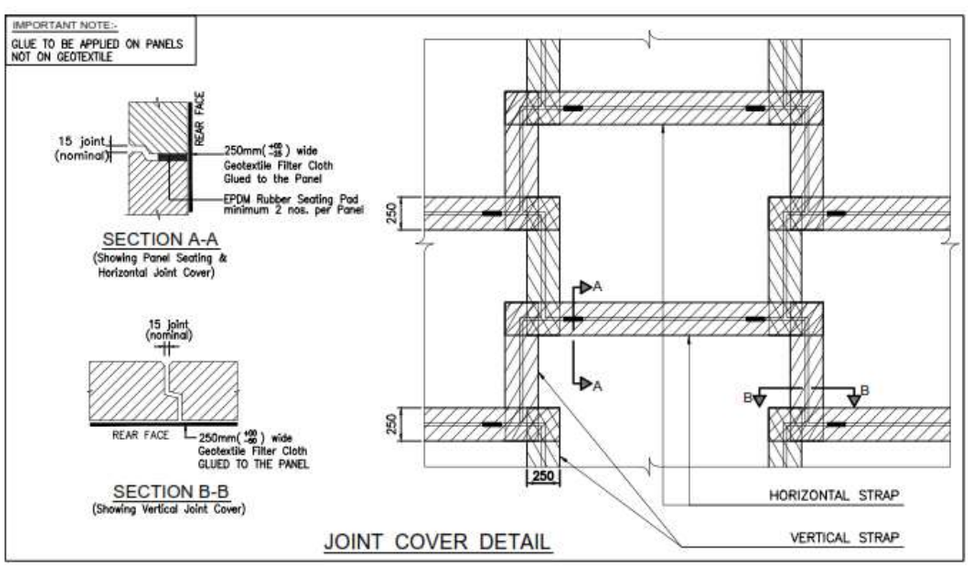
This work shall consist of Design and Construction of Reinforced Soil Retaining Wall/ Mechanically Stabilized Earth Wall (MSE Wall) consisting of precast discrete T-Shape concrete facing panel, high adherence polymeric strap reinforcement to generate high pull-out resistance, synthetic mechanical connection, with accessories / fittings and backfill material in accordance with the specifications and in reasonably close conformity with the lines, grades, design and dimensions shown on the approved drawings. No other Polymeric soil reinforcement in the form of strip, grid, mesh, mat, sheet etc. and no other connection system in the form of friction based, metallic dowels and hooks etc. shall be allowed for construction.

Design and construction of reinforced soil retaining wall shall include loading from traffic or pavement depending on application, seismic activity, drainage subsystem, and precast RCC discrete facia panel element of nominal 180mm thickness, PCC leveling pad, etc., necessary foundation treatment, to ensure meeting deficiency of the bearing capacity of the underlying soil for adequate and uniform distribution of the stresses induced due to the incident loads to the base, besides ensuring stability against slip circle failure. The contractor shall need to assess and substantiate the availability and design adequacy of soil safe bearing capacity under the location of the wall before execution of the work.

B. MATERIALS

(i) Precast Concrete Facing

R.C.C Precast concrete facing elements shall be of Tee shape (minimum 1.5m height) and the dimensions shall conform to the details shown on the final construction drawing. However, all dimensions shown below are typical. Concrete shall be of grade shown on the drawings and shall conform to the requirements specified in section 1700 "Structural Concrete" of MORT&H. The grade of Facia concrete shall be M-35. The thickness of RCC Panel shall be minimum 180mm including architectural finish of maximum 20mm depth with adequate steel reinforcement as per typical system requirement.



Precast modular blocks as facing of Reinforced Soil structure shall not acceptable.

PARTICULAR SPECIFICATION (CONT'D…../-)

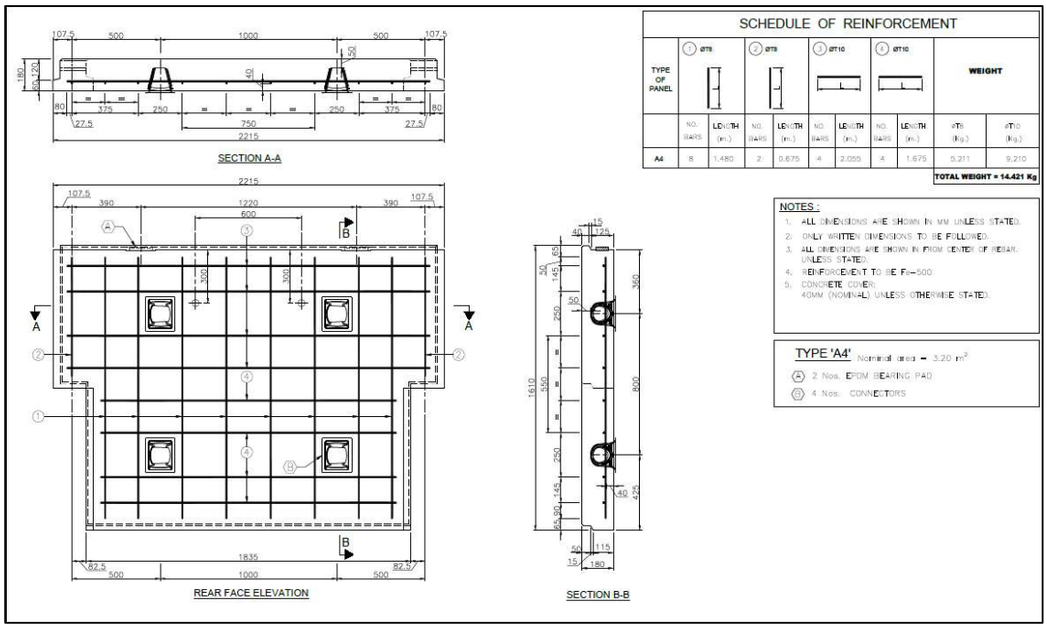


Figure 2: Typical Fascia Panel with connectors and reinforcement detail

(a.) Casting

The elements shall be cast on flat area of a specially prepared mould. Connectors and lifting devices shall be cast in place to the dimensions and tolerances shown on approved System drawing and connectors shall be set on the rear face, prior to casting. The concrete in each unit shall be placed without interruption and shall be compacted using an approved vibrator supplemented by such hand-tamping as may be necessary to ensure that the concrete reaches into the corners of the forms and prevent formation of stone pockets or cleavage planes. Clear form oil of the approved and only from single manufacturing source shall be used throughout the casting operations to avoid colour variation. There shall not be any steel component in the connection system, neither encased nor coated inside the connector for connecting the polymeric soil reinforcement.

The connection strength shall be derived based on third party test conducted on all types of facing panels (as per the proposed shape). The third-party test reports shall be produced for all types of connections. No other mechanical connection system where the connection strength cannot be justified by laboratory testing shall be allowed.

Friction based connection system shall not be allowed.

(b.) Curing

The precast elements shall be cured for a sufficient length of time as approved by Engineer so that the concrete develops the required compressive strength. Only fresh potable water shall be used for curing.

(c.) Removal of forms

The forms shall remain in place until they can be removed without damaging the elements. The scheme of removal of form work shall be as per relevant MORT&H specifications.

(d.) Scribing

The date of manufacture and batch number shall be clearly scribed on the rear face of each unit.

(e.) Concrete finish

The front (exposed) face of the elements shall have the finish approved by the Engineer-in-charge. The rear face shall have the finish of unformed surface and shall be roughly screened to eliminate open pockets of aggregates.

(f.) Tolerances

PARTICULAR SPECIFICATION (CONT'D…../-)

All elements shall be manufactured within the following tolerances:

- Evenness of the front face : ± 5 mm over 1500mm
- Difference between lengths of two diagonals: ±10 mm max.
- Thickness : ±15 mm

(g.) Handling, Storage and Transporting

All elements shall be handled, stored and transported in such manner as to eliminate the danger of chipping, cracks, fracture and excessive bending stresses. Elements in storage shall be supported on firm wooden / rubber blocks/pads placed adjacent to the Dowel bars to avoid bending.

(h.) Acceptability

Acceptability of the precast elements shall be determined on the basis of compression tests, as per MORT&H. Specifications and visual inspection. A minimum of one sample of 6 cubes shall be taken for each lot of 5 cum or 12 cubes above 5 cum per day. Elements shall be acceptable for placement in the structure if the strength at 10 days, or before, exceeds 75% of the 28 days requirements.

(i.) Rejection

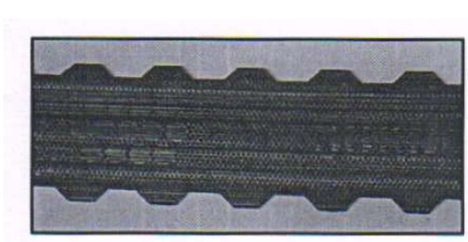
Elements shall be subject to rejection in case of failure to meet any of the requirements specified above. In addition, defects, which indicate imperfect moulding, or defects indicating honeycombed or open textured concrete, shall be sufficient cause for rejection.

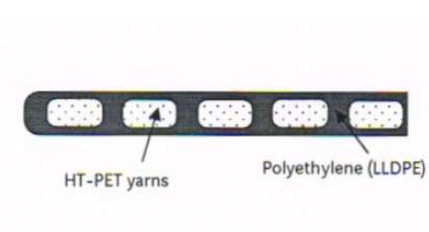
(ii) Leveling Pad

A PCC leveling pad shall be provided under walls as seat beam just to rest the facing panel on leveled surface. Concrete shall have grade M-15. Maximum size of aggregates shall be 20 mm. The pad shall be cured for at least 05 days prior to placing panels. The top surface shall be leveled at ±3mm tolerance.

(iii) The specification of reinforcing element

High adherence polymeric strap consists of discrete channels of closely packed high tenacity polyester yarns encased in Linear Low-Density Polyethylene sheath and having trapezoidal groove like rib (lateral teeth) on both side of the strap to generate a high coefficient of friction. The reinforcement layers shall be designed using strips with lateral teeth to improve friction and interaction with selected backfill soil, which is particularly important to improve the behaviour and performance of the reinforced soil structure under high vibration. The coefficient of apparent friction at the topmost layer of soil reinforcement shall not be less than 1.50. The vertical spacing between two layers of soil reinforcement shall be limited to maximum 800mm.





a) Specification

Type	High Adherence Polymeric Strap
Grade	50
Minimum Width (mm)	49.5±0.5
Characteristic Strength (kN)	50

(iv) Joint Fillers

PARTICULAR SPECIFICATION (CONT'D…../-)

a) Bedding material between the horizontal joints (excluding the joint between the PCC base and the bottom most panel) of the panels shall consist of EPDM (Ethylene Propylene Diene Monomer) material.

b) Sealing material for filling joint gaps, if required, other than bedding joints shall consist of non- woven polypropylene geotextile of unit weight not less than 130 g/m^2. For facia drainage and prevention of soil leaching wherever necessary, non-woven polypropylene geotextile strips shall be provided against the vertical and horizontal joints on the back face of the facia, as shown in the approved drawing.

(v) Properties of the fill

a) The properties of wall fill, subsoil, the retained fill and their interface media shall be based on assumed representative soil data at the design stage, but the values shall be verified at the time of the construction.

b) Fill in the wall shall be well graded wholly frictional non-cohesive free draining material. Stratified layers of combined frictional and cohesive-frictional fill shall not be used without specific design check. The use of soft chalk, unburnt colliery shale and unsuitable material shall not be permitted. The pH value shall be confirmed before using such material. Polyester strap shall be used as main reinforcement if 3<pH<9.

c) The angle of internal friction of the reinforced and retained fill shall not be less than 32°. Fill for reinforced soil structures shall be well graded selected material as specified. The fill must be so designed to allow dissipation of pore water pressure and shall have free draining characteristics or by providing vertical drainage galleries in front of reinforced soil wall. The association of drainage bay or interface drains shall be connected properly to the gradient required and shall be maintained during compaction in layers.

d) Backfill materials used in the reinforced soil volume shall be reasonably free from organic or otherwise deleterious materials and shall conform to the following mechanical and physical-chemical requirements. Such drainage parameters can be achieved with backfill materials having Plasticity index (PI) ≤6 and coefficient of uniformity (Cu) ≥ 2.

- *Mechanical requirements*

Sieve size	Percent passing
150mm	100%
100mm	more than 75%
75 micron	less than 15%

Acceptance limits for materials with more than 15% passing 75 microns are related to the percentage of particles smaller than 15 microns as follows:

- materials with more than 15% passing 75 micron sieve but less than 10% of particles smaller than 15 microns are acceptable.
- materials with more than 15% passing 75 micron sieve and more than 10% of particles smaller than 15 microns are inadequate and shall not be used.

- *Quality Control*

Initially at approval of source and subsequent at each change of source, backfill is to be checked for gradation, pH and angle of internal friction. The results will indicate what further tests are needed, if any.

- *Water for Compaction*

Water of minimum resistivity exceeding 700Ωcm shall be used for compaction.

- *Materials not conforming to the above*

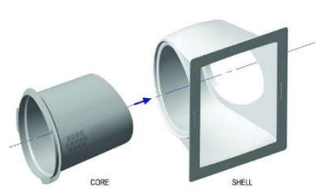
Materials not conforming to the above requirements may be used with the written consent of Engineer after carrying out test. Such materials shall be tested for their functional

PARTICULAR SPECIFICATION (CONT'D…../-)

properties to assure that they are consistent with the parameters used in the design calculations.

(vi) Mechanical Connection

The elements shall be cast on flat area of a specially prepared mould. Connectors shall be cast in place to the dimensions and tolerances shown on approved system drawing and connectors shall be set on the rear face, prior to casting. No other type of connection system shall be used for durability and long-term performance purpose. The concrete in each unit shall be placed without interruption and shall be compacted using an approved vibrator supplemented by such hand-tamping as may be necessary to ensure that the concrete reaches into the corners of the forms and prevent formation of stone pockets or cleavage planes. Clear form oil of the approved and only from single manufacturing source shall be used throughout the casting operations to avoid colour variation. There shall not be any steel component in the connection system, neither encased nor coated inside the connector contributing to connection strength.



The connection strength shall be derived based on third party test conducted on all types of facing panels (as per the proposed shape). The third-party test reports shall be produced for all types of connections.

(vii) Geosynthetic Clay Liner:

The top and slope portion shall be covered with one layer of Geosynthetic Clay Liner (GCL) to mitigate ingress of any water into the reinforced soil structure. Geosynthetic clay liners shall provide impermeable lining having Bentonite mass per unit area (at 0% moisture) = 4,200 grams/sqm , Bentonite mass per unit, GCL mass per unit area (at 12% moisture) = 5,185 grams/sqm, Laminated (HDPE/LLDPE) should be minimum 200 Microns sheet and Hydraulic conductivity (k20) = 5.0 x 10-12 m/sec and Index flux (qi) = 1 x 10-9 m3/m2/sec max as per the technical specification including manpower, tools, tackles, all complete. Geosynthetic clay liners (GCLs) should be factory manufactured hydraulic barriers consisting of a layer of bentonite or other very low-permeability (Minimum 1 x 10-7 m/s) material supported by geotextile combinations, mechanically held together by needling, stitching, or chemical adhesives.

(viii) Erosion Control Blanket

Erosion Control Blanket shall be laid on the sloped portion. It shall be be government approved (any one of FRI, CRRI, CSWCRTI, BARC, CMRI, NHAI and Indian Railway) mechanically bonded erosion control blanket made from bio-degradable, non-woven coconut fibre reinforced with closely woven very fine HDPE nets on one side having (a) minimum aperture size of not greater than 1.5 mm, (b) minimum tensile strength as per ASTM D: 4595-2005 of 5kN/m (MD) and 2.5kN/m (CD), (c) elongation at break as 35.5% (MD) and 41% (CD), (d) minimum CBR puncture strength as per ASTM D 6241 as 751.4 N and (e) minimum density of 600 grams per sqm (+/-10%).

C. CONSTRUCTION REQUIREMENTS

1. Orientation

The reinforcing Polymeric Strap shall be placed at right angles to the face of the wall, with greater cross-sectional dimension in the horizontal plane. The reinforcement shall be placed in the cross-sectional slope of 2.5 to 5% away from the facing towards the tail end to drain-out storm water away from facing element. Clear vertical spacing between the reinforcement shall not be less than 200mm except for cross walls / return walls and in general reinforcement spacing shall not exceed 800m.

2. Facing batter

PARTICULAR SPECIFICATION (CONT'D…../-)

With panel RCC facia initial batter (or set batter) shall be necessary for practical installation as per proprietary design guidelines.

3. Drainage

Drainage shall be very strictly followed as per drawing and specifications in detail. The reinforced soil backfill is considered a self-draining media, having sufficient permeability to relive hydrostatic pressures. However, water logging in the reinforced fill increases the pore pressure coefficient thereby resulting in tensile forces in reinforced soil structure, reducing the stability of the structure. The reinforced fill shall have drainage geo-composite for drainage purpose behind the rear face of reinforced soil fascia panel and multi linear drainage geo-composite placed horizontally near ground level.

Geo-composite drains shall be made from high-density polyethylene (HDPE) geonet core wrapped with needle-punched staple fibre non-woven geotextile made of polypropylene having unit weight greater than equal to 142 grams/sqm (MARV), grab tensile strength greater than equal to 0.533kN (MARV), minimum water flow 5500 l/sqm/min and maximum AOS of 0.212mm (MARV), heat bonded on both sides of geonet to form the geo-composite drains, which along with non-woven geotextile shall have unit mass greater than equal to 710 grams per sqm, thickness @ 2kPa greater than equal to 4.5 mm, tensile strength greater than equal to 24/24 (MD/CD) kN/m, CBR puncture strength greater than equal to 3000 N and in-plane permeability (Hydraulic Gradient, i=1 at @ 200 kPa Pressure) greater than equal to 0.45 l/m.sec.

The multi linear drainage geo-composite shall consists of a filter layer, a drainage mat and mini pipes. The components are joined together by needle punching. Mini pipes are 100% polypropylene made of granules and perforated. Mini pipes shall have outside diameter of is 20mm and placed at 1m centre to centre spacing. The mini pipes have two perforations per corrugation at 180˚ and alternating at 90˚, at a gradient of i=1 having in plane flow capacity of 1L/s/m under 400 kPa during 100h testing. The multi linear drainage geo-composite shall not have any long-term creep and/ or shall not allow any intrusion of the geotextile filter, which shall impact the long-term drainage capacity.

4. Excavations

Excavations shall be in accordance with the requirements of General and Special Specifications and in reasonably close conformity to the limits and construction stages shown on the drawings.

4.1 Foundation Preparation

The foundation for Reinforced Soil Structures shall be graded level for a width equal to or exceeding the length of reinforcing element or as shown on the drawings. Prior to wall construction, the foundation shall be compacted with a smooth wheel vibratory roller. Foundation soils shall be removed and replaced with selected fill for up to 2m depth from ground level. A layer of Bi-planer anisotropic soil reinforcement shall be used on the entire plan area of the stretch. i.e., entire length of reinforced soil wall. This treatment is in addition to the necessary foundation improvement requires for the proposed Reinforced Soil structure. Bi-planer anisotropic soil reinforcement geosynthetics shall have minimum tensile strength of 200 kN/m and made from 50mm/70mm/90mm (±1mm tolerance) co-extruded high tensile polymeric strap soil reinforcement consisting discrete channels of closely packed high tenacity Polyvinyl Acetate (PVA) yarns encased in Linear Low-Density Polyethylene (LLDPE) sheath having trapezoidal groove like rib (lateral teeth) on both sides of the strap to generate high coefficient of friction and together weft welded to cross strips made from low shrinkage yarns to generate an orthogonal structure.

5. Erection

PARTICULAR SPECIFICATION (CONT'D…../-)

Precast concrete panels shall be placed vertically with the aid of a compatible light crane. For erection, panels are handled by means of lifting devices set into the upper edge of the panels. Panels shall be placed in successive horizontal lifts in the sequence shown on the drawings as backfill placement proceeds. As fill materials is placed behind a panel, the panels shall be maintained in vertical position by means of temporary wooden wedges placed in the joint at the junction of the two adjacent top rows of panels during construction. As construction proceeds, and a fourth row is erected, the lowest row of wedges can be removed and so on.

External bracing shall also be provided for the initial lift. However, bracings shall be placed in an area not more than 2.0 meter wide beyond the outer face of panels.

Vertical tolerances (plumb ness) and horizontal alignment tolerance shall not exceed 50mm when measured along a 4 meters straight edge. The maximum allowable offset in any panel joint shall be 25mm.

6. Laying and Compaction

Backfill placement shall follow closely the erection of each lift of panels. At each reinforcing element level, backfill should be roughly leveled before placing and connecting the reinforcing elements. Reinforcing elements shall be placed normal to the face of the wall or as shown on the drawings. The maximum layer thickness shall not exceed 200 mm. The Contractor shall decrease this thickness if that is necessary to obtain the specified density.

At the end of each day's operation, the Contractor shall shape the last level of backfill as to permit runoff of rainwater away from the wall face. Backfill shall be compacted in accordance with the project specifications for embankment to a minimum required compaction of 95% of modified proctor density in the entire width of the embankment, except up to 1.5m from the rear face of wall, where the required compaction shall be minimum 90 % of maximum density and shall determine by the standard test and that the moisture content shall be plus or minus 2% of the optimum Moisture Content as determined. Backfill compaction shall be accomplished without disturbance or distortion of reinforcing elements and panels. Compaction in a strip of the 1.5meter wide adjacent to the backside of the wall facing, shall be achieved by the use of a manually operated vibrating compactor, such that adverse edge stresses are not transferred to the facing panels during construction.

D. TECHNICAL GUIDANCE

A technical representative of the system provider shall provide technical guidance during the casting and installation phases to ensure that the quality of the works performed by the contractor is in accordance with the specifications.

E. MODE OF MEASUREMENT

After completion of the structure the measurement for payment shall be done as per the area of the front elevation wall on inclined plane. The measurement shall be done from the top of levelling pad to top of RCC panel facia. Typically, the measurement is done in square meter.

F. PAYMENT TERMS

Item No.	Description	Amount
1	Upon mobilization of equipment and setting up casting yard.	10%
2	Excavation of soft soil and filling up of selected fill material on prorata basis.	15%
3	Casting of RCC panels on prorata basis.	25%
4	Secured advance on supply of soil reinforcing material and other specialized items as per approved Bill of Material (BOM), on prorata basis.	30%

PARTICULAR SPECIFICATION (CONT'D…../-)

5	Upon filling and compaction for MSE Wall of selected fill on prorata basis.	20%

G. SECURED ADVANCE

For materials brought to site by the Contractor, the Engineer may allow (Interest free) payment @ 75% cost of major materials brought to site for use in the works as secured advance which will normally be paid along with next on account payment. The payment of secured advance shall be made without any bank guarantee but on written request of the Contractor along with indemnity bond indemnifying Employer/Engineer against any loss and /or damages to the materials for which secured advance is sought by the Contractor. The indemnity bond shall be submitted on a non-judicial stamp paper of minimum value of Rs 10/- duly notarised as per the format approved by the Engineer. The Contractor should supply necessary vouchers etc as evidence that payment has been made by the Contractor for all the materials brought to site for which secured advance is sought by the Contractor. Secured advance shall be paid at the rates derived from the accepted rate of the item (s) for which the materials are to be consumed.

PARTICULAR SPECIFICATION (CONT'D…../-)

40. QUALITY CONTROL PLAN

(To be submitted by contractor within 15 days of commencement of contract)

PART I

1	Contract Agreement Reference No	
2.	CPM Net work prepared and approved by GE	
3.	Resource scheduling done base on CPM	
4.	Site Laboratory (with equipments) set up as per Contract Agreement (CA)	
5.	Concrete mix design submitted and approved.	
6.	Preliminary works completed to standard engineering practice	
7.	Arrangements for water made (including testing of water).	
8.	Arrangement for electric supply made	
9.	Materials	

Sl No	Item	Source as per CA	Contractor's plan of Sourcing	Refer to testing clause	Agency for testing	Responsibility for testing

10.	List of all T & P, make and numbers that the contractor would deploy at site of work.	
11.	Name of person nominated by contractor for exercising quality control.	
12.	Qualifications / Experience of person at Serial No. 11 above.	
13.	Names of supervisors with their qualifications / experience employed by contractor.	
14.	Confirmation that contract requirements relating to quality of all materials and quality standards of workmanship and finishes and acceptance criteria are explained and understood by all.	
15.	Confirmation that requirement of tests to be conducted on materials before approval of samples and during execution, tests on workmanship, tests before acceptance including the testing procedure, sampling techniques, frequency and agencies responsible for testing are understood and shall be complied with.	
16.	Method to be adopted for maintaining records of test result.	
17.	Certificate that contractor shall maintain a log of all materials received at site as per the following format :-	

Sl No	Date	Material	Quantity received	Source	Whether as per approved sample or not	Test carried out by supplier	Tests to be carried out before incorporation

18	General Remarks by contractor of his plan of actions to ensure that quality standards.	

PARTICULAR SPECIFICATION (CONT'D…../-)

PART-II

(To be completed by GE before forwarding for approval by CWE)

1.	Verification of Serial No 2 to 8 of Part I.	
2.	Verification of Serial No 9 to 18 of Part I.	
3.	Confirmation that Stage Passing Register laying down the stages and authority responsible for approving the same has been prepared, shown to contractor and kept at site.	
4.	Confirmation that all sites as required by contract had been handed over to contractor on the date fixed in the Work Order No 01.	
5.	Confirmation that arrangements for Govt liability in supply of water electricity have been made and no hold up on this account is expected.	

(Signature of GE)

Approved by CWE

Signature of Contractor

AAD Contracts)
For Accepting Officer

PARTICULAR SPECIFICATION (CONT'D…../-)

LIST OF DRAWINGS

NAME OF WORK: PROVN OF RCC SHELTER ALONGWITH ASSOCIATED WORK SERVICES AT AF STN BAREILLY

Srl No	Description of Drgs	Drg No	Sht No.	Date of Drg	Date of revision
1.	2.	3.	4.	5.	6.
PROVN OF TWO SINGLE ENTRY ORP BLAST PENS ALONGWITH ASSOCIATED WORK SERVICES AT AF STN BAREILLY					
SITE PLAN					
1.	SITE PLAN SHOWING EXTERNAL B/R SERVICES	CE(AF)PRZ/1203/SP	1/2	19.12.24	
2.	SITE PLAN SHOWING EXTERNAL E/M SERVICES	CE(AF)PRZ/1203/SP	2/2	19.12.24	
BLAST PEN					
3.	PLAN	CE(AF)PRZ/1203/AR-BLP	1/2	19.12.24	
4.	ELEVATION SECTIONS & DETAIL	CE(AF)PRZ/1203/AR-BLP	2/2	19.12.24	
5.	E/M PLAN AND SCHEDULE OF FINISHES	CE(AF)PRZ/1203/EM&SOF-BLP	1/1	19.12.24	
6.	FOUNDATION PLAN & PLINTH BEAM LAYOUT	CE(AF)PRZ/1203/STR-BLP	1/18	18.12.24	
7.	DETAIL OF COL & FOOTINGS, PLINTH BEAM DETAIL	CE(AF)PRZ/1203/STR-BLP	2/18	18.12.24	
8.	RCC DETAILS OF TRAVERSE WALL FOR BLAST PEN	CE(AF)PRZ/1203/STR-BLP	3/18	18.12.24	
9.	DETAILS OF RCC WALL OF BLAST PEN & TOE WALL OF BLAST PEN	CE(AF)PRZ/1203/STR-BLP	4/18	18.12.24	
10.	ROOF PLAN OF BLAST PEN, PLAN OF RCC VERTICAL WALL	CE(AF)PRZ/1203/STR-BLP	5/18	18.12.24	
11.	RCC & DETAIL, TYPICAL SECTION FOR RB1	CE(AF)PRZ/1203/STR-BLP	6/18	18.12.24	
12.	RCC & DETAIL, SECTIONAL DETAIL OF ROOF BEAM	CE(AF)PRZ/1203/STR-BLP	7/18	18.12.24	
13.	RCC SECTIONAL DETAIL OF ROOF SLAB, RCC DETAIL	CE(AF)PRZ/1203/STR-BLP	8/18	18.12.24	

Srl No	Description of Drgs	Drg No	Sht No.	Date of Drg	Date of revision

PARTICULAR SPECIFICATION (CONT'D…../-)

1.	2.	3.	4.	5.	6.
ORP ANNEXE & GARAGE					
14.	PLAN AND ROOF PLAN	CE(AF)PRZ/1203/AR-ANNEXE	1/4	19.12.24	
15.	ELEVATION AND DETAIL	CE(AF)PRZ/1203/AR-ANNEXE	2/4	19.12.24	
16.	SECTION AND DETAIL	CE(AF)PRZ/1203/AR-ANNEXE	3/4	19.12.24	
17.	TOILET DETAILS	CE(AF)PRZ/1203/AR-ANNEXE	4/4	19.12.24	
18.	EM PLAN	CE(AF)PRZ/1203/EM-ANNEXE	1/1	19.12.24	
19.	SCHEDULE OF FINISHES	CE(AF)PRZ/1203/SOF-ANNEXE	1/1	19.12.24	
20.	FOUNDATION PLAN, SCHEDULE OF FOOTINGS & COLUMN	CE(AF)PRZ/1203/STR-ANNEXE	1/3	18.12.24	
21.	PLINTH BEAM LAYOUT, SCHEDULE AND DETAIL	CE(AF)PRZ/1203/STR-ANNEXE	2/3	18.12.24	
22.	GROUND SLAB LAYOUT, BEAM SCHEDULE, LINTEL DETAILS AND OTHER RCC DETAILS	CE(AF)PRZ/1203/STR-ANNEXE	3/3	18.12.24	
23.	NOTES & GEN STR DETAIL	CE(AF)PRZ/1203/STR-GEN-ANNEXE	1/1	18.12.24	
RIGID PAVEMENT LAYERS					
24.	CROSS SECTION AND DETAILS OF JOINTS IN RIGID PAVEMENTS	CE(AF)PRZ/1203/STR-RIGPAV	1/1	18.12.24	
MSE TRAVERSE WALL, MSE DEFLECTOR WALL					
25.	PLAN	CE(AF)PRZ/1203/STR-BLP	9/18	18.12.24	
26.	ELEVATION OF MSE WALL	CE(AF)PRZ/1203/STR-BLP	10/18	18.12.24	
27.	ELEVATION OF MSE WALL (CLOSING WALL)	CE(AF)PRZ/1203/STR-BLP	11/18	18.12.24	
28.	CROSS SECTION WITH MECHANICALLY STABILIZED EARTH WALL (1-1')	CE(AF)PRZ/1203/STR-BLP	12/18	18.12.24	
29.	CROSS SECTION WITH MECHANICALLY	CE(AF)PRZ/1203/STR-BLP	13/18	18.12.24	

PARTICULAR SPECIFICATION (CONT'D…../-)

	STABILIZED EARTH WALL (2-2')				
30.	SECTIONAL PLAN (A-A')	CE(AF)PRZ/1203/STR-BLP	14/18	18.12.24	
31.	SECTIONAL PLAN (B-B')	CE(AF)PRZ/1203/STR-BLP	15/18	18.12.24	
32.	PRECAST RCC PANEL DETAIL (FOR HIGH ADHERENCE POLYMERIC STRAPS)	CE(AF)PRZ/1203/STR-BLP	16/18	18.12.24	
33.	PRECAST RCC PANEL DETAIL (FOR HIGH ADHERENCE STEEL STRIPS)	CE(AF)PRZ/1203/STR-BLP	17/18	18.12.24	
34.	TYPICAL DETAIL	CE(AF)PRZ/1203/STR-BLP	18/18	18.12.24	
CONSTRUCTION OF TWO SINGLE ENTRY ORP BLAST PENS AT RUNWAY ENDS ALONGWITH ASSOCIATED WORK SERVICES AT AF STN BAREILLY					
SITE PLAN					
35.	SITE PLAN SHOWING EXTERNAL B/R SERVICES	CE(AF)PRZ/1202/SP	1/2	19.12.24	
36.	SITE PLAN SHOWING EXTERNAL E/M SERVICES	CE(AF)PRZ/1202/SP	2/2	19.12.24	
ORP ANNEXE & GARAGE					
37.	PLAN AND ROOF PLAN	CE(AF)PRZ/1202/AR-ANNEXE	1/4	19.12.24	
38.	ELEVATION & DETAILS	CE(AF)PRZ/1202/AR-ANNEXE	2/4	19.12.24	
39.	SECTION & DETAILS	CE(AF)PRZ/1202/AR-ANNEXE	3/4	19.12.24	
40.	TOILET DETAILS	CE(AF)PRZ/1202/AR-ANNEXE	4/4	19.12.24	
41.	E/M PLAN	CE(AF)PRZ/1202/E/M-ANNEXE	1/1	19.12.24	
42.	SCHEDULE OF FINISHES	CE(AF)PRZ/1202/SOF-ANNEXE	1/1	19.12.24	
43.	FOUNDATION PLAN, SCHEDULE OF FOOTINGS AND COLUMN	CE(AF)PRZ/1202/STR-ANNEXE	1/3	18.12.24	
44.	PLINTH BEAM LAYOUT, SCHEDULE AND DETAIL	CE(AF)PRZ/1202/STR-ANNEXE	2/3	18.12.24	
45.	GROUND SLAB LAYOUT, BEAM SCHEDULE, LINTEL	CE(AF)PRZ/1202/STR-ANNEXE	3/3	18.12.24	

PARTICULAR SPECIFICATION (CONT'D…../-)

	DETAILS & OTHER RCC DETAILS				
46.	NOTES & GEN STR. DETAIL	CE(AF)PRZ/1202/STR/GEN/ANNEXE	1/1	18.12.24	
BLAST PEN					
47.	PLAN	CE(AF)PRZ/1202/AR-BLP	1/2	19.12.24	
48.	ELEVATIONS SECTIONS, ROOF PLAN & DETAIL	CE(AF)PRZ/1202/AR-BLP	2/2	19.12.24	
49.	INTERNAL E/M AND SCHEDULE OF FINISHES	CE(AF)PRZ/1202/E/M & SOF	1/1	19.12.24	
50.	LAYOUT OF FOUNDATION COLUMN, STRUCTURAL GEN NOTES AND CROSS SECTION OF FOOTING OF DIFFENT THICKNESS	CE(AF)PRZ/1202/STR-BLP	1/17	18.12.24	
51.	DETAILS OF COLUMN, PLINTH BEAM AND TYP DETAILS OF COLUMN AND PLINTH BEAM REINFORCEMENT	CE(AF)PRZ/1202/STR-BLP	2/17	18.12.24	
52.	STRUCTURAL DETAILS OF TRAVERSE WALL & TOE WALL WITH FILTER MEDIA AND TYP DETAIL OF CONSTRUCTION JOINT	CE(AF)PRZ/1202/STR-BLP	3/17	18.12.24	
53.	STRUCTURAL DETAILS OF RCC WALL AND COLUMN, BP WALL AND TRAVERSE WALL JUNCTION	CE(AF)PRZ/1202/STR-BLP	4/17	18.12.24	
54.	STRUCTURAL PLAN OF ROOF BEAM, ROOF SLAB AND DETAILS OF WALL REINFORCEMENT	CE(AF)PRZ/1202/STR-BLP	5/17	18.12.24	
55.	DETAILS OF ROOF BEAM	CE(AF)PRZ/1202/STR-BLP	6/17	18.12.24	
56.	DETAILS OF ROOF SLAB, ROOF PROJECTION, FILLING, BEAL COLUMN JUNCTION AND MISC DETAILS	CE(AF)PRZ/1202/STR-BLP	7/17	18.12.24	
MSE TRAVERSE WALL, MSE DEFLECTOR WALL					
57.	PLAN	CE(AF)PRZ/1202/STR-BLP	8/17	18.12.24	
58.	ELEVATION OF MSE WALL	CE(AF)PRZ/1202/STR-BLP	9/17	18.12.24	
59.	ELEVATION OF MSE WALL (CLOSING WALL)	CE(AF)PRZ/1202/STR-BLP	10/17	18.12.24	
60.	CROSS SECTION WITH MECHANICALLY	CE(AF)PRZ/1202/STR-BLP	11/17	18.12.24	

PARTICULAR SPECIFICATION (CONT'D…../-)

	STABILIZED EARTH WALL (1-1')				
61.	CROSS SECTION WITH MECHANICALLY STABILIZED EARTH WALL (2-2')	CE(AF)PRZ/1202/STR-BLP	12/17	18.12.24	
62.	SECTIONAL PLAN (A-A')	CE(AF)PRZ/1202/STR-BLP	13/17	18.12.24	
63.	SECTIONAL PLAN (B-B')	CE(AF)PRZ/1202/STR-BLP	14/17	18.12.24	
64.	PRECAST RCC PANEL DETAIL (FOR HIGH ADHERENCE POLYMERIC STRAPS)	CE(AF)PRZ/1202/STR-BLP	15/17	18.12.24	
65.	PRECAST RCC PANEL DETAIL (FOR HIGH ADHERENCE STEEL STRIPS)	CE(AF)PRZ/1202/STR-BLP	16/17	18.12.24	
66.	TYPICAL DETAIL	CE(AF)PRZ/1202/STR-BLP	17/17	18.12.24	
RIGID PAVEMENT LAYERS					
67.	CROSS SECTION AND DETAILS OF JOINTS IN RIGID PAVEMENTS	CE(AF)PRZ/1202/STR-RIGPAV	1/1	18.12.24	
TD DRAWING					
68.	TYPICAL SCHEDULE FOF FINISHES	CE/ALD/TD/303	1/2	18-04-22	
69.	TYPICAL SCHEDULE FOF FINISHES	CE/ALD/TD/303	2/2	18-04-22	
70.	GENERAL NOTES ON ARCHITECTURAL DRAWINGS, GENERAL SCHEDULE OF FINISHES & STANDARD 4 SPECIFICATIONS & LIST OF COMMON TD DRAWINGS.	CE/ALD/TD/199	1/1	29-02-12	28-01-14
71.	SOLID PANEL PVC DOOR	CE/ALD/TD-214	1/1	07-05-12	
72.	DETAILS OF ALUMINIUM SLIDING WINDOW	CE/ALD/TD-194 (R)	1/2	29.2.12	
73.	DETAILS OF ALUMINIUM SLIDING WINDOW	CE/ALD/TD-194 (R)	2/2	29.2.12	

PARTICULAR SPECIFICATION (CONT'D…../-)

74.	DEATIL OF ALUMINIUM DOOR & PARTITION WALL	CE/ALD/TD-224	1/1	17-10-12	
75.	TYP. DEATIL OF PLINTH PROTECTION, BK STEP, BK DRAIN, RAMP, RCC JALI & XPM PARTITION	CE/ALD/TD-196 (R)	1/1	29-02-12	27.1.14
76.	TYPICAL LOCATION OF, DETAILS, FITTINGS & FIXTURES OF TOILET	CE/ALD/TD-204	1/1	29-02-12	
77.	DETAIL OF FIXING OF EXHAUST FAN	CE/ALD/TD-197	1/1	29-02-12	
78.	DETAIL OF RWP, CI GRATING, BK PARAPET WALL	CE/ALD/TD-198	1/1	29-02-12	
79.	DETAIL OF XPM PARTITION, RAMP, BK STEPS, PLINTH PROTECTION AND RCC JALI	CE/ALD/TD-196 (R1)	1/1	29-2-12	27-1-14
80.	TYPICAL DETAILS OF FLOOR TRAP/NAHANI TRAP AND GULLY TRAP	CE/ALD/TD-205	1/1	29-2-12	
81.	TYPICAL DETAILS OF COOKING PLATFORM AND SERVICE COUNTER	CEAF/ALD/TD/203(R)	1/1	2-07-12	
82.	TYPICAL DETAILS OF 2-TIER RCC SHELF	CE/ALD/TD/240	1/1	01-4-14	
83.	DETAIL OF CB-01, CUB BOARD, CB, LOFT DOOR FWD RUNGS & HAND RAIL AT ROO LEVEL.	CE/ALD/TD/022	1/1	17.01.95	07.01.97
84.	PREVENTION OF LEAKAGE, SEEPAGE & DAMPNESS TO BUILDING.	CEAF/TD/31	1/1	5.10.95	20.11.98
85.	DETAIL OF COUNTER TYPE WASH HAND BESIN.	CE/ALD/TD/301-AR	1/1	12.03.22	
86.	DETAIL OF RWP AND SPLASH STONE	STD 121.1.2	1/1	22.7.85	
87.	GENERAL NOTES OF ARCH DRG. SCHEDULE OF FIISHES STANDARD SPECF.	CE/ALD/TD/146	1/1	26.5.05	31.7.6
88.	DETAILS OF WALL PEGS, CURTIAN ROD AND	TD-566 (R)	1/1	25.2.89	

PARTICULAR SPECIFICATION (CONT'D…../-)

	PELMET BOX				
89.	TYPICAL DETAIL FORLT DISTIBUTION POLE & STREET LIGHT FITTING ETC	CEAF/TD/E/M/08	1/1	16.3.93	01.11.03
90.	EARTHING PIT	CE/ALD/TD/E/M/23	1/1	09.06.03	24.06.03
91.	DETAIL FIXING OF EXHAUST FAN & DETAIL OF FAN HOOK BOX	CE/ALD/TD/197	1/1	29.02.12	
92.	TYPICAL DETAIL OF VALVE PIT	CEAF/TD/E/M/01	1/1	16.03.93	16.12.97
93.	DETAIL OF FIRE POINT	TD/228	1/1	17.06.13	
94.	PCC ROAD AND HARD STANDING	TD-252-ST	1/1	19.02.15	
95.	SEC THRU ROADS, PATH & DETAIL OF CULVERTS PCC/HUME PIPE	TD-575	1/2	16.06.89	04.10.89
96.	BRICK DRAIN (TYPICAL)	CE/ALD/TD/130	1/1	04.08.03	
97.	TYPICAL DETAIL OF SEPTIC TANK (BRICK CONST)	TD-217	1/2	25.01.68	07.03.03
98.	TYPICAL DETAIL & SCHEDULE OF SEPTIC TANKS (BRICK CAST)	TD-217	2/2	25.01.68	07.03.03
99.	TYPICAL SOAK WELL (PLAN, SECTION & DETAIL)	TD-576	1/2	16.6.89	
100	TYPICAL SOAK WELL (PLAN, SECTION & DETAIL)	TD-576	2/2	16.6.89	
101	TYPICAL DETAIL BK MANHOLE	TD-222	2/2	19.4.85	07.03.03
102	TYPICAL DETAIL OF MANHOLE WITH RCC SLAB & RCC COVER	TD-571 (R)	1/1	20.05.91	07.03.03
103	STANDARD ROAD CROSS SECTION FOR PLAN	MICS/CPT/01/2011	1/2	08.09.11	
104	STANDARD ROAD CROSS SECTION FOR PLAN	MICS/CPT/01/2011	2/2	08.09.11	
105	TYPICAL DRAIN	TD-577	1/1	16.06.89	
106	PCC HARD STANDING FOR GARRAGES	CE/ALD/TD/249-ST	1/1	03.12.04	

PARTICULAR SPECIFICATION (CONT'D…../-)

107	DETAIL OF RWP, CI GRATING AND BK PARAPET WALL	CE/ALD/TD/198	1/1	29.2.12	
108	TYPES OF CONCRETE BEDDINGFOR SEWERS	CE/ALD/TD/002	1/1	12.02.93	
109	RCC DRAIN COVER SLAB	CEAF/ALD/TD/47	1/1	4.8.97	
110	BRICK DRAIN (TYPICAL)	CEAF/ALD/TD-130	1/1	04.08.03	
111	GARBAGE BIN (RECTANGULAR)	CEAF/ALD/TD-136	1/1	4.12.13	
112	STATIC WATER TANK 110 KL CAPACITY	CE/ALD/TD/-253-ST	1/1	19.02.15	
113	DETAILS OF DRAIN AND DRAIN COVER AT RUNWAYS	CE/ALD/TD/-245-ST	1/2	31.05.14	
114	DETAILS OF DRAIN AND DRAIN COVER AT RUNWAYS	CE/ALD/TD/-245-ST	2/2	31.05.14	
115	SEPTIC TANK IN RCC FOR 50 TO 500 USERS	TD/750	1/2	21.03.18	
116	SEPTIC TANK IN RCC FOR 50 TO 500 USERS	TD/750	2/2	21.03.18	

Signature of Contractor

AAD Contracts)
For Accepting Officer

PARTICULAR SPECIFICATION (CONT'D…../-)

Appendix "A"

APPROVED MAIN PRODUCERS/MANUFACTURERS OF CEMENT

Sl No	Company Name	Brand Name	Address	Type/ Grade of cement
1	2	3	4	5
1.	The Associated Cement Companies Ltd	ACC	414-421, Splendor Forum (4th Floor), 3, District Centre, Jasola, New Delhi-110044 Ph: 011-46583600	All
2.	Ultra Tech Cement	Ultratech	'B' Wing, 2nd Floor, Mahakali Caves Road, Andheri (East), Mumbai-400093	All
3.	The India Cement	-	Dhun Building, 827, Anna Salai, Chennai-600002	All
4.	Dalmia Cement (Bharat) Ltd	Dalmia Infra Pro	Dalmiapuram, Distt: Tiruchirappalli, Tamil Nadu – 621651	All
5.	Century Cements	Century	Industry House, 159, Chruch gate, Reclamation, Mumbai-400020 Ph-022-22023936	All
6.	Saurashtra Cement	Saurashtra	Gala No A-1, Ground Floor, UdhyogSadan No 3 MIDC, Central Road, Andheri (East), Mumbai-400093, Ph 022 32955557/67 MO 9320290081	All
7.	The Ramco Cements Ltd. (Formerly Madras Cement)	Ramco	Auras Corporate Centre, 98-A, Dr. Radhakrishnan Salai, Mylapore, Chennai-600004 Ph-044 28478666	All
8.	Mangalam Cement Ltd	Mangalam	PO-Adityanagar,Morak,Dist-Kota, Rajasthan-326520 Mob-9351468076	All
9.	Birla Corporation Ltd	Birla	Birla Building (3rd & 4th Floor) 9/1 R.N Mukherjee Road Kolkata-700001, Ph-033-30573700	All
10.	Orient Cement	Orient	5-9-22/57/D, 2nd and 3rd Floor, GP Birla Centre, Adarsh Nagar, Hyderabad-500063 Ph-04423688600	All
11.	Nuvoco Vistas Corporation Ltd. (Formerly Lafarge Cement)	Nuvoco	Equinox Business Park, Tower-3, East Wing, 4th Floor ,LBS Marg, Kurla (West),Kurla Mumbai, Maharashtra-400070	All
12.	Shree Cement	Shree	Bangur Nagar, Beawar, Distt-Ajmer, Rajasthan-305901 Ph-01462228101/06	All
13.	J K Cement	JK	KamlaTower , Kanpur 208001	All
14.	JK Lakshmi Cement Ltd	J K Lakshmi	Jaykaypuram, Dist-Sirohi, Rajashthan Ph-02971 244409/10	All

PARTICULAR SPECIFICATION (CONT'D…../-)

Appendix "A" (Contd….)

APPROVED MAIN PRODUCERS/MANUFACTURERS OF CEMENT

Sl No	Company Name	Brand Name	Address	Type/ Grade of cement
1	2	3	4	5
15.	Jaypee Rewa Cement	Jaypee	Japee Nagar, P.O. JapeeNagar,Rewa-486450(M.P.)	All
16.	Ambuja Cement Ltd	Ambuja	Kodinar, PO-Ambujanagar, Taluka-Kodinar, Distt :Junagadh,Gujrat-362715 Ph-02795237000	
17.	M/s Shree Guru Kripa Cement (Pvt) Ltd.	Sartaj	4/4 ,Trikuta Nagar Jammu-180012 Ph. 0191-2472043	OPC 43 & PPC
18.	M/s Parasakti Cement Industries Ltd.	Parasakti Cement	123/3 RT, Plot # 8-3-214/21, Srinivasa Nagar Colony (West) Hyderabad-500038 Tel-040-44119100/200, Fax-040-23747562	OPC 43 & PPC
19.	M/s Chettinad Cement Corporation Ltd.	Chettinad Cement	4th Floor, Rani Seethai Hall Building, 603, Anna Salai Hall, Chennai-600006, Tel No. 044-42951800 Fax No. 044-28291558 Email:info@chettinadcement.com Website:www.chettinadcement.com	OPC 43 & PPC
20.	M/s JSW Cement Ltd AP	JSW OPC 43 JSW OPC 53 JSW PSC	JSW Centre, Opp MIMRDA Ground, Bandra Kurla Complex, Bandra East, Mumbai -400051, Tele -91-22-4286-5047, Fax :-91-22-2650-2001 Website : www.jswcement.in	OPC 43, OPC 53 & PSC
21.	M/s Sagar Cements Ltd	Sagar	M/s Sagar Cements Ltd , Plot No.111 , Road No. 10 , Jubilee Hills , Hyderabad-500033, Tele : 91-40-23351571 Fax : 91-40-2335673, Email:- info@sagarcements.in Website : www.sagarcements.in	OPC 43, OPC 53, PPC & PSC
22.	M/s Sanghi Industries Ltd.	Sanghi	10th Floor, Kataria Arcade, Off SG Highway, PO-Makarba, Dist-Ahmedabad,Pin-380051, Mob : 08238093065, Tel: 07926838000 Fax No: 07926838111 Website : www.sanghicement.com	OPC 43 & PPC
23.	M/s My Home Industries Ltd	Maha Cement	9th Floor,Block-3,My Home Hub, Madhapur,Hydrabad-500081 Ph-040-6692-9696 Fax -040-66929797/98 Email:- corp@myhomegroup.in Website: : www.myhomeindustries.com	OPC 43, PPC & PSC
24.	Wonder Cement Ltd.	Wonder Cement	17, Old Fatehpura Seva Mandir Road, Udaipur-13004 Rajsthan(India), Tel : +91-294-33991133, Fax: +91-294-3006333 Email:corp.office@wondercement.com	OPC 43, OPC 53 & PPC

Note:- Contractor can procure cement from those makes/ brands as approved by the AHQ E-in-C's Branch even during currency of work.

PARTICULAR SPECIFICATION (CONT'D…../-)

Appendix 'B'

CEMENT SUPPLY & ACCEPTANCE REGISTER

1. C. A. No. & Name of Work :
2. Control No. :
3. Name of Manufacturer/Brand Name/Grade of Cement :
   (A) Manufacture :
   (B) Brand :
   (C) Grade :
4. Quantity of cement & Lot No./Week No. (In Bags) :
   (a) Quantity :
   (b) Lot No./Week No. :
5. Manufacturer's test certificates No. :
6. Random test Details:
   (a) Physical test report from_____ vide their letter No.:
   (b) Chemical test report form_____ vide their letter No. :

7. Details of Physical & Chemical properties

	Physical Requirements (As per IS-4031)										Chemical Requirements (As per IS-4032)							
	Specific Surface (Sq.m/Gram)	Soundness by Le Chatlier Expansion (%)	Soundness by Auto Clave Expansion	Initial setting time (Minutes)	Final setting time (Minutes)	Compressive Strength 03 Days (Mpa)	Compressive Strength 07 Days (Mpa)	Compressive Strength 28 Days (Mpa)	Temp during testing 0C	Standard Consistency (%)	Lime saturation Factor (Ratio)	Alumina Ion Ratio (Ratio)	Insoluble Residue (%)	Sulphuric Anhydride (%)	Loss on ignition (%)	Alkalis (%)	Chlorides (%)	Magnesium (%)
As per Relevant IS																		
As per manufacturer's test certificate																		
As per random test certificate																		

Remarks with signature :

Junior Engineer (Civil) :
Engineer-in-Charge :
Contractor :
Accepted/Rejected :

Garrison Engineer :

Remarks of BOO/Inspecting Officer/CE Zone :

PARTICULAR SPECIFICATION (CONT'D…../-)

Appendix "C"

Sl No.	Name of materials	Location of Query	Condition of crusher	Remarks
1.	Crushed stone aggregate 20 mm & below conforming to IS :383 PQC/RCC/PCC/DLC/ WMM/GSB	Jhansi (UP)/Kalpi/Haldwani	With Valid mining License issued by competent authority and should have equipped with vertical shaft impact machine (VSI)	Copies of documents of mining licensend and VSI installation shall be submitted at the time of submission of sample.
2.	Crushed stone aggregate 40 mm above & Hardcore conforming to IS :383 PQC/RCC/PCC/DLC/ WMM/GSB	Jhansi (UP)/Kalpi/Haldwani	With Valid mining License issued by competent authority	Copies of documents of mining licensed shall be submitted at the time of submission of sample
3.	Fine aggregate (River sand) for RCC/PCC/PQC/DLC/ Brick work	Kalpi/Banda		Shall confirm to Zone II grading
4.	Coarse sand	Kalpi/Banda		Shall confirm to Zone II grading

NOTES :-

(a) Sources of materials shall be as given in Srl No 1 to 4 above or in the vicinity thereof. The tenderer shall ascertain the actual position/exact location of source before submitting his tender and no additional payment shall be made on account of misunderstanding or its distance from site of works.

(b) The tenderer shall amongst other things also ascertain all information such as royalties, taxes, duties and other charges etc, on the materials and no additional payment shall be made on account of the foregoing.

(c) However, if due to any ban imposed by the state Govt. authority on obtaining the materials from the sources specified above and/or specified materials is not available at the sources mentioned above, these materials conforming to specifications as given in particular specifications will be procured from any other place/source after getting the same approved from GE in writing and without any extra cost to Govt.

PARTICULAR SPECIFICATION (CONT'D…../-)

Appendix "D"

APPROVED MAIN PRODUCERS OF STEEL FOR REINFORECEMT (TMT BARS)

Sl No	Company Name	Brand Name	Address
1	2	3	4
1.	RashtriyaIspat Nigam Limited (RINL),	RINL	Visakhapatnam Steel Plant, Visakapatnam-530031, India Tel: (91891) 518226, 518376 Fax: (91891) 518316 E-mail: cmdvsp@itpvis.ap.nic.in
2.	Tata Iron & Steel Company (TISCO, or Tata Steel),	TATA	Bombay House,2, 4, Homimodi Street, Mumbai-400001, India, Tel: (9122) 2049131 & Fax: (9122) 2049522, 2770840 E-mail: corpcomm@jsr.tatasteel.com (Br office for North: Jeevan Tara Bldg, Patel Chowk, New Delhi)
3.	Steel Authority of India Limited (SAIL)	SAIL	Central Marketing Organization, Northern Region 17th Floor, scope Minar, Laxmi Nagar Distt Centre, Delhi-110092
4.	M/S Jai Balaji Industries Ltd.,	Balaji Shakti	5 Bentek Street, Kolkata – 700001 Delhi Office 510, Block-B, Navraung House, 21 Kasturba Gandhi Marg, New Delhi-110001 Tel: (011) 43620219, 43620220 Mob: 7838272772, 9958936103 E-mail: info@jaibalajigroup.com
5.	M/S Shyam Steel Industries Ltd	Shyam	Shyam Towers EN-32, Sector-V, Salt Lake, Kolkata-700091 Tel: (033) 40074007 Fax: (033) 40074010 E-mail: **marketing@shyamsteel.com**
6.	M/S SPS Steel Rolling Mills Ltd	Elegant TMT	"Diamond Prestige",41A, A.J.C Bose Road, 8th Floor, Room No. 801, Kolkata-700017 Mob-9831055568/8910797649 Telefax: 033-66255252 Email: info@spsgroup.co.in
7.	M/S Steel Exchange India Ltd.,	Simhadri TMT	My Home Laxminivas Apartments, Ameerpet, Hyderabad-500016, A.P. Tel: (040) 23403725 Fax: (040) 23413267, E-mail: **info@seil.co.in**
8.	M/S Jindal Steel and Power Ltd.	Jindal Panther	OP Jindal Road, Hissar , Haryana -125005, Tel: +91 166 2222471-84 Fax: +91 166 220476
9.	M/S SRMB Srijan Ltd.,	SRMB	SRMB House, 7, Khetra das lane, Kolkata-700012 Tel: 033-6600 6600 & Fax: 033- 22110483
10.	Shri NakodaIspat Ltd	Nakoda TMT	Near Railway Crossing ,Mowa , PO Shankar Nagar , Raipur-492007 (CG), Tele -91-771-2282130, Website www.nakodagroup.com
11.	M/S Shri Bajrang Power &Ispat Ltd	Goel TMT	Vill- Borjhara, Urla Industrial Area, Raipur-493 221, Chhattisgarh, Tel: 0771 4288019/29/39 Fax : 0771-2323601/602,4288123 Email- infobir@goelgroup.co.in & commercial.bir@goelgroup.co.in

PARTICULAR SPECIFICATION (CONT'D…../-)

Appendix "D" (Contd….)

APPROVED MAIN PRODUCERS OF STEEL FOR REINFORECEMT (TMT BARS)

Sl No	Company Name	Brand Name	Address
1	2	3	4
12.	M/S JSW Steel Ltd.,	Neosteel	JSW Centre,Bandra Kurla Complex, Bandra (Eat) Mumbai- 400051, Maharashtra, Phone :022-42861000 Fax: 022-42863000
13.	M/S ShyamSel& Power	SEL	SS Chambers , S.C.R Avenue, Kolkata - 700046 Ph: 91 3340111000 Website: www.shyamgroup.com
14.	M/S Kamachi Industries Ltd.	Kamachi	ABC Trade Centre, 3rd Floor (Inside Devi Theatre Complex) Old No. 50, New No. 39, Anna Salai, Chennai- 600002, India Tel: +91-044-42961100 Fax: +91-044-42961122 E-mail: sales@kamachitmt.com Website: **www.kamachitmt.com**
15.	M/S Gallantt Metal Ltd	Gallantt TMX	Ward 10BC, Plot No. 123, Ground Floor, Gandhi Dham Kutch, Gujarat - 370201 Tel: +91-2836-228164 & Fax: +91-2836-235787 E-mail: gml@gallantt.com& Website: www.gallantt.com
16.	M/S Rashmi Metaliks Ltd	Rashmi TMT	Premlata Building, 39, Shakespeare Sarani, 6th Floor, Kolkata - 700017 Tel: 033-22894255/56 & Fax: 033-22894254 E-mail: mkt.domesticdip@rashmigroup.com Website: www.rashmigroup.com
17.	M/S Real Ispat& Power ltd	G K TMT	Vrindavan, Near IDBI Bnk Civil Lines Raipur - 492001, C.G. Tel: +91-771-4224000 Fax: +91-771-4224010 E-mail: real@realispat.com Website: www.realispat.com
18.	M/S Super Smelters Ltd. Kolkata	Super Shakti	Premlata, 39, Shakespeare Sarani, 3rd Floor, Kolkata - 700017 Tel/Fax: + 91-33-2289-2734/36 E-mail: info@supershakti.in Website: www.supershakti.in

Note: (i) Contractor can procure TMT bars from the primary producer approved by the AHQ E- in-Cs Branch even during currency of the contract work.
(ii) The documents in support of purchase of steel shall be furnished to GE for verification. The particulars of the manufacturers of steel shall be furnished by the contractor for every lot of steel separately as per the Proforma given

PARTICULAR SPECIFICATION (CONT'D…../-)

Appendix "E"

APPROVED MAIN PRODUCERS OF STRUCTURAL STEEL

Sl No	Company name	Brand name	Address	
1.	M/S RashtriyaIsp at Nigam Limited (RINL)	RINL	Visakhapatnam Steel Plant Visakhapatnam - 530 031, India Tel: (91 891) 518226,518376 Fax: (91 891) 518316 E-mail: cmdvsp@itpvis.ap.nic.in	
2.	M/S Tata Iron & Steel Company (TISCO, or Tata Steel)	TATA	Bombay House, 2, 4 Homi Modi Street Mumbai - 400 001, India Tel: (91 22) 204 9131 Fax: (91 22) 204 9522, 287 0840 E-mail: corpcomm@jsr.tatasteel.com (Br office for North : Jeevan Tara Bldg, Patel Chowk, New Delhi)	
3.	M/S Steel Authority of India Limited (SAIL)	SAIL	Central Marketing Organization, Northern Region 17th Floor, Scope Minar, Laxmi Nagar Distt. Centre, Delhi - 110092	
4.	M/S Jindal Steels and Power Ltd.	JINDAL	Jindal Centre, Plot No 2, Sector- 32, Gurgaon- 122001, Haryana Tel: 0124 661 2000 Fax: 0124 661 2125 Website: www.jindalsteelpower.com	

PARTICULAR SPECIFICATION (CONT'D…../-)

Appendix 'F'

STEEL SUPPLY/ACCEPTANCE FORM

Contract No. :
Name of Work :
Control No. :
Details of Purchase :
(i) Particulars of Manufacturers :
(ii) Details of Suppliers, if any :
Details of Test Certificate :
(i) No. and Date :
(ii) Particulars of Issuing Authority :

Serial No.	Nomenclature and size of steel	IS Reference	Quantity (Tons)		Physical properties					Remarks
					UTS (N/Sq.mm)		Elongation (%)		Bend Test	
			Actual	Conversion	IS Test Sheet	Random Test*	IS Test Sheet	Random Test*		
1	2	3	4	5	6	7	8	9	10	11

* As per frequency of testing given vide clause 10.3.6 above

Remarks with signature :
Junior Engineer (Civil) :
Engineer-in-Charge :
Contractor :
Accepted/Rejected :
Garrison Engineer :
Remarks of BOO/Inspecting Officer/CE Zone :

TEST CERTIFICATE

No. of Test :
Name of Articles(s) :
Samples submitted by :
Reference to the letter No/Date :
Test No.

Serial No	Sample (s)	Carbon (C%)	Sulphur (S%)	Phosphorus (P%)	Manganese (Mn%)	Silicon (Si%)	Chromium (Cr%)	Nickel (Ni%)	Copper (Cu%)	Tin (Sn%)	Lead (Pb%)	Ferrous (Fe%)	Zinc (Za%)	Aluminium (Al%)	Impurities %	Wt of Za	UTS	% Elongation	Proof Stress MPa

Remarks with signature :
Junior Engineer (Civil) :
Engineer-in-Charge :

Contractor :
Accepted/Rejected :
Garrison Engineer :

Remarks of BOO/Inspecting Officer/CE Zone :

PARTICULAR SPECIFICATION (CONT'D…../-)

Appendix 'F'(Contd….)

TEST REPORT

Name of Article material :

Particulars Identification Mark :

Name of the firm manufacturer :

This is to certify that the article, the particulars of which are given has been tested in this centre another test results are as follows:

Chemical tests on metal (Percentage by Weight)								
Ferrous	C	S	P	Mn	Si	Ni	Cu	Mo

Non Ferrous	Sn	Pb	Cu	Fe	Zn	Others

Serial No.	Weight/Metre	Weight/Metric	Mechanical/Physical Tests		Elongation	Bend Test	Re-Bend Test	Miscellaneous	Remarks
			Yield/Proof Stress (N/Sq.mm)	Tensile Strength (N/Sq.mm)					

Remarks with signature :

Junior Engineer (Civil) :

Engineer-in-Charge :

Contractor :

Accepted/Rejected :

Garrison Engineer :

Remarks of BOO/Inspecting Officer/CE Zone :

PARTICULAR SPECIFICATION (CONT'D…../-)

APPENDIX 'G'

(MATERIALS SHALL BE ISI MARKED)

APPROVED NAMES/MAKES FOR PRODUCTS TO BE INCORPORATED IN WORKS:

(a) In addition, makes/brands of materials/items required in the work can also be of Group I as approved in any of the Command Make list.

(b) If no make is available for any item, the same shall be procured of ISI marked. However, no ISI marked materials/Items are manufactured, the same shall be confirming to IS specification.

Sl No.	Item Name	Makes
1.	2.	3.
1.	Non skid/Glazed Ceramic and Vitrified tiles	Kajaria Johnson Orient Bell Ltd (Brand –Orient Bell) Somany Ceramics Ltd (Brand: Somany) RAK Ceramics India Pvt Ltd (Brand- RAK Ceramic)
2.	Wooden flooring	HIL Ltd Greenpanel Industries Ltd Armstrong Action Tesa
3.	Exterior Paints	Asian Paints Ltd Berger Paints India Ltd Kansai Nerolac Paints Ltd
4.	Interior Paint	Asian Paints Ltd Berger Paints India Ltd Kansai Nerolac Paints Ltd
5.	Synthetic enamel /Oil Paint	Asian Paints Ltd (Apcolite Premium Gloss) Berger Paints India Ltd (Luxol Hi-Gloss) Kansai Nerolac Paints Ltd (Synthetic Hi-gloss Enamel)
6.	Cement base paint	Super Snowcem Duracem Plus Acro Paints Ltd
7.	High Build Textured & Wall finishes	Asian Paints Ltd Berger Paints India Ltd Kansai Nerolac Paints Ltd
8.	UPVC Shutters for doors, windows , ventilators & Frames	Fenesta Koemmerling Aluplast Prominance NCL Veka Ltd (Brand : NCL Veka)
9.	Factory made Wooden flush door Panel/Gauge shutters for doors & widows	Rama Woods Crafts Satna A-1 Teak Products Pvt Ltd Greenply Industries Ltd Greenpanel Industries Ltd Century Plyboard (India) Ltd (Brand-Century Doors) Jain wood Industries
10.	Pre-painted Steel Windows, Ventilators & Door Frames	Tata Pravesh NCL Buildtek Limited Elixir Met Form Pvt Ltd Ashish Industries
11.	Metal False Ceiling	PR Ceiling Products Manraj Ceiling Products Pvt Ltd (Brand: Royal Kraft) Armstrong
12.	Gypsum Board false ceiling/Gypsum Plaster	Everest Industries Ltd PR Ceiling Products Ltd Gyproc Armstrong USG Boral

PARTICULAR SPECIFICATION (CONT'D…../-)

		Yamuna Interiors Pvt Ltd (Brand : Black cobra) JK Cements Ltd
13.	Non-Asbestos Fibre Cement Board (Confirming to IS-14862-2000) False ceiling	Everest Industries Ltd Shera Kalsi Prima Duragreen
14.	Glass reinforced Gypsum (GRG) False ceiling	Gyproc Armstrong USG Boral
15.	Plain Particle board/ Particle Board ceiling, Pre-laminated Particle board	HIL Ltd Kitply Greenpanel Industries Ltd Century Ply Greenply
16.	Aluminium composite Panels (ACP)	Aludecor Alstrong Alucobond Alstone Viva Ashoka Foam Multiplast Ltd (Brand: Alutuff ACP & Alutuff AHPL) Virgo Laminates Ltd Alexia Panels (Brand : Alex) Alutech Industries, New Delhi Vemicol Aluminium Tech Industries
17.	Laminates/Veneers	Marino Laminates Royale Touche Laminates Greenlam Laminates Century Laminates Kitply Sunmica Formica Greenpanel Industries Ltd Virgo Laminates Ltd Alexia Panels (Brand : Alex)
18.	Plywood for General purpose	Century Ply Greenply Kitply Greenpanel Industries Ltd Jain wood Industries
19.	CP Bath fittings and accessories	Jaquar & Co Pvt Ltd (Brand: Jaquar) Kajaria Bathware Pvt Ltd Cera sanitaryware Ltd Grohe Kohler
20.	Vitreous China Sanitary ware WHB/WC/Urinal	Jaquar& Co Pvt Ltd (Brand : Jaquar) Kajaria Bathware Pvt Ltd Cera Sanitaryware Ltd
21.	Metropole & concealed flushing cistern	Jaquar Grohe Kohler
22.	Seat Cover for EWC	Kajaria Bathware Pvt Ltd Cera Sanitaryware Ltd Jaquar Grohe Kohler
23.	PVC Flushing Cistern	Kajaria Bathware Pvt Ltd Cera sanitaryware Ltd Parryware
24.	Stainless Steel Sinks	Johnson Nirali

PARTICULAR SPECIFICATION (CONT'D…../-)

		Hindware
25.	CPVC pipes and fittings	Astral Ltd (Brand : CPVC Pro) Ashirvad Supreme RC Plasto Tanks & Pipes Pvt Ltd HIL Ltd Savour Fair Manufacturing Co Pvt Ltd (Brand- SFMC)
26.	UPVC pipes and fittings	Astral Ltd (Brand : Aquasafe & Aquarius) Ashirvad Supreme RC Plasto Tanks & Pipes Pvt Ltd HIL Ltd Savour Fair Manufacturing Co Pvt Ltd (Brand- SFMC)
27.	HDPE pipe /MDPE Pipe/ PVC pipes and fittings	Finolex Astral Ltd (Brand : HDPE Pro) RC Plasto Tanks & Pipes Pvt Ltd
28.	PPR pipes and fittings	Prince Finolex Supreme Savour Fair Manufacturing Co Pvt Ltd (Brand- SFMC)
29.	ERW MS pipe	Surya Roshini Ltd Tata Jindal
30.	PVC Soil, Waste , Rainwater (SWR) pipes and fittings [Confirming to IS-13592 for pipes & IS-14735 for fittings]	Astral Ltd (Brand : Drain Master) HIL Ltd Supreme
31.	PVC UG drainage pipes Confirming to IS-16098 (Part-I)	Astral Ltd (Brand : Foam Core) Prince Finolex
32.	DWC UG drainage pipes Confirming to IS-16098 (Part-II)	Astral Ltd (Brand : D-rex) Prince Finolex
33.	DI pipe and fittings	TATA Electro Steel Jindal Saw Ltd Electrotherm India Ltd (Brand-Electrotherm)
34.	GI Pipe and fittings	TATA Jindal Hisar Surya Roshini Ltd
35.	Ready Mix concrete	Ultra Tech ACC JK Lashmi Cement Lafarge Ramco
36.	AAC Block	Ultra Tech ACC Birla Aerocon Ascolite Siporex
37.	BUILDERS HARDWARE	Novapan GLACIAR Brands Classic New Delhi M/S Shri Balaji Industries Sarswati Round Road, Ajamabad Mathura (UP). DP Garg & Co. Mepro M/S Elte Enterprises C/6 Shalimar Hardware 133,

PARTICULAR SPECIFICATION (CONT'D…../-)

		Jarg Mahal, Dhobitalao, Bombay M/S Mohan Metal Ind.
38.	Mortice lock	Godrej Harrison Spider Metal Product Ltd Dorset India Jainson Links Lock
39.	Drapery Rod	Vista Levolor MAC-Decor DECO Windows Dorset India Jainson LUXAFLEX
40.	Venetian Blinds	Vista Levolor Aerolux MAC Johnson Blinds Classic Blind LUXAFLEX
41.	Barbed wire/Chain link/ Concertina coil/Razor Wire	A-1 Fence Product Company Pvt Ltd Shankar Weldmesh Pvt Ltd Tata Steel Global Wires Secure Fencing, Delhi Eco Netting Nagpur OM Wires and Wire Product, Kolkatta Oswal wire mesh Co., Bangalore Mysore Wire netting Co Pvt Ltd, Bangalore Asian Fence Wire Product, Hyderabad
42.	Plain/ Corrugated Galvalume sheet	M/S Crill colour Roof (India)Ltd M/S Shiv Shakti Fibre udyog Faridabad (UP) M/S Ispat Industries Ltd M/S Met Form Corpn (Unit of Thakkar Group) Lloyed deck, lloyed insulation (India) M/S Japan Metal Building Systems Pvt Ltd M/S Kamdhenu Ltd TATA Blue Scope JSW COLOURON+ Kirby Building Solutions (I) Ltd.
43.	Non-Asbestos Plain/ Corrugated Sheets	M/s Ramco AC Sheets Charminar M/S Eternit Everest Ltd SWASTIK Visaka UP Asbestos (UPAL) RBL
44.	Adhasive	Fevicol Pidilite Vemicol Araldite Anabond
45.	Wall Care Putty	Birla white wall putty JK wall putty Asian wall putty Berger Dulux Nerolac wall putty Shalimar Jenson & Nicolson
46.	Sheet Glass/ Frosted Glass	Modi Guard Saint gobain AIS Atul Glass Industries Gold Fish

PARTICULAR SPECIFICATION (CONT'D…../-)

		Trupti
47.	Oil Putty	Shalimar Hardware Gold Mohar Atul Dyes and Chenicals UK Paint Industries Rajdoot Berger Jenson & Nicolson
48.	Brass Float Valve	Leader *Zoloto* *BIR Jalandhar* *Vardhman* Upadhyay Kirloskar Sant Bombay Metal & alloy Co, Mumbai
49.	PTMT Float Valve	Prayag Polymers Polytuf Plasto Vectus
50.	HDPE Water Storage Tanks	Sintex Polycon Rolex Infra JS Polyplast Dhara Vectus Plasto Supreme
51.	Hydraulic Door closer	Godrej Hafele Harrison Ozone Kich Link Locks
52.	Floor Door Closer	Godrej Hafele Harrison Ozone Kich
53.	Transparent Polycarbonate Roofing Sheets	Everest Saraswati Plastotech India Pvt Ltd Palram Industries Ltd
54.	Tile adhesive	JK Lakshmi Cement Tile Adhesive Myk Laticrete Tile Adhesive Berger Tile Adhesive Asian paints tile adhesive Dulux tile adhesive BASF India Ltd tile adhesive Saint-Gobain Weber tile adhesive Fosroc India tile adhesive
55.	Road Furniture	Dark Eye Green Eye 3M Automat Shakti
56.	Branded Furniture	Godrej Durian Euroacoustic Products Ltd Usha Lexus
57.	Modular Kitchen	Godrej Interio Kohler Hafele Hettachi India Pvt ltd Johnson

PARTICULAR SPECIFICATION (CONT'D…../-)

		Anchor by Panasonic Sleek Kitchen
58.	Single Component UV Resistance PU coating for roof & wall	Smart Care from Asian Paints Dr Fixit SIKKA Berger Dulux
59.	PU MAGNUM	Smart Care from Asian Paints Dr Fixit SIKKA
60.	SEWER BRICKS	M/S Priya Clay Products Pvt Ltd, New Delhi, M/S Ritu Bricks Pvt Ltd, Bhilwara Rajasthan, M/S Priya Bricks Pvt Ltd, Kolkata. M/S Red Ceramics, Limited, Kolkata. M/S Modern Bricks & Engg works, Bassi, Patiala M/S Prayag Bricks, Varanasi. M/S Maruthi Bricks works1 9/7,5thfloor, 'MaruthimansionCunninghamRoad,Banglore-560052,Karnatka
61.	Lifts/Elevators	Otis Elevator company India Ltd Kone India Schindler India Pvt Ltd Hitachi Lifts India Pvt. Ltd
62.	Engine	Kirloskar Cummins Greeves Cotton Caterpillar Eicher Engines
63.	Alternator	Kirloskar Stemford Crompton Jyoti
64.	DG Set assembled with sound proof canopy	Kirloskar Cummins Greeves Cotton Sudhir
65.	Single/Double Girder EOT Cranes, Goliath Cranes, Jib Cranes	Safex Energy Pvt. Ltd Pioneer Cranes & Elevators Pvt Ltd Hercules Hoists Ltd
66.	Compact Sub Station	Seimens Schneider Crompton ABB Indian Transformers & Electricals Pvt Ltd. (Brand- ITE Gurgaon)
67.	Transformer 11/0.433 KV upto 500 KVA copper wound	Crompton Schneider Indian transformers & electricals Pvt Ltd [Brand : ITE Gurgaon] Transtron Electricals Pvt Ltd
68.	Transformer stepdown 11/0.433 KV above 500 KVA copper wound	Crompton Schneider Indian transformers & electricals Pvt Ltd [Brand : ITE Gurgaon] Transtron Electricals Pvt Ltd
69.	Transformer stepdown 33/11 KV upt 10 MVA & 33/0.433 KV copper wound	Crompton Schneider Indian transformers & electricals Pvt Ltd [Brand : ITE Gurgaon] Transtron Electricals Pvt Ltd
70.	HT Current & Potential Transformer	Schneider Crompton Greaves Siemens

PARTICULAR SPECIFICATION (CONT'D…../-)

		ABB
71.	Chiller for Central AC Plant	Voltas Carrier Bluestar Lloyd Corporation Ltd
72.	Cooling Coil/AHU	Voltas Blue Star Zeco Accel Carrier
73.	Cooling Tower	Paharpur Mihir Delta Advance
74.	Expansion Valves/Thermostat/Humidistat	Danfoss Sporlon Indfoss Honeywell
75.	Package type Compressor/Condenser/Cooling coil	Bluestar Hitachi Daikin Voltas
76.	Split Type AC and Window Type AC	O General Mitsubishi Electric Hitachi Daikin Voltas Carrier
77.	VCB/SF6 Circuit breakers/ HT Switchgears panels upto 36 KV	Schneider ABB Siemens Crompton Indian Transformers & Electricals Pvt Ltd. (Brand- ITE Gurgaon)
78.	HT Ring Main Units (RMU) upto 36 KV	Schneider ABB Siemens Crompton
79.	APFC relay	L&T Schneider Ducati
80.	Power factor improvement capacitor	L&T MD XL Ultra Schneider Siemens ABB Neptune Havells
81.	APFC panels (Fabricators)	Shalabh India Industries Roorkee Indo Asian Power Systems and Control Jalandhar REW Constructions Pvt Ltd Indian Transformers & Electricals Pvt Ltd. (Brand- ITE Gurgaon)
82.	HT XLPE Cable upto 33 KV as per IS	Havells India Ltd Polycab Wires Pvt Ltd Gloster Cables Ltd KEI industries Ltd
83.	Modular Switch/ Socket/Fan regulator/ Bell push & mounting accessories	Havells India Ltd (Brand : Carbtree) Anchor Electrical Pvt Ltd Legrand Schneider Zencelo
84.	Non- Modular Switch/ Socket/Fan regulator/ Lamp holder/ Ceiling	Havells India Ltd (Brand : Havells Reo) Anchor Penta PC

PARTICULAR SPECIFICATION (CONT'D…../-)

	rose/Bell push/Call bell (Polycarbonate body only)	SSK
85.	Ceiling Fans ISI marked & not less than 4 BEE star rating / Exhaust Fans/ Air circulators/ Wall Mounting Fans/Cabin fans/Fresh Air fans	Havells Bajaj Orient Usha Crompton Khaitan
86.	BLDC ceiling fan	Havells Bajaj Orient Atomberg Crompton Khaitan
87.	LED Lights & light Fittings for Indoor and Outdoor use	Signify Innovations India Ltd (Brand-Philips & Ecolink) Orient Bajaj Havells Halonix Crompton
88.	Solar Based Street Light fittings	Havells Halonix Philips Bajaj Crompton
89.	POTENTIAL TRANSFORMER (PTs) AND CURRENT TRANSFORMER (CTs) VOLTAGE CLASS 33 KV & 11 KV	Crompton Greaves Kappa ABB Pragati
90.	ELECTRO MECHANICAL/ SOLID STATE MICRO PROCESSOR BASED RELAYS	ABB Siemens Schneider L&T C&S
91.	LT XLPE cable 1100V grade (as per IS 7098/1988 Part-I	Gloster Havells Polycab Grandlay Shalabh Plaza SS Cables KEI Gemscab Bentec (Brand : Benlo) HPL
92.	DIGITAL RELAY	L&T JVS ABB Ashida C&S CG
93.	Air insulator trunk bus bars	C&S Legrand Schneider L&T
94.	COLD/HEAT SHRINKAGE CABLE JOINTS 33KV/11KV GRADE & BELOW	Cabseal 3M Denson Reychem
95.	LT PANELS/ FEEDERPILLAR BOX	Power system and control Sahlabh Indo Asian REW Construction

PARTICULAR SPECIFICATION (CONT'D…../-)

		HPL Bentec (Brand : Benlo) PSC Indian Transformers & Electricals Pvt Ltd. (Brand- ITE Gurgaon)
96.	AIR CIRCUIT BREAKERS (ACBS)	L&T Siemens Crompton Greaves Legrand ABB Schneider
97.	MOULDED CASE CIRCUIT BREAKERS (MCCBs), SFU	Schneider L&T Siemens Legrand ABB HPL Bentec (Brand : Benlo)
98.	CHANGE OVER SWITCHES	L&T Legrand C&G Change Over Switch Bentec (Brand : Benlo) HPL
99.	Distribution Board (SPN/TPN)	Schneider Legrand ABB L&T GE HPL Bentec (Brand : Benlo)
100.	MCB/MCB PLUG SOCKET/ELCB/ RCCB & MCB DBs WITH PLUG-SOCKET	ABB Legrand L&T Schneider Hager HPL Bentec (Brand : Benlo)
101.	ACSR CONDUCTOR	Jaipuria Brothers ALIND Apar Industries Oswal Cables Mahavir Transmission
102.	SOLAR WATER HEATER	Supreme V-Guard Havells Recold Venus
103.	GANG OPERATED AIR BREAK ISOLATORS & DO FUSES FOR 11 KA.33KV SYSTEM	Jaipuria Brothers Electrolites Power (PG) Rashtriya Haryana Elect Industries PACTIL BHEL
104.	INSULATORS HT/LT	Jaipuria Brothers WS Insulators Jayshree Insulators Modern Insulators Indian Potteries Aditya Birla Insulators Allied Ceramics
105.	LIGHTENING ARRESTOR/ SURGE PROTECTION SYSTEM	Jaipuria Brothers Electrolites

PARTICULAR SPECIFICATION (CONT'D…../-)

		CG Oblum Schneider PACTIL
106.	STEEL TUBULAR POLES	Utkarsh National Tubing Co., Kanpur BCW Bharat Conduits and Steelwork Roll Tubes Ltd (PVVNL)
107.	MULTI STRANDED FLAT CABLES/ CONTROL CABLES/ SUBMERSIBLE CABLES	HPL Havells Finolex Gloster Polycab Grandlay RR Kable KEI RPG Asian Plaza Shalabh Anchor SS Cables
108.	TIMERS (TIMES SWITCHES)	L&T Hager BCH Havells Schneider Indo Asian GE ABB Legrand HPL
109.	ROTARY/ SELECTOR SWITCHES	C&S HPL ABB Legrand BCH Siemens Altos Kaycee GE L&T
110.	AUTOMATIC/ ELECTRONIC VOLTAGE STABILIZERS	AE ITE Gurgaon Aplab Blue Bird Sinetrac M/S Sai Electricals Microtech V-Guard Premier SAR Zenith Volina
111.	ELECTRICAL INSULATION RUBBER MAT	Jyoti Jaipuria Brothers Duratuf (ISI marked)
112.	PCC POLES/RCC POLES	Concrete Udyog Jhansi Mohan Concrete Udyog Lucknow Kusum Industries (MP) (MP SEB) SJ Enterprises (PVVNL approved)
113.	VOLTMETER/AMETER MF METER	Havells L&T

PARTICULAR SPECIFICATION (CONT'D…../-)

		AE BCH Schneider
114.	UPS (INDUSTRIAL TYPE)	Hitachi ABB Aplab Siemens Schneider
115.	HIGH MAST LIGHT SYSTEM	Bajaj Wipro Philips CG
116.	PVC insulated copper conductor multistranded cable/ wires for house wiring	Havells Polycab Grandlay Anchor KEI HPL SS Cables
117.	Conduit Steel/ERW Steel	Kalinga Jindal AKG Asian Bharat Steel tube Finolex MK India TATA Precision Steel Kraft
118.	Conduit Rigid PVC and Accessories	Jindal AKG Kalinga Premium Finolex Ploycab Asian Presto Plast Plaza National Harsh
119.	Laminated Sheet	Anchor Hylam National
120.	FR PVC Casing Capping	Prestoplast National Plaza AKG Paramount
121.	DOL/STAR Delta starter	Hevells ABB BCH C&S Indo Asian Crompton Schneider Seco Bentec (Brand : Benlo) Sinetrac Standard HPL Siemens L&T
122.	Heating Elements	Daspass Airex electrical

PARTICULAR SPECIFICATION (CONT'D…../-)

		Anupam
123.	AC Contractors 6-1200A	BCH ABB Siemens C&S L&T GE
124.	Smart energy meter	Siemens L&T Secure Schneider Elmeasure Bentec Brand : Benlo HPL
125.	UPS < 5KVA	V Guard Sukam Microtek Luminous Amaron
126.	Static Frequency Convertor	Toshiba Siemens ABB Mitsubishi
127.	CENTRIFUGAL PUMP SET	Beacon Kirloskar Jyoti Crompton Siemens Grundfos KSB BCH Maxflow Wasp Shenider L&T NGF ABB
128.	SUBMERSIBLE PUMP	Kirloskar KSB Calama Shakti V-Guard Jyoti Tullu Grandfos CRI
129.	VERTICAL TURBINE PUMPS	Jyoti KSB Orthington Kirloskar Johnshon Crompton
130.	HORIZONTAL PUMPA	Tullu CRI Shakti Kirloskar KSB Jyoti Grundfos Calama V-Guard
131.	SLUICE VALVE/ REFLUX/NRV/AIR RELEASE/FOOT VALVE	Kirloskar Leader

PARTICULAR SPECIFICATION (CONT'D…../-)

		BIR Kartar Venus Kejriwal Zoloto AUDCO VC Upadhyay L&T
132.	STOP VALVES/ GATE VALVES	Zoloto Leader Audoc Venus Kartar BIR
133.	UPVC/ HDPE CONDUIT	Finolex Supreme Industries Durlon Reliance Astral Triupati Prince Vectus Neelkamal
134.	AIR COMPRESSOR	BAC Ingersoll ELGC equipments ELGI Indo Air
135.	BATTERY	Exide Okaya Amaron Luminous AMCO
136.	FIRE PUMP/ JOCKEY PUMP	Kirloskar KSB Grundfos CRI
137.	WATER HEATER	Racold CG Havells Jaguar Usha
138.	WATER LEVEL INDICATOR (MECHANICAL)	RMCO Kabir Flowtech DB Instruments
139.	Air Diffuser/ Grill	Carrier MHP MAPRO ATE
140.	Galvanized Steel sheet for ducting	TATA SAIL Jindal
141.	Thermostat, Humidistat	Danfoss Sporion Indfoss Honeywell
142.	Air Purge/ Air vent valve	Anergy Rapid control Oventrop
143.	Pressure Gauge	H Guru Scientific Zenith

PARTICULAR SPECIFICATION (CONT'D…../-)

		Fiebig
144.	Temperature Gauge	H Guru Scientific Precision
145.	Motorised Butterfly Valve	Belimo Siemens Honeywell
146.	3 Way flow control valve	Belimo Honeywell, Anergy Rapid Control
147.	Flow switch	Danfoss Anergy Honeywell
148.	HP/LP control	Danfoss Indfoss Johnson
149.	Non return valve	Kirloskar Leader Zoloto Audco Upadhyay
150.	Pot/Y Strainer	Sant Rapid control Emerold, Leader
151.	Balancing Valve	Leader Sant Rapid control
152.	Butterfly/ Ball valve	Kirloskar Leader Zoloto, Audco Upadhyay, IVC
153.	Nitrite Rubber Insulation	BASF (India) Ltd Beardsell Armacell Lloyd
154.	Expanded Polystyrene Insulation	Styrene BASF (India) Ltd Armacell, Lloyd
155.	MS/ERW pipe	TATA Jindal SAIL
156.	Pump Monoblock/ Centrifugal/ Submersible	Kirloskar CG, Jyoti KSB Kalama
157.	Hot water Generator	Thermax Rapid Coll Zenith
158.	PVC Insulated cable 1100 V grade ISI marked FRLS/FRLSH internal wiring cables with copper conductor	Havells Polycab SS Cable Industries (Brand : SCI) Anchor

Signature of Contractor

AAD (Contracts)
For Accepting Officer

PARTICULAR SPECIFICATION (CONT'D…../-)

Appendix 'H'

GUARANTEE TO BE EXECUTED BY THE CONTRACTOR FOR REMOVAL OF DEFECTS AFTER COMPLETION IN RESPECT OF WATER PROOFING WORKS

This agreement made this _____day of_____ two thousand_____between of____ _____(son of _____of_____ (hereinafter called the guarantor on the one part) and the PRESIDENT OF INDIA (hereinafter called the Government on other part).

WHEREAS THIS agreement is supplementary to a contract hereinafter called the contract) dated _____ and made between the 'GUARANTOR' of the one part and the 'GOVERNMENT' on the other part where by the Contractor interalia, undertook render the buildings and structures in the said contract recited completely water and leak proof.

AND WHEREAS THE GUARANTOR agreed to give a guarantee to effect that the said structures will remain water leak proof for ten years from the certified date of completion.

NOW THE GUARANTOR hereby guarantees that water proofing treatment given by him will render the structures completely leak proof and the minimum life of such water proofing treatment shall be Ten years to be reckoned from the certified date of completion.

Provided that the guarantor will not be responsible for leakages caused by earthquakes or structural defects or misuse of roof and sunken floor slabs or alterations and for such purpose.

Misuse of and sunken floor slabs shall mean by operation which will damage roofing treatment, like chopping of firewood and things of the same nature which might cause damage to the roof.

Alteration shall mean construction of any additional storey or any part of roof or construction adjoining to existing roof, where by roofing treatment is removed in parts. And removal of finishes of sunken portion of slabs.

The decision of the Garrison Engineer with regard to cause of leakage shall be final and binding.
During this period of guarantee the guarantor shall make good all defects and in case of any defects being found, render the building water proof and finish the disturbed surfaces as existed to the satisfaction of Garrison Engineer at his cost and shall commence the work for such rectification within seven days from the date of issue of the notice from the Garrison Engineer calling upon him to rectify the defects, failing which the work shall be got done by the department by some other Contractor at the GUARANTOR'S risk and cost. The decision of the Garrison Engineer as to the cost, payable by the Guarantor shall be final and binding

That if the Guarantor fails to execute the water proofing or commit breach there under then the Guarantor will indemnify the Principal and his successors against all loss, damage, cost expense or otherwise which may be incurred by him by reason of any default on the part of GUARANTOR in performance and observing of this supplementary agreement. As to the amount of loss and/or damage and/or cost incurred by the Government, the decision of the Garrison Engineer will be final and binding on the parties.

IN WITNESS WHEREOF these presents have been executed by the obligator _____ and by _____and for and on behalf of the PRESIDENT OF INDIA on the day month and year first above written.
SIGNED sealed and delivered by OBLIGATOR in the presence of:
1.
2.
SIGNED ON BEHALF OF THE PRESIDENT OF INDIA BY _____in the presence of:
1.
2.